*[21] AML Software Version

(b) Table 202 defines the EEC software version per the Service Bulletin.

NOTE: If "77 EEC AVM Part Number" shows "NOT LOADED", keep EEC power on, exit the EEC System Configuration Page, and wait five minutes before accessing again. If "NOT LOADED" is still present, check for maintenance messages 77-4498X or 77-0900X. If either maintenance message is present, follow the fault isolation instructions in the IFIM. If the maintenance message is not present, replace the affected EEC at a convenient maintenance opportunity (refer to CFM LEAP-1B Service Bulletin 73–0034 for additional information).

Table 202/73-21-00-993-809-G00 EEC Software Version Reference Table

EEC Software Version	CFM Service Bulletin (Boeing Service Bulletin)	73 EEC OPS Part Number (73 EEC OPS Version) [CFM Part Number]	77 EEC AVM Part Number (77 EEC AVM Version) [CFM Part Number]	77 EEC EHM Part Number (77 EEC EHM Version) [CFM Part Number]	77 EEC AML Part Number (77 EEC AML Version) [CFM Part Number]
4.5	73-0002 (N/A)	GEC46-2628-8607 (0450) [2628M86P07]	GEC4D-2628-8802 (0305) [2628M88P02]	GEC47-2628-8707 (0450) [2628M87P07]	GEC42-2697-8302 (0200) [2697M83P02]
5.2	73-0011 (737-73-1017)	GEC48-2628-8609 (0520) [2628M86P09]		GEC49-2628-8709 (0520) [2628M87P09]	
6.1	73-0015 (737-73-1018)	GEC40-2628-8610 (0610) [2628M86P10]		GEC41-2628-8710 (0610) [2628M87P10]	
6.5	73-0016 (737-73-1019)	GEC41-2628-8611 (0650) [2628M86P11]		GEC40-2628-8711 (0650) [2628M87P11]	
6.52	73–0022 (737–73–1020)			GEC43-2628-8712 (0652) [2628M87P12]	
6.7	73-0025 Issue 1 (737-73-1021)	GEC42-2628-8612 (0670) [2628M86P12]		GEC42-2628-8713 (0670) [2628M87P13]	
6.72	73-0025 Issue 2 (737-73-1023)			GEC45-2628-8714 (0672) [2628M87P14]	
6.73	73-0029 (737-73-1024)			GEC44-2628-8715 (0673) [2628M87P15]	
7.1	73–0040 (737-73-1026)	GEC43-2628-8613 (0710) [2628M86P13]		GEC47-2628-8716 (0710) [2628M87P16]	

NOTE: EEC software version 6.52 and previous are not allowed.

(c) Table 203 gives the definition of the menu hardware mode.

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Table 203/73-21-00-993-810-G00 Menu Mode Hardware Codes

Menu Mode Hardware Code	EEC Channel B Accelerometer Input
00	Number One Bearing (NOB) Accelerometer Active (Default Selection)
01	Disabled (No Input Selected)
-	Invalid (Reserved for Future Use)

(d) Table 204 gives the definition of the auto-detected hardware configuration code.

Table 204/73-21-00-993-808-G00 Auto-detected Hardware Configuration Code

Auto-detected Hardware Configuration Code	MTC System Configuration	EEC Blower (NOTE)	Fan Overheat Sensors	PSS Heater Configured	RBS Configured
00000	Dual	Not Installed	Connected	Not Configured	Not Configured
00001	Single				
00002	Dual	Installed			
00003	Single				
00004	Dual	Not Installed	Not Connected		
00005	Single				
00006	Dual	Installed			
00007	Single				
00008	Dual	Not Installed	Connected	Configured	
00009	Single				
00010	Dual	Installed			
00011	Single				
00012	Dual	Not Installed	Not Connected		
00013	Single				
00014	Dual	Installed			
00015	Single				

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Table 204/73-21-00-993-808-G00 Auto-detected Hardware Configuration Code (Continued)

Auto-detected Hardware Configuration Code	MTC System Configuration	EEC Blower (NOTE)	Fan Overheat Sensors	PSS Heater Configured	RBS Configured
00016	Dual	Not Installed	Connected	Not Configured	Configured
00017	Single				
00018	Dual	Installed			
00019	Single				
00020	Dual	Not Installed	Not Connected		
00021	Single				
00022	Dual	Installed			
00023	Single				
00024	Dual	Not Installed	Connected	Configured	
00025	Single				
00026	Dual	Installed			
00027	Single				
00028	Dual	Not Installed	Not Connected		
00029	Single				
00030	Dual	Installed			
00031	Single				

SUBTASK 73-21-00-970-016-G00

(2) Make a record of the EEC System Configuration Page for your airline's records.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 73-21-00-860-075-G00

(1) If the maintenance laptop was used and is no longer necessary, stop the ONS maintenance browser and disconnect the laptop.

——— END OF TASK ———

TASK 73-21-00-470-802-G00

3. EEC Shop Faults Function Check

(Figure 201 and Figure 202)

A. General

(1) This task gives the instructions for the Electronic Engine Control (EEC) shop fault function check and correlate the Fault IDs in the report to maintenance messages on the Onboard Maintenance Function (OMF).

(2) The shop fault function displays the faults saved in the EEC.

NOTE: The flight leg (LEG) in the EEC shop faults report is calculated separately from the aircraft flight leg, and may not match with the corresponding aircraft flight leg on the OMF. Use the date (MM/DD) and time (HH:MM) of when the fault code was set to compare to the maintenance messages recorded by the OMF.

(3) There are two ways that you can verify the system configuration:

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(a) Use a Portable Maintenance Device (PMD) such as a laptop.

(b) Use the Portable Software Data Loader (PSDL).

(4) The EEC shop faults check can be used to determine the EEC fault(s) that set a flight deck indication, status message, or Scheduled Maintenance Task (SMT) message where there is no corresponding maintenance message present in the OMF memory.

(5) Faults that are economic in its impact can be corrected at operator's discretion.

B. References

Reference	Title
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Prepare for the Check

SUBTASK 73-21-00-860-015-G00

(1) Make sure that the airplane is in the ground mode (TASK 32-09-00-860-802).

SUBTASK 73-21-00-860-014-G00

(2) Clear the test inhibit conditions.

SUBTASK 73-21-00-860-018-G00

(3) Select the left or right Multi Function Display (MFD).

(a) Push the SYS button, on the P9 MFD control panel.

(b) Push the SEL button on SELECTOR knob, on the P9 MFD control panel.

SUBTASK 73-21-00-860-017-G00

(4) Set the applicable EEC MAINT POWER switch on the MISC SYSTEM CTRLS page on the MFD to the TEST position (TASK 73-21-00-800-801-G00).

NOTE: Master Caution will come on.

SUBTASK 73-21-00-860-019-G00

(5) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-21-00-860-020-G00

(6) Set the GND TESTS switch, on the P61 maintenance panel, to the ENABLE position.

SUBTASK 73-21-00-860-021-G00

(7) Set the applicable thrust lever to idle.

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E. EEC Shop Faults Function Check (with Maintenance Laptop)

SUBTASK 73-21-00-860-076-G00

(1) Connect the maintenance laptop to the Onboard Network System (ONS) port, on the P61 maintenance panel, in the flight deck (TASK 46-13-00-480-801).

SUBTASK 73-21-00-860-022-G00

(2) Make sure that the EEC Data Load and Trim Balance Functions are inactive.

SUBTASK 73-21-00-860-077-G00

(3) Use the OMF screen to do the EEC shop faults as follows:

(a) Make these selections on the OMF main function menu:

1) ONBOARD NETWORK SYSTEM
2) OTHER FUNCTIONS
3) SHOP FAULTS
4) 71-80 Left Engine or 71- 80 Right Engine
5) EEC - 1A or EEC - 1B
6) CONTINUE
7) REPORT.

a) On the REPORT popup screen, select the steps that follow:

<1> Portable Maintenance Device
<2> SEND
<3> STOP
<4> Wait until Complete is shown.

F. EEC Shop Faults Function Check (with Portable Maintenance Devices)

SUBTASK 73-21-00-480-001-G00

(1) Connect the PMD to the PSDL RJ45 port and to the ONS port, on the P61 aft overhead maintenance panel, in the flight deck.

SUBTASK 73-21-00-860-023-G00

(2) Use the screen to do the EEC shop faults as follows:

(a) Make these selections on the OMF main function menu:

1) ONBOARD MAINTENANCE
2) OTHER FUNCTIONS
3) SHOP FAULTS
4) 71-80 Left Engine or 71- 80 Right Engine
5) EEC - 1A or EEC - 1B
6) CONTINUE
7) REPORT.

a) On the REPORT popup screen, select the steps that follow:

<1> Portable Maintenance Device
<2> SEND
<3> STOP
<4> Wait until Complete is shown.

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G. EEC Shop Faults Maintenance Message Correlation

SUBTASK 73-21-00-970-008-G00

(1) Use the Fault ID in the EEC Shop Fault report to find the associated OMF maintenance message.

NOTE: Maintenance messages generated by the EECs are in a specific format based on the associated fault ATA chapter, Fault ID (EEC fault code), and which EEC channel(s) detected the fault.

NOTE: All economic faults have no time limits and can be corrected at operator's discretion.

(a) An example of maintenance message with format description is shown in Figure 201.

SUBTASK 73-21-00-970-009-G00

(2) Use Table 205 to find the message number, description, and dispatch group for each Fault ID.

NOTE: The EEC fault codes indicated with a "*" are fault codes related to EEC internal faults. When one or more of these faults are detected the ONS collects these faults into a single maintenance message 77-0900X - ENGINE-Y EEC CHANNEL-Z HAS AN INTERNAL FAULT where X is the EEC channel location (1 through 4), Y is the engine position (1 or 2), and Z is channel position (A or B).

Table 205/73-21-00-993-805-G00 EEC Shop Faults Correlation Table

EEC Shop Fault Report Fault ID (EEC Fault Code)	OMF Maintenance Message Number	OMF Maintenance Message Text	Dispatch Group
100	77-4100X	ACCELEROMETER SENSOR IS OUT OF RANGE	EEC Channel A (TCF): D EEC Channel B (NOB): H
101	77-4101X	ACCELEROMETER SENSOR IS OUT OF RANGE EXTERNAL TO THE EEC	EEC Channel A (TCF): D EEC Channel B (NOB): H
102	77-4102X	ACCELEROMETER SENSOR IS OUT OF RANGE INTERNAL TO THE EEC	EEC Channel A (TCF): D EEC Channel B (NOB): H
103	73-4103X	ADIRU-L DATA FROM DPC-1 IS INVALID	D
104	73-4104X	ADIRU-R DATA FROM DPC-2 IS INVALID	D
105	73-4105X	ALTERNATE MODE AND DPC-1 SIGNALS DO NOT AGREE	C
106	73-4106X	ALTERNATE MODE AND DPC-2 SIGNALS DO NOT AGREE	C
107*	77-4107X (77-0900X)	ALTERNATE MODE RELAY IS FAILED CLOSED INTERNAL TO THE EEC	D
108*	77-4108X (77-0900X)	ALTERNATE MODE LIGHT RELAY IS FAILED OPEN INTERNAL TO THE EEC	B
109*	73-4109X (77-0900X)	DATA LOADER CHECKSUM ERROR	H
110	73-4110X	ENGINE-X EEC CHANNEL-X HAS NO ARINC-429 INPUT FROM DPC-1	C
111	73-4111X	ENGINE-X EEC CHANNEL-X HAS NO ARINC-429 INPUT FROM DPC-2	C

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Table 205/73-21-00-993-805-G00 EEC Shop Faults Correlation Table (Continued)

EEC Shop Fault Report Fault ID (EEC Fault Code)	OMF Maintenance Message Number	OMF Maintenance Message Text	Dispatch Group
112	73-4112X	ENGINE-X EEC CHANNEL-X HAS NO ARINC-429 INPUT FROM DPC-1 OR DPC-2	G
113*	73-4113X (77-0900X)	ENGINE-X EEC CHANNEL-X HAS AN INTERNAL FAULT	F
114*	73-4114X (77-0900X)	ENGINE-X EEC CHANNEL-X HAS AN INTERNAL FAULT	F
115*	73-4115X (77-0900X)	DUAL NON-VOLATILE MEMORY CHECKSUM FAULT	C
116*	73-4116X (77-0900X)	SINGLE NON-VOLATILE MEMORY CHECKSUM FAULT	H
117	73-4117X	CROSS CHANNEL DATA LINK DISCRETE INPUTS AND BACKUP SERIAL CROSS CHANNEL DATA LINK DO NOT AGREE	B
118	73-4118X	CROSS CHANNEL DATA LINK SERIAL DATA IS INVALID	A, D
119	79-4119X	OIL DEBRIS MONITORING SYSTEM HAS DETECTED DEBRIS	D
120	73-4120X	DATA ENTRY PLUG PROGRAM PINS ARE INCORRECT	A
121	73-4121X	DATA ENTRY PLUG FUSE IS INCORRECT	A
122	73-4122X	DATA ENTRY PLUG N1 MODIFIER PARITY FAULT	C
123	75-4123X	DATA ENTRY PLUG CHANNEL-X AND CHANNEL-Y DISCRETES DO NOT AGREE	A
124	75-4124X	BLEED DATA FROM DPC-1 AND DPC-2 DO NOT AGREE	B
126	71-4126X	FLAP CONFIGURATION FROM DPC-1 AND DPC-2 DO NOT AGREE	C
127	74-4127X	LEFT AND RIGHT IGNITER SELECTION SIGNALS FROM DPC-1 ARE INVALID	C
128	71-4128X	LEFT AND RIGHT MAIN LANDING GEAR SIGNALS FROM DPC-1 DO NOT AGREE	C
129	77-4129X	NO LOW-RANGE RADIO ALTIMETER (LRRA-1) INPUT RECEIVED FROM DPC-1	H
130	77-4130X	NO LOW-RANGE RADIO ALTIMETER (LRRA-2) INPUT RECEIVED FROM DPC-1	H
132	71-4132X	FLAP DATA FROM DPC-1 OR DPC-2 IS INVALID	C
133	74-4133X	LEFT AND RIGHT IGNITER SELECTION SIGNALS FROM DPC-2 ARE INVALID	C

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Table 205/73-21-00-993-805-G00 EEC Shop Faults Correlation Table (Continued)

EEC Shop Fault Report Fault ID (EEC Fault Code)	OMF Maintenance Message Number	OMF Maintenance Message Text	Dispatch Group
134	71-4134X	LEFT AND RIGHT MAIN LANDING GEAR SIGNALS FROM DPC-2 DO NOT AGREE	C
135	77-4135X	NO LOW-RANGE RADIO ALTIMETER (LRRA-1) INPUT RECEIVED FROM DPC-2	H
136	77-4136X	NO LOW-RANGE RADIO ALTIMETER (LRRA-2) INPUT RECEIVED FROM DPC-2	H
137	73-4137X	DPC-1 HAS NO EEC CHANNEL A INPUT ON ARINC-429	C
138	73-4138X	DPC-2 HAS NO EEC CHANNEL A INPUT ON ARINC-429	C
139	73-4139X	ENGINE-X EEC CHANNEL-X HAS NO ARINC-429 INPUT ON RECEIVER 1	C
140	73-4140X	ENGINE-X EEC CHANNEL-X HAS NO ARINC-429 INPUT ON RECEIVER 2	C
141*	73-4141X (77-0900X)	ENGINE-X EEC CHANNEL-X HAS AN ARINC-429 OUTPUT FAULT	E
142	73-4142X	DPC-1 HAS NO EEC CHANNEL B INPUT ON ARINC-429	C
143	73-4143X	DPC-2 HAS NO EEC CHANNEL B INPUT ON ARINC-429	C
144	76-4144X	BLOWER INDICATES OFF WHEN COMMANDED ON	C
145	76-4145X	BLOWER INDICATES ON WHEN COMMANDED OFF	D
146*	73-4146X (77-0900X)	ENGINE-X EEC CHANNEL-X HAS AN INTERNAL FAILURE	F
149	73-4149X	EEC DISCRETES FROM CHANNEL-X OR CHANNEL-Y ARE INVALID	C
150	73-4150X	EEC OVERVOLTAGE TEST FAILED	E
151*	73-4151X (77-0900X)	ENGINE-X EEC CHANNEL-X HAS AN INTERNAL FAILURE	F
152	77-4152X	EEC IS OVERHEATED	F
153	73-4153X	ONBOARD MAINTENANCE DATA CHANNEL-X AND CHANNEL-Y DISCRETES DO NOT AGREE	B
154	73-4154X	PRESSURE SUBSYSTEM (PSS) IS NOT POWERED	E
155*	73-4155X (77-0900X)	PRESSURE SUBSYSTEM POWER SUPPLY FAULT INTERNAL TO THE EEC	H
156	73-4156X	AIRCRAFT POWER VOLTAGE IS LOW	C

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Table 205/73-21-00-993-805-G00 EEC Shop Faults Correlation Table (Continued)

EEC Shop Fault Report Fault ID (EEC Fault Code)	OMF Maintenance Message Number	OMF Maintenance Message Text	Dispatch Group
157	73-4157X	EEC CHANNEL SYNCHRONIZATION IS INVALID	A, D
158*	77-4158X (77-0900X)	VIBRATION MONITORING FAULT	H
159	77-4159X	EXHAUST GAS TEMPERATURE (EGT) PROBE 1 VALUE DOES NOT AGREE WITH THE AVERAGE OF ALL OTHER VALID EGT PROBE VALUES	H
160	77-4160X	EXHAUST GAS TEMPERATURE (EGT) PROBE 1 SIGNAL IS OUT OF RANGE	H
161	77-4161X	EXHAUST GAS TEMPERATURE (EGT) PROBE 2 VALUE DOES NOT AGREE WITH THE AVERAGE OF ALL OTHER VALID EGT PROBE VALUES	H
162	77-4162X	EXHAUST GAS TEMPERATURE (EGT) PROBE 2 SIGNAL IS OUT OF RANGE	H
163	77-4163X	EXHAUST GAS TEMPERATURE (EGT) PROBE 3 VALUE DOES NOT AGREE WITH THE AVERAGE OF ALL OTHER VALID EGT PROBE VALUES	H
164	77-4164X	EXHAUST GAS TEMPERATURE (EGT) PROBE 3 SIGNAL IS OUT OF RANGE	H
165	77-4165X	EXHAUST GAS TEMPERATURE (EGT) PROBE 4 VALUE DOES NOT AGREE WITH THE AVERAGE OF ALL OTHER VALID EGT PROBE VALUES	H
166	77-4166X	EXHAUST GAS TEMPERATURE (EGT) PROBE 4 SIGNAL IS OUT OF RANGE	H
167	77-4167X	EXHAUST GAS TEMPERATURE (EGT) PROBE 5 VALUE DOES NOT AGREE WITH THE AVERAGE OF ALL OTHER VALID EGT PROBE VALUES	H
168	77-4168X	EXHAUST GAS TEMPERATURE (EGT) PROBE 5 SIGNAL IS OUT OF RANGE	H
169	77-4169X	EXHAUST GAS TEMPERATURE (EGT) PROBE 6 VALUE DOES NOT AGREE WITH THE AVERAGE OF ALL OTHER VALID EGT PROBE VALUES	H
170	77-4170X	EXHAUST GAS TEMPERATURE (EGT) PROBE 6 SIGNAL IS OUT OF RANGE	H
171	77-4171X	EXHAUST GAS TEMPERATURE (EGT) PROBE 7 VALUE DOES NOT AGREE WITH THE AVERAGE OF ALL OTHER VALID EGT PROBE VALUES	H
172	77-4172X	EXHAUST GAS TEMPERATURE (EGT) PROBE 7 SIGNAL IS OUT OF RANGE	H
173	77-4173X	EXHAUST GAS TEMPERATURE (EGT) PROBE 8 VALUE DOES NOT AGREE WITH THE AVERAGE OF ALL OTHER VALID EGT PROBE VALUES	H

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Table 205/73-21-00-993-805-G00 EEC Shop Faults Correlation Table (Continued)

EEC Shop Fault Report Fault ID (EEC Fault Code)	OMF Maintenance Message Number	OMF Maintenance Message Text	Dispatch Group
174	77-4174X	EXHAUST GAS TEMPERATURE (EGT) PROBE 8 SIGNAL IS OUT OF RANGE	H
175	77-4175X	ALL EXHAUST GAS TEMPERATURE (EGT) PROBES HAVE FAILED	A, D
176	71-4176X	EXHAUST GAS TEMPERATURE (EGT) AMBER LIMIT EXCEEDED	D
177	77-4177X	6 OR 7 TOTAL EXHAUST GAS TEMPERATURE (EGT) PROBES HAVE FAILED	B
178	71-4178X	EXHAUST GAS TEMPERATURE (EGT) REDLINE LIMIT EXCEEDED	D
179	71-4179X	EXHAUST GAS TEMPERATURE (EGT) START REDLINE LIMIT EXCEEDED	D
180	79-4180X	ENGINE OIL PRESSURE (PEO) IS BELOW THE AMBER LIMIT	D
181	79-4181X	ENGINE OIL PRESSURE (PEO) IS BELOW THE REDLINE LIMIT	D
182	79-4182X	ENGINE OIL PRESSURE (PEO) SENSOR CHANNEL-X AND CHANNEL-Y SIGNALS DO NOT AGREE	C
183	79-4183X	ENGINE OIL PRESSURE (PEO)SENSOR IS OUT OF RANGE	G
184	79-4184X	ENGINE OIL PRESSURE (PEO) SENSOR IS OUT OF RANGE EXTERNAL TO THE EEC	G
185*	79-4185X (77-0900X)	ENGINE OIL PRESSURE (PEO) SENSOR IS OUT OF RANGE INTERNAL TO THE EEC	G
186	73-4186X	ENGINE POSITION DISCRETE IS INVALID	C
187	71-4187X	N2 SPEED IS APPROACHING REDLINE LIMIT	A, D
188	79-4188X	ENGINE OIL QUANTITY IS HIGH	B
194	73-4194X	ELECTRONIC OVERSPEED TEST FAILED	F
195	73-4195X	ENGINE CORE (N2) OVERSPEED EVENT	D
196	73-4196X	FUEL COOLED OIL COOLER (FCOC) BYPASS	D
198	73-4198X	FUEL COOLED OIL COOLER (FCOC) DELTA PRESSURE SENSOR CHANNEL-X AND CHANNEL-Y SIGNALS DO NOT AGREE	C
199	73-4199X	FUEL COOLED OIL COOLER (FCOC) DELTA PRESSURE SENSOR IS OUT OF RANGE	C
200	73-4200X	FUEL COOLED OIL COOLER (FCOC) DELTA PRESSURE SENSOR IS OUT OF RANGE EXTERNAL TO THE EEC	C

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Table 205/73-21-00-993-805-G00 EEC Shop Faults Correlation Table (Continued)

EEC Shop Fault Report Fault ID (EEC Fault Code)	OMF Maintenance Message Number	OMF Maintenance Message Text	Dispatch Group
201*	73-4201X (77-0900X)	FUEL COOLED OIL COOLER (FCOC) DELTA PRESSURE SENSOR IS OUT OF RANGE INTERNAL TO THE EEC	C
202	77-4202X	AIR/GROUND SYSTEM 1 AND SYSTEM 2 DO NOT AGREE	C
203	77-4203X	AIR/GROUND SYSTEM DATA FROM DPC-1 IS INVALID	C
204	77-4204X	AIR/GROUND SYSTEM DATA FROM DPC-2 IS INVALID	C
205	77-4205X	AIR/GROUND SYSTEM 1 DATA IS INVALID	C
208	77-4208X	AIR/GROUND SYSTEM 2 DATA IS INVALID	C
209	73-4209X	FUEL METERING VALVE (FMV) TORQUE MOTOR CURRENT IS OUT OF RANGE	F
210	73-4210X	FUEL METERING VALVE (FMV) TORQUE MOTOR CURRENT IS OUT OF RANGE EXTERNAL TO THE EEC	F
211	73-4211X	FUEL METERING VALVE (FMV) TORQUE MOTOR CURRENT IS OUT OF RANGE INTERNAL TO THE EEC	F
212	73-4212X	FUEL METERING VALVE (FMV) IS NOT IN COMMANDED POSITION	A
213	73-4213X	FUEL METERING VALVE (FMV) POSITION SENSOR CHANNEL-X AND CHANNEL-Y DO NOT AGREE	C
214	73-4214X	FUEL METERING VALVE (FMV) POSITION SENSOR IS OUT OF RANGE	G
215	73-4215X	FUEL METERING VALVE (FMV) POSITION SENSOR IS OUT OF RANGE EXTERNAL TO THE EEC	G
216*	73-4216X (77-0900X)	FUEL METERING VALVE (FMV) POSITION SENSOR IS OUT OF RANGE INTERNAL TO THE EEC	G
217	73-4217X	FUEL FILTER BYPASS	D
218	73-4218X	FUEL FILTER APPROACHING BYPASS	H
219	73-4219X	FUEL FILTER IMPENDING BYPASS	D
220	73-4220X	FUEL FILTER DELTA PRESSURE (FFDP) SENSOR CHANNEL-X AND CHANNEL-Y SIGNALS DO NOT AGREE	C
221	73-4221X	FUEL FILTER DELTA PRESSURE (FFDP) SENSOR IS OUT OF RANGE	C

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Table 205/73-21-00-993-805-G00 EEC Shop Faults Correlation Table (Continued)

EEC Shop Fault Report Fault ID (EEC Fault Code)	OMF Maintenance Message Number	OMF Maintenance Message Text	Dispatch Group
222	73-4222X	FUEL FILTER DELTA PRESSURE (FFDP) SENSOR IS OUT OF RANGE EXTERNAL TO THE EEC	C
223*	73-4223X (77-0900X)	FUEL FILTER DELTA PRESSURE (FFDP) SENSOR IS OUT OF RANGE INTERNAL TO THE EEC	C
224	73-4224X	FUEL MANIFOLD PRESSURE SENSOR CHANNEL-X AND CHANNEL-Y SIGNALS DO NOT AGREE	C
225	73-4225X	FUEL MANIFOLD PRESSURE SENSOR IS OUT OF RANGE	G
226	73-4226X	FUEL MANIFOLD PRESSURE SENSOR IS OUT OF RANGE EXTERNAL TO THE EEC	G
227*	73-4227X (77-0900X)	FUEL MANIFOLD PRESSURE SENSOR IS OUT OF RANGE INTERNAL TO THE EEC	G
228	73-4228X	FUEL STRAINER BYPASS	D
229	73-4229X	FUEL STRAINER IMPENDING BYPASS	D
230	73-4230X	FUEL STRAINER DELTA PRESSURE (FSDP) SENSOR CHANNEL-X AND CHANNEL-Y SIGNALS DO NOT AGREE	C
231	73-4231X	FUEL STRAINER DELTA PRESSURE (FSDP) SENSOR IS OUT OF RANGE	C
232	73-4232X	FUEL STRAINER DELTA PRESSURE (FSDP) SENSOR IS OUT OF RANGE EXTERNAL TO THE EEC	C
233*	73-4233X (77-0900X)	FUEL STRAINER DELTA PRESSURE (FSDP) SENSOR IS OUT OF RANGE INTERNAL TO THE EEC	C
234	73-4234X	ADIRU-L AND ADIRU-R GROUND SPEED SIGNALS DO NOT AGREE	H
235	73-4235X	FUEL METERING UNIT SHUTOFF VALVE IS FAILED CLOSED	A, D
236	73-4236X	FUEL METERING UNIT SHUTOFF VALVE IS FAILED OPEN	A, D
237	75-4237X	HIGH PRESSURE TURBINE ACTIVE CLEARANCE CONTROL (HPTACC) TORQUE MOTOR CURRENT IS OUT OF RANGE	C
238	75-4238X	HIGH PRESSURE TURBINE ACTIVE CLEARANCE CONTROL (HPTACC) TORQUE MOTOR CURRENT IS OUT OF RANGE EXTERNAL TO THE EEC	C

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Table 205/73-21-00-993-805-G00 EEC Shop Faults Correlation Table (Continued)

EEC Shop Fault Report Fault ID (EEC Fault Code)	OMF Maintenance Message Number	OMF Maintenance Message Text	Dispatch Group
239	75-4239X	HIGH PRESSURE TURBINE ACTIVE CLEARANCE CONTROL (HPTACC) TORQUE MOTOR CURRENT IS OUT OF RANGE INTERNAL TO THE EEC	C
240	75-4240X	HIGH PRESSURE TURBINE ACTIVE CLEARANCE CONTROL (HPTACC) VALVE IS FAILED CLOSED	D
241	75-4241X	HIGH PRESSURE TURBINE ACTIVE CLEARANCE CONTROL (HPTACC) VALVE IS FAILED OPEN	A, D
242	75-4242X	HIGH PRESSURE TURBINE ACTIVE CLEARANCE CONTROL (HPTACC) VALVE CHANNEL-X AND CHANNEL-Y POSITION DO NOT AGREE	C
243	75-4243X	HIGH PRESSURE TURBINE ACTIVE CLEARANCE CONTROL VALVE (HPTACC) POSITION SENSOR IS OUT OF RANGE	C
244	75-4244X	HIGH PRESSURE TURBINE ACTIVE CLEARANCE CONTROL VALVE (HPTACC) POSITION SENSOR IS OUT OF RANGE EXTERNAL TO THE EEC	C
245*	75-4245X (77-0900X)	HIGH PRESSURE TURBINE ACTIVE CLEARANCE CONTROL VALVE (HPTACC) POSITION SENSOR IS OUT OF RANGE INTERNAL TO THE EEC	C
246	77-4246X	ALTERNATE MODE CHANNEL-X AND CHANNEL-Y SIGNALS DO NOT AGREE	B
247	74-4247X	LEFT IGNITER RELAY IS FAILED CLOSED INTERNAL TO THE EEC	B
248	74-4248X	LEFT IGNITER RELAY IS FAILED OPEN INTERNAL TO THE EEC	C
249	74-4249X	LEFT IGNITION SYSTEM IS FAILED	D
250	74-4250X	LEFT IGNITION SYSTEM IS POWERED WITH THE ENGINE START LEVER IN CUTOFF	C
251	74-4251X	LEFT IGNITION SYSTEM IS NOT POWERED WITH ENGINE START LEVER AT IDLE	D
252	74-4252X	RIGHT IGNITER RELAY IS FAILED CLOSED INTERNAL TO THE EEC	B
253	74-4253X	RIGHT IGNITER RELAY IS FAILED OPEN INTERNAL TO THE EEC	C
254	74-4254X	RIGHT IGNITION SYSTEM IS FAILED	D
255	74-4255X	RIGHT IGNITION SYSTEM IS POWERED WITH THE ENGINE START LEVER IN CUTOFF	C
256	74-4256X	RIGHT IGNITION SYSTEM IS NOT POWERED WITH ENGINE START LEVER AT IDLE	D

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EEC Shop Fault Report Fault ID (EEC Fault Code)	OMF Maintenance Message Number	OMF Maintenance Message Text	Dispatch Group
257	74-4257X	IGNITER POWER MONITOR IS FAILED	G
258	80-4258X	ENGINE START FAILED DUE TO LOW FAN SPEED	D
259	77-4259X	STARTER AIR PRESSURE IS LOW	H
260	75-4260X	LOW PRESSURE TURBINE ACTIVE CLEARANCE CONTROL (LPTACC) TORQUE MOTOR CURRENT IS OUT OF RANGE	C
261	75-4261X	LOW PRESSURE TURBINE ACTIVE CLEARANCE CONTROL (LPTACC) TORQUE MOTOR CURRENT IS OUT OF RANGE EXTERNAL TO THE EEC	C
262	75-4262X	LOW PRESSURE TURBINE ACTIVE CLEARANCE CONTROL (LPTACC) TORQUE MOTOR CURRENT IS OUT OF RANGE INTERNAL TO THE EEC	C
263	75-4263X	LOW PRESSURE TURBINE ACTIVE CLEARANCE CONTROL (LPTACC) VALVE IS FAILED CLOSED	D
264	75-4264X	LOW PRESSURE TURBINE ACTIVE CLEARANCE CONTROL (LPTACC) VALVE IS FAILED OPEN	A, D
265	75-4265X	LOW PRESSURE TURBINE ACTIVE CLEARANCE CONTROL (LPTACC) VALVE CHANNEL-X AND CHANNEL-Y POSITION DO NOT AGREE	C
266	75-4266X	LOW PRESSURE TURBINE ACTIVE CLEARANCE CONTROL (LPTACC) VALVE POSITION SENSOR IS OUT OF RANGE	C
267	75-4267X	LOW PRESSURE TURBINE ACTIVE CLEARANCE CONTROL (LPTACC) VALVE POSITION SENSOR IS OUT OF RANGE EXTERNAL TO THE EEC	C
268*	75-4268X (77-0900X)	LOW PRESSURE TURBINE ACTIVE CLEARANCE CONTROL (LPTACC) VALVE POSITION SENSOR IS OUT OF RANGE INTERNAL TO THE EEC	C
269	77-4269X	LOW RANGE RADIO ALTIMETER (LRRA) SIGNALS DO NOT AGREE	H
270	73-4270X	ENGINE-X EEC CHANNEL-Y AIRCRAFT POWER IS NOT AVAILABLE	A, D
271	73-4271X	ENGINE-X EEC CHANNEL-Y IS NOT AVAILABLE BY: ENGINE-X EEC CHANNEL-X	A, D
272	73-4272X	MODULATED TURBINE COOLING (MTC) VALVE 1 IS FAILED CLOSED	B
273	73-4273X	MODULATED TURBINE COOLING (MTC) VALVE 1 IS FAILED OPEN	C
274	73-4274X	MODULATED TURBINE COOLING (MTC) VALVE 1 POSITION SENSOR IS OUT OF RANGE	B

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EEC Shop Fault Report Fault ID (EEC Fault Code)	OMF Maintenance Message Number	OMF Maintenance Message Text	Dispatch Group
275	73-4275X	MODULATED TURBINE COOLING (MTC) VALVE 1 POSITION SENSOR IS OUT OF RANGE EXTERNAL TO THE EEC	B
276	73-4276X	MODULATED TURBINE COOLING (MTC) VALVE 1 POSITION SENSOR IS OUT OF RANGE INTERNAL TO THE EEC	B
277	73-4277X	MODULATED TURBINE COOLING (MTC) VALVE 2 IS FAILED CLOSED	B
278	73-4278X	MODULATED TURBINE COOLING (MTC) VALVE 2 IS FAILED OPEN	C
279	73-4279X	MODULATED TURBINE COOLING (MTC) VALVE 2 POSITION SENSOR IS OUT OF RANGE	B
280	73-4280X	MODULATED TURBINE COOLING (MTC) VALVE 2 POSITION SENSOR IS OUT OF RANGE EXTERNAL TO THE EEC	B
281	73-4281X	MODULATED TURBINE COOLING (MTC) VALVE 2 POSITION SENSOR POSITION SENSOR IS OUT OF RANGE INTERNAL TO THE EEC	B
282	73-4282X	MODULATED TURBINE COOLING (MTC) TORQUE MOTOR CURRENT IS OUT OF RANGE	B
283	73-4283X	MODULATED TURBINE COOLING (MTC) TORQUE MOTOR CURRENT IS OUT OF RANGE EXTERNAL TO THE EEC	B
284	73-4284X	MODULATED TURBINE COOLING (MTC) TORQUE MOTOR CURRENT IS OUT OF RANGE INTERNAL TO THE EEC	B
285	79-4285X	OIL DEBRIS MONITORING SYSTEM IS FAILED	D
286	79-4286X	OIL DEBRIS MONITORING SYSTEM HAS DETECTED DEBRIS IN A SINGLE FLIGHT	D
287	79-4287X	OIL FILTER BYPASS	D
288	79-4288X	OIL FILTER APPROACHING BYPASS	H
289	79-4289X	OIL FILTER IMPENDING BYPASS	D
290	79-4290X	OIL FILTER DELTA PRESSURE (OFDP) SENSOR CHANNEL-X AND CHANNEL-Y SIGNALS DO NOT AGREE	C
291	79-4291X	OIL FILTER DELTA PRESSURE (OFDP) SENSOR IS OUT OF RANGE	C
292	79-4292X	OIL FILTER DELTA PRESSURE (OFDP) SENSOR IS OUT OF RANGE EXTERNAL TO THE EEC	C

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EEC Shop Fault Report Fault ID (EEC Fault Code)	OMF Maintenance Message Number	OMF Maintenance Message Text	Dispatch Group
293*	79-4293X (77-0900X)	OIL FILTER DELTA PRESSURE (OFDP) SENSOR IS OUT OF RANGE INTERNAL TO THE EEC	C
294	73-4294X	ENGINE (P0) AND ADIRU AMBIENT PRESSURE SIGNALS DO NOT AGREE	B
296	73-4296X	AMBIENT PRESSURE (P0) SENSOR CHANNEL-X AND CHANNEL-Y SIGNALS DO NOT AGREE	C
297	73-4297X	AMBIENT PRESSURE (P0) SENSOR IS OUT OF RANGE	E
298	73-4298X	BLEED PRESSURE (P3B) SENSOR CHANNEL-X AND CHANNEL-Y SIGNALS DO NOT AGREE	C
299	73-4299X	BLEED PRESSURE (P3B) SENSOR IS OUT OF RANGE	E
300	73-4300X	BLEED PRESSURE (P3B) SENSOR AND COMPRESSOR DISCHARGE PRESSURE (PS3) SENSOR SIGNALS DO NOT AGREE	B
301	77-4301X	FAN SPEED (N1) SENSOR CHANNEL-X AND CHANNEL-Y SIGNALS DO NOT AGREE	B
302	77-4302X	FAN SPEED (N1) SENSOR IS OUT OF RANGE	A
303	77-4303X	FAN SPEED (N1) SENSOR IS OUT OF RANGE EXTERNAL TO THE EEC	A
304	77-4304X	FAN SPEED (N1) SENSOR IS OUT OF RANGE INTERNAL TO THE EEC	A
305	77-4305X	FAN SPEED (N1) REDLINE LIMIT EXCEEDED	D
306	77-4306X	CORE SPEED (N2) SENSOR CHANNEL-X AND CHANNEL-Y SIGNALS DO NOT AGREE	C
307	77-4307X	CORE SPEED (N2) SENSOR IS OUT OF RANGE	G
308	77-4308X	CORE SPEED (N2) SENSOR IS OUT OF RANGE EXTERNAL TO THE EEC	G
309	77-4309X	CORE SPEED (N2) SENSOR IS OUT OF RANGE INTERNAL TO THE EEC	G
310	77-4310X	N2 SPEED REDLINE LIMIT EXCEEDED	D
311	80-4311X	BOWED ROTOR OVERSPEED	H
312	73-4312X	ADIRU-L PROBE HEAT STATUS SIGNAL FROM DPC-1 IS INVALID	C
313	73-4313X	ADIRU-R PROBE HEAT STATUS SIGNAL FROM DPC-2 IS INVALID	C
314	73-4314X	PERMANENT MAGNET ALTERNATOR (PMA) CIRCUIT IS OPEN OR SHORTED	F

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EEC Shop Fault Report Fault ID (EEC Fault Code)	OMF Maintenance Message Number	OMF Maintenance Message Text	Dispatch Group
315	73-4315X	PERMANENT MAGNET ALTERNATOR (PMA) N2 SPEED SENSOR CHANNEL-X AND CHANNEL-Y SIGNALS DO NOT AGREE	C
316	73-4316X	PERMANENT MAGNET ALTERNATOR (PMA) N2 SPEED SIGNAL IS OUT OF RANGE	G
317	73-4317X	PERMANENT MAGNET ALTERNATOR (PMA) N2 SPEED SIGNAL IS OUT OF RANGE EXTERNAL TO THE EEC	G
318	73-4318X	PERMANENT MAGNET ALTERNATOR (PMA) N2 SPEED SIGNAL IS OUT OF RANGE INTERNAL TO THE EEC	G
319	73-4319X	PILOT METERING VALVE (PMV) TORQUE MOTOR CURRENT IS OUT OF RANGE	F
320	73-4320X	PILOT METERING VALVE (PMV) TORQUE MOTOR CURRENT IS OUT OF RANGE EXTERNAL TO THE EEC	F
321	73-4321X	PILOT METERING VALVE (PMV) TORQUE MOTOR CURRENT IS OUT OF RANGE INTERNAL TO THE EEC	F
322	73-4322X	PILOT METERING VALVE (PMV) IS FAILED CLOSED	A, D
323	73-4323X	PILOT METERING VALVE (PMV) IS FAILED OPEN	A, D
324	73-4324X	PILOT METERING VALVE (PMV) POSITION SENSOR CHANNEL-X AND CHANNEL-Y DO NOT AGREE	C
325	73-4325X	PILOT METERING VALVE (PMV) POSITION SENSOR IS OUT OF RANGE	G
326	73-4326X	PILOT METERING VALVE (PMV) POSITION SENSOR IS OUT OF RANGE EXTERNAL TO THE EEC	G
327*	73-4327X (77-0900X)	PILOT METERING VALVE (PMV) POSITION SENSOR IS OUT OF RANGE INTERNAL TO THE EEC	G
328	73-4328X	PITOT PROBE HEAT INDICATES OFF	D
329	73-4329X	COMPRESSOR DISCHARGE PRESSURE (PS3) SENSOR CHANNEL-X AND CHANNEL-Y SIGNALS DO NOT AGREE	B
330	73-4330X	COMPRESSOR DISCHARGE PRESSURE (PS3) SENSOR IS OUT OF RANGE	F
331	73-4331X	POWER SUPPLY REGULATOR HAS A FAULT INTERNAL TO THE EEC	B

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EEC Shop Fault Report Fault ID (EEC Fault Code)	OMF Maintenance Message Number	OMF Maintenance Message Text	Dispatch Group
332	75-4332X	PRESSURE SUBSYSTEM (PSS) POWER IS NOT AVAILABLE	E
333	75-4333X	PRESSURE SUBSYSTEM CROSS CHANNEL DATA LINK IS INVALID	H
334	75-4334X	ENGINE-X EEC CHANNEL-X HAS NO INPUT FROM PRESSURE SUBSYSTEM (PSS)	E
335	75-4335X	PRESSURE SUBSYSTEM (PSS) CHANNEL INDICATION IS INVALID	H
336	75-4336X	PRESSURE SUBSYSTEM (PSS) IS OVERHEATED	E
337	75-4337X	PRESSURE SUBSYSTEM (PSS) POWER MONITOR SIGNAL IS INVALID	H
339	75-4339X	PRESSURE SUB-SYSTEM (PSS) INTERNAL TEMPERATURE SENSOR DATA IS INVALID	C
340	73-4340X	ADIRU-L TOTAL PRESSURE SIGNAL FROM DPC-1 IS INVALID	D
341	73-4341X	ADIRU-R TOTAL PRESSURE SIGNAL FROM DPC-2 IS INVALID	D
342	73-4342X	ADIRU-L AND ADIRU-R TOTAL PRESSURE SIGNALS DO NOT AGREE	D
343	73-4343X	EEC RESET FAULT	G
344	80-4344X	STARTER AIR PRESSURE (SAP)SENSOR CHANNEL-X AND CHANNEL-Y DO NOT AGREE	H
345	80-4345X	STARTER AIR PRESSURE (SAP)SENSOR IS OUT OF RANGE	C
346	80-4346X	STARTER AIR PRESSURE (SAP)SENSOR IS OUT OF RANGE EXTERNAL TO THE EEC	C
347*	80-4347X (77-0900X)	STARTER AIR PRESSURE (SAP)SENSOR IS OUT OF RANGE INTERNAL TO THE EEC	C
348	80-4348X	STARTER AIR VALVE (SAV) SOLENOID HAS AN ELECTRICAL FAULT	C
349	80-4349X	STARTER AIR VALVE (SAV) SOLENOID HAS AN ELECTRICAL FAULT EXTERNAL TO THE EEC	C
350	80-4350X	STARTER AIR VALVE (SAV) SOLENOID HAS AN ELECTRICAL FAULT INTERNAL TO THE EEC	C
351	80-4351X	STARTER AIR VALVE (SAV) IS FAILED OPEN	D
352	75-4352X	START BLEED VALVE BOOSTER ANTI-ICE (SBV/BAI) TORQUE MOTOR CURRENT IS OUT OF RANGE	E

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Table 205/73-21-00-993-805-G00 EEC Shop Faults Correlation Table (Continued)

EEC Shop Fault Report Fault ID (EEC Fault Code)	OMF Maintenance Message Number	OMF Maintenance Message Text	Dispatch Group
353	75-4353X	START BLEED VALVE BOOSTER ANTI-ICE (SBV/BAI)TORQUE MOTOR CURRENT IS OUT OF RANGE EXTERNAL TO THE EEC	E
354	75-4354X	START BLEED VALVE BOOSTER ANTI-ICE (SBV/BAI)TORQUE MOTOR CURRENT IS OUT OF RANGE INTERNAL TO THE EEC	E
355	75-4355X	START BLEED VALVE BOOSTER ANTI-ICE (SBV/BAI) VALVE IS FAILED CLOSED	A, D
356	75-4356X	START BLEED VALVE BOOSTER ANTI-ICE (SBV/BAI) VALVE IS FAILED OPEN	A, D
357	75-4357X	START BLEED VALVE BOOSTER ANTI-ICE (SBV/BAI) POSITION SENSOR CHANNEL-X AND CHANNEL-Y DO NOT AGREE	C
358	75-4358X	START BLEED VALVE BOOSTER ANTI-ICE (SBV/BAI) POSITION SENSOR IS OUT OF RANGE	G
359	75-4359X	START BLEED VALVE BOOSTER ANTI-ICE (SBV/BAI) POSITION SENSOR IS OUT OF RANGE EXTERNAL TO THE EEC	G
360*	75-4360X (77-0900X)	START BLEED VALVE BOOSTER ANTI-ICE (SBV/BAI) POSITION SENSOR IS OUT OF RANGE INTERNAL TO THE EEC	G
361	73-4361X	ENGINE START LEVER AND DPC-1 SIGNALS DO NOT AGREE	C
362	73-4362X	ENGINE START LEVER AND DPC-2 SIGNALS DO NOT AGREE	C
364	73-4364X	CHANNEL-X AND CHANNEL-Y ANALOG ENGINE START LEVER SIGNALS DO NOT AGREE	C
365	80-4365X	STARTER AIR PRESSURE (SAP) IS NOT AVAILABLE	H
366	71-4366X	HIGH EXHAUST GAS TEMPERATURE (EGT) DURING START	D
367	71-4367X	STARTER DUTY CYCLE IS EXCEEDED	H
368	80-4368X	START FAILED DUE TO STARTER FAILURE	H
369	71-4369X	ENGINE STALLED DURING START	D
370	73-4370X	N2 SPEED IS BELOW IDLE WITH FUEL SWITCH ON	D
371	73-4371X	EEC HAS INCORRECT SOFTWARE CONFIGURATION	A
372	73-4372X	EEC CHANNEL-X AND CHANNEL-Y SOFTWARE IS INCOMPATIBLE	A

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EEC Shop Fault Report Fault ID (EEC Fault Code)	OMF Maintenance Message Number	OMF Maintenance Message Text	Dispatch Group
373	73-4373X	FAN INLET TEMPERATURE (T12) SENSOR CHANNEL-X AND CHANNEL-Y SIGNALS DO NOT AGREE	C
374	73-4374X	FAN INLET TEMPERATURE (T12) SENSOR IS OUT OF RANGE	G
375	73-4375X	FAN INLET TEMPERATURE (T12) SENSOR IS OUT OF RANGE EXTERNAL TO THE EEC	G
376*	73-4376X (77-0900X)	FAN INLET TEMPERATURE (T12) SENSOR IS OUT OF RANGE INTERNAL TO THE EEC	G
377	73-4377X	COMPRESSOR INLET TEMPERATURE (T25) SENSOR CHANNEL-X AND CHANNEL-Y SIGNALS DO NOT AGREE	B
378	73-4378X	COMPRESSOR INLET TEMPERATURE (T25) SENSOR IS OUT OF RANGE	E
379	73-4379X	COMPRESSOR INLET TEMPERATURE (T25) SENSOR IS OUT OF RANGE EXTERNAL TO THE EEC	E
380*	73-4380X (77-0900X)	COMPRESSOR INLET TEMPERATURE (T25) SENSOR IS OUT OF RANGE INTERNAL TO THE EEC	E
381	73-4381X	COMPRESSOR DISCHARGE TEMPERATURE (T3) SENSOR CHANNEL-X AND CHANNEL-Y SIGNALS DO NOT AGREE	B
382	73-4382X	COMPRESSOR DISCHARGE TEMPERATURE (T3) SENSOR IS OUT OF RANGE	E
383	73-4383X	ADIRU-L AND ADIRU-R TOTAL AIR TEMPERATURE (TAT) SIGNALS DO NOT AGREE	C
384	73-4384X	ADIRU-L AND ADIRU-R TOTAL AIR TEMPERATURE (TAT) SIGNALS ARE INVALID	C
385	73-4385X	ENGINE FAN INLET (T12) TEMPERATURE SENSOR AND CALCULATED ADIRU AIR TEMPERATURE SIGNALS DO NOT AGREE	H
386	73-4386X	CALCULATED ENGINE FAN INLET (T12) TEMPERATURE SENSOR AND CALCULATED ADIRU AIR TEMPERATURE SIGNALS DO NOT AGREE	B
387	73-4387X	ENGINE FAN INLET (T12) TEMPERATURE SENSOR AND ADIRU AIR TEMPERATURE SIGNALS DO NOT AGREE	B
388	75-4388X	TRANSIENT BLEED VALVE (TBV) TORQUE MOTOR CURRENT IS OUT OF RANGE	E

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EEC Shop Fault Report Fault ID (EEC Fault Code)	OMF Maintenance Message Number	OMF Maintenance Message Text	Dispatch Group
389	75-4389X	TRANSIENT BLEED VALVE (TBV) TORQUE MOTOR CURRENT IS OUT OF RANGE EXTERNAL TO THE EEC	E
390	75-4390X	TRANSIENT BLEED VALVE (TBV) TORQUE MOTOR CURRENT IS OUT OF RANGE INTERNAL TO THE EEC	E
391	75-4391X	TRANSIENT BLEED VALVE (TBV) IS FAILED CLOSED	A, D
392	75-4392X	TRANSIENT BLEED VALVE (TBV) IS FAILED OPEN	A, D
393	75-4393X	TRANSIENT BLEED VALVE (TBV) POSITION SENSOR CHANNEL-X AND CHANNEL-Y DO NOT AGREE	C
394	75-4394X	TRANSIENT BLEED VALVE (TBV) POSITION SENSOR IS OUT OF RANGE	G
395	75-4395X	TRANSIENT BLEED VALVE(TBV) POSITION SENSOR IS OUT OF RANGE EXTERNAL TO THE EEC	G
396*	75-4396X (77-0900X)	TRANSIENT BLEED VALVE POSITION (TBV) SENSOR IS OUT OF RANGE INTERNAL TO THE EEC	G
397*	73-4397X (77-0900X)	TEMPERATURE COLD JUNCTION HAS A FAULT INTERNAL TO THE EEC	G
398	73-4398X	ELECTRONIC OVERSPEED THRUST CONTROL MALFUNCTION ACCOMMODATION TEST FAILED (SINGLE CHANNEL)	B
399	73-4399X	ELECTRONIC OVERSPEED THRUST CONTROL MALFUNCTION ACCOMMODATION TEST FAILED (DUAL CHANNEL)	A, D
400	80-4400X	ELECTRONIC OVERSPEED THRUST CONTROL MALFUNCTION ACCOMMODATION TEST DID NOT START	E
401	73-4401X	THRUST CONTROL MALFUNCTION EVENT	A, D
402*	73-4402X (77-0900X)	MULTIPLE EEC INTERNAL TEMPERATURE SENSOR SIGNALS ARE OUT OF RANGE	B
403*	73-4403X (77-0900X)	EEC INTERNAL TEMPERATURE SENSOR SIGNAL IS OUT OF RANGE	H
404	79-4404X	ENGINE OIL TEMPERATURE (TEO) EXCEEDS AMBER LIMIT	D
405	79-4405X	ENGINE OIL TEMPERATURE (TEO) EXCEEDS REDLINE	D

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EEC Shop Fault Report Fault ID (EEC Fault Code)	OMF Maintenance Message Number	OMF Maintenance Message Text	Dispatch Group
406	79-4406X	ENGINE OIL TEMPERATURE (TEO) SENSOR CHANNEL-X AND CHANNEL-Y SIGNALS DO NOT AGREE	C
407	79-4407X	ENGINE OIL SUPPLY TEMPERATURE (TEO) SENSOR IS OUT OF RANGE	G
408	79-4408X	ENGINE OIL SUPPLY TEMPERATURE (TEO) SENSOR IS OUT OF RANGE EXTERNAL TO THE EEC	G
409*	79-4409X (77-0900X)	ENGINE OIL SUPPLY TEMPERATURE (TEO) SENSOR IS OUT OF RANGE INTERNAL TO THE EEC	G
412	73-4412X	DUAL CHANNEL THRUST RESOLVER ANGLE (TRA) SENSOR IS OUT OF RANGE	A
413	78-4413X	THRUST REVERSER AIR/GROUND SIGNALS DO NOT AGREE	D
414	73-4414X	THRUST RESOLVER ANGLE (TRA) SENSOR IS OUT OF RANGE	B
415	78-4415X	EEC THRUST REVERSER INTERLOCK RELAY IS FAILED CLOSED	D
416	78-4416X	EEC THRUST REVERSER INTERLOCK RELAY IS FAILED OPEN	D
417	78-4417X	EEC THRUST REVERSER INTERLOCK RELAY POWER IS NOT AVAILABLE	D
418	73-4418X	THRUST RESOLVER ANGLE (TRA) CHANNEL-X AND CHANNEL-Y SENSORS DO NOT AGREE	A, D
419	78-4419X	THRUST REVERSER DIRECTIONAL CONTROL VALVE (DCV) IS FAILED IN THE DEPLOY POSITION	D
421	78-4421X	EEC DIRECTIONAL CONTROL VALVE (DCV) RELAY IS FAILED CLOSED	D
422	78-4422X	EEC DIRECTIONAL CONTROL VALVE (DCV) RELAY IS FAILED OPEN	D
424	78-4424X	THRUST REVERSER DIRECTIONAL CONTROL VALVE (DCV) IS FAILED IN THE STOW POSITION	D
425	78-4425X	THRUST REVERSER HYDRAULIC ISOLATION VALVE (HIV) IS FAILED CLOSED	D
426	78-4426X	THRUST REVERSER HYDRAULIC SYSTEM INDICATES PRESSURIZED WHEN COMMANDED STOWED	D

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EEC Shop Fault Report Fault ID (EEC Fault Code)	OMF Maintenance Message Number	OMF Maintenance Message Text	Dispatch Group
427	78-4427X	THRUST REVERSER DIRECTIONAL CONTROL VALVE (DCV) STOW SOLENOID POWER IS NOT AVAILABLE	D
428	78-4428X	THRUST REVERSER HYDRAULIC ISOLATION VALVE POWER IS NOT AVAILABLE	D
429	78-4429X	EEC HYDRAULIC ISOLATION VALVE (HIV) RELAY IS FAILED CLOSED	D
430	78-4430X	EEC HYDRAULIC ISOLATION VALVE (HIV) RELAY IS FAILED OPEN	D
432	78-4432X	THRUST REVERSER LEFT SLEEVE IS NOT IN THE COMMANDED POSITION	D
433	78-4433X	THRUST REVERSER LEFT SLEEVE POSITION SENSOR SIGNAL CHANNEL-X AND CHANNEL-Y DO NOT AGREE	B
434	78-4434X	THRUST REVERSER LEFT SLEEVE POSITION SENSOR SIGNAL IS OUT OF RANGE	B
435	78-4435X	THRUST REVERSER LEFT SLEEVE POSITION SENSOR SIGNAL IS OUT OF RANGE EXTERNAL TO THE EEC	B
436	78-4436X	THRUST REVERSER CENTER ACTUATOR (CH A = LEFT/ CH B = RIGHT) IS FAILED IN THE LOCKED POSITION	D
437	78-4437X	THRUST REVERSER CENTER ACTUATOR (CH A = LEFT/ CH B = RIGHT)PROXIMITY SWITCH INDICATES LOCKED WITH THRUST REVERSER DEPLOYED	D
438	78-4438X	THRUST REVERSER CENTER ACTUATOR (CH A = LEFT/ CH B = RIGHT) PROXIMITY SWITCH INDICATES UNLOCKED WITH THRUST REVERSER STOWED	D
439	78-4439X	THRUST REVERSER CENTER ACTUATOR (CH A = LEFT/ CH B = RIGHT) PROXIMITY SWITCH SIGNAL IS OUT OF RANGE	D
440	78-4440X	THRUST REVERSER UPPER (CH A = LEFT/ CH B = RIGHT) ACTUATOR IS FAILED IN THE LOCKED POSITION	D
441	78-4441X	THRUST REVERSER UPPER (CH A = LEFT/ CH B = RIGHT) ACTUATOR PROXIMITY SWITCH INDICATES LOCKED WITH THRUST REVERSER DEPLOYED	D

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Table 205/73-21-00-993-805-G00 EEC Shop Faults Correlation Table (Continued)

EEC Shop Fault Report Fault ID (EEC Fault Code)	OMF Maintenance Message Number	OMF Maintenance Message Text	Dispatch Group
442	78-4442X	THRUST REVERSER UPPER (CH A = LEFT/ CH B = RIGHT) ACTUATOR PROXIMITY SWITCH INDICATES UNLOCKED WITH THRUST REVERSER STOWED	D
443	78-4443X	THRUST REVERSER UPPER ACTUATOR (CH A = LEFT/ CH B = RIGHT) PROXIMITY SWITCH SIGNAL IS OUT OF RANGE	D
444	78-4444X	THRUST REVERSER RIGHT SLEEVE IS NOT IN THE COMMANDED POSITION	D
445	78-4445X	THRUST REVERSER RIGHT SLEEVE POSITION SENSOR SIGNAL CHANNEL-X AND CHANNEL-Y DO NOT AGREE	B
446	78-4446X	THRUST REVERSER RIGHT SLEEVE POSITION SENSOR SIGNAL IS OUT OF RANGE	B
447	78-4447X	THRUST REVERSER RIGHT SLEEVE POSITION SENSOR SIGNAL IS OUT OF RANGE EXTERNAL TO THE EEC	B
448	78-4448X	THRUST REVERSER SYNC LOCK (CH A = LEFT/ CH B = RIGHT) IS FAILED IN THE LOCKED POSITION	D
449	78-4449X	THRUST REVERSER SYNC LOCK (CH A = LEFT/ CH B = RIGHT) POWER CIRCUIT IS FAILED OPEN	D
450	78-4450X	THRUST REVERSER SYNC LOCK (CH A = LEFT/ CH B = RIGHT) POWER CIRCUIT IS FAILED CLOSED	D
451	75-4451X	VARIABLE BLEED VALVE (VBV) TORQUE MOTOR CURRENT IS OUT OF RANGE	F
452	75-4452X	VARIABLE BLEED VALVE (VBV) TORQUE MOTOR CURRENT IS OUT OF RANGE EXTERNAL TO THE EEC	F
453	75-4453X	VARIABLE BLEED VALVE (VBV) TORQUE MOTOR HAS AN ELECTRICAL FAULT INTERNAL TO THE EEC	F
454	75-4454X	VARIABLE BLEED VALVE (VBV) IS FAILED CLOSED	A
455	75-4455X	VARIABLE BLEED VALVE (VBV) IS FAILED OPEN	A
456	75-4456X	POSITION SIGNALS FROM VARIABLE BLEED VALVE (VBV) ACTUATORS LEFT AND RIGHT DO NOT AGREE	C
457	75-4457X	VARIABLE BLEED VALVE (VBV) POSITION SENSOR IS OUT OF RANGE	G

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Table 205/73-21-00-993-805-G00 EEC Shop Faults Correlation Table (Continued)

EEC Shop Fault Report Fault ID (EEC Fault Code)	OMF Maintenance Message Number	OMF Maintenance Message Text	Dispatch Group
458	75-4458X	VARIABLE BLEED VALVE (VBV) POSITION SENSOR IS OUT OF RANGE EXTERNAL TO THE EEC	G
459*	75-4459X (77-0900X)	VARIABLE BLEED VALVE (VBV) POSITION SENSOR IS OUT OF RANGE INTERNAL TO THE EEC	G
460	77-4460X	ADIRU-L AND ADIRU-R CALIBRATED AIRSPEED SIGNALS DO NOT AGREE	H
461	77-4461X	ENGINE N1 VIBRATION IS APPROACHING EXCEEDANCE LIMIT	H
462	77-4462X	ENGINE N1 VIBRATION LIMIT EXCEEDED	D
463	73-4463X	FAN SPEED (N1) SENSOR LOW TOOTH SIGNAL IS INVALID	H
464	77-4464X	ENGINE N2 VIBRATION IS APPROACHING EXCEEDANCE LIMIT	H
465	77-4465X	ENGINE N2 VIBRATION LIMIT EXCEEDED	D
466	75-4466X	VARIABLE STATOR VANE (VSV) TORQUE MOTOR CURRENT IS OUT OF RANGE	F
467	75-4467X	VARIABLE STATOR VANE (VSV) TORQUE MOTOR CURRENT IS OUT OF RANGE EXTERNAL TO THE EEC	F
468	75-4468X	VARIABLE STATOR VANE (VSV) TORQUE MOTOR MOTOR CURRENT IS OUT OF RANGE INTERNAL TO THE EEC	F
469	75-4469X	VARIABLE STATOR VANE (VSV) IS FAILED CLOSED	A
470	75-4470X	VARIABLE STATOR VANE (VSV) IS FAILED OPEN	A
471	75-4471X	POSITION SIGNALS FROM VARIABLE STATOR VANE (VSV) ACTUATORS LEFT AND RIGHT DO NOT AGREE	C
472	75-4472X	VARIABLE STATOR VANE (VSV) POSITION SENSOR IS OUT OF RANGE	G
473	75-4473X	VARIABLE STATOR VANE (VSV) POSITION SENSOR IS OUT OF RANGE EXTERNAL TO THE EEC	G
474*	75-4474X (77-0900X)	VARIABLE STATOR VANE (VSV) POSITION SENSOR IS OUT OF RANGE INTERNAL TO THE EEC	G
475	73-4475X	FUEL FLOW METER (WFM) SENSOR CHANNEL-X AND CHANNEL-Y SIGNALS DO NOT AGREE	C

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Table 205/73-21-00-993-805-G00 EEC Shop Faults Correlation Table (Continued)

EEC Shop Fault Report Fault ID (EEC Fault Code)	OMF Maintenance Message Number	OMF Maintenance Message Text	Dispatch Group
476	73-4476X	FUEL FLOW METER (WFM) SENSOR SIGNAL IS OUT OF RANGE	C
477	73-4477X	MODULATED TURBINE COOLING (MTC) VALVES 1 AND 2 ARE NOT IN COMMANDED POSITION	B
478*	74-4478X (77-0900X)	ENGINE-X EEC CHANNEL-X HAS A POWER SUPPLY CAPACITOR CHARGE FAULT INTERNAL TO THE EEC	C
479	73-4479X	EEC IS POWERED WHEN NOT COMMANDED	H
480	73-4480X	FUEL FLOW METER (WFM) SENSOR SIGNAL IS OUT OF RANGE EXTERNAL TO THE EEC	C
481	73-4481X	THRUST INSTABILITY EVENT (TIE) ON FUEL METERING VALVE (FMV)	H
482	75-4482X	AIRCRAFT MODEL AND ENGINE RATING DO NOT AGREE	A
483	73-4483X	FUEL FLOW METER (WFM) SENSOR SIGNAL IS OUT OF RANGE INTERNAL TO THE EEC	C
484	73-4484X	EEC CHANNEL-X IS IN TEST CELL MODE	A
485	73-4485X	OFF-NOMINAL ADJUSTMENT CONFIGURATION	A
486	78-4486X	THRUST REVERSER PRESSURE SWITCH SIGNAL AND THRUST REVERSER COMMANDED POSITION DO NOT AGREE	D
488	77-4488X	ONS AUTHENTICATION IS INVALID	H
489	77-4489X	ONS AUTHORIZATION IS INVALID	H
490	73-4490X	THRUST INSTABILITY EVENT (TIE) ON FUEL METERING VALVE (FMV) AT TAKEOFF	A
491	78-4491X	THRUST REVERSER LEFT SLEEVE POSITION SENSOR SIGNAL IS OUT OF RANGE INTERNAL TO THE EEC	B
492	78-4492X	THRUST REVERSER RIGHT SLEEVE POSITION SENSOR SIGNAL IS OUT OF RANGE INTERNAL TO THE EEC	B
493	73-4493X	THRUST RESOLVER ANGLE (TRA) SENSOR IS OUT OF RANGE INTERNAL TO THE EEC	B
494	73-4494X	THRUST RESOLVER ANGLE (TRA) SENSOR IS OUT OF RANGE EXTERNAL TO THE EEC	B
495	77-4495X	ONS ETHERNET RECEIVER IS INVALID	H
496	77-4496X	ONS ETHERNET HARDWARE IS FAILED	H
497	77-4497X	ONS ETHERNET ARP IS INVALID	H

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Table 205/73-21-00-993-805-G00 EEC Shop Faults Correlation Table (Continued)

EEC Shop Fault Report Fault ID (EEC Fault Code)	OMF Maintenance Message Number	OMF Maintenance Message Text	Dispatch Group
498	77-4498X	EEC OPS AND EEC EHM HAVE INCOMPATIBLE SOFTWARE VERSIONS	H
499	73-4499X	AIRCRAFT MODEL DISCRETES ARE NOT VALID	G
500	75-4500X	UPPER FAN COMPARTMENT TEMPERATURE SENSOR IS OUT OF RANGE	C
501	75-4501X	UPPER FAN COMPARTMENT TEMPERATURE SENSOR IS OUT OF RANGE EXTERNAL TO THE EEC	C
502	75-4502X	UPPER FAN COMPARTMENT TEMPERATURE SENSOR IS OUT OF RANGE INTERNAL TO THE EEC	C
503	75-4503X	LOWER FAN COMPARTMENT TEMPERATURE SENSOR IS OUT OF RANGE	C
504	75-4504X	LOWER FAN COMPARTMENT TEMPERATURE SENSOR IS OUT OF RANGE EXTERNAL TO THE EEC	C
505	75-4505X	LOWER FAN COMPARTMENT TEMPERATURE SENSOR IS OUT OF RANGE INTERNAL TO THE EEC	C
506	75-4506X	UPPER FAN COMPARTMENT TEMPERATURE SENSOR CHANNEL-X AND CHANNEL-Y SIGNALS DO NOT AGREE	C
507	75-4507X	LOWER FAN COMPARTMENT TEMPERATURE SENSOR CHANNEL-X AND CHANNEL-Y SIGNALS DO NOT AGREE	C
508	75-4508X	UPPER FAN COMPARTMENT TEMPERATURE REDLINE LIMIT EXCEEDED	D
509	75-4509X	LOWER FAN COMPARTMENT TEMPERATURE REDLINE LIMIT EXCEEDED	D
510	74-4510X	DUAL IGNITER RELAY IS FAILED	D
511	73-4511X	DPC1 DATA IS NOT AVAILABLE	C
512	78-4512X	THRUST REVERSER INTERLOCK FAILED IN BLOCKING POSITION	D
513	78-4513X	THRUST REVERSER INTERLOCK FAILED IN NON-BLOCKING POSITION	D
514	73-4514X	DPC2 DATA IS NOT AVAILABLE	C
515	77-4515X	ENGINE N2 VIBRATION LIMIT EXCEEDED SUB-IDLE	A, D
516	77-4516X	ENGINE N1 HIGH VIBRATION DURING ICE CRYSTAL ICING CONDITIONS	H

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Table 205/73-21-00-993-805-G00 EEC Shop Faults Correlation Table (Continued)

EEC Shop Fault Report Fault ID (EEC Fault Code)	OMF Maintenance Message Number	OMF Maintenance Message Text	Dispatch Group
533	73-4533X	ENGINE-X EEC CHANNEL-Y IS IN CONTINUOUS RESET CONDITION	A, D
534	75-4534X	PRESSURE SUBSYSTEM (PSS) INTERNAL TEMPERATURE SENSOR CHANNEL-X AND CHANNEL-Y SIGNALS DO NOT AGREE	C
535	75-4535X	PRESSURE SUBSYSTEM (PSS) HEATER CONFIGURATION IS INVALID	A
536	75-4536X	SWITCHED OUTPUT POWER 2 FOR PRESSURE SUBSYSTEM HEATER IS FAULED OPEN INTERNAL TO THE EEC	E
537	75-4537X	SWITCHED OUTPUT POWER 2 FOR PRESSURE SUBSYSTEM HEATER IS FAILED CLOSED INTERNAL TO THE EEC	A
538	75-4538X	PRESSURE SUBSYSTEM (PSS) HEATER IS FAILED	E
539	75-4539X	PRESSURE SUBSYSTEM (PSS) HEATER FUNCTIONALITY IS DISABLED	A
540	73–4540X	THRUST INSTABILITY EVENT (TIE) ON PILOT METERING VALVE (PMV) AT TAKEOFF	A
541	73–4541X	THRUST INSTABILITY EVENT (TIE) ON PILOT METERING VALVE (PMV)	H
542	75-4542X	REVERSE BLEED SYSTEM COOLING NOT AVAILABLE	B
543	75-4543X	REVERSE BLEED SYSTEM NVM PARTITION FAULT	C
544	75-4544X	REVERSE BLEED SYSTEM NVM PARTITION DUAL CHANNEL FAULT	B

SUBTASK 73-21-00-970-010-G00

(3) The dispatch group in Table 205 is associated to the dispatch level and associated flight deck indications. The EEC fault dispatch group summary is shown in Table 206.

Table 206/73-21-00-993-806-G00 Dispatch Group

Dispatch Group	Dispatch Level Determination	Associated Flight Deck Indication
A	No Dispatch	ENGINE CONTROL Alert (direct) ENG X CONTROL Status (direct)
B	Short-Time Dispatch	ENG X EEC C1 SMT (direct)
C	Long-Time Dispatch	ENG X EEC C2 SMT (direct)

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Table 206/73-21-00-993-806-G00 Dispatch Group (Continued)

Dispatch Group	Dispatch Level Determination	Associated Flight Deck Indication
E	Long-Time Dispatch OR No Dispatch (if any other Group E or F fault is set on other channel or if Active Channel is in SCO)	ENG X EEC C2 SMT (direct) OR ENGINE CONTROL Alert (indirect) ENG X CONTROL Status (indirect)
F	Short-Time Dispatch OR No Dispatch (if any other Group E or F fault is set on other channel or if Active Channel is in SCO)	ENG X EEC C1 SMT (direct) OR ENGINE CONTROL Alert (indirect) ENG X CONTROL Status (indirect)
G	Long-Time Dispatch OR No Dispatch (if same fault is set on other channel or if Active Channel is in SCO)	ENG X EEC C2 SMT (direct) OR ENGINE CONTROL Alert (indirect) ENG X CONTROL Status (indirect)
D	Dedicated Flight Deck Effect (FDE) Dispatch per the MMEL or SMT procedures	Dedicated Alert/Status/SMT (direct)
A, D	Same as Group A (with dedicated Alert/Status/SMT Message)	ENGINE CONTROL Alert (direct) ENG X CONTROL Status (direct) AND Dedicated Alert/Status/SMT (direct)
H	Economic Dispatch	NONE (Maintenance Message only)

NOTE: A “direct” flight deck indication results solely via setting of an individual fault on a single EEC channel. An “indirect” flight deck indication requires multiple faults to be set (either the same fault on both channels or a combination of different faults).

Some faults in Dispatch Groups B, C, E, F, and G are indirect contributors to dedicated flight deck indications.

(a) The EEC assigns the final dispatch limitation level based on the EEC detected faults as shown in Figure 202.

SUBTASK 73-21-00-970-011-G00

(4) Use the EEC fault dispatch group summary to find the dispatch level and associated flight deck indications for the Fault IDs (Table 206 and Figure 202).

H. Put the Airplane Back to Its Usual Condition

SUBTASK 73-21-00-080-001-G00

(1) If the maintenance device is used, disconnect the PMD (or laptop) from the ONS port, on the P61 aft overhead maintenance panel, in the flight deck.

SUBTASK 73-21-00-860-025-G00

(2) Set the applicable EEC MAINT POWER switch on the MISC SYSTEM CTRLS page on the MFD to the NORM position (TASK 73-21-00-800-801-G00).

SUBTASK 73-21-00-860-026-G00

(3) Set the GND TESTS switch, on the P61 aft overhead maintenance panel, to the NORM position.

SUBTASK 73-21-00-860-027-G00

(4) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

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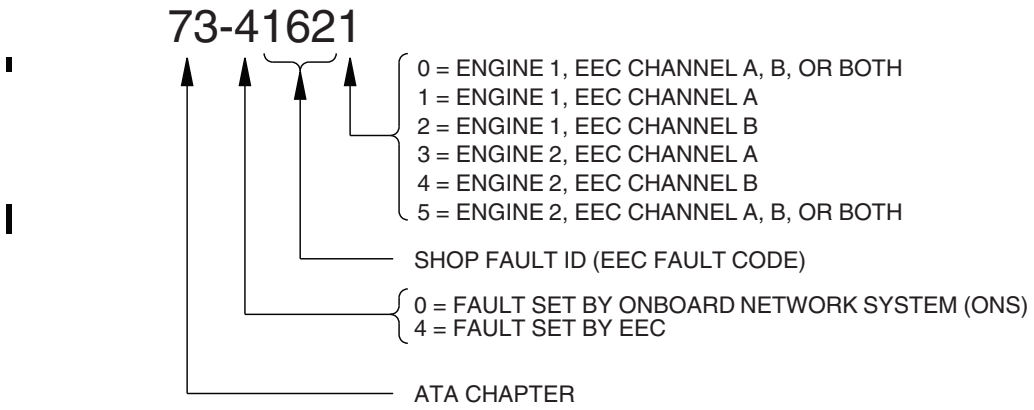
(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

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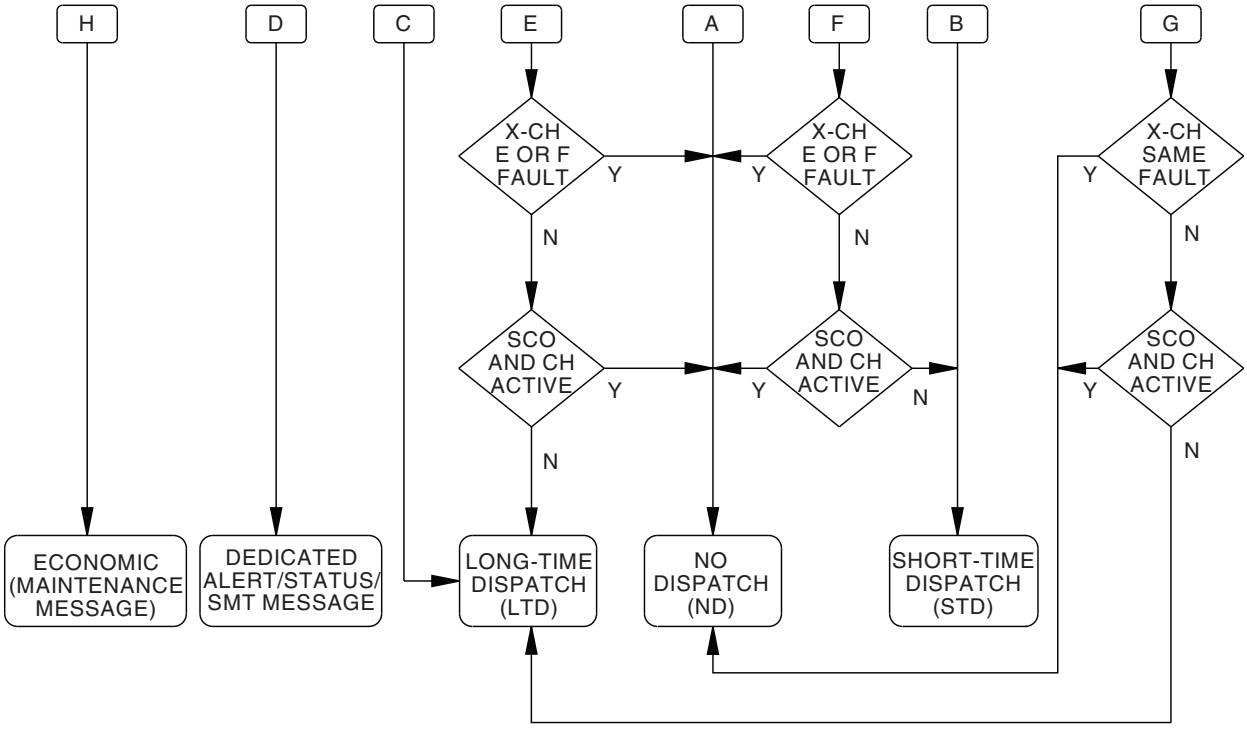
EXAMPLE MESSAGE NUMBER:



5041442-00
2786092 S0000631956_V3

Maintenance Message Format
Figure 201/73-21-00-990-809-G00

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NOTE:

ACRONYMS ARE DEFINED AS FOLLOWS:

1) DCO (DUAL-CHANNEL OPERATION) - FAULT AND DISPATCH DATA FROM BOTH CHANNELS IS AVAILABLE.
2) SCO (SINGLE-CHANNEL OPERATION) - THE CHANNEL IS NOT RECEIVING SERIAL DATA FROM THE OPPOSITE CHANNEL. EITHER THE LOCAL CHANNEL IS HEALTHIER AND MADE ACTIVE ("CHANNEL IS ACTIVE") OR THE LOCAL CHANNEL IS KEPT INACTIVE BECAUSE THE OPPOSITE CHANNEL IS HEALTHIER ("CHANNEL IS NOT ACTIVE").
3) ND (NO-DISPATCH) - A "NO DISPATCH" LIMITATION LEVEL EXISTS.
4) STD (SHORT-TIME DISPATCH) - A "SHORT-TIME DISPATCH" LIMITATION LEVEL EXISTS.
5) LTD (LONG-TIME DISPATCH) - A "LONG-TIME DISPATCH" LIMITATION LEVEL EXISTS.

5041443-00
2786112 S0000631957_V1

Dispatch Limitation Level
Figure 202/73-21-00-990-810-G00

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TASK 73-21-00-470-803-G00

4. EEC Onboard Software Installation

A. General

(1) This task gives the instructions to upload the Electronic Engine Control (EEC) software.

(2) These are the ways that you can install the software in the system:

(a) Use the MAX Display System (MDS) screen.

(b) Use the universal maintenance device, SPL-11321.

(3) There are 4 EEC software part numbers (Operational Program Software (OPS), Engine Health Management (EHM), AVM, and AML).

(a) Only OPS and EHM can be loaded from Onboard Network System (ONS).

(b) To update the AVM or AML software part numbers, do this task: EEC Software Installation Using a PMAT, TASK 73-21-00-470-805-G00.

B. References

Reference	Title
31-65-00-750-801	Fault Interrogation for a Maintenance Message (P/B 201)
46-13-00-470-801	LRU Software Parts Installation (P/B 201)
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)
46-13-00-860-804	Mass Storage Device Software Parts Check (P/B 201)
71-00-00-700-802-G00	Test No. 26 - Debris Monitoring System Channel B Special Functions Test (P/B 501)
71-00-00-700-810-G00	Test No. 34 - Menu Mode Hardware Code Entry Special Functions Test (P/B 501)
71-00-00-700-813-G00	Test No. 37 - Engine Serial Number Entry Special Functions Test (P/B 501)
71-00-00-710-806-G00	Test No. 17 - Electronic Engine Control (EEC) System Test (P/B 501)
71-00-00-720-805-G00	Test No. 11 - Low Pressure System (Fan) Trim Balance (With ONS) (P/B 501)
71-00-00-970-801-G00	Maintenance Data Pages from the Engine Propulsion Control System (EPCS) (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

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E. Prepare for the EEC Onboard Software Installation (with MDS/MD)

SUBTASK 73-21-00-200-002-G00

(1) Make sure that the new software part, or parts, are available on the Mass Storage Device (MSD) (TASK 46-13-00-860-804).

SUBTASK 73-21-00-860-029-G00

(2) Select the left or right Multi Function Display (MFD).

(a) Push the SYS button on the MFD control panel, P9.

(b) Push the SEL button on SELECTOR knob on the MFD control panel, P9.

SUBTASK 73-21-00-860-028-G00

(3) Set the applicable EEC MAINT POWER switch on the MISC SYSTEM CTRLS (MCS) page on the MFD to the TEST position (TASK 73-21-00-800-801-G00).

NOTE: The EEC will do a self-test when you set the EEC MAINT POWER switch to the TEST position.

(a) Wait for a minimum of 30 seconds to let the EEC complete the test.

SUBTASK 73-21-00-860-030-G00

(4) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-21-00-860-031-G00

(5) Set the GND TEST switch on the maintenance bite panel, P61-4, to the ENABLE position.

SUBTASK 73-21-00-860-032-G00

(6) Set the applicable thrust lever to idle.

SUBTASK 73-21-00-970-001-G00

(7) Make a record of the current fan trim balance weight configuration (TASK 71-00-00-720-805-G00).

NOTE: With EEC SW7.1 or higher, fan trim balance weight configurations are not erased from EEC NVM. If only one EEC channel was replaced, Test No. 26 (DMS Special Function) is not required. The EEC channel that was not replaced will automatically transfer the fan trim balance weight configurations to the newly installed EEC channel after the Engine Serial Number (ESN) is entered to the new EEC channel (Test No. 37) and the engine is started and at stable ground idle.

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SUBTASK 73-21-00-970-012-G00

(8) Make a record of the menu mode hardware code and current EEC software version installed (TASK 73-21-00-470-801-G00).

NOTE: If you reprogram the EEC software, it erases the menu mode hardware code stored in the EEC Non Volatile Memory (NVM).

NOTE: With EEC SW7.1 or higher, the menu mode hardware code is not erased from EEC NVM. If only one EEC channel was replaced, Test No. 26 (DMS Special Function) is not required. The EEC channel that was not replaced will automatically transfer the menu mode hardware code to the newly installed EEC channel after the ESN is entered to the new EEC channel (Test No. 37) and the engine is started and at stable ground idle.

SUBTASK 73-21-00-910-003-G00

(9) Make a record of the current engine cumulative oil debris chip count (Debris Monitoring System (DMS)) on Engine Propulsion Control System (EPCS) page 2 (TASK 71-00-00-970-801-G00).

NOTE: If you reprogram the EEC software, it erases the cumulative oil debris chip count stored in the EEC NVM.

NOTE: With EEC SW7.1 or higher, the cumulative oil debris chip count is not erased from EEC NVM. If only one EEC channel was replaced, Test No. 26 (DMS Special Function) is not required. The EEC channel that was not replaced will automatically transfer the cumulative oil debris chip count to the newly installed EEC channel after the ESN is entered to the new EEC channel (Test No. 37) and the engine is started and at stable ground idle.

SUBTASK 73-21-00-910-004-G00

(10) Before you reprogram the EEC with new software, refer to the applicable Boeing Service Bulletin.

NOTE: It is important that the new EEC software version agrees with the Boeing interchangeability requirements.

F. EEC Onboard Software Installation (with MDS or Maintenance Device)

SUBTASK 73-21-00-860-107-G00

(1) Use the ONS to install the EEC software (TASK 46-13-00-470-801).

NOTE: There are 4 EEC software part numbers (OPS, EHM, AVM, and AML), but only OPS and EHM can be loaded from the ONS. The software parts loadable from the ONS are formatted as follows where XX is unique to the applicable software version:

- 73 EEC OPS part number: GEC4X-2628-86XX
- 77 EEC EHM part number: GEC4X-2628-87XX.

NOTE: Only one software part can be loaded at a time. If you load the OPS and EHM software parts, it is recommended to load OPS before EHM.

NOTE: If the AVM or AML software part numbers needs to be updated, use the EEC Software Installation Using a PMAT, TASK 73-21-00-470-805-G00.

NOTE: It is recommended to install the EEC software on one EEC channel at a time.

EEC-1A is the left engine CH A and EEC-1B is the left engine CH B.

EEC-2A is the right engine CH A and EEC-2B is the right engine CH B.

(a) If the software part reports “FAIL” in the Status field, do the steps that follow:

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1) Make sure that the applicable EEC(s) are powered on (TASK 73-21-00-800-801-G00).

2) Operate the GND TEST switch on the maintenance bite panel, P61-4, from ENABLE to NORM, then back to ENABLE.

3) Try to install the EEC software again.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 73-21-00-840-001-G00

(1) If it is necessary, do these steps after the EEC software installation:

(a) Erase these status messages set by the EEC software installation procedure (TASK 31-65-00-750-801):

NOTE: ENG X CONTROL, ENG X REVERSER, and ENG X FAN COMP OVHT SENSOR status messages may be set due to EEC reprogramming, where "X" is specific to the engine reporting the fault. No fault isolation is required for these maintenance messages when set due to EEC software installation procedure.

1) Erase the ENG X CONTROL, ENG X REVERSER.

2) Erase the ENG X FAN COMP OVHT SENSOR.

(b) Cycle power to the EECs with the EEC MAINT POWER on the MISC SYSTEM CTRLS page on the MFD (TASK 73-21-00-800-801-G00).

(c) Confirm the status messages are no longer present (TASK 31-65-00-750-801).

(d) If the status message(s) is or are still active, do the IFIM task(s) for the associated maintenance message(s).

SUBTASK 73-21-00-970-002-G00

(2) If the OPS software was installed, do this task to input the Engine Serial Number (ESN): Test No. 37 - Engine Serial Number Entry Special Functions Test, TASK 71-00-00-700-813-G00.

NOTE: The next engine start after one of the tasks that follow can have a longer pre-start motoring time than typical of cold engines. This is because the engine off time has been erased from EEC NVM.

(a) Install new EEC software.

(b) Enter new Engine Serial Number (ESN).

(c) Install the engine on a new aircraft tail number.

(d) Install the engine on the opposite engine position on the same aircraft tail number.

(e) Install a data entry plug with new thrust rating, N1 trim, or engine configuration.

SUBTASK 73-21-00-970-013-G00

(3) If the OPS software was installed, input the menu mode hardware configuration code (TASK 71-00-00-700-810-G00).

SUBTASK 73-21-00-710-001-G00

(4) Cycle power to the EECs with the EEC MAINT POWER switch on the MISC SYSTEM CTRLS page on the MFD (TASK 73-21-00-800-801-G00).

NOTE: This will make sure that the data entry plug information is read and stored correctly to the EEC NVM.

SUBTASK 73-21-00-860-105-G00

(5) Do this task: EEC Onboard Software Installation Verification, TASK 73-21-00-470-804-G00.

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SUBTASK 73-21-00-970-003-G00

(6) If the OPS software was installed, input the fan trim balance weight configuration, as necessary.

(a) After saving a Weights Configuration change, wait 75 seconds before making additional Weights Configuration changes.

NOTE: If you do not wait 75 seconds, the EEC will not save the Weight Configuration to memory.

(b) If only one EEC channel was reprogrammed, you do not need to manually enter the engine trim balance weight configuration.

NOTE: The EEC channel that was not replaced will automatically transfer the engine trim balance weight configuration to the newly installed EEC channel, after the engine is started and at stable ground idle.

(c) If both EEC channels were reprogrammed, input the fan trim balance weight configuration.

NOTE: This can be done on the next maintenance schedule, but the balance weight configuration will not be correct when you try to do an engine fan trim balance, or when you view the EHM vibration reports.

1) Connect the Portable Maintenance Device (PMD) to the ONS (TASK 46-13-00-480-801).

2) Make these selections on the PMD:

a) OTHER FUNCTIONS

b) EHM

c) Engine Position: 1 or Engine Position: 2

d) Select OK on the CONFIRM BALANCE WEIGHTS pop up window

e) Select ADD.

<1> Select the part number (P0x) column and configuration hole/blade (yy) column.

<2> Input the previously recorded weights.

<3> Select SAVE.

(d) If EEC software version prior to installing the new software part number was 6.7 or later (post LEAP-1B Service Bulletin 73-0025), the fan trim balance weight configuration was preserved through the EEC software installation.

NOTE: It is not necessary to input the fan trim balance weight configuration.

SUBTASK 73-21-00-860-102-G00

(7) If the OPS software was installed, input the engine cumulative oil debris chip count, as necessary.

(a) If only one EEC channel was reprogrammed, you do not need to perform Test No. 26 (DMS Reset) special function.

1) The EEC channel that was not replaced will automatically transfer the engine oil debris cumulative chip count to the newly installed EEC channel, after the engine is started and at stable ground idle.

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(b) If both EEC channels were reprogrammed, check the cumulative oil debris chip count.

NOTE: If EEC software version that was installed before the new software part number (Step E.(8)) was 6.7 or later (post LEAP-1B Service Bulletin 73-0025), the cumulative oil debris chip count was preserved through the EEC software installation. It is not necessary to input the cumulative oil debris chip count.

1) Confirm the engine cumulative oil debris chip count (DMS) on EPCS page 2 is the same as what was recorded in Step E.(9) (TASK 71-00-00-970-801-G00).

2) If the chip count is not correct, input the correct count (TASK 71-00-00-700-802-G00).

SUBTASK 73-21-00-730-001-G00

(8) Do this task: Test No. 17 - Electronic Engine Control (EEC) System Test, TASK 71-00-00-710-806-G00.

SUBTASK 73-21-00-860-103-G00

(9) Set the applicable EEC MAINT POWER switch on the MISC SYSTEM CTRLS (MSC) page on the MFD to the NORM position (TASK 73-21-00-800-801-G00).

SUBTASK 73-21-00-860-104-G00

(10) Set the GND TEST switch on the maintenance bite panel, P61-4, to the NORM position.

SUBTASK 73-21-00-860-055-G00

(11) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

TASK 73-21-00-470-805-G00

5. EEC Software Installation Using a PMAT

(Figure 203)

A. General

(1) This task gives the instructions to install software on the Electronic Engine Control (EEC) using a Portable Maintenance Access Terminal (PMAT).

(a) Portable data loaders (PDLs) compatible with the LEAP-1B EECs are an acceptable alternative to the PMAT. If you use a PDL, follow the steps in this task but refer to the PDL user manual for task steps specific to the PMAT interface. Contact CFM if you have any questions on PDL compatibility or task procedures.

(2) These tools are necessary to retrieve Non Volatile Memory (NVM) data from the EEC:

(a) PMAT 2000, ARINC 615/615A data loader, COM-4963, or ARINC 645-1 compliant portable data loader, COM-21223.

(b) CFM recommends the reprogramming kit Cable Assembly, SPL-14903, (reprogramming box, reprogramming box cable, and Ethernet cable).

(c) Ethernet adapter (Teledyne controls cable), CIS-MS/FSM boot-loader breakout kit, COM-13929.

(d) ARINC 615A-3 software image update, part number 80719-3-UPG or newer, to upgrade the PMAT 2000.

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(3) Modern software loaders that meet the security requirements defined in ARINC 645-1 or newer are recommended to use. The procedures for modern software loaders will be marked as preferred for now and will be the only procedures available after removing floppy disk loaders procedures in the future. Please check with your Software Loader Supplier for current capabilities since this is still maturing across current and future software loaders.

(4) This procedure can generate the Onboard Maintenance Function (OMF) maintenance messages that follow as both EECs are powered and only one EEC is connected to the PMAT:

(a) 73-4117X 'CROSS CHANNEL DATA LINK DISCRETE INPUTS AND BACKUP SERIAL CROSS CHANNEL DATA LINK DO NOT AGREE'

(b) 73-4270X 'ENGINE-Y EEC CHANNEL-Z AIRCRAFT POWER IS NOT AVAILABLE'

(c) 73-4271X 'ENGINE-Y EEC CHANNEL-Z IS NOT AVAILABLE'.

NOTE: Where X refers to EEC channel detecting the fault (0 - Engine 1 Channel A and B, 1 - Engine 1 Channel A, 2 - Engine 1 Channel B, 3 - Engine 2 Channel A, 4 - Engine 2 Channel B, 5 - Engine 2 Channel A and B), Y refers to engine position (1-Left or 2-Right) and Z refers to EEC channel (A or B).

NOTE: No fault isolation is necessary for these maintenance messages when they are set because of the performance of this procedure.

B. References

Reference	Title
31-65-00-750-801	Fault Interrogation for a Maintenance Message (P/B 201)
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)
71-00-00-700-810-G00	Test No. 34 - Menu Mode Hardware Code Entry Special Functions Test (P/B 501)
71-00-00-700-813-G00	Test No. 37 - Engine Serial Number Entry Special Functions Test (P/B 501)
71-00-00-710-806-G00	Test No. 17 - Electronic Engine Control (EEC) System Test (P/B 501)
71-00-00-720-805-G00	Test No. 11 - Low Pressure System (Fan) Trim Balance (With ONS) (P/B 501)
71-00-00-970-801-G00	Maintenance Data Pages from the Engine Propulsion Control System (EPCS) (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-4963	Loader - Data, ARINC 615/615A (Model is PMAT 2000) Part #: PXS-LDS-10 Supplier: 98571 Opt Part #: 80000-05 Supplier: 0BAW0 Opt Part #: P2K-615A-06 Supplier: 0BAW0
COM-13929	Kit - Breakout, CIS-MS/FSM Boot-loader Part #: 80080-1 Supplier: 0BAW0

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(Continued)

Reference	Description
COM-21223	Data Loader - ARINC 615 (with ARINC 645-1 Compliant) Part #: 403474 Supplier: CG603 Part #: 403562 Supplier: CG603 Part #: AE-615-MPDL Supplier: CD624 Part #: AE-DL-SMD Supplier: CD624 Part #: PDL-615-XT Supplier: 89Q20 Part #: PXS-LDS-12 Supplier: 98571
SPL-14903	Cable Assembly - PMAT (LEAP-1B Engine) Part #: 9471M34G02 Supplier: 58828
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
414	Engine 1 - Fan Cowl, Right
424	Engine 2 - Fan Cowl, Right

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

F. Prepare for the Software Installation

SUBTASK 73-21-00-470-002-G00

(1) Install the EEC software on the PMAT:

(a) Make sure that the power is applied to the PMAT.

(b) If the new EEC software version is not installed on the PMAT, transfer it to the PMAT C:\datafiles\NEWLOADS directory.

(c) If not already active, select the "PMAT Main Menu" icon on the PMAT desktop.

(d) Select the LOADSTAR box.

NOTE: The LOADSTAR box can be in a different location on a different PMAT.

(e) When LOADSTAR opens, select the LIBRARY tab on the lower right edge of the screen.

(f) Select the INSTALL SOFTWARE bar at the top of the screen.

(g) Select the NEW LOAD DIRECTORY radio button.

(h) Select the specified software to load.

NOTE: Multiple items can be selected.

(i) Select NEXT.

(j) When the new INSTALL SOFTWARE Window shows, select NEXT.

NOTE: If you select more than one part, an INSTALL SOFTWARE window will show for each subsequent part after the previous part is installed.

(k) When the INSTALLATION COMPLETE window shows, after you install all the parts, select OK.

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(l) Select LOGOFF LOADSTAR.

NOTE: The selection is on the lower left corner of the display.

(m) Close LOADSTAR.

SUBTASK 73-21-00-860-114-G00

(2) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-21-00-970-007-G00

(3) Make a record of the fan trim balance weight configuration (TASK 71-00-00-720-805-G00).

NOTE: Reprogramming the EEC software erases the fan trim balance weight configuration stored in the EEC NVM.

SUBTASK 73-21-00-970-014-G00

(4) Make a record of the menu mode hardware code and current EEC software version installed (TASK 73-21-00-470-801-G00).

NOTE: If you reprogram the EEC software, it erases the menu mode hardware code stored in the EEC NVM.

SUBTASK 73-21-00-910-001-G00

(5) Make a record of the current engine cumulative oil debris chip count ("Debris Monitoring System (DMS)") on Engine Propulsion Control System (EPCS) page 2 (TASK 71-00-00-970-801-G00).

NOTE: If you reprogram the EEC software, it erases the cumulative oil debris chip count stored in the EEC NVM.

SUBTASK 73-21-00-970-017-G00

(6) Before you reprogram the EEC with new software, refer to the applicable Boeing Service Bulletin.

NOTE: It is important that the new EEC software version agrees with the Boeing interchangeability requirements.

SUBTASK 73-21-00-210-021-G00

(7) Make sure that the applicable EEC MAINT POWER on the MISC SYSTEM CTRLS page is set to NORM (TASK 73-21-00-800-801-G00).

SUBTASK 73-21-00-700-001-G00

(8) Test the battery charge and cable continuity on the Cable Assembly, SPL-14903:

(a) Remove the dust cap from the Ground Support Equipment (GSE) VIBRATION DSP ETHERNET port on the reprogramming box.

(b) Connect one end of the Ethernet cable to the GSE VIBRATION DSP ETHERNET port.

(c) Remove the dust cap from the GSE ETHERNET port on the reprogramming box.

(d) Connect the other end of the Ethernet cable to the GSE ETHERNET PORT.

(e) Make sure that the BATTERY FULLY CHARGED light is on.

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1) If the BATTERY FULLY CHARGED light is not on, replace the standard 9V battery on the reprogramming box.

(f) Disconnect both ends of the Ethernet cables.

(g) Put the dust caps on GSE VIBRATION DSP ETHERNET and GSE ETHERNET ports on the reprogramming box.

(h) Disconnect the dust cap on the TESTER RECEPTACLE on the reprogramming box.

(i) Connect the end of the reprogramming box cable to the TESTER RECEPTACLE on the reprogramming box.

1) Connect the end of the reprogramming box cable used for connection to the EEC J9.

(j) Make sure that the CABLE OK light is illuminates.

1) If the CABLE OK light does not illuminate, replace the battery and do the test again.

2) If the CABLE OK light still does not illuminate, replace the Cable Assembly, SPL-14903, and do the test again.

(k) Replace the dust cap on the TESTER RECEPTACLE.

SUBTASK 73-21-00-860-115-G00

(9) Make sure that the engines are shut down at least 5 minutes before you start the software installation.

SUBTASK 73-21-00-010-005-G00

(10) Do this task: Open the Fan Cowl Panels (Selection), TASK 71-11-04-010-801-G00.

(a) On the applicable engine, open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

SUBTASK 73-21-00-470-003-G00

(11) Connect the PMAT to the EEC:

(a) Remove the dust cap on the J9 connector.

(b) Connect the end of the EEC programming Cable Assembly, SPL-14903, to the EEC J9 connector.

(c) Connect the Ethernet adapter (Teledyne controls cable), CIS-MS/FSM boot-loader breakout kit, COM-13929, to the reprogramming port of the PMAT.

(d) Connect the EEC programming Cable Assembly, SPL-14903, to the PMAT reprogramming Ethernet adapter cable port with a “straight” Ethernet cable at either:

NOTE: If you use a ”cross-over” Ethernet cable, the PMAT will not be able to find an EEC target to connect with for the software upload.

1) The GSE ETHERNET port to reprogram Control & System Configuration (Operational Program Software (OPS)) software, Engine Health Management (EHM) Reporting & Onboard Network System (ONS) Functions (EHM) software, or Application Memory Loader (AML) software.

2) The GSE VIBRATION DSP ETHERNET port to reprogram the Advanced Vibration Analysis (AVM) OMAP software.

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SUBTASK 73-21-00-860-120-G00

(12) Set the GND TESTS switch on the auxiliary panel, P61, to the ENABLE position.

G. Electronic Engine Control (EEC) Software Installation Using a PMAT

SUBTASK 73-21-00-470-004-G00

(1) Install the EEC software using a PMAT:

(a) Set the applicable EEC MAINT POWER on the MISC SYSTEM CTRLS page to TEST (TASK 73-21-00-800-801-G00).

(b) Power on the PMAT.

(c) Select “PMAT MAIN MENU” icon on the PMAT desktop to open the loading software.

(d) Select “ARINC 615A DATALOADER” (Figure 203 (Sheet 5)).

(e) Enter USER NAME into box 1 (Figure 203 (Sheet 6)).

(f) In box 2, SELECT A TAIL NUMBER and then select OK.

1) If a tail number is not present in box 2, select OTHER and enter a tail number.

2) Then select OK.

(g) In box 3, SELECT CONNECTION TYPE.

(h) In box 4, select CONTINUE.

(i) Select “Upload” (Figure 203 (Sheet 7)).

(j) Wait for the screen to show that a target was found, then select that target (Figure 203 (Sheet 8)).

NOTE: If reprogramming the field loadable OMAP software, there will be a delay of approximately 40 seconds between EEC power-on and when the target appears.

NOTE: If you will use the GSE ETHERNET port when reprogramming, the screen will display the target name as: “GEC-EECCPU_A” or “GEC-EECCPU_B” depending on the channel the PMAT will connect to (A or B).

NOTE: If you will use the GSE VIBRATION DSP ETHERNET port when reprogramming, the screen will display the target name as: “GEC-EECDSP_A” or “GEC-EECDSP_B” depending on the channel the PMAT will connect to (A or B).

(k) Select CONTINUE (Figure 203 (Sheet 8)).

(l) Select a tail number (Figure 203 (Sheet 9)).

(m) Scroll through the list of parts and select the part to be loaded as defined by the release documentation.

(n) Select CONTINUE (Figure 203 (Sheet 8)).

(o) Select CONTINUE “UPLOAD” (Figure 203 (Sheet 9)).

(p) Monitor the software loading progress bar on the PMAT display.

NOTE: A complete successful load is indicated by a green bar on the PMAT display.

(q) If the software load is successful, then do the steps that follow:

1) Select “Close” and “GO BACK”.

2) If installing another software using the same Ethernet port on the EEC programming Cable Assembly, SPL-14903, do this step:

a) Select “PMAT Main Menu” icon on the PMAT desktop to open the loading software.

b) Do the steps again for installation of the software.

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3) If no additional software load, select “SHUTDOWN” on the PMAT MAIN MENU.

4) Set the applicable EEC MAINT POWER on the MISC SYSTEM CTRLS page is set to NORM (TASK 73-21-00-800-801-G00).

SUBTASK 73-21-00-860-116-G00

(2) If the software installation was not successful, do these steps:

(a) Make sure that the EEC power is on (TASK 73-21-00-800-801-G00).

(b) Make sure that the installation of the cables between the PMAT and the EEC is correct.

(c) Make sure that you tighten cable connections to the proper torque.

(d) Make sure that these circuit breakers are closed:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A
B	3	C01312	ENGINE 1 RUN/PWR

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	5	C01313	ENGINE 2 RUN/PWR
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	3	C04018	ENG 1 STBY ALTN PWR RELAY CHAN A
E	4	C04019	ENG 1 STBY ALTN PWR CHAN A

(e) Make sure that the PMAT is properly configured.

(f) Do the procedure again at least three times to load the software.

1) If the software installation fails after three or more attempts, do a repair action.

SUBTASK 73-21-00-000-001-G00

(3) Disconnect the PMAT from the EEC:

(a) Disconnect the EEC programming Cable Assembly, SPL-14903, from the EEC J9 connector.

(b) Disconnect the Ethernet cable from the applicable GSE Ethernet port on the EEC programming Cable Assembly, SPL-14903, and the PMAT Ethernet port.

(c) Install the dust cap on the J9 connector.

SUBTASK 73-21-00-470-005-G00

(4) Connect the PMAT to the other EEC channel or to load the other EEC software:

(a) Remove the dust cap on the J9 connector.

(b) Connect the end of the EEC programming Cable Assembly, SPL-14903, to the EEC J9 connector.

(c) Connect the Ethernet adapter (Teledyne controls cable), CIS-MS/FSM boot-loader breakout kit, COM-13929, to the reprogramming port of the PMAT.

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(d) Connect the EEC programming Cable Assembly, SPL-14903, to the PMAT reprogramming Ethernet adapter cable port with a "straight" Ethernet cable at either:

NOTE: If you use a "cross-over" Ethernet cable, the PMAT will not be able to find an EEC target to connect with for the software upload.

1) The GSE ETHERNET port to reprogram Control & System Configuration (OPS) software, EHM Reporting & ONS Functions (EHM) software, or Application Memory Loader (AML) software.

2) The GSE VIBRATION DSP ETHERNET port to reprogram the Advanced Vibration Analysis (AVM) OMAP software.

SUBTASK 73-21-00-470-006-G00

(5) Do these steps again to continue the loading of the software on the other EEC channel:

(a) Set the applicable EEC MAINT POWER on the MISC SYSTEM CTRLS page to TEST (TASK 73-21-00-800-801-G00).

(b) Power on the PMAT.

(c) Select “PMAT MAIN MENU” icon on the PMAT desktop to open the loading software.

(d) Select “ARINC 615A DATALOADER” (Figure 203 (Sheet 5)).

(e) Enter USER NAME into box 1 (Figure 203 (Sheet 6)).

(f) In box 2, SELECT A TAIL NUMBER and then select OK.

1) If a tail number is not present in box 2, select OTHER and enter a tail number.

2) Select OK.

(g) In box 3, SELECT CONNECTION TYPE.

(h) In box 4, select CONTINUE.

(i) Select “Upload” (Figure 203 (Sheet 7)).

(j) Wait for the screen to show that a target was found, then select that target (Figure 203 (Sheet 8)).

NOTE: If reprogramming the field loadable OMAP software, there will be a delay of approximately 40 seconds between EEC power-on and when the target appears.

NOTE: If you will use the GSE ETHERNET port when reprogramming, the screen will display the target name as: “GEC-EECCPU_A” or “GEC-EECCPU_B” depending on the channel the PMAT will connect to (A or B).

NOTE: If you will use the GSE VIBRATION DSP ETHERNET port when reprogramming, the screen will display the target name as: “GEC-EECDSP_A” or “GEC-EECDSP_B” depending on the channel the PMAT will connect to (A or B).

(k) Select CONTINUE (Figure 203 (Sheet 8)).

(l) Select a tail number (Figure 203 (Sheet 9)).

(m) Scroll through the list of parts and select the part to be loaded as defined by the release documentation.

(n) Select CONTINUE (Figure 203 (Sheet 8)).

(o) Select CONTINUE “UPLOAD” (Figure 203 (Sheet 9)).

(p) Monitor the software loading progress bar on the PMAT display.

NOTE: A complete successful load is indicated by a green bar on the PMAT display.

(q) If the software load is successful, then do these steps:

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1) Select “Close” and “GO BACK”.

2) If installing another software using the same Ethernet port on the EEC programming Cable Assembly, SPL-14903, do this step:

a) Select “PMAT Main Menu” icon on the PMAT desktop to open the loading software.

b) Do the steps again for installation of the software.

3) If no additional software load, select “SHUTDOWN” on the PMAT MAIN MENU.

4) Set the applicable EEC MAINT POWER on the MISC SYSTEM CTRLS page is set to NORM (TASK 73-21-00-800-801-G00).

H. Put the Airplane Back to its Usual Condition

SUBTASK 73-21-00-410-006-G00

(1) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

(a) On the applicable engine, close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

SUBTASK 73-21-00-860-117-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

SUBTASK 73-21-00-210-022-G00

(3) Do a check on the OMF for the EEC maintenance messages that follows:

(a) 73-41170, 73-41175, 73-42701, 73-42702, 73-42703, 73-42704, 73-42711, 73-42712, 73-42713, and 73-42714.

NOTE: No fault isolation is required for these maintenance messages when set due to EEC connection to the PMAT.

1) If the maintenance messages are displayed, cycle the power to the EECs with the EEC MAINT POWER on the MISC SYSTEM CTRLS page on the Multi Function Display (MFD) (TASK 73-21-00-800-801-G00).

2) If the maintenance messages clear, confirm that no STATUS or SMT messages are on display (TASK 31-65-00-750-801).

SUBTASK 73-21-00-470-007-G00

(4) If the OPS software was installed, input the Engine Serial Number (ESN) (TASK 71-00-00-700-813-G00).

NOTE: The next engine start after one of the tasks that follow can have a longer pre-start motoring time than typical of cold engines. This is because the engine off time has been erased from EEC NVM.

(a) Install new EEC software.

(b) Enter new Engine Serial Number (ESN).

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(c) Install the engine on a new aircraft tail number.

(d) Install the engine on the opposite engine position on the same aircraft tail number.

(e) Install a data entry plug with new thrust rating, N1 trim, or engine configuration.

SUBTASK 73-21-00-970-015-G00

(5) If the OPS software was installed, input the menu mode hardware configuration code (TASK 71-00-00-700-810-G00).

SUBTASK 73-21-00-910-002-G00

(6) Cycle power to the EECs with the EEC MAINT POWER switch on the MISC SYSTEM CTRLS page on the MFD (TASK 73-21-00-800-801-G00).

NOTE: If you cycle power to the EECs, it will make sure that the data entry plug information is read and stored correctly to EEC NVM.

SUBTASK 73-21-00-700-002-G00

(7) Do this task: EEC Onboard Software Installation Verification, TASK 73-21-00-470-804-G00.

SUBTASK 73-21-00-470-008-G00

(8) If the OPS software was installed, input the fan trim balance weight configuration, as necessary.

(a) After saving a Weights Configuration change, wait 75 seconds before making additional Weights Configuration changes.

NOTE: If you do not wait 75 seconds, the EEC will not save the Weight Configuration to memory.

(b) If only one EEC channel was reprogrammed, you do not need to manually enter the engine trim balance weight configuration.

NOTE: The EEC channel that was not replaced will automatically transfer the engine trim balance weight configuration to the newly installed EEC channel, after the engine is started and at stable ground idle.

(c) If both EEC channels were reprogrammed, input the fan trim balance weight configuration.

NOTE: This can be done on the next maintenance schedule, but the balance weight configuration will not be correct when you do an engine fan trim balance, or when you view the EHM vibration reports.

1) Connect the Portable Maintenance Device (PMD) to the ONS (TASK 46-13-00-480-801).

2) Make these selections on the PMD:

a) OTHER FUNCTIONS

b) EHM

c) Engine Position: 1 or Engine Position: 2

d) Select OK on the CONFIRM BALANCE WEIGHTS pop up window.

e) Select ADD.

<1> Select the part number (P0x) column and configuration hole/blade (yy) column.

<2> Input the previously recorded weights.

<3> Select SAVE.

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(d) If EEC software version prior to installing the new software part number was 6.7 or later (post LEAP-1B Service Bulletin 73-0025), the fan trim balance weight configuration was preserved through the EEC software installation.

NOTE: It is not necessary to input the fan trim balance weight configuration.

SUBTASK 73-21-00-860-118-G00

(9) If the OPS software was installed, input the engine cumulative oil debris chip count, as necessary.

(a) If only one EEC channel was reprogrammed, you do not need to perform Test No. 26 (DMS Reset) special function.

NOTE: The EEC channel that was not replaced will automatically transfer the engine oil debris cumulative chip count to the newly installed EEC channel, after the engine is started and at stable ground idle.

(b) If both EEC channels were reprogrammed, check the cumulative oil debris chip count.

(c) If EEC software version that was installed before the new software part number (Step E. (8)) was 6.7 or later (post LEAP-1B Service Bulletin 73–0025), the cumulative oil debris chip count was preserved through the EEC software installation.

NOTE: It is not necessary to input the cumulative oil debris chip count.

1) Confirm the engine cumulative oil debris chip count (DMS) on EPCS page 2 is the same as what was recorded in Step E.(9) (TASK 71-00-00-970-801-G00).

2) If the chip count is not correct, input the correct count (TASK 71-00-00-970-801-G00).

SUBTASK 73-21-00-860-121-G00

(10) Set the GND TESTS switch on the auxiliary panel, P61, to the NORM position.

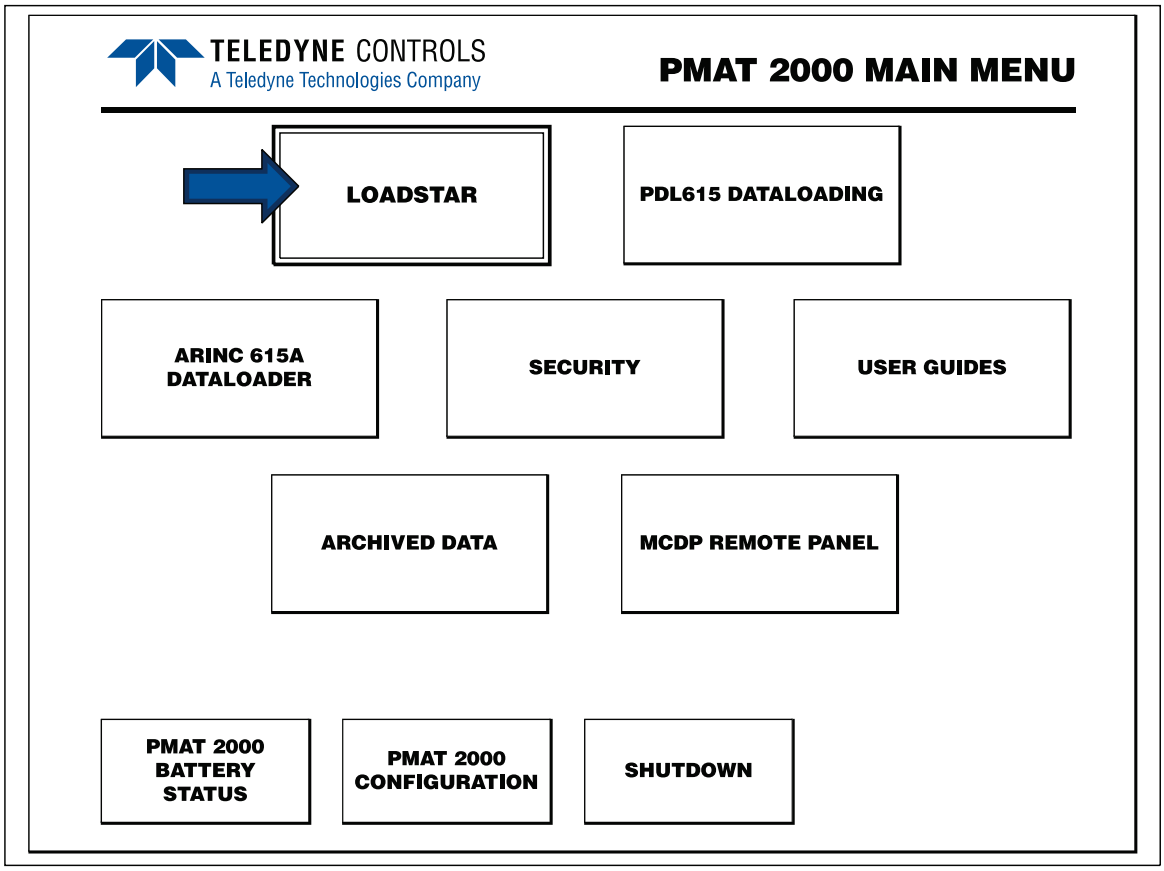
I. Electronic Engine Control (EEC) Software Installation Test

SUBTASK 73-21-00-730-002-G00

(1) Do this task: Test No. 17 - Electronic Engine Control (EEC) System Test, TASK 71-00-00-710-806-G00.

——— END OF TASK ———

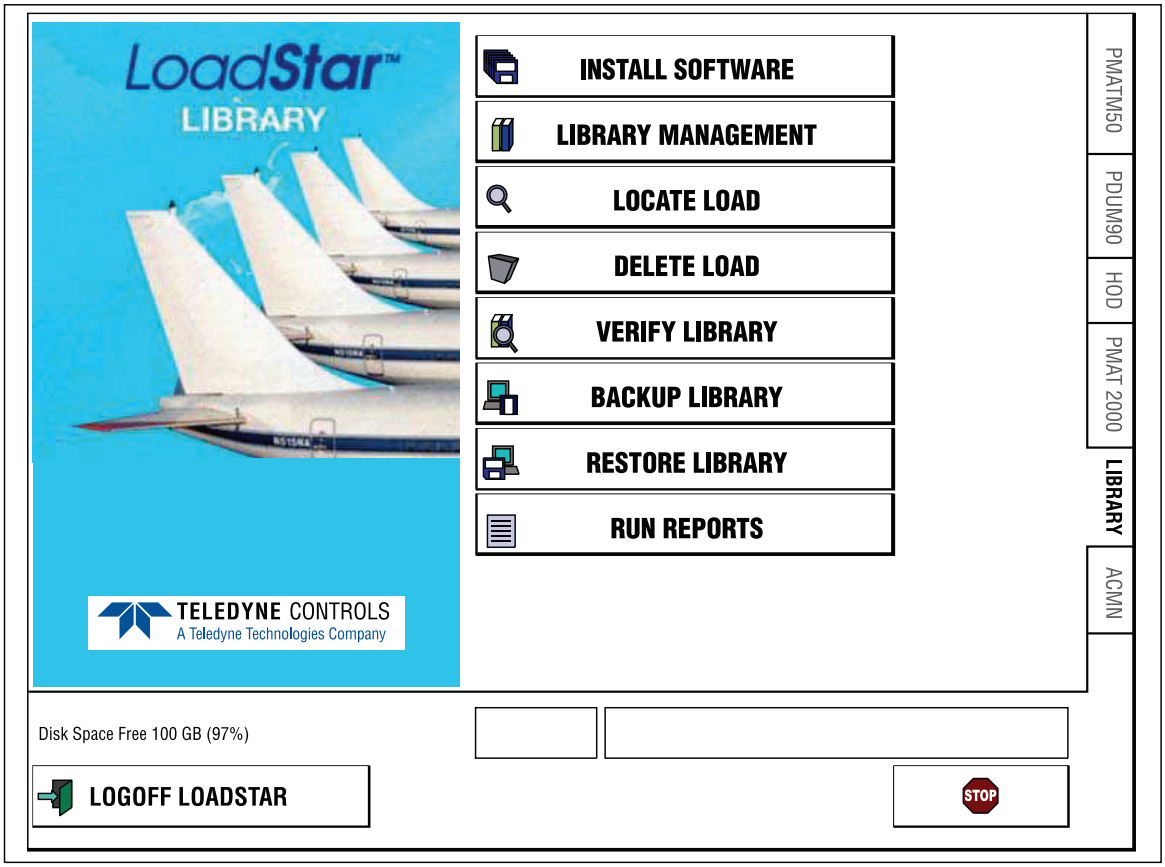
ECCN 9E991 BOEING PROPRIETARY - See title page for details



5035499-00
2568012 S0000614809_V1

Electronic Engine Control (EEC) Software Installation Using a PMAT
Figure 203/73-21-00-990-808-G00 (Sheet 1 of 9)

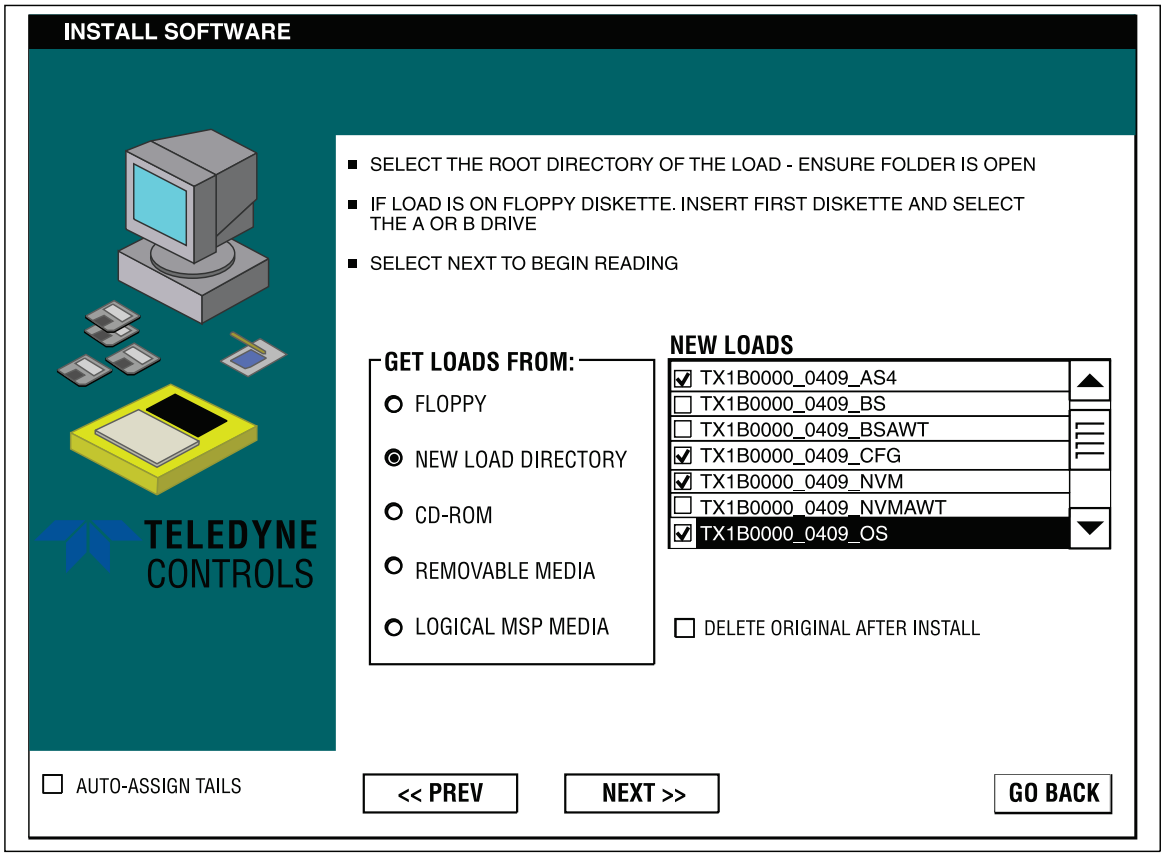
ECCN 9E991 BOEING PROPRIETARY - See title page for details



5035500-00
2568013 S0000614810_V1

Electronic Engine Control (EEC) Software Installation Using a PMAT
Figure 203/73-21-00-990-808-G00 (Sheet 2 of 9)

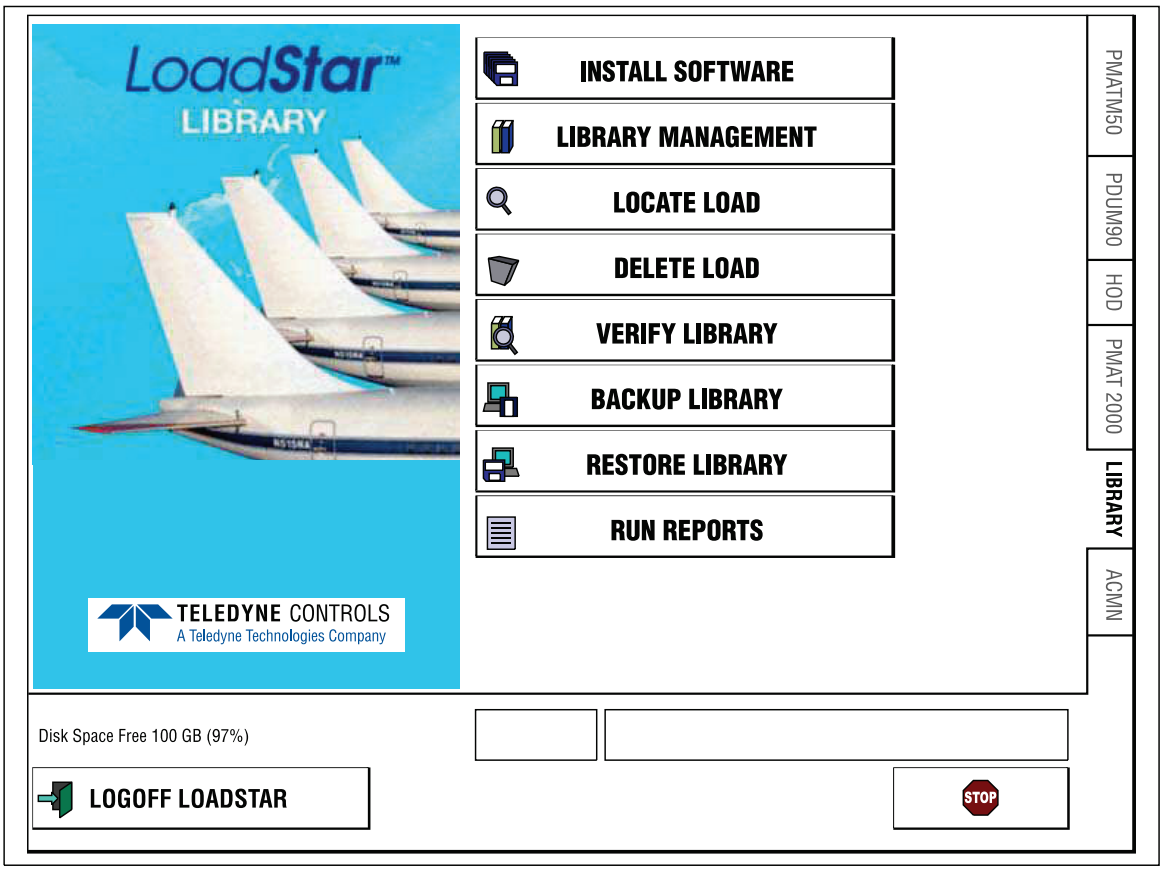
ECCN 9E991 BOEING PROPRIETARY - See title page for details



5035501-00
2568014 S0000614811_V1

Electronic Engine Control (EEC) Software Installation Using a PMAT
Figure 203/73-21-00-990-808-G00 (Sheet 3 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

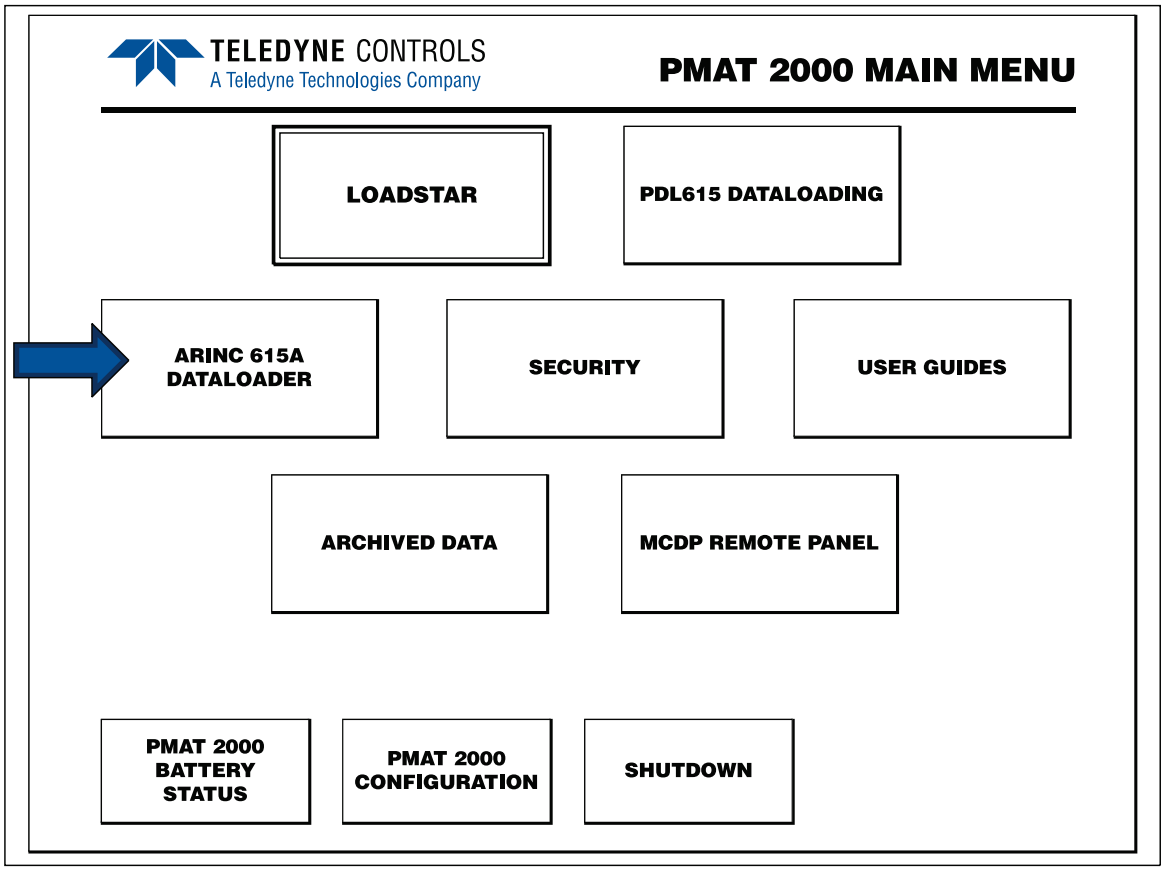


5035502-00

2568019 S0000614812_V1

Electronic Engine Control (EEC) Software Installation Using a PMAT
Figure 203/73-21-00-990-808-G00 (Sheet 4 of 9)

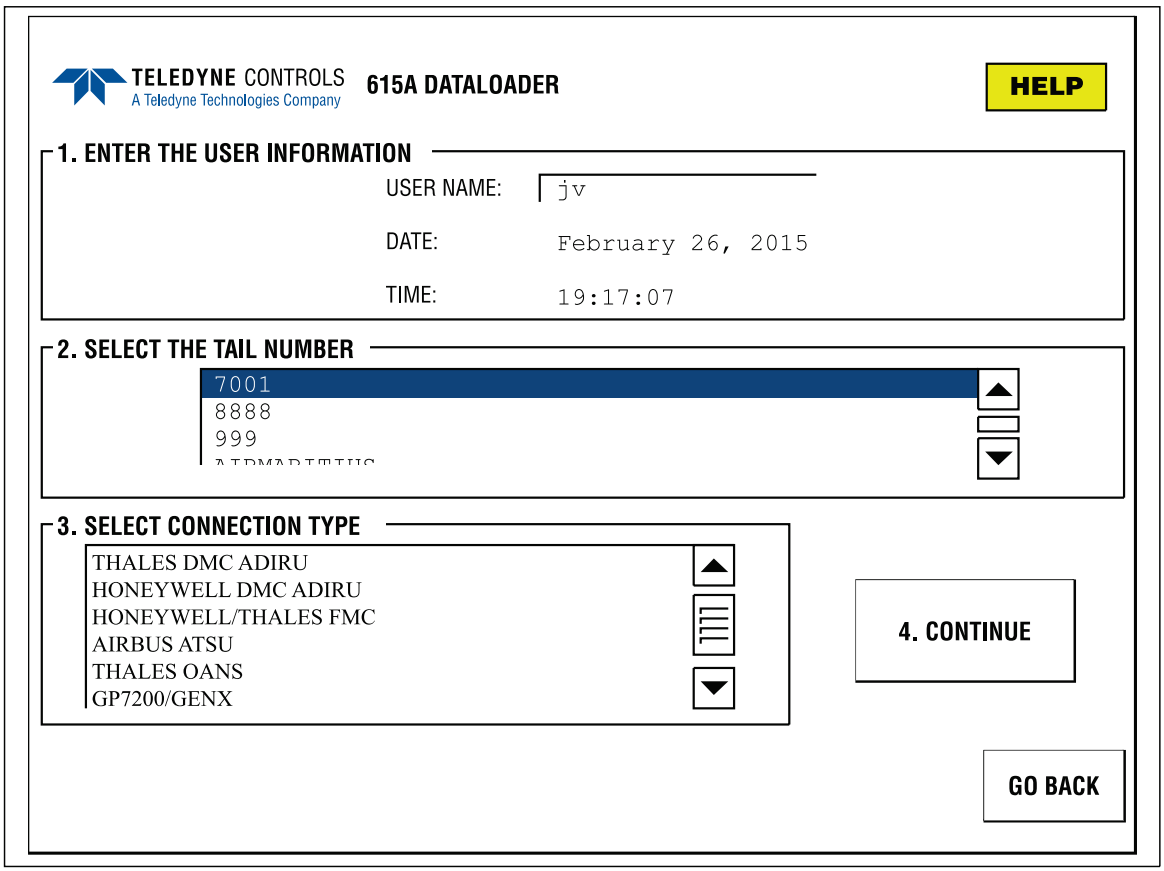
ECCN 9E991 BOEING PROPRIETARY - See title page for details



5035503-00
2568017 S0000614813_V1

Electronic Engine Control (EEC) Software Installation Using a PMAT
Figure 203/73-21-00-990-808-G00 (Sheet 5 of 9)

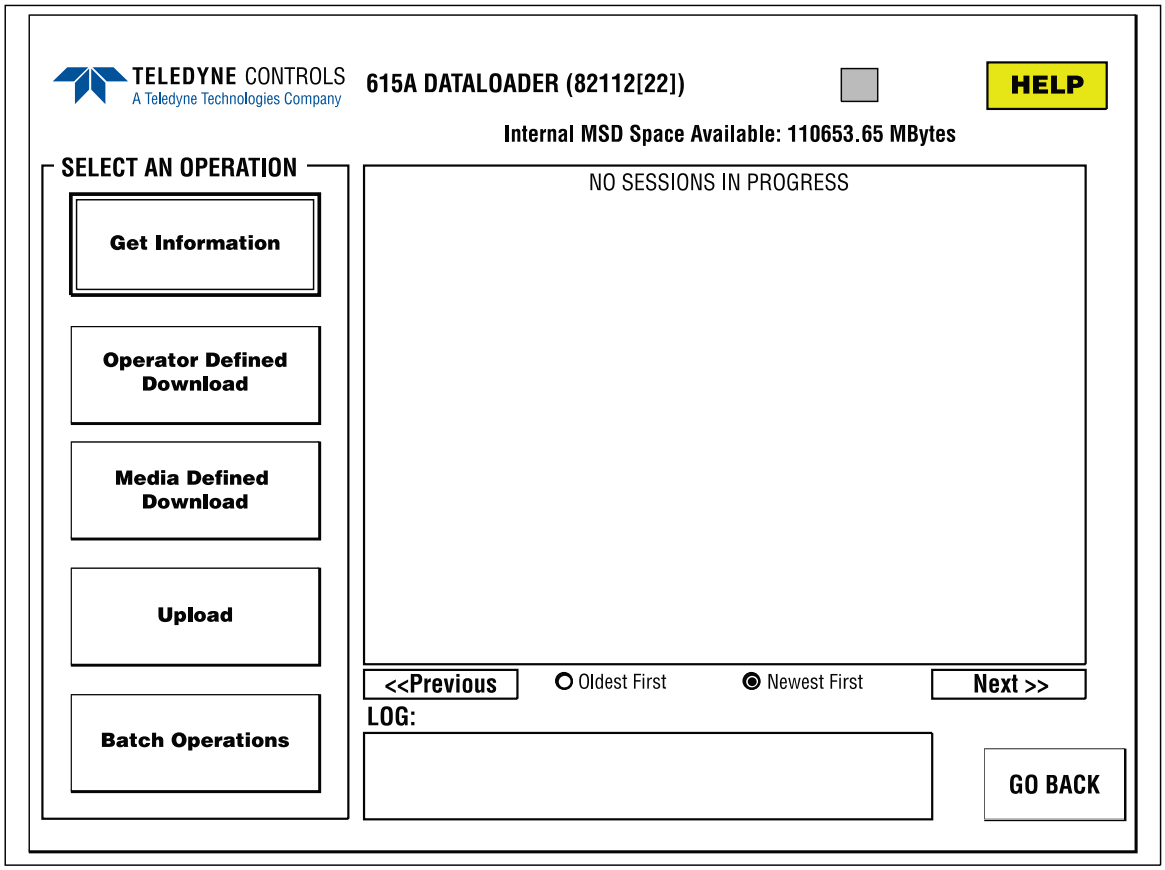
ECCN 9E991 BOEING PROPRIETARY - See title page for details



5035504-00
2568020 S0000614814_V1

Electronic Engine Control (EEC) Software Installation Using a PMAT
Figure 203/73-21-00-990-808-G00 (Sheet 6 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

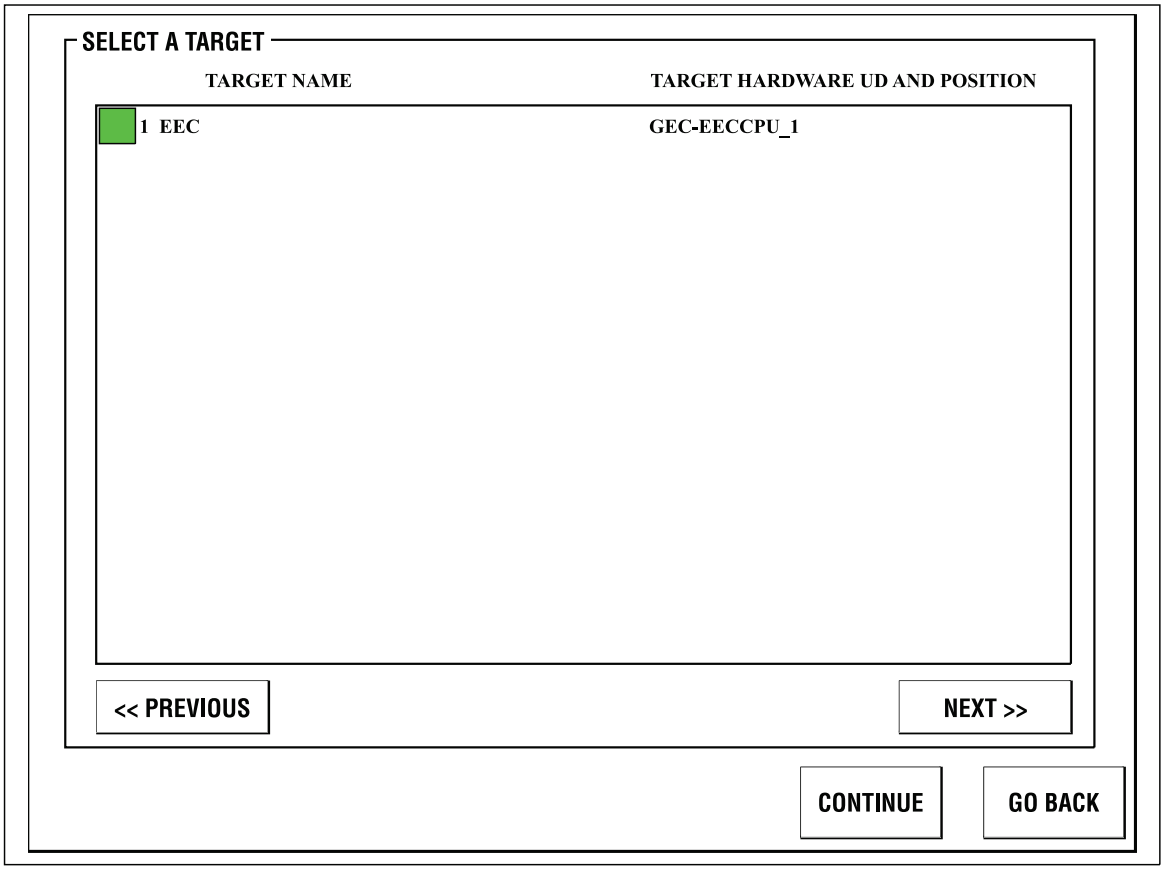


5035505-00

2568021 S0000614815_V1

Electronic Engine Control (EEC) Software Installation Using a PMAT
Figure 203/73-21-00-990-808-G00 (Sheet 7 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

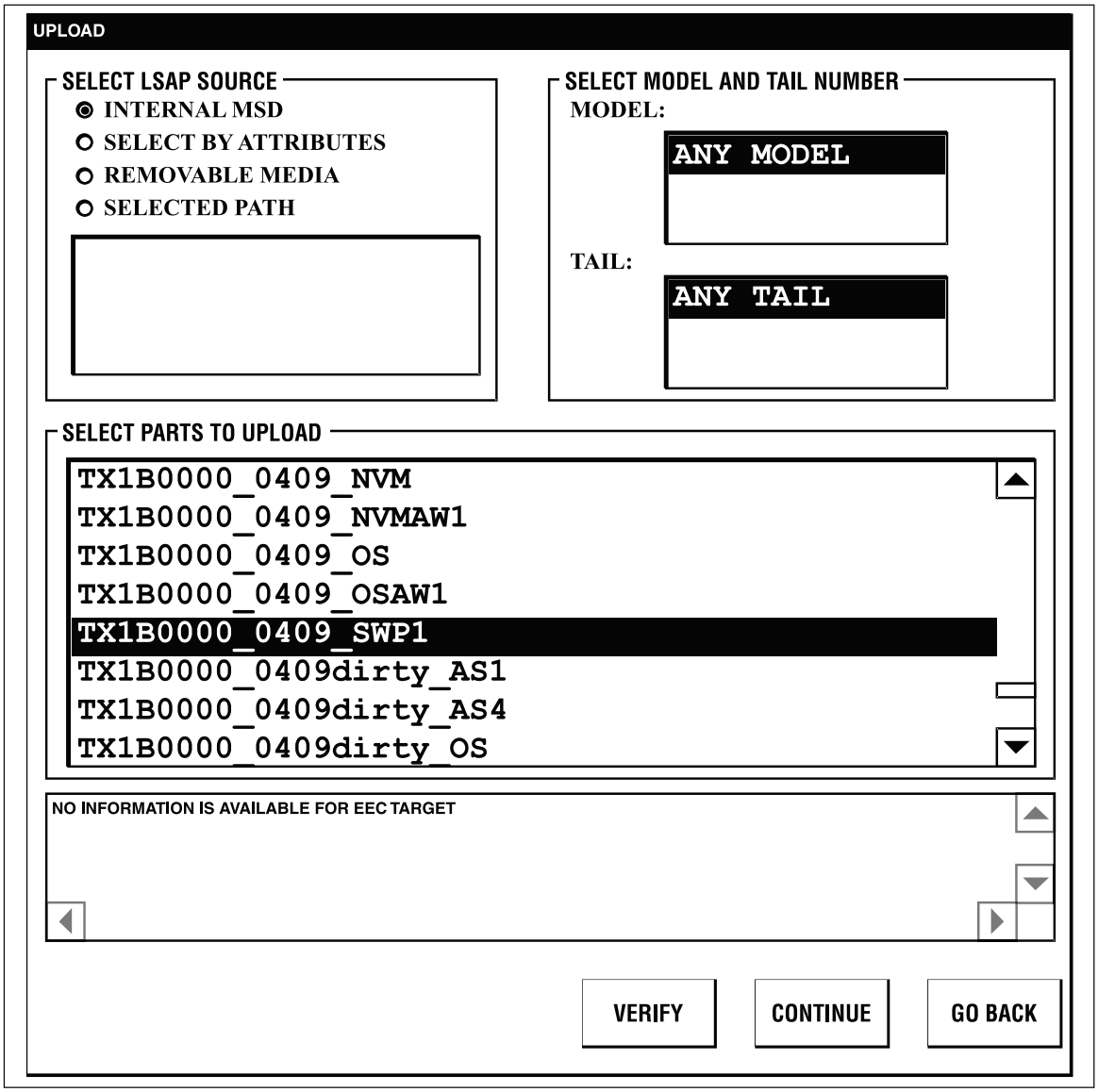


5035506-00

2568022 S0000614816_V1

Electronic Engine Control (EEC) Software Installation Using a PMAT
Figure 203/73-21-00-990-808-G00 (Sheet 8 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



5035507-00

2568023 S0000614817_V1

Electronic Engine Control (EEC) Software Installation Using a PMAT
Figure 203/73-21-00-990-808-G00 (Sheet 9 of 9)

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TASK 73-21-00-470-804-G00

6. EEC Onboard Software Installation Verification

A. General

(1) This task gives the instructions to do the EEC onboard software verification.

(2) This task verifies the software residing in the Onboard Network System's LRU.

(3) There are two ways that you can verify the software in the system:

(a) Use the Max Display System (MDS) screen.

(b) Use the universal maintenance device, SPL-11321.

B. References

Reference	Title
46-13-00-710-802	System Configuration Check (P/B 201)
71-00-00-700-810-G00	Test No. 34 - Menu Mode Hardware Code Entry Special Functions Test (P/B 501)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Prepare for the EEC Software Installation Verification (with MDS/MD)

SUBTASK 73-21-00-860-057-G00

(1) Select the left or right Multi-Function Display (MFD).

(a) Push the SYS button on the MFD control panel, P9.

(b) Push the SEL button on SELECTOR knob on the MFD control panel, P9.

SUBTASK 73-21-00-860-056-G00

(2) Set the applicable EEC MAINT POWER switch on the MISC SYSTEM CTRLS (MSC) page on the MFD to the TEST position (TASK 73-21-00-800-801-G00).

NOTE: The EEC will do a self-test when you set the EEC MAINT POWER switch to the TEST position. Wait for a minimum of 30 seconds to let the EEC complete the test.

F. EEC Onboard Software Installation Verification (with MDS or Maintenance Device)

SUBTASK 73-21-00-860-108-G00

(1) Use the Onboard Network System (ONS) to check the EEC software part numbers.

(a) Do this task: System Configuration Check, TASK 46-13-00-710-802.

NOTE: EEC-1A is the left engine CH A and EEC-1B is the left engine CH B.
EEC-2A is the right engine CH A and EEC-2B is the right engine CH B.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Make sure that the system EEC software part numbers are displayed.

NOTE: If the 77 EEC AVM Part Number shows "NOT LOADED", exit the System Configuration report, wait five minutes, and try to access the report again.

1) If the software part numbers are not correct, do this task: EEC Onboard Software Installation, TASK 73-21-00-470-803-G00.

NOTE: Do the steps again for the software installation to make sure it is correct.

(c) Make sure that you check the menu mode hardware code and auto-detected hardware configuration code (TASK 73-21-00-470-801-G00).

(d) If the menu mode hardware code is not correct, do this task: Test No. 34 - Menu Mode Hardware Code Entry Special Functions Test, TASK 71-00-00-700-810-G00.

(e) If the auto-detected hardware configuration code is not correct, do the fault isolation for the component or system that is reporting the incorrect configuration as follows:

1) Examine the applicable system electrical connections.

2) Make sure that the installed hardware configuration is correct.

3) Do another check to make sure that the auto-detected hardware configuration code is correct (TASK 73-21-00-470-801-G00).

(f) Do these steps to exit:

1) Select LINE MAINTENANCE

2) Select EXIT.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 73-21-00-860-106-G00

(1) If the MFD is used, follow the instructions on the MFD screen to go back to the ONBOARD MAINTENANCE screen.

SUBTASK 73-21-00-860-084-G00

(2) If the Maintenance Device (MD) is used, stop the ONS maintenance browser and disconnect the MD.

SUBTASK 73-21-00-860-065-G00

(3) Set the applicable EEC MAINT POWER switch on the MISC SYSTEM CTRLS (MSC) page on the MFD to the NORM position (TASK 73-21-00-800-801-G00).

——— END OF TASK ———

TASK 73-21-00-800-801-G00

7. EEC Maintenance Power Selection

(Figure 204)

A. General

(1) This task provides the instructions on how to apply maintenance power to the electronic engine control system when the engines are off. Maintenance power is necessary to do engine maintenance functions such as engine ground tests, engine special functions, engine data loading, engine trim balance, functional tests, display engine configuration and data and download the engine shop fault data.

(2) To access the EEC maintenance functions, you must use two control switches on the flight deck. The EEC MAINT power switch is a virtual switch accessed through the multi-functional panel control (MFP) to supply maintenance ground power to the EECs. The GND TEST switch is a physical switch that must be operated for EEC maintenance functions.

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(3) The electronic engine control maintenance power is referred to as the EEC MAINT power. The EEC MAINT power selection has two positions: NORM and TEST. The NORM selection enables normal EEC operation. The TEST selection energizes the EEC when the engine is not in operation.

(4) All EEC controlled ground tests, special functions, data load, or trim balance operations can not be done if the GND TEST switch is not set to ENABLE.

(5) You use the controls on the P9-82 multi-function display (MFD) panel which is located on the forward instrument panel and the P61-4 MAINT/BITE side panel in the flight compartment.

(a) The forward instrument panel has two rotary cursor control knobs, each with a center selector button SEL to select the applicable function. One knob is the CAPT SELECTOR for the L or C display unit position. The other knob is the FO SELECTOR for the C or R display unit position. There are four display units. The captain has the outboard-left display unit and an inboard-center display unit. The first officer has an inboard-center display unit and the outboard-right display unit.

(b) The P61-4 MAINT/BITE panel has one rotary cursor control knob with two positions for MAINT SELECTOR, L or R and a center selector button SEL to select the applicable function. This panel also has the GND TEST switch with the ENABLE and NORM positions.

(c) The rotary cursor control knob is also known as the “Tabber” which moves and highlights the function tabs that you see on the display.

NOTE: If the airplane power is removed while the EEC Maintenance Power is selected, ENG CONTROL and ENG REVERSER status messages will be set without a correlated maintenance message. Refer to this task: Fault Interrogation for a Maintenance Message, TASK 31-65-00-750-801.

B. References

Reference	Title
31-65-00-750-801	Fault Interrogation for a Maintenance Message (P/B 201)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Prepare for the Procedure

SUBTASK 73-21-00-860-086-G00

(1) Make sure that the engine 1 and 2 start levers are in the CUTOFF position.

SUBTASK 73-21-00-860-087-G00

(2) Make sure that the engine 1 and 2 start switches are in the OFF position.

SUBTASK 73-21-00-860-088-G00

(3) Make sure that the engine 1 and 2 forward thrust levers are in the IDLE position.

SUBTASK 73-21-00-860-089-G00

(4) Make sure that the engine 1 and 2 reverse thrust levers are in the forward and down (stowed) position.

E. Electronic Engine Control (EEC) Maintenance Power Selection

SUBTASK 73-21-00-860-001-G00

(1) Turn the rotary cursor control knob on the multi-functional panel control (MFP) to select the CAPT SELECTOR L or C display unit or the FO SELECTOR C or R display unit.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Push the SYS button on the P9-82 multi-function panel control (MFP).

NOTE: The captains outboard left or inboard left display unit or the first officers inboard right or outboard right display unit will show the system's main menu page when you select the SYS button.

SUBTASK 73-21-00-860-090-G00

(2) Turn the rotary cursor control knob clockwise or counter-clockwise until the MAINT function is highlighted.

(a) Push the center selector button SEL to select the MAINT function.

SUBTASK 73-21-00-860-091-G00

(3) Turn the rotary cursor control knob clockwise or counter-clockwise until the MAINT CTRL PGS function is highlighted at top right corner.

(a) Push the center selector button SEL to select the MAINT CTRL PGS function.

SUBTASK 73-21-00-860-092-G00

(4) Turn the rotary cursor control knob clockwise until the MISC SYSTEM CTRLS (MSC) function is highlighted.

(a) Push the center selector button SEL to select the MISC SYSTEM CTRLS (MSC) function.

SUBTASK 73-21-00-860-093-G00

(5) Turn the rotary cursor control knob clockwise until the TEST function diamond is highlighted for ENG 1 or ENG 2 in the EEC MAINT POWER box.

(a) Push the center selector button SEL to select the TEST function to power the EEC for ENG 1 or the ENG 2.

SUBTASK 73-21-00-860-094-G00

(6) Make sure that the STATUS: PWR is shown below ENG 1 or ENG 2.

SUBTASK 73-21-00-860-095-G00

(7) Set the GND TEST switch on the P61-4 MAINT/BITE panel to ENABLE.

NOTE: All EEC controlled ground tests, special functions, data load, or trim balance operations can not be done if the GND TEST switch is not set to ENABLE.

NOTE: You can do the engine ground tests and engine special functions in any order after this step.

F. Put the Airplane Back To the Usual Condition

SUBTASK 73-21-00-860-085-G00

(1) When you have completed the engine ground tests and special functions, set the GND TEST switch on the P61-4 MAINT/BITE panel to the NORM position to turn off the power on the applicable EEC MAINT POWER selection.

SUBTASK 73-21-00-860-096-G00

(2) Turn the rotary cursor control knob clockwise or counter-clockwise until the MAINT CTRL PGS function is highlighted.

(a) Push the center selector button SEL to select the MAINT CTRL PGS function.

SUBTASK 73-21-00-860-097-G00

(3) Turn the rotary cursor control knob clockwise until the MISC SYSTEM CTRLS (MSC) function is highlighted.

(a) Push the center selector button SEL to select the MISC SYSTEM CTRLS (MSC) function.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 73-21-00-860-098-G00

(4) Turn the rotary cursor control knob clockwise until the NORM function diamond is highlighted for ENG 1 or ENG 2 in the EEC MAINT POWER box.

(a) Push the center selector button SEL to select the NORM function to turn off the EEC maintenance power for ENG 1 or ENG 2.

SUBTASK 73-21-00-860-099-G00

(5) Make sure that the STATUS: OFF is shown below ENG 1 or ENG 2.

SUBTASK 73-21-00-860-100-G00

(6) Turn the rotary cursor control knob clockwise or counter-clockwise until the SYS MENU function is highlighted.

(a) Push the center selector button SEL to select SYS MENU screen.

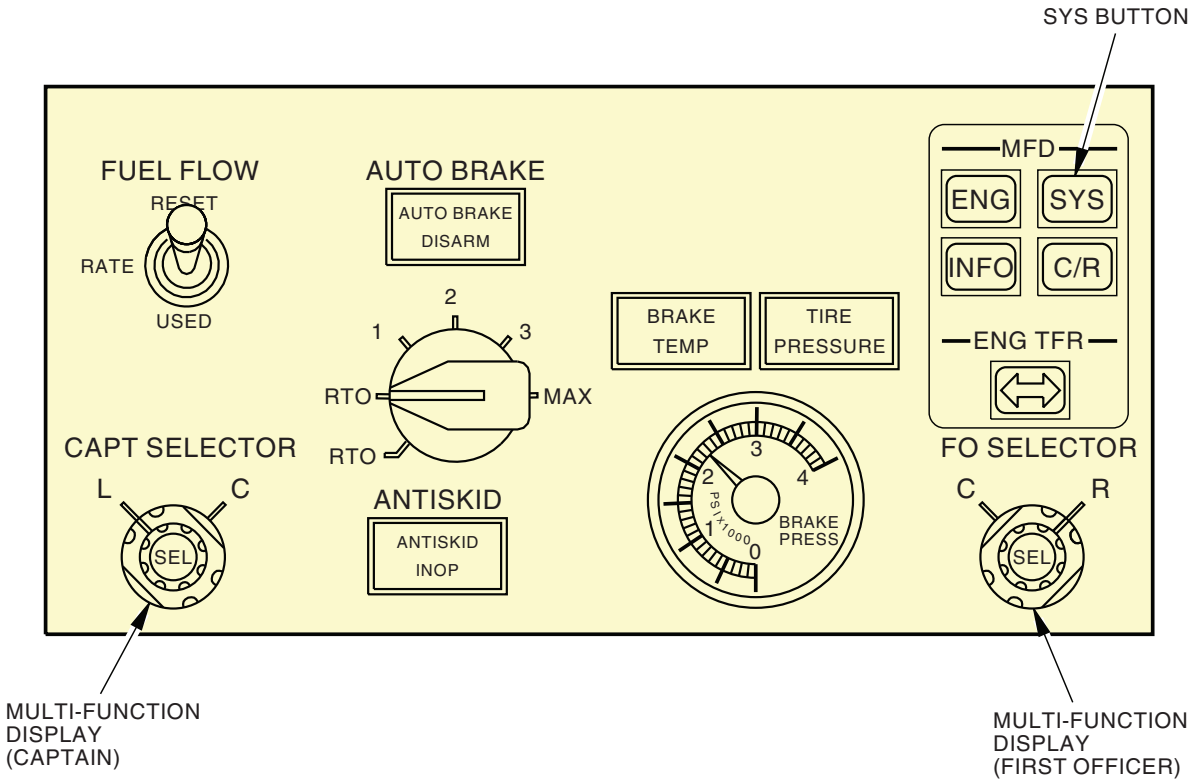
SUBTASK 73-21-00-860-101-G00

(7) Make sure that the MAINT function is highlighted.

(a) Push the SYS button on the multi-functional panel control (MFP) to EXIT.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



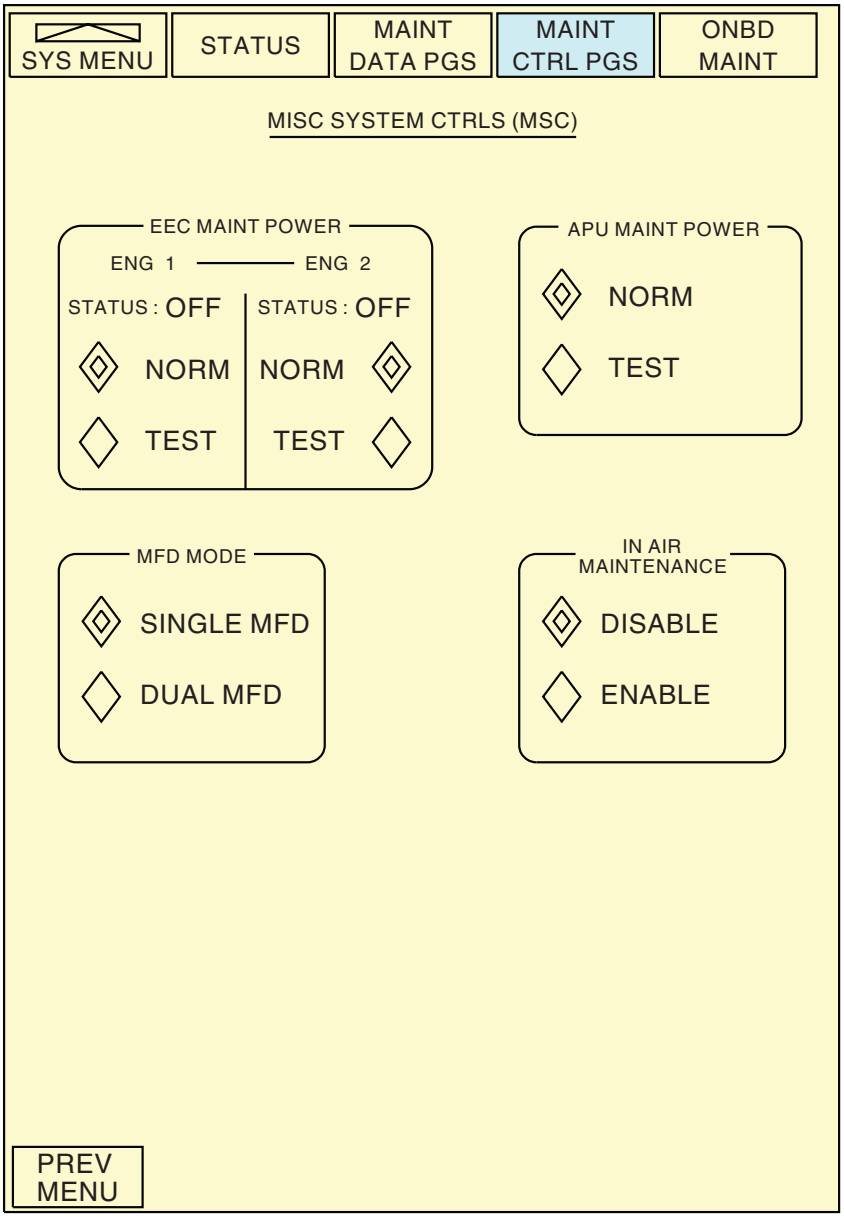
ENGINE DISPLAY CONTROL PANEL

A

2354821 S00061515986_V2

Electronic Engine Control (EEC) Maintenance Power Selection
Figure 204/73-21-00-990-801-G00 (Sheet 1 of 2)

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MISCELLANEOUS SYSTEM CONTROL PAGE

B

2354822 S00061515987_V2

Electronic Engine Control (EEC) Maintenance Power Selection
Figure 204/73-21-00-990-801-G00 (Sheet 2 of 2)

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FUEL CONTROL SYSTEM - INSPECTION/CHECK

TASK 73-21-00-200-801-G00

1. Fuel Control System Inspection

A. General

(1) This task provides the references to inspect the fuel control system components.

B. References

Reference	Title
73-11-06-210-801-G00	Split Control Unit Inspection (P/B 601)
73-11-09-210-801-G00	FMU Fuel Strainer Differential Pressure Transducer Inspection (P/B 601)
73-11-10-210-801-G00	Fuel Filter Differential Pressure Transducer Inspection (P/B 601)
73-11-12-210-801-G00	Main Fuel/Oil Heat Exchanger Differential Pressure Sensor Inspection (P/B 601)
73-21-04-200-801-G00	Fuel Metering Unit (FMU) Inspection (P/B 601)
73-21-05-210-801-G00	Electronic Engine Control Inspection (P/B 601)
73-21-06-210-801-G00	T12 Sensor Inspection (P/B 601)
73-21-08-210-801-G00	Permanent Magnet Alternator (PMA) Inspection (P/B 601)
73-21-09-211-801-G00	Data Entry Plug Inspection (P/B 601)
73-21-11-210-801-G00	Pressure Sub-System Inspection (P/B 601)
73-21-63-210-801-G00	T25 Sensor Inspection (P/B 601)
73-21-64-210-801-G00	T3 Sensor Inspection (P/B 601)
73-21-67-210-801-G00	Fuel Manifold Pressure Transducer Inspection (P/B 601)
73-31-01-210-801-G00	Fuel Flow Transmitter Inspection (P/B 601)

C. Fuel Control System Inspection

SUBTASK 73-21-00-210-023-G00

(1) To inspect the Split Control Unit, do this task: Split Control Unit Inspection, TASK 73-11-06-210-801-G00.

SUBTASK 73-21-00-210-024-G00

(2) To inspect the FMU Strainer Differential Pressure Transducer, do this task: FMU Fuel Strainer Differential Pressure Transducer Inspection, TASK 73-11-09-210-801-G00.

SUBTASK 73-21-00-210-025-G00

(3) To inspect the Fuel Filter Differential Pressure Transducer, do this task: Fuel Filter Differential Pressure Transducer Inspection, TASK 73-11-10-210-801-G00.

SUBTASK 73-21-00-210-026-G00

(4) To inspect the Main Fuel/Oil Heat Exchanger Differential Pressure Sensor, do this task: Main Fuel/Oil Heat Exchanger Differential Pressure Sensor Inspection, TASK 73-11-12-210-801-G00.

SUBTASK 73-21-00-210-027-G00

(5) To inspect the Fuel Metering Unit (FMU), do this task: Fuel Metering Unit (FMU) Inspection, TASK 73-21-04-200-801-G00.

SUBTASK 73-21-00-210-028-G00

(6) To inspect the Electronic Engine Control (EEC), do this task: Electronic Engine Control Inspection, TASK 73-21-05-210-801-G00.

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SUBTASK 73-21-00-210-029-G00

(7) To inspect the T12 Sensor, do this task: T12 Sensor Inspection, TASK 73-21-06-210-801-G00.

SUBTASK 73-21-00-210-030-G00

(8) To inspect the Permanent Magnet Alternator (PMA), do this task: Permanent Magnet Alternator (PMA) Inspection, TASK 73-21-08-210-801-G00.

SUBTASK 73-21-00-210-031-G00

(9) To inspect the Data Entry Plug, do this task: Data Entry Plug Inspection, TASK 73-21-09-211-801-G00.

SUBTASK 73-21-00-210-032-G00

(10) To inspect the Pressure Sub-System, do this task: Pressure Sub-System Inspection, TASK 73-21-11-210-801-G00.

SUBTASK 73-21-00-210-033-G00

(11) To inspect the T25 Sensor, do this task: T25 Sensor Inspection, TASK 73-21-63-210-801-G00.

SUBTASK 73-21-00-210-034-G00

(12) To inspect the T3 Sensor, do this task: T3 Sensor Inspection, TASK 73-21-64-210-801-G00.

SUBTASK 73-21-00-210-035-G00

(13) To inspect the Fuel Manifold Pressure Transducer, do this task: Fuel Manifold Pressure Transducer Inspection, TASK 73-21-67-210-801-G00.

SUBTASK 73-21-00-210-036-G00

(14) To inspect the Fuel Flow Transmitter, do this task: Fuel Flow Transmitter Inspection, TASK 73-31-01-210-801-G00.

——— END OF TASK ———

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FUEL METERING UNIT - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Fuel metering unit removal

(2) Fuel metering unit installation.

TASK 73-21-04-000-801-G00

2. Fuel Metering Unit Removal

(Figure 401 and Figure 402)

A. General

(1) This task provides the instructions on how to remove the Fuel Metering Unit (FMU).

(2) A minimum of two persons are required to handle the FMU if the jack stand and the FMU adapter support equipment are not used.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
73-11-01-000-801-G00	Main Fuel Pump Removal (P/B 401)
73-11-02-000-801-G00	Fuel Filter Cartridge Removal (P/B 401)
73-11-04-000-801-G00	Fuel Metering Unit Strainer Removal (P/B 401)
73-11-09-000-801-G00	FMU Fuel Strainer Differential Pressure Transducer Removal (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14500	Stand - Jack Part #: 9471M17P01 Supplier: 58828
SPL-14528	Adapter - FMU Part #: 956A1613G01 Supplier: 58828
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Prepare for the Removal

SUBTASK 73-21-04-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-21-04-860-002-G00

(2) Open the applicable circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 73-21-04-010-001-G00

(3) On the applicable engine, open the left fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

G. Fuel Metering Unit Removal

SUBTASK 73-21-04-680-003-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

CAUTION A LARGE VOLUME OF FUEL WILL DRAIN FROM THE FUEL SUPPLY SYSTEM. TO PREVENT A FUEL SPILL, YOU MUST USE A MINIMUM OF TWO CONTAINERS TO DRAIN THE FUEL SYSTEM.

(1) Drain the main fuel pump (TASK 73-11-01-000-801-G00).

SUBTASK 73-21-04-680-004-G00

(2) Drain the FMU strainer (TASK 73-11-04-000-801-G00).

SUBTASK 73-21-04-680-005-G00

(3) Drain the main fuel filter housing (TASK 73-11-02-000-801-G00).

SUBTASK 73-21-04-010-002-G00

(4) Remove the FMU fuel strainer DP transducer [2] (TASK 73-11-09-000-801-G00).

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SUBTASK 73-21-04-030-001-G00

CAUTION PUT A TAG ON THE CONNECTOR IF YOU REMOVE MORE THAN ONE CONNECTOR IN THE SAME AREA. IF YOU INCORRECTLY CONNECT THE CONNECTORS, IT CAN CAUSE DAMAGE TO THE EQUIPMENT.

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(5) Use a teflon-jawed pliers, STD-664, to disconnect the electrical connectors from the FMU [1] as follows (TASK 70-00-01-910-803-G00):

(a) Disconnect the electrical connector (HJ5B) [4].

(b) Disconnect the electrical connector (HJ5A) [3].

(c) Disconnect the electrical connector (W300) [5].

(d) Install the protection covers on the electrical connectors and the electrical receptacles on the FMU [1].

SUBTASK 73-21-04-010-003-G00

(6) Disconnect the fuel tube [6] from the interface PB2 line on the FMU [1] as follows:

(a) Remove the four bolts [7] from the fuel tube [6].

(b) Disconnect the fuel tube [6].

(c) Remove and discard the gasket [8].

(d) Install the protection covers on the fuel tube [6] and the FMU [1] open port.

SUBTASK 73-21-04-030-002-G00

(7) Disconnect the fuel tube [9] from the interface PHS line on the FMU [1] as follows:

(a) Loosen the coupling nut [10] from the pipe coupling spherical [11].

(b) Disconnect the fuel tube [9].

(c) Install the protection covers on the fuel tube [9] and the FMU [1] open port.

SUBTASK 73-21-04-030-003-G00

(8) Disconnect the fuel tube [12] from the interface PB line on the FMU [1] as follows:

(a) Remove the four bolts [13] from the fuel tube [12].

(b) Disconnect the fuel tube [12].

(c) Remove and discard the gasket [14].

(d) Install the protection covers on the fuel tube [12] and the FMU [1] open port.

SUBTASK 73-21-04-010-004-G00

(9) Disconnect the fuel tube [15] from the interface PS line on the FMU [1] as follows:

(a) Remove the four bolts [16] from the fuel tube [15].

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(b) Disconnect the fuel tube [15].

(c) Remove and discard the gasket [17].

(d) Install the protection covers on the fuel tube [15] and the FMU [1] open port.

SUBTASK 73-21-04-010-005-G00

(10) Disconnect the fuel tube [18] from the interface P22 line on the FMU [1] as follows:

(a) Remove the four bolts [19] from the fuel tube [18].

(b) Disconnect the fuel tube [18].

(c) Remove and discard the gasket [20].

(d) Install the protection covers on the fuel tube [18] and the FMU [1] open port.

SUBTASK 73-21-04-490-001-G00

WARNING BE CAREFUL WHEN YOU MOVE THE FUEL METERING UNIT WITHOUT AID. THE FUEL METERING UNIT IS HEAVY. IF YOU DO NOT OBEY, INJURIES TO PERSONNEL OR DAMAGE TO THE FUEL METERING UNIT CAN OCCUR.

(11) Remove the FMU [1] as follows:

NOTE: The fuel metering unit weighs approximately 34.0 lb (15.4 kg).

NOTE: If the two-person alternative method is used, you do not have to install the adapter, SPL-14528, and jack stand, SPL-14500.

(a) If necessary, move the jack stand, SPL-14500, to the correct position.

PTF ALL PRE SB 737-CFM-LEAP-1B-72-0181

(b) If necessary, install the adapter, SPL-14528, on the jack stand, SPL-14500.

(c) If necessary, install the four captive bolts [21] of the adapter, SPL-14528, to the bolt holes of the FMU [1].

(d) Remove the three bolts [22] and the bolt [23] with integrated washers from the FMU [1].

(e) Remove nut [24] from the FMU [1].

(f) Install the protection covers on the FMU [1] open ports.

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(g) Remove the three bolts [22] and the bolt [23] with integrated washers from the FMU [1].

(h) Remove the FMU [1] from the engine.

(i) If necessary, remove the four captive bolts [21] of the adapter, SPL-14528, from the bolt holes of the FMU [1].

(j) If necessary, remove the FMU [1] from the adapter, SPL-14528.

(k) Install the protection covers on the FMU [1] open ports.

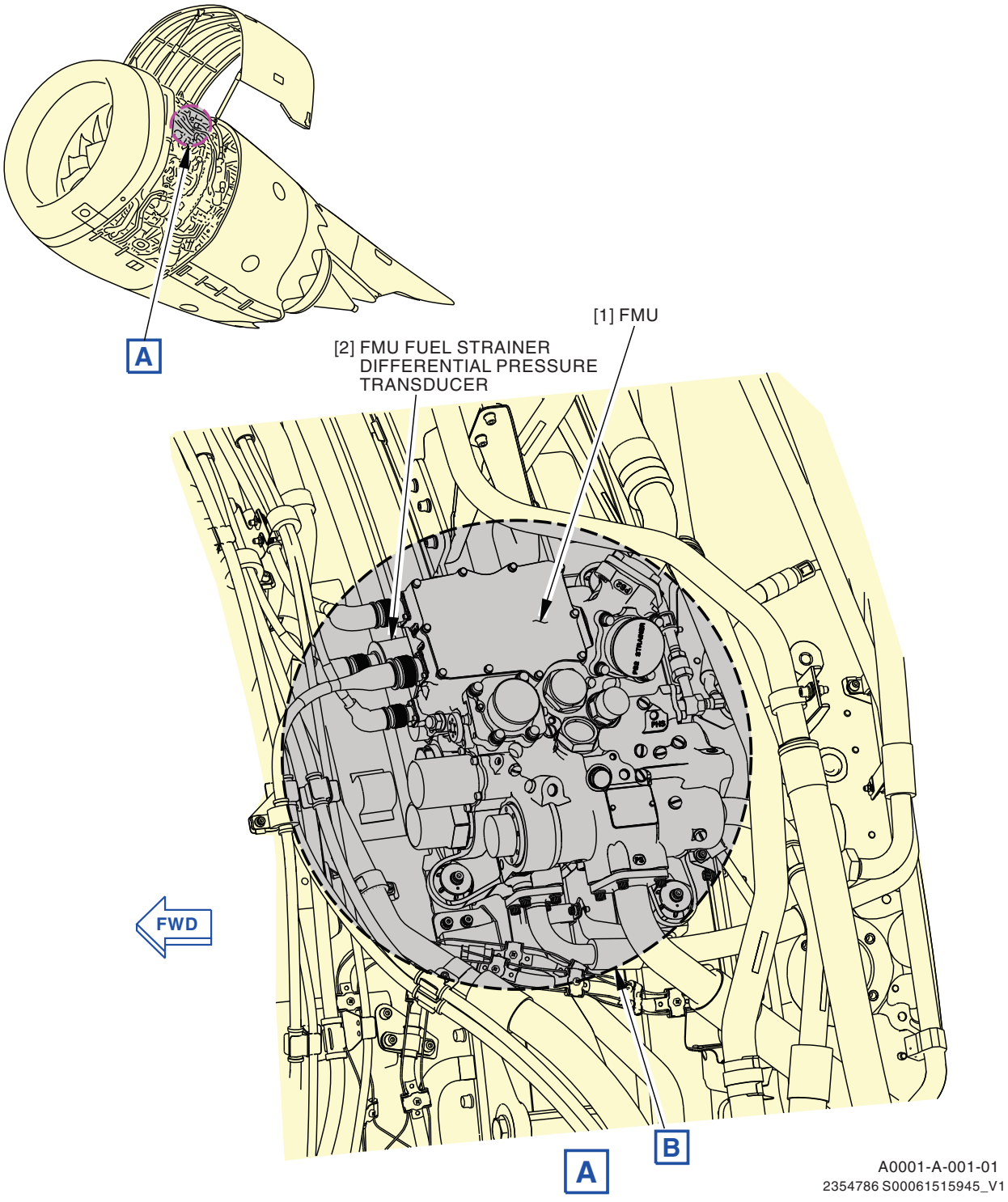
PTF ALL

SUBTASK 73-21-04-560-001-G00

(12) If necessary, move the jack stand, SPL-14500, and the FMU [1] away from the engine.

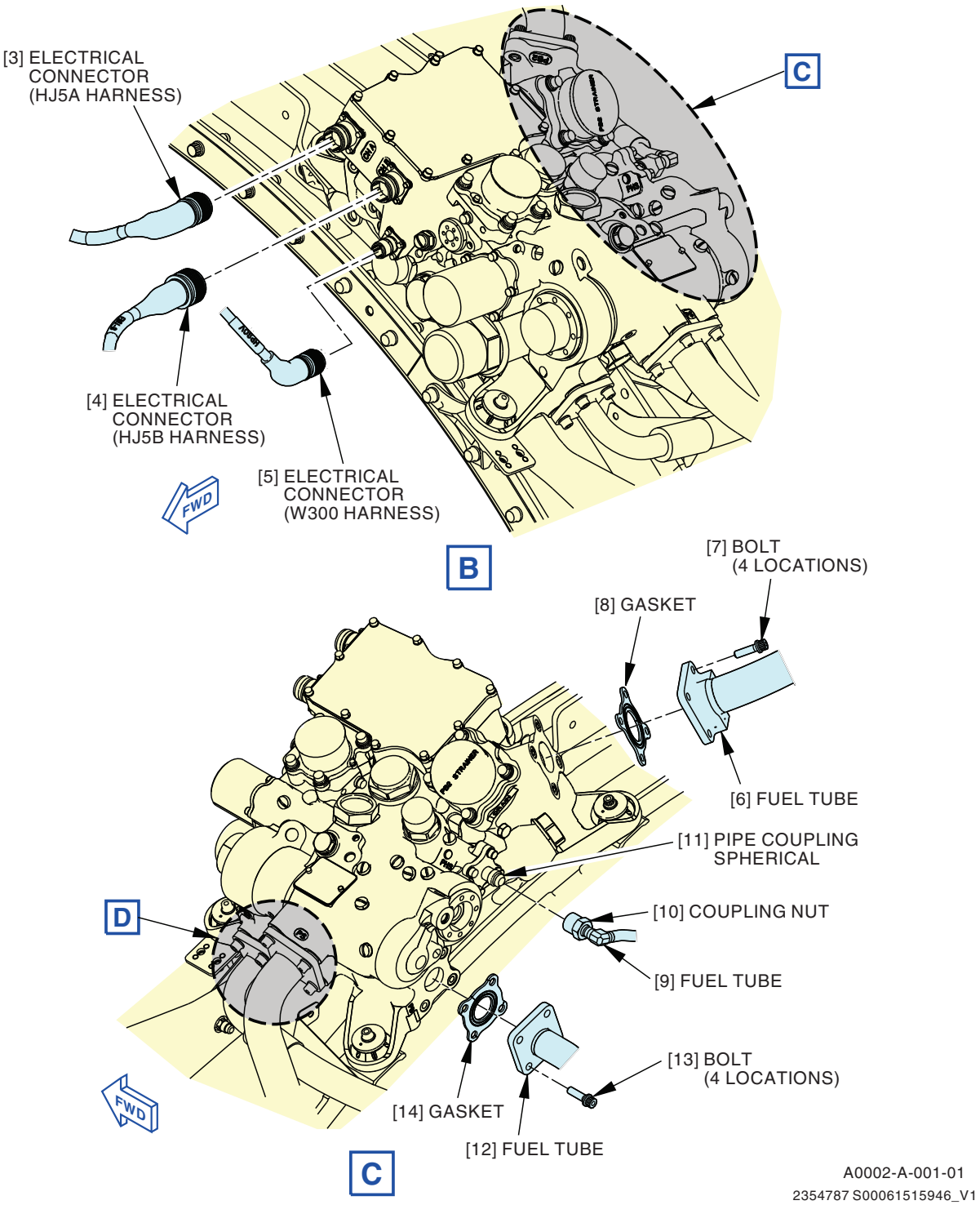
END OF TASK

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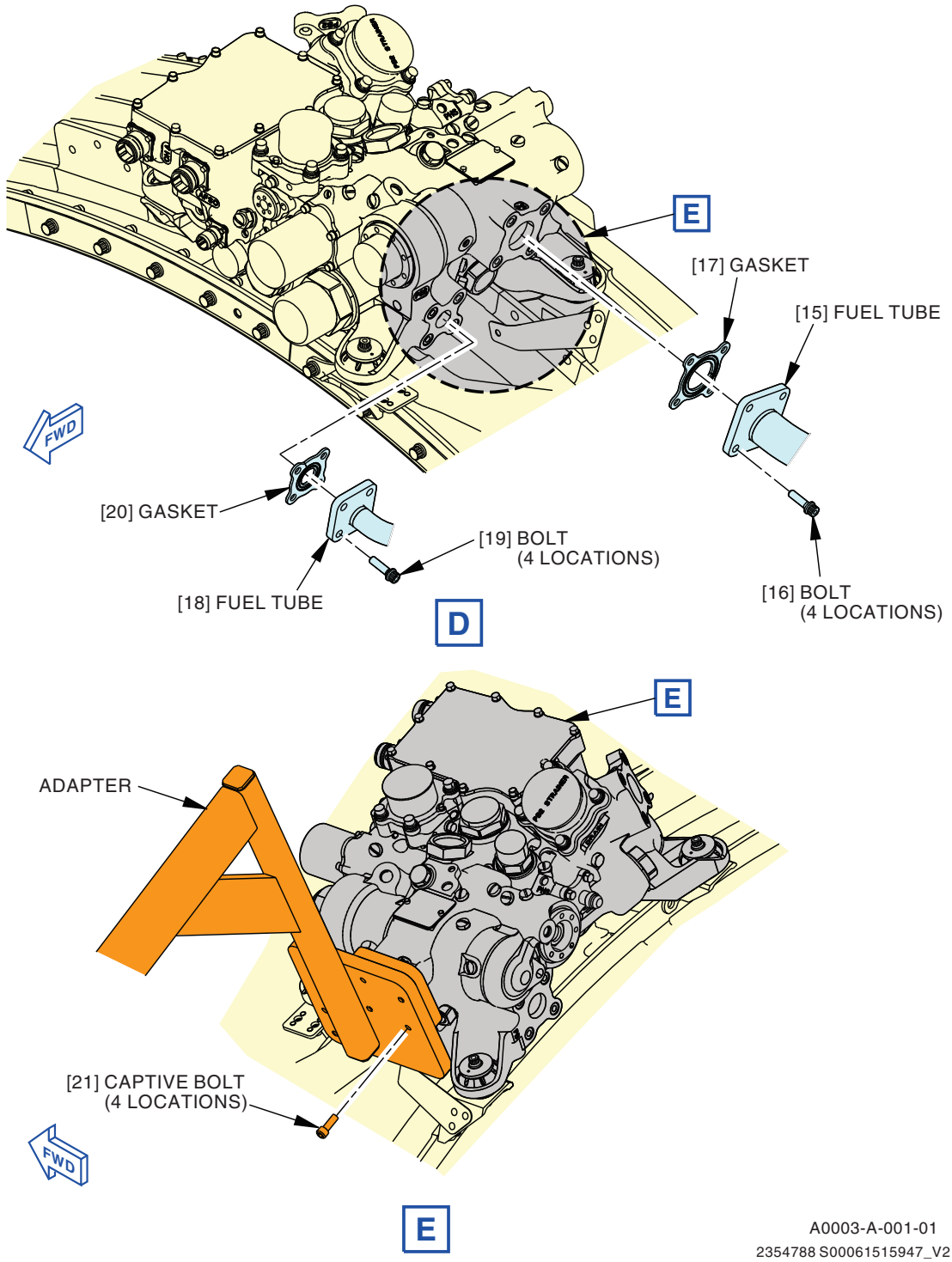
Fuel Metering Unit Installation
Figure 401/73-21-04-990-802-G00 (Sheet 1 of 3)

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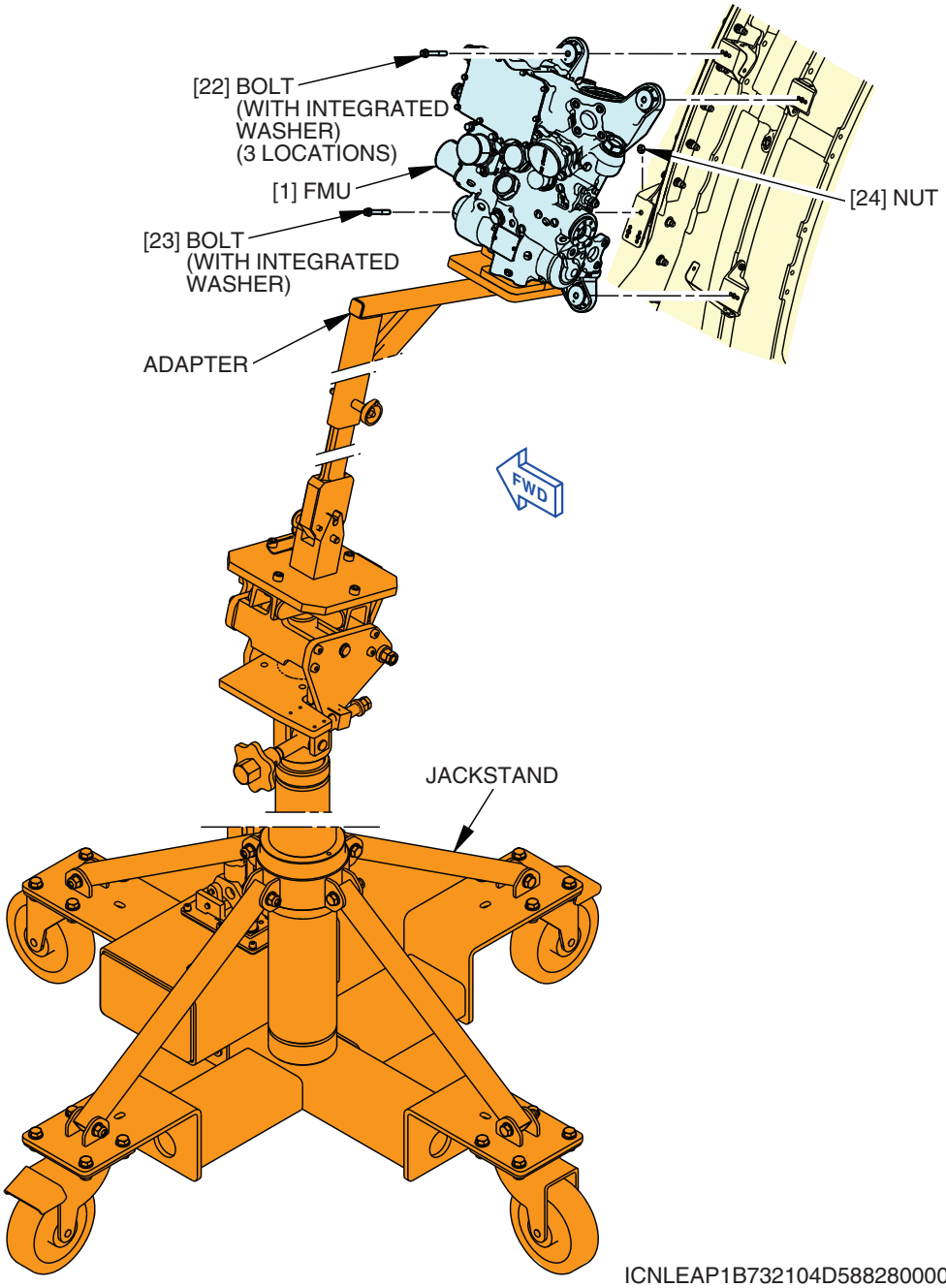
Fuel Metering Unit Installation
Figure 401/73-21-04-990-802-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



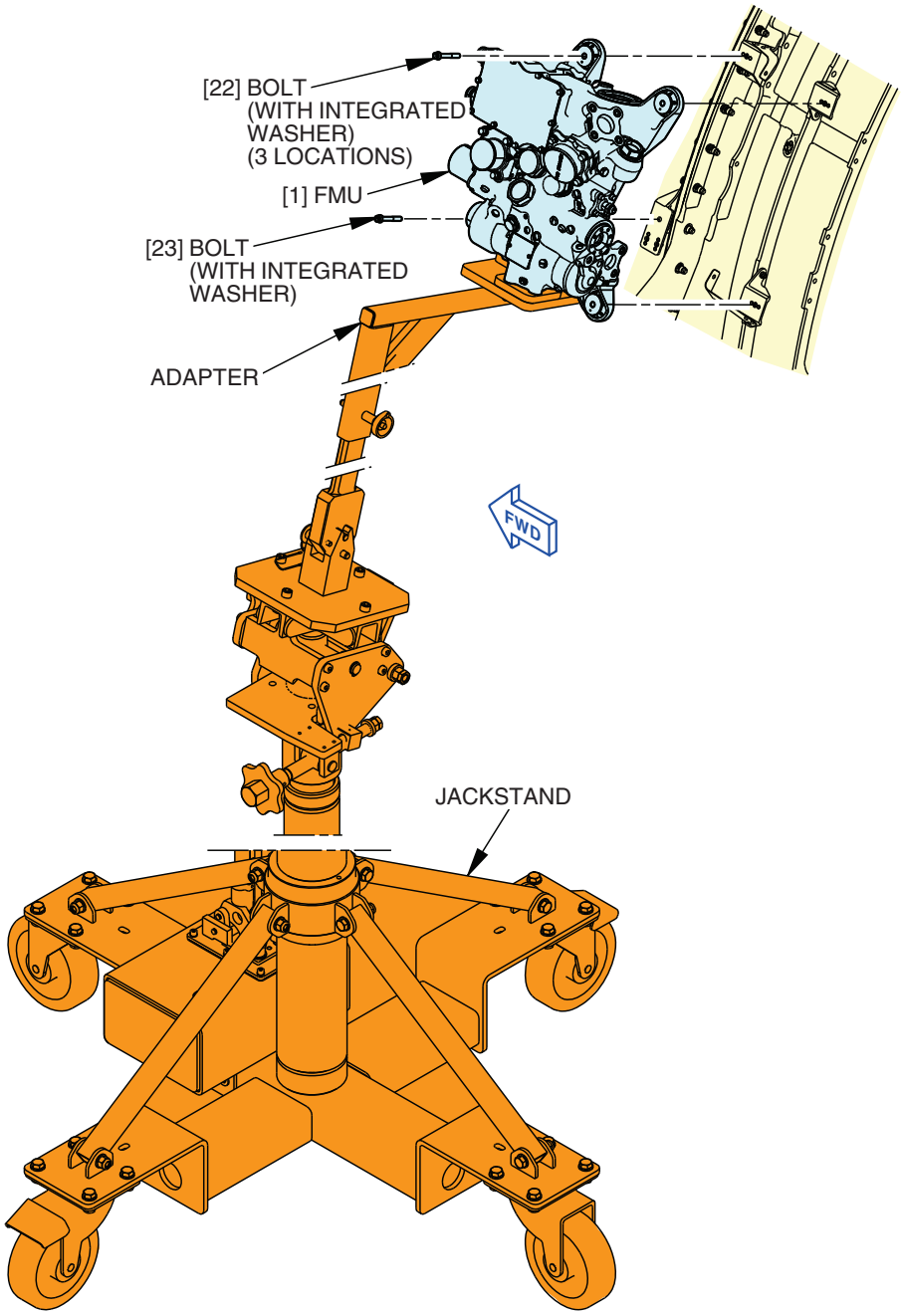
Fuel Metering Unit Installation
Figure 401/73-21-04-990-802-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FMU Line Replaceable Unit Fixture Installation
Figure 402/73-21-04-990-803-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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3112627 S0000851128_V1

FMU Line Replaceable Unit Fixture Installation
Figure 402/73-21-04-990-803-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 73-21-04-400-801-G00

3. Fuel Metering Unit Installation

(Figure 401 and Figure 402)

A. General

(1) This task gives the instructions on how to install the Fuel Metering Unit (FMU).

(2) You must do the tests in the Power Plant Test Reference Table after you install the FMU.

(3) A minimum of two persons are required to handle the FMU if the jack stand and the FMU adapter support equipment are not used.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-00-00-910-802-G00	Start the Engine (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
73-11-09-400-801-G00	FMU Fuel Strainer Differential Pressure Transducer Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14500	Stand - Jack Part #: 9471M17P01 Supplier: 58828
SPL-14528	Adapter - FMU Part #: 956A1613G01 Supplier: 58828
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	
D00601 [CP2101]	Vaseline - Graphite Mineral	
D00672 [CP5070]	Vaseline - Pure Mineral	V V-P-236
G51335 [CP2674]	Lint-Free Cloth	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	FMU	73-21-04-01-010	PTF ALL
8	Gasket	73-21-04-01-070	PTF ALL
14	Gasket	73-21-04-01-100	PTF ALL
17	Gasket	73-21-04-01-120	PTF ALL
20	Gasket	73-21-04-01-140	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine

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(Continued)

Zone	Area
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

H. Fuel Metering Unit Installation

SUBTASK 73-21-04-490-002-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

WARNING BE CAREFUL WHEN YOU MOVE THE FUEL METERING UNIT WITHOUT AID. THE FUEL METERING UNIT IS HEAVY. IF YOU DO NOT OBEY, INJURIES TO PERSONNEL OR DAMAGE TO THE FUEL METERING UNIT CAN OCCUR.

(1) If necessary, install the FMU [1] on the jack stand, SPL-14500, as follows:

NOTE: The FMU weighs approximately 34.0 lb (15.4 kg).

NOTE: If the two-person alternative method is used, you do not have to install the adapter, SPL-14528, and jack stand, SPL-14500.

(a) Make sure that the mating faces are clean and clear of unwanted materials.

(b) Install the adapter, SPL-14528, on the jack stand, SPL-14500.

(c) Put the FMU [1] into its position on the jack stand, SPL-14500.

(d) Install the four captive bolts [21] of the adapter, SPL-14528, to the bolt holes of the FMU [1].

(e) Move the jack stand, SPL-14500, and the FMU [1] into the correct position.

PTF ALL PRE SB 737-CFM-LEAP-1B-72-0181

SUBTASK 73-21-04-410-001-G00

(2) Install the FMU [1] on the engine as follows:

(a) Remove the protection covers from the FMU [1] ports.

(b) Lubricate the threads and pressure faces of the three bolts [22] and the nut [24] with engine oil, D00599 [CP2442].

(c) Align the FMU [1] with the bolt holes on the engine.

(d) Install the three bolts [22], the bolt [23], and the nut [24] to attach the FMU [1] to the engine.

NOTE: The bolts have integrated washers.

(e) Tighten the three bolts [22] and bolt [23] in a criss-cross pattern to 67.45 in-lb (7.6 N·m) - 74.55 in-lb (8.4 N·m).

(f) Tighten the nut [24] to 109.25 in-lb (12.34 N·m) - 120.75 in-lb (13.64 N·m).

(g) Remove the unwanted engine oil, D00599 [CP2442].

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PTF ALL PRE SB 737-CFM-LEAP-1B-72-0181 (Continued)

(h) If necessary, remove the four captive bolts [21] of the adapter, SPL-14528, from the bolt holes of the FMU [1].

PTF ALL POST SB 737-CFM-LEAP-1B-72-0181

SUBTASK 73-21-04-410-009-G00

(3) Install the FMU [1] on the engine as follows:

(a) Remove the protection covers from the FMU [1] ports.

(b) Lubricate the threads and the pressure faces of the three bolts [22] and the bolt [23] with engine oil, D00599 [CP2442].

(c) Align the FMU [1] with the bolt holes on the engine.

(d) Install the three bolts [22] and the bolt [23] to attach the FMU [1] to the engine.

NOTE: The bolts have integrated washers.

(e) Tighten the three bolts [22] and bolt [23] in a criss-cross pattern to 67.45 in-lb (7.6 N·m) - 74.55 in-lb (8.4 N·m).

(f) Remove the unwanted engine oil, D00599 [CP2442].

PTF ALL

SUBTASK 73-21-04-160-001-G00

(4) Connect the fuel tube [18] to the interface P22 line on the FMU [1] as follows:

(a) Remove the protection covers from the fuel tube [18] and the FMU [1] port.

(b) Make sure that the mating faces are clean and clear of unwanted materials.

(c) Lubricate the threads and the pressure faces of the four bolts [19] with graphite mineral vaseline, D00601 [CP2101].

(d) Lubricate each side of the elastomer part of a new gasket [20] with vaseline, D00672 [CP5070], or engine oil, D00599 [CP2442].

(e) Install the new gasket [20] and the fuel tube [18] on the FMU [1].

(f) Install the four bolts [19] on the fuel tube [18].

(g) Tighten the four bolts [19] in a criss-cross pattern to 109.3 in-lb (12.35 N·m) - 120.8 in-lb (13.65 N·m).

(h) Remove the unwanted vaseline, D00672 [CP5070], and the graphite mineral vaseline, D00601 [CP2101], with a lint-free cloth, G51335 [CP2674].

SUBTASK 73-21-04-410-002-G00

(5) Connect the fuel tube [15] to the interface PS line on the FMU [1] as follows:

(a) Remove the protection covers from the fuel tube [15] and the FMU [1] port.

(b) Make sure that the mating faces are clean and clear of unwanted materials.

(c) Lubricate the threads and the pressure faces of the four bolts [16] with graphite mineral vaseline, D00601 [CP2101].

(d) Lubricate each side of the elastomer part of a new gasket [17] with vaseline, D00672 [CP5070], or engine oil, D00599 [CP2442].

(e) Install the new gasket [17] and the fuel tube [15] on the FMU [1].

(f) Install the four bolts [16] on the fuel tube [15].

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(g) Tighten the four bolts [16] in a criss-cross pattern to 109.3 in-lb (12.35 N·m) - 120.8 in-lb (13.65 N·m).

(h) Remove the unwanted vaseline, D00672 [CP5070], and the graphite mineral vaseline, D00601 [CP2101], with a lint-free cloth, G51335 [CP2674].

SUBTASK 73-21-04-410-003-G00

(6) Connect the fuel tube [12] to the interface PB line on the FMU [1] as follows:

(a) Remove the protection covers from the fuel tube [12] and the FMU [1] port.

(b) Make sure that the mating faces are clean and clear of unwanted materials.

(c) Lubricate the threads and the pressure faces of the four bolts [13] with graphite mineral vaseline, D00601 [CP2101].

(d) Lubricate each side of the elastomer part of a new gasket [14] with vaseline, D00672 [CP5070], or engine oil, D00599 [CP2442].

(e) Install the new gasket [14] and the fuel tube [12] on the FMU [1].

(f) Install the four bolts [13] on the fuel tube [12].

(g) Tighten the four bolts [13] in a criss-cross pattern to 109.3 in-lb (12.35 N·m) - 120.8 in-lb (13.65 N·m).

(h) Remove the unwanted vaseline, D00672 [CP5070], and the graphite mineral vaseline, D00601 [CP2101], with a lint-free cloth, G51335 [CP2674].

SUBTASK 73-21-04-410-004-G00

(7) Connect the fuel tube [9] to the interface PHS line on the FMU [1] as follows:

(a) Remove the protection covers from the fuel tube [9] and the FMU [1] port.

(b) Make sure that the mating faces are clean and clear of unwanted materials.

(c) Lubricate the threads of the pipe coupling spherical [11] with engine oil, D00599 [CP2442].

(d) Engage the coupling nut [10] on the pipe coupling spherical [11].

(e) Tighten the coupling nut [10] to 270.0 in-lb (30.5 N·m) - 300.0 in-lb (33.9 N·m).

SUBTASK 73-21-04-410-005-G00

(8) Connect the fuel tube [6] to the interface PB2 line on the FMU [1] as follows:

(a) Remove the protection covers from the fuel tube [6] and the FMU [1] port.

(b) Make sure that the mating faces are clean and clear of unwanted materials.

(c) Lubricate the threads and the pressure faces of the four bolts [7] with graphite mineral vaseline, D00601 [CP2101].

(d) Lubricate each side of the elastomer part of a new gasket [8] with vaseline, D00672 [CP5070], or engine oil, D00599 [CP2442].

(e) Install the new gasket [8] and the fuel tube [6] on the FMU [1].

(f) Install the four bolts [7] on the fuel tube [6].

(g) Tighten the four bolts [7] in a criss-cross pattern to 109.3 in-lb (12.35 N·m) - 120.8 in-lb (13.65 N·m).

(h) Remove the unwanted vaseline, D00672 [CP5070], and the graphite mineral vaseline, D00601 [CP2101], with a lint-free cloth, G51335 [CP2674].

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SUBTASK 73-21-04-430-001-G00

CAUTION PUT A TAG ON THE CONNECTOR IF YOU REMOVE MORE THAN ONE CONNECTOR IN THE SAME AREA. IF YOU INCORRECTLY CONNECT THE CONNECTORS, IT CAN CAUSE DAMAGE TO THE EQUIPMENT.

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION DO NOT USE PLIERS THAT HAVE METAL JAWS TO TIGHTEN THE ELECTRICAL CONNECTOR. DAMAGE TO THE ELECTRICAL CONNECTOR CAN OCCUR.

(9) Use a teflon-jawed pliers, STD-664, to connect the electrical connectors to the FMU [1] as follows (TASK 70-00-01-910-803-G00):

NOTE: It is physically possible to connect the electrical connector (HJ5A) or electrical connector (HJ5B) to the FMU Fuel Strainer Differential Pressure Transducer location. If these connectors are not installed at the correct locations, the following maintenance messages will be set when you do the test Fuel Metering Unit Installation Test:

- 73-4209X - FUEL METERING VALVE (FMV) TORQUE MOTOR CURRENT IS OUT OF RANGE

- 73-4214X - FUEL METERING VALVE (FMV) POSITION SENSOR IS OUT OF RANGE

- 73-4231X - FUEL STRAINER DELTA PRESSURE (FSDP) SENSOR IS OUT OF RANGE

- 73-4194X - ELECTRONIC OVERSPEED TEST FAILED

where X indicates the Electronic Engine Control (EEC) channel detecting the fault. If these faults are set during the FMU installation test, do a check the harness plug installed locations before fault isolation.

(a) Remove the protection covers from the electrical connectors..

(b) Remove the protection covers from the electrical receptacles on the FMU [1].

(c) Connect the electrical connector (W300) [5] to the FMU [1].

(d) Connect the electrical connector (HJ5A) [3] to the FMU [1].

(e) Connect the electrical connector (HJ5B) [4] to the FMU [1].

SUBTASK 73-21-04-410-006-G00

(10) Do this task: FMU Fuel Strainer Differential Pressure Transducer Installation, TASK 73-11-09-400-801-G00.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 73-21-04-410-007-G00

(1) On the applicable engine, close the left fan cowl panel (TASK 71-11-04-410-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

SUBTASK 73-21-04-860-003-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

SUBTASK 73-21-04-860-004-G00

(3) Remove the safety tags and close the applicable circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

J. Fuel Metering Unit Installation Test

SUBTASK 73-21-04-730-002-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

NOTE: If the FMU installed has not previously operated on the engine, the first engine start after FMU installation may require special start procedures due to residual air trapped within the FMU EOS fuel circuit.

SUBTASK 73-21-04-800-001-G00

(2) If the engine fails to start and the maintenance message 73-4235X - FUEL METERING UNIT SHUTOFF VALVE IS FAILED CLOSED is set, do these steps:

(a) Do the engine start procedure up to when the START LEVER is moved to IDLE (TASK 71-00-00-910-802-G00).

(b) When the engine is at 25% N2 or the maximum motoring speed and 'MOTORING' is no longer displayed on the N2 dial, toggle the START LEVER switch between IDLE and CUTOFF for five seconds (approximately 8 cycles) then put the START LEVER to IDLE.

(c) If the engine did not start, repeat the procedure.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FUEL METERING UNIT - INSPECTION/CHECK

1. General

A. This procedure has one task:

(1) Fuel Metering Unit (FMU) inspection.

TASK 73-21-04-200-801-G00

2. Fuel Metering Unit (FMU) Inspection

(Figure 601)

A. General

(1) This task provides the instructions on how to examine the Fuel Metering Unit (FMU).

(2) The FMU is located on the fan case at the 11:00 o'clock position.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
73-21-04-000-801-G00	Fuel Metering Unit Removal (P/B 401)
73-21-04-400-801-G00	Fuel Metering Unit Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

F. Prepare for the Inspection

SUBTASK 73-21-04-860-005-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

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SUBTASK 73-21-04-860-006-G00

(2) For the left engine, open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

SUBTASK 73-21-04-860-007-G00

(3) For the right engine, open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 73-21-04-010-006-G00

(4) On the applicable engine, open the left fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

G. Fuel Metering Unit (FMU) Inspection

SUBTASK 73-21-04-210-001-G00

(1) Look for signs of fuel leakage between the FMU and the fuel tubes.

(a) If you see signs of fuel leakage, replace the gaskets between the FMU and the fuel tubes as necessary (TASK 73-21-04-000-801-G00 and TASK 73-21-04-400-801-G00).

SUBTASK 73-21-04-280-001-G00

(2) Make sure that the FMU is not loose.

(a) If it is necessary, tighten the bolts that attach the FMU to the engine (TASK 73-21-04-400-801-G00).

SUBTASK 73-21-04-210-002-G00

(3) Look for cracks on the FMU.

(a) If you see cracks, replace the FMU (TASK 73-21-04-000-801-G00 and TASK 73-21-04-400-801-G00).

SUBTASK 73-21-04-280-002-G00

(4) Make sure that the electrical connectors are not loose.

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(a) If it is necessary, use a teflon-jawed pliers, STD-664 to tighten a loose connector.

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H. Put the Airplane Back to Its Usual Condition

SUBTASK 73-21-04-410-008-G00

(1) On the applicable engine, close the left fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

SUBTASK 73-21-04-860-008-G00

(2) For the left engine, remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

SUBTASK 73-21-04-860-009-G00

(3) For the right engine, remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 73-21-04-860-010-G00

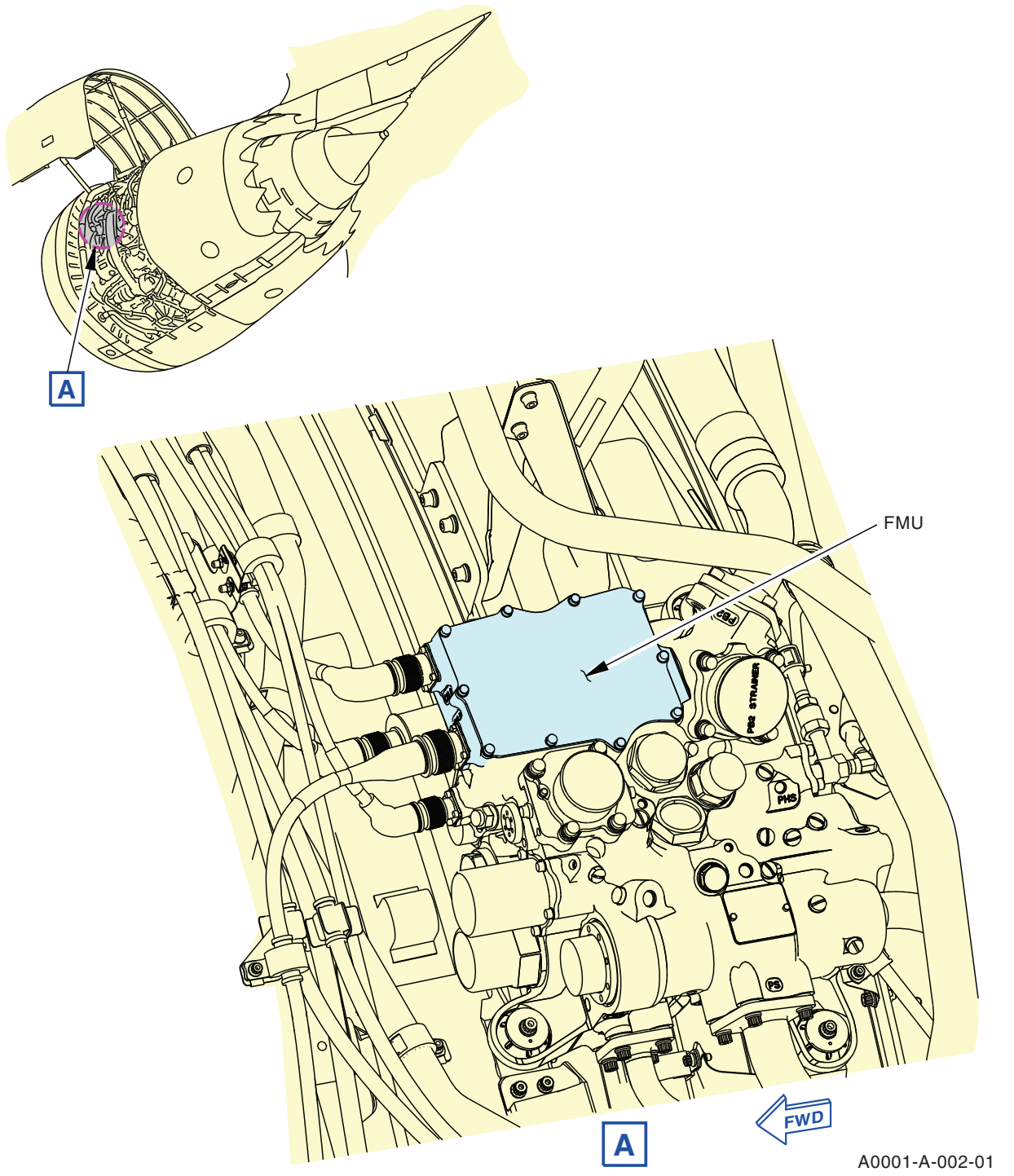
(4) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

END OF TASK

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Fuel Metering Unit Inspection
Figure 601/73-21-04-990-804-G00

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ELECTRONIC ENGINE CONTROL (EEC) - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

NOTE: Do either task to download engine data from the Electronic Engine Controls (EECs).

(1) EEC data download (on-wing) using the Onboard Network System (ONS)

NOTE: This task is recommended. It does not require opening the engine fan cowl panels.

(2) EEC data download (on-wing) using a Portable Maintenance Access Terminal (PMAT)

NOTE: Use this task if the ONS is not able to connect to the EECs.

(3) EEC data download (on wing) using the Engine Health Management (EHM) function.

NOTE: This task is recommended for the download of detailed fault data.

B. For general information for software installation times and data loaders, do this task: On-Airplane Software Installation, TASK 20-15-11-470-801.

TASK 73-21-05-970-803-G00

2. Electronic Engine Control (EEC) Data Download (On-Wing)(Selection)

A. General

(1) This procedure gives the mechanics the task selection for electronic engine control (EEC) data download on-wing using different methods.

B. Electronic Engine Control (EEC) Data Download, On-Wing (Selection)

SUBTASK 73-21-05-970-005-G00

(1) Do one of these tasks to download the electronic engine control (EEC) data from an on-wing engine:

(a) Do this task: Electronic Engine Control (EEC) Data Download (On-Wing) Using a PMAT, TASK 73-21-05-970-801-G00.

(b) Do this task: Electronic Engine Control (EEC) Data Download (On Wing) Using the ONS, TASK 73-21-05-970-802-G00.

(c) Do this task: Electronic Engine Control (EEC) Data Download (On Wing) Using the Engine Health Management (EHM) Function, TASK 73-21-05-970-804-G00.

——— END OF TASK ———

TASK 73-21-05-970-802-G00

3. Electronic Engine Control (EEC) Data Download (On Wing) Using the ONS

(Figure 201)

A. General

(1) This task gives the instructions on how to download the EEC data with the Onboard Network System (ONS).

(2) You will use the Onboard Maintenance Function (OMF) download manager to transfer engine data files to the Portable Maintenance Device (PMD).

(3) The download manager lets the mechanic select engine data files to transfer from ONS to the PMD. You can also delete the files from the ONS. Files can be filtered by application and file type. Engine data files available on the ONS are as follows:

(a) EEC Generated Snapshot and Event History Data

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Continuous Engine Operation Data (CEOD)

(c) Bearing Detailed Report

(d) Debugging Log Files

(e) Fault History, Flight History Vibe, and Engine Balancing Data.

(4) The Download Manager function is only able to download the EEC data stored on the ONS, which typically comes from the active EEC channel.

(a) If data is needed from a specific EEC channel for a particular flight, you may have to download the data directly from the EEC using a PMAT or PDL.

1) If this is the case, refer to Electronic Engine Control (EEC) Data Download (On-Wing) Using a PMAT, TASK 73-21-05-970-801-G00.

(b) Some data types may be transferred from the EECs to the ONS as an alternative, and are then accessible via the Download Manager function.

1) If this is the case, refer to Electronic Engine Control (EEC) Data Download (On Wing) Using the Engine Health Management (EHM) Function, TASK 73-21-05-970-804-G00.

(5) Do this task when you get access to the Onboard Maintenance Function (OMF) with the maintenance laptop or other type of approved Portable Maintenance Device (PMD).

(6) This task is not available when you access the OMF with the two inboard Multi-function Displays (MFD's).

(7) If attempting to get data immediately following a flight or engine ground run, wait at least 10 minutes after engine shutdown before accessing OMF Download Manager.

NOTE: It can take up to 10 minutes for a data file to become available in the Download Manager list after the file is completed by the EEC. A similar delay in file availability is possible when transferring files from the EECs using the EHM function.

B. References

Reference	Title
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Prepare for the Data Download

SUBTASK 73-21-05-480-001-G00

(1) Do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801.

SUBTASK 73-21-05-970-004-G00

(2) Use the universal maintenance device, SPL-11321, to access the OMF menu.

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F. Electronic Engine Control (EEC) Data Download (On Wing) Using the ONS

SUBTASK 73-21-05-860-023-G00

(1) From the OMF main menu, select LINE MAINTENANCE > DOWNLOAD MANAGER and do these steps:

(a) From the FILTERS tab, select one or more of the engine-related data files.

NOTE: Make sure that a green check mark shows adjacent to your selection.

Engine Related Data Files

APPLICATION FILE TYPE *[1]	DATA TYPE
EECx:ENxACARS	Engine ACARS reports
EECx:ENxFFD	Continuous Engine Operation Data (CEOD)
EECx:ENxEVENT	Engine event reports
EECx:ENxVIBE	Detailed bearing reports
EECx:ENxSHOPFLT	EEC NVM, engine shop faults, and debugging files
EECx:ENxBOVIBE	Vibration data for Boeing engine trim balance
EECx:ENxUDREPCONF	Uplink/downlink report files
EECx:ENxUPLINKREP	Uplink notifications, uplinked report file
EHM:REPORT	Engine trim balance reports, including balance weight configuration and trim solution data

*[1] The x corresponds engine position (1 for left, 2 for right).

(b) Select CONTINUE.

(c) Select the files.

NOTE: Make sure that a green check mark shows adjacent to your selection.

NOTE: To see the files arranged by the most recent, toggle the DATE/TIME header in the third column.

NOTE: If you see a check mark in the TRANSFERRED column, this indicates files that have been downloaded.

NOTE: If you select DETAILS, you can see the number and size of the files.

(d) If directed by the engineering department to delete the files, select the DELETE icon.

(e) Select the REPORT icon to download the files.

(f) Make sure that the destination “Maintenance Device” is selected.

(g) Select the SEND button.

(h) When the Windows popup shows, make your location selection on the maintenance laptop the location where the data will be downloaded.

NOTE: This procedure recommends that you use the auto-created file name. The file name is created automatically. The naming convention is airplane identification/tail ID, filter type (FFD or DIS, for example), date and time.

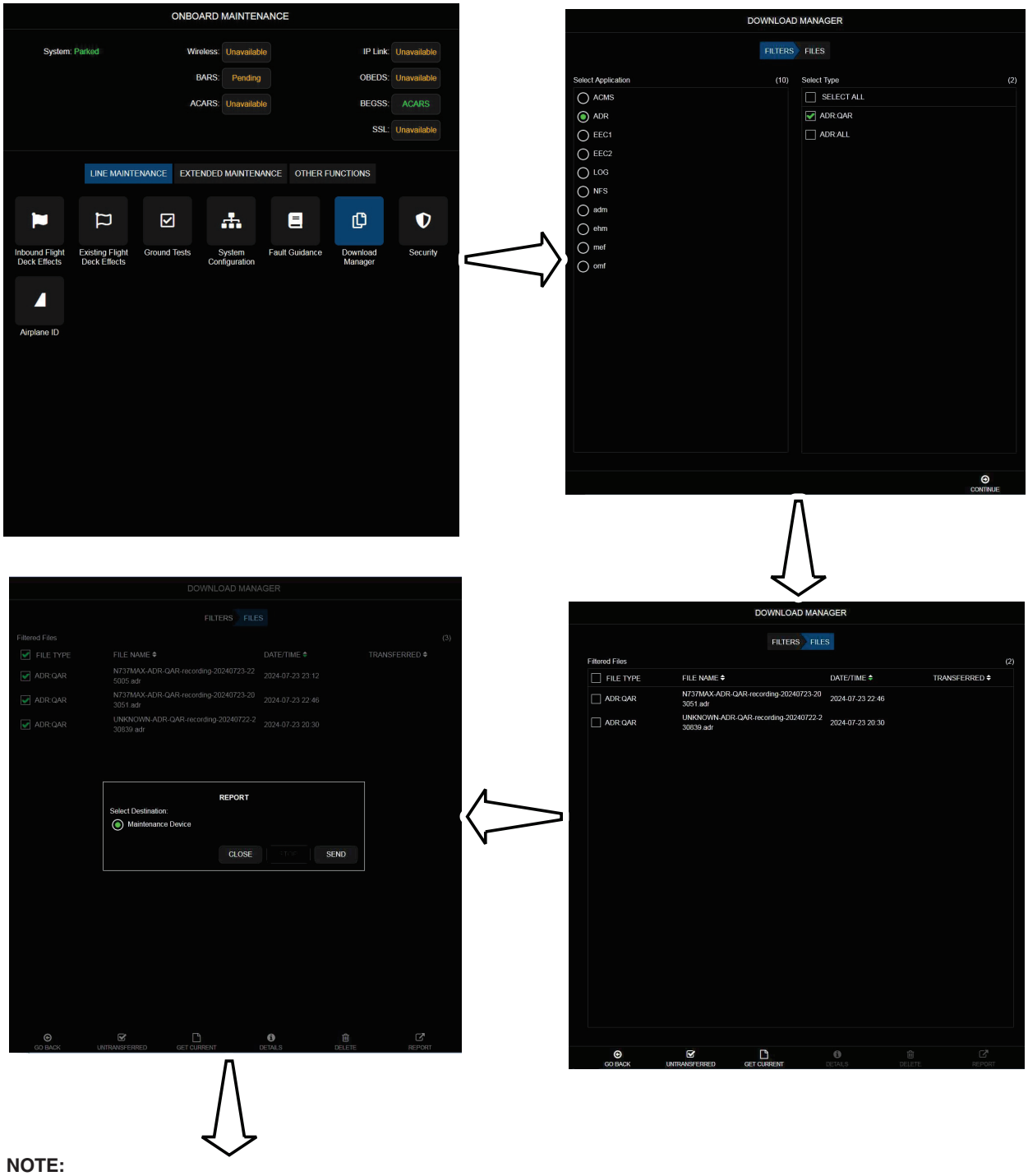
G. Put the Airplane Back to Its Usual Condition

SUBTASK 73-21-05-080-001-G00

(1) If it is no longer necessary, stop the OMF access and disconnect the universal maintenance device, SPL-11321.

END OF TASK

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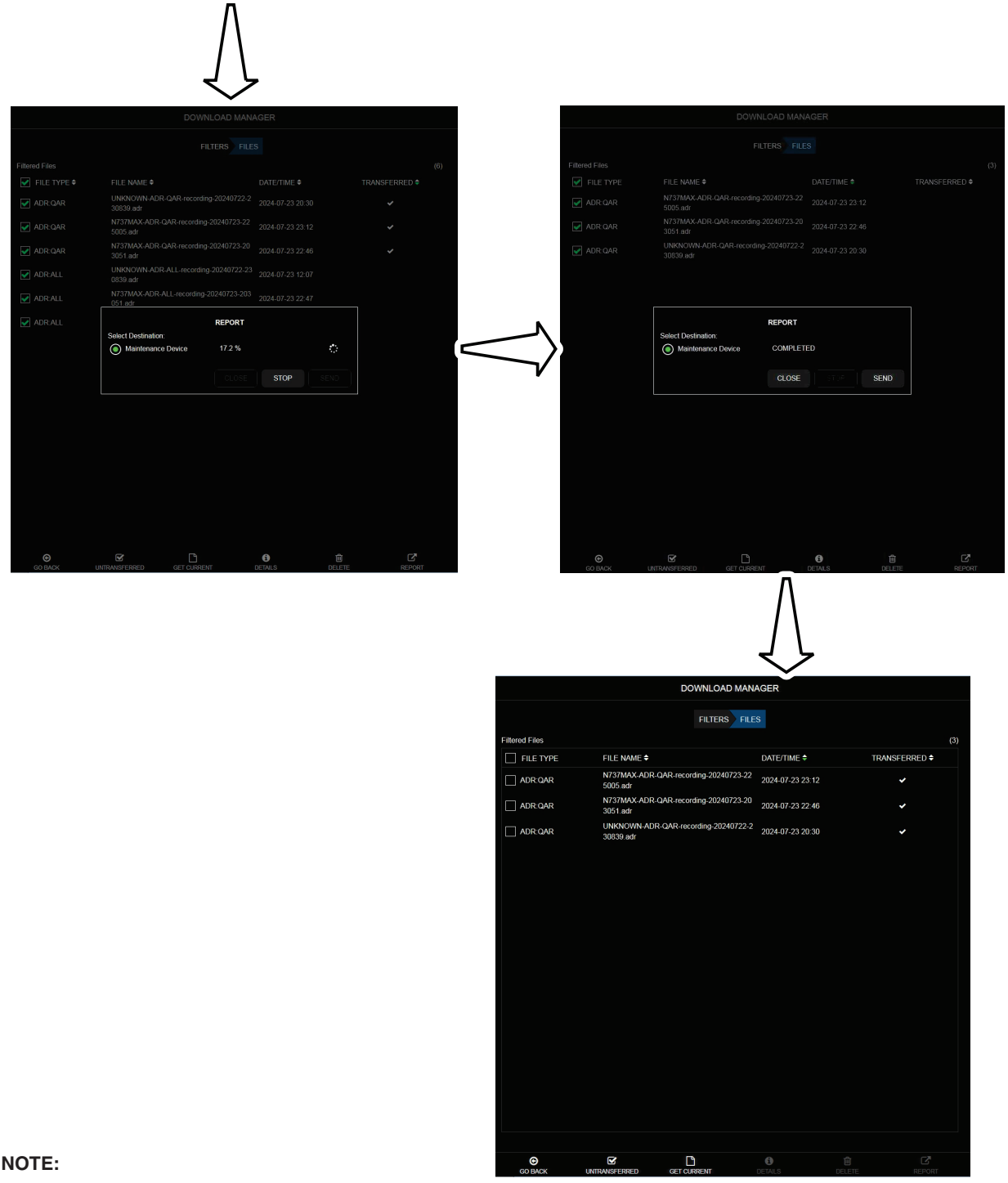


NOTE:
THE DATA ON THE DISPLAY IS ONLY AN EXAMPLE.

2479773 S0000582248_V2

Download Manager
Figure 201/73-21-05-990-814-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

THE DATA ON THE DISPLAY IS ONLY AN EXAMPLE.

2479851 S0000582250_V2

Download Manager
Figure 201/73-21-05-990-814-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 73-21-05-970-801-G00

4. Electronic Engine Control (EEC) Data Download (On-Wing) Using a PMAT

(Figure 202)

A. General

(1) This task gives the instructions to retrieve Non-Volatile Memory (NVM) data from the Electronic Engine Control (EEC).

(2) Retrieving the Non-Volatile Memory (NVM) data from EEC is also referred to as downloading NVM.

(3) Portable data loaders (PDLs) compatible with the LEAP-1B EECs are an acceptable alternative to the PMAT. If you use a PDL, follow the steps in this task but refer to the PDL user manual for task steps specific to the PMAT interface. Contact CFM if you have any questions on PDL compatibility or task procedures.

(4) These tools are necessary when you retrieve NVM data from the EEC:

(a) PMAT 2000, ARINC 615/615A data loader, COM-4963, or ARINC 645-1 compliant portable data loader, COM-21223.

(b) CFM recommends the reprogramming kit, Cable Assembly, SPL-14903 (reprogramming box, reprogramming box cable, and Ethernet cable).

(c) Ethernet adapter (Teledyne controls cable), CIS-MS/FSM boot-loader breakout kit, COM-13929.

(d) ARINC 615A-3 software image update, part number 80719-3-UPG or newer, to upgrade the PMAT 2000.

(5) Modern software loaders that meet the security requirements defined in ARINC 645-1 or newer are recommended to use. The procedures for modern software loaders will be marked as preferred for now and will be the only procedures available after removing floppy disk loaders procedures in the future. Please check with your Software Loader Supplier for current capabilities since this is still maturing across current and future software loaders.

(6) This procedure can generate the OMF maintenance messages that follow as both EECs are powered and only one EEC is connected to the PMAT:

(a) 73-4117X 'CROSS CHANNEL DATA LINK DISCRETE INPUTS AND BACKUP SERIAL CROSS CHANNEL DATA LINK DO NOT AGREE'

(b) 73-4270X 'ENGINE-Y EEC CHANNEL-Z AIRCRAFT POWER IS NOT AVAILABLE'

(c) 73-4271X 'ENGINE-Y EEC CHANNEL-Z IS NOT AVAILABLE'.

NOTE: Where X refers to EEC channel detecting the fault (0 - Engine 1 Channel A and B, 1 - Engine 1 Channel A, 2 - Engine 1 Channel B, 3 - Engine 2 Channel A, 4 - Engine 2 Channel B, 5 - Engine 2 Channel A and B), Y refers to engine position (1-Left or 2-Right) and Z refers to EEC channel (A or B).

No fault isolation is necessary for these maintenance messages when they are set because of the performance of this procedure.

B. References

Reference	Title
31-65-00-750-801	Fault Interrogation for a Maintenance Message (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
73-00-00-700-803-G00	Maintenance Messages Check for EEC Faults (P/B 501)

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(Continued)

Reference	Title
73-21-00-800-801-G00	EEC Maintenance Power Selection (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-4963	Loader - Data, ARINC 615/615A (Model is PMAT 2000) Part #: PXS-LDS-10 Supplier: 98571 Opt Part #: 80000-05 Supplier: 0BAW0 Opt Part #: P2K-615A-06 Supplier: 0BAW0
COM-13929	Kit - Breakout, CIS-MS/FSM Boot-loader Part #: 80080-1 Supplier: 0BAW0
COM-21223	Data Loader - ARINC 615 (with ARINC 645-1 Compliant) Part #: 403474 Supplier: CG603 Part #: 403562 Supplier: CG603 Part #: AE-615-MPDL Supplier: CD624 Part #: AE-DL-SMD Supplier: CD624 Part #: PDL-615-XT Supplier: 89Q20 Part #: PXS-LDS-12 Supplier: 98571
SPL-14903	Cable Assembly - PMAT (LEAP-1B Engine) Part #: 9471M34G02 Supplier: 58828
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
414	Engine 1 - Fan Cowl, Right
424	Engine 2 - Fan Cowl, Right

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

F. Prepare for the Data Download

SUBTASK 73-21-05-860-016-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

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SUBTASK 73-21-05-860-043-G00

(2) Set the GND TESTS switch on the auxiliary panel, P61, to the ENABLE position.

SUBTASK 73-21-05-010-008-G00

(3) Do this task: Open the Fan Cowl Panels (Selection), TASK 71-11-04-010-801-G00.

(a) On the applicable engine, open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

SUBTASK 73-21-05-940-001-G00

(4) Do these steps to make sure that you have all the necessary tools to download the data from the EEC:

NOTE: You can retrieve the data directly from the EEC using the PMAT 2000 kit, ARINC 615/615A data loader, COM-4963, reprogramming kit, Cable Assembly, SPL-14903, and Ethernet adapter, CIS-MS/FSM boot-loader breakout kit, COM-13929.

NOTE: To download the data using an older PMAT, the CIS-MS/FSM boot-loader breakout kit, COM-13929, will have to be procured separately.

(a) Make sure that the ARINC 615/615A data loader, COM-4963, has the ARINC 615A-3 Data Loading Software (P/N: 80719-3-UPG or newer).

NOTE: To update the ARINC 615/615A data loader, COM-4963, with ARINC 615A-3 Data Loading Software, refer the Software upgrade procedure from the PMAT Users Guide.

(b) Use only the ARINC 615/615A data loader, COM-4963, and the CIS-MS/FSM boot-loader breakout kit, COM-13929, for this procedure.

NOTE: The ARINC 615/615A data loader, COM-4963, contains PMAT 2000, PMAT cable, Ethernet cables (labelled "Straight" & "Crossover"), floppy disk drive, power cable and Ethernet adapter (Teledyne controls cable). The plug is a United States-110 Volt type; therefore, a local type plug/adapter may be required based on the region.

NOTE: Do not use the PMAT cable, Ethernet cable labelled "Crossover" and floppy disk drive.

NOTE: Use Ethernet cable labelled "Straight". If you use Ethernet cable labelled "Crossover", the PMAT will not be able to find an EEC target to connect with for data download.

(c) Use the port GSE Vibration DSP Ethernet on the left hand side of the reprogramming box to download these data:

NOTE: The Cable Assembly, SPL-14903, contains reprogramming box cable connected to a reprogramming box and an Ethernet cable. The reprogramming box contains 2 Ethernet ports, the tester receptacle, and two indicator lights.

1) Continuous Engine Operational Data (CEOD)

2) EEC generated snapshot data

3) Event history and bearing detailed report.

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(d) Use the port GSE Ethernet on the right hand side of the reprogramming box to download these data:

NOTE: The CIS-MS/FSM boot-loader breakout kit, COM-13929, is connected on the ARINC 615/615A data loader, COM-4963, and interfaces with the reprogramming box using the Ethernet cable.

NOTE: The EEC when powered ON must be in the programming enabled state, in order to enable the retrieval of the data. Once the EEC is powered ON, and connected to the ARINC 615/615A data loader, COM-4963, the downloading application of the ARINC 615/615A data loader, COM-4963, can be used to retrieve the data from the EEC.

1) Fault history and EEC control learning data.

SUBTASK 73-21-05-860-018-G00

(5) Make sure that the engines are shutdown and the most recent shutdown occurred at least 5 minutes before you start this procedure.

SUBTASK 73-21-05-710-001-G00

(6) Do these steps to test the battery charge and cable continuity on the Cable Assembly, SPL-14903:

(a) Remove the dust cap from the GSE VIBRATION DSP ETHERNET port on the left side of the reprogramming box.

(b) Connect the one end of the Ethernet cable to the GSE VIBRATION DSP ETHERNET port.

(c) Remove the dust cap from the GSE ETHERNET port on the right side of the reprogramming box.

(d) Connect the other end of the Ethernet cable to the GSE ETHERNET port.

(e) Make sure that the BATTERY FULLY CHARGED light is on.

NOTE: If the BATTERY FULLY CHARGED light is not on, replace the standard 9V battery on the reprogramming box.

(f) Disconnect the dust cap on the TESTER RECEPTACLE on the reprogramming box.

(g) Connect the end of the reprogramming box cable (used for connecting to the EEC J9 connector) to the TESTER RECEPTACLE on the reprogramming box.

(h) Make sure that the CABLE OK light is illuminates.

NOTE: If the CABLE OK light does not illuminate, replace the battery, and do the test again. If it is necessary, replace the Cable Assembly, SPL-14903, and do the test again.

(i) Replace the dust cap on the TESTER RECEPTACLE.

G. Electronic Engine Control (EEC) Data Download (On-Wing) Using a PMAT

SUBTASK 73-21-05-970-001-G00

(1) Do these steps to download data from the EEC channel A:

(a) Make sure that the engine is shutdown.

(b) Make sure that the applicable EEC MAINT POWER on the MISC SYSTEM CTRLS page is set to NORM (TASK 73-21-00-800-801-G00).

(c) Disconnect the dust cap from the J9 connector on EEC.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CAUTION

MAKE SURE THAT YOU REMOVE ALL ELECTRICAL CHARGES FROM THE EEC. DAMAGE CAN OCCUR.

(d) Touch the connector-coupling nut of the cable to the J9 connector body on the EEC and hold it for at least 1 second.

NOTE: This will remove all electrical charges on the EEC.

(e) Connect the reprogramming box cable to the J9 connector on the EEC.

(f) Disconnect the dust cap on the ARINC 615/615A data loader, COM-4963, reprogramming port.

(g) Disconnect the dust cap on the CIS-MS/FSM boot-loader breakout kit, COM-13929.

CAUTION

MAKE SURE THAT YOU REMOVE ALL ELECTRICAL CHARGES FROM THE EEC. DAMAGE CAN OCCUR.

(h) Touch the connector-coupling nut of the CIS-MS/FSM boot-loader breakout kit, COM-13929, cable to the connector-coupling nut of the ARINC 615/615A data loader, COM-4963, and hold it for at least 1 second.

NOTE: This will remove all electrical charges on the EEC.

(i) Connect the CIS-MS/FSM boot-loader breakout kit, COM-13929, to the ARINC 615/615A data loader, COM-4963, reprogramming port.

(j) Connect one end of the Ethernet cable to the GSE VIBRATION DSP ETHERNET port on the reprogramming box and the other end to the CIS-MS/FSM boot-loader breakout kit, COM-13929, already connected to ARINC 615/615A data loader, COM-4963.

NOTE: Use the GSE VIBRATION DSP ETHERNET port to download the Continuous Engine Operational Data (CEOD), EEC generated snapshot data, Event History, and bearing detailed report.

NOTE: Do not connect the Ethernet cable to the Ethernet port on the ARINC 615/615A data loader, COM-4963.

NOTE: Use Ethernet cable labelled "Straight". Do not use Ethernet cable labelled "Crossover". If you use Ethernet cable labelled "Crossover", the PMAT will not be able to find an EEC target to connect with for data download.

(k) Turn on the ARINC 615/615A data loader, COM-4963, or ARINC 645-1 compliant portable data loader, COM-21223.

(l) Set the EEC MAINT POWER for the applicable engine on the MISC SYSTEM CTRLS to TEST (TASK 73-21-00-800-801-G00).

NOTE: The EEC will do a self-test when you set the EEC MAINT POWER to TEST position. Wait for a minimum of 30 seconds to let the EEC complete the test.

(m) Select the applicable ARINC 615A DATALOADER from the PMAT Main Menu.

(n) Enter a USERNAME, in box 1.

(o) Select the Tail Configuration, by entering the Tail Number, in box 2. Then click "OK".

(p) Select the Connection Type, in box 3.

1) Select “EEC CH.B DSP” or the applicable LOADER STANDARD.

2) Select "4. Continue".

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(q) Set the EEC MAINT POWER for the applicable engine on the MISC SYSTEM CTRLS to TEST (TASK 73-21-00-800-801-G00).

NOTE: The EEC will do a self-test when you set the EEC MAINT POWER to TEST position. Wait for a minimum of 30 seconds to let the EEC complete the test.

(r) Select Operator Defined Download in the select an operation box.

(s) Wait for the screen to show that a target has been found, then select that target.

NOTE: You should see the target "GEC-EECDSP_A". It may take 1 to 2 minutes to appear on the screen.

(t) Select Continue.

(u) Select the applicable files contain data as given in Table 201.

Table 201/73-21-05-993-801-G00 Channel A Directories

File Name(s)	Data type
When connected to Channel A	
EECDSP_DIR1_A.RTV	EEC generated Snapshot data, Event History
EECDSP_DIR2_A.RTV	Continuous Engine Operational Data (CEOD)
EECDSP_DIR3_A.RTV	High frequency engine vibration data
EECDSP_DIR4_A.RTV	EEC-generated Ethernet (ONS-related) log files
EECDSP_DIR5_A.RTV	Unused Currently
EECDSP_DIRg_A.RTV	EEC OMAP activity log files
EECDSP_DIRy_A.RTV	Unused Currently

1) Select one file name from Table 201/73-21-05-993-801-G00 in the Download List window for the appropriate data type you will download.

NOTE: Download each file one at a time. If you download multiple files at the same time, it can cause a failed download.

NOTE: It is recommended to download DIR1 first as it downloads quickly and will make sure that the PMAT system is correctly connected and configured.

2) Select Start.

3) After a few seconds, make sure that a progress bar appears and throughput indication starts to update (in KBytes/Sec).

NOTE: When you download DIR2, it can take up to 25 minutes for the progress bar and throughput to update. Wait at least 25 minutes before you abort the download procedure. DIR2 file size is up to 1.5 GB and can take 45 minutes to fully download.

4) The progress status bar will turn green after the download completes correctly.

NOTE: If the download for DIR2 has taken approximately 45 minutes and the progress bar turns red and shows failed, check the size of the file downloaded. If the size of the DIR2 .RTV file is approximately 1.5 GB, the download has completed correctly.

5) If you will download another directory, do this step again:

a) Select Operator Defined Download in the select an operation box.

(v) Click Close on each of the individual status window to properly terminate the download session.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(w) Click Go Back to exit the current window and display the PMAT Main Menu.

(x) Disconnect the Ethernet cable from the GSE Vibration DSP Ethernet port.

(y) Connect the Ethernet cable to the GSE Ethernet port (right hand side port) on the reprogramming box.

NOTE: Use the GSE Ethernet port to download the Flight History Vibe data, Fault History, and balancing data.

(z) Cycle power to the applicable engine EECs. Set the EEC MAINT POWER for the applicable engine on the MISC SYSTEM CTRLS to NORM then to TEST (TASK 73-21-00-800-801-G00).

NOTE: The EEC will do a self-test when you set the EEC MAINT POWER to TEST position. Wait for a minimum of 30 seconds to let the EEC complete the test.

(aa) Select ARINC 615A DATALOADER from the PMAT Main Menu.

(ab) Enter a USERNAME, in box 1.

(ac) Select the Tail configuration, by entering the Tail number, in box 2 and then click "OK".

1) Select “EEC CH.B DSP” or “615A LOADER STANDARD (IP ADDRESS = 192.168.1.2)".

(ad) Select Continue.

(ae) Select Operator Defined Download in the select operation box.

(af) Wait for the screen to show that a target has been found, then select that target.

NOTE: You should see the target "GEC-EECCPU_A”. It may take 1 to 2 minutes to appear on the screen.

(ag) Select Continue.

(ah) Select the file by using select all in the Download List window.

(ai) Do these steps to download CPU NVM:

1) Select directory EECCPU_NVM_A.RTV as shown in Table 202.

Table 202/73-21-05-993-802-G00 Channel A CPU File

File Name(s)	Data type
When connected to Channel A	
EECCPU_NVM_A.RTV	Fault history and EEC control learning data

2) Select Start, after a few seconds a progress bar should appear and throughput indication should begin updating (in KBytes/Sec).

NOTE: While selecting the directory, if this step is not completed within 10 seconds, the PMAT will have a timeout error. You will need select the operator defined download in the select operation box again. Wait for the screen to show that a target has been found and then select that target (you should see the target "GEC-EECCPU_A" in 1 to 2 minutes). Then select continue.

3) The progress status bar will turn green after the download is completed successfully.

(aj) Click Close on each of the individual status window to properly terminate the download session.

(ak) Click Go Back to exit the current window and display the PMAT Main Menu.

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(al) Set the EEC MAINT POWER for the applicable engine on the MISC SYSTEM CTRLS to NORM (TASK 73-21-00-800-801-G00).

(am) Disconnect the reprogramming box cable from the J9 connector on EEC and connect the dust cap.

SUBTASK 73-21-05-970-002-G00

(2) Do these steps to download data from the EEC channel B:

(a) Make sure that the engine is shutdown.

(b) Make sure that the EEC MAINT POWER on the MISC SYSTEM CTRLS page are in NORM (TASK 73-21-00-800-801-G00).

(c) Disconnect the dust cap from the J9 connector on EEC.

CAUTION

MAKE SURE THAT YOU REMOVE ALL ELECTRICAL CHARGES FROM THE EEC. DAMAGE CAN OCCUR.

(d) Touch the connector-coupling nut of the cable to the J9 connector body on the EEC and hold it for at least 1 second.

NOTE: This will remove all electrical charges on the EEC.

(e) Connect the reprogramming box cable to the J9 connector on the EEC.

(f) Disconnect the dust cap on the ARINC 615/615A data loader, COM-4963, reprogramming port.

(g) Disconnect the dust cap on the CIS-MS/FSM boot-loader breakout kit, COM-13929.

CAUTION

MAKE SURE THAT YOU REMOVE ALL ELECTRICAL CHARGES FROM THE EEC. DAMAGE CAN OCCUR.

(h) Touch the connector-coupling nut of the CIS-MS/FSM boot-loader breakout kit, COM-13929, cable to the connector-coupling nut of the ARINC 615/615A data loader, COM-4963, and hold it for at least 1 second.

NOTE: This will remove all electrical charges on the EEC.

(i) Connect the CIS-MS/FSM boot-loader breakout kit, COM-13929, to the ARINC 615/615A data loader, COM-4963, reprogramming port.

(j) Connect one end of the Ethernet cable to the GSE Vibration DSP Ethernet port (left hand side port) on the reprogramming box and the other end to the CIS-MS/FSM boot-loader breakout kit, COM-13929, already connected to ARINC 615/615A data loader, COM-4963.

NOTE: Use the GSE VIBRATION DSP ETHERNET port to download the Continuous Engine Operational Data (CEOD), EEC generated snapshot data, event History, and bearing detailed report.

NOTE: Do not connect the Ethernet cable to the Ethernet port on the ARINC 615/615A data loader, COM-4963.

NOTE: Use Ethernet cable labelled "Straight". Do not use Ethernet cable labelled "Crossover". If you use Ethernet cable labelled "Crossover", the PMAT will not be able to find an EEC target to connect with for data download.

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(k) Turn on the ARINC 615/615A data loader, COM-4963, or ARINC 645-1 compliant portable data loader, COM-21223.

(l) Set the EEC MAINT POWER for the applicable engine on the MISC SYSTEM CTRLS to TEST (TASK 73-21-00-800-801-G00).

NOTE: The EEC will do a self-test when you set the EEC MAINT POWER to TEST position. Wait for a minimum of 30 seconds to let the EEC complete the test.

(m) Select the applicable ARINC 615A DATALOADER from the PMAT Main Menu.

(n) Enter a USERNAME, in box 1.

(o) Select the Tail Configuration, by entering the Tail Number, in box 2. Then click "OK".

(p) Select the Connection Type, in box 3.

1) Select the applicable LOADER STANDARD.

2) Select "4. Continue".

(q) Select Operator Defined Download in the select an operation box.

(r) Wait for the screen to show that a target has been found, then select that target.

NOTE: You should see the target "GEC-EECDSP_B". It may take 1 to 2 minutes to appear on the screen.

(s) Select Continue.

(t) Select the applicable files contain data as given in Table 203.

Table 203/73-21-05-993-803-G00 Channel B Directories

File Name(s)	Data type
When connected to Channel B	
EECDSP_DIR1_B.RTV	EEC generated Snapshot data, Event History
EECDSP_DIR2_B.RTV	Continuous Engine Operational Data (CEOD)
EECDSP_DIR3_B.RTV	High frequency engine vibration data
EECDSP_DIR4_B.RTV	EEC-generated Ethernet (ONS-related) log files
EECDSP_DIR5_B.RTV	Unused Currently
EECDSP_DIRg_B.RTV	EEC OMAP activity log files
EECDSP_DIRy_B.RTV	Unused Currently

1) Select one file name from Table 201/73-21-05-993-801-G00 in the Download List window for the appropriate data type you will download.

NOTE: Download each file one at a time. If you download multiple files at the same time, it can cause a failed download.

NOTE: It is recommended to download DIR1 first as it downloads quickly and will make sure that the PMAT system is correctly connected and configured.

2) Select Start.

3) After a few seconds, make sure that a progress bar appears and throughput indication starts to update (in KBytes/Sec).

NOTE: When you download DIR2, it can take up to 25 minutes for the progress bar and throughput to update. Wait at least 25 minutes before you abort the download procedure. DIR2 file size is up to 1.5 GB and can take 45 minutes to fully download.

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4) The progress status bar will turn green after the download completes correctly.

NOTE: If the download for DIR2 has taken approximately 45 minutes and the progress bar turns red and shows failed, check the size of the file downloaded. If the size of the DIR2 .RTV file is approximately 1.5 GB, the download has completed correctly.

5) If you will download another directory, do this step again:

a) Select Operator Defined Download in the select an operation box.

(u) Click Close on each of the individual status window to properly terminate the download session.

(v) Click Go Back to exit the current window and display the PMAT Main Menu.

(w) Disconnect the Ethernet cable from the GSE VIBRATION DSP ETHERNET port.

(x) Connect the Ethernet cable to the GSE ETHERNET port (right port) on the reprogramming box.

NOTE: Use the GSE ETHERNET port to download the flight history vibe data, fault history, and balancing data.

(y) Cycle power to the applicable engine EECs. Set the EEC MAINT POWER for the applicable engine on the MISC SYSTEM CTRLS to NORM then to TEST (TASK 73-21-00-800-801-G00).

NOTE: The EEC will do a self-test when you set the EEC MAINT POWER to TEST position. Wait for a minimum of 30 seconds to let the EEC complete the test.

(z) Select ARINC 615A DATALOADER from the PMAT Main Menu.

(aa) Enter a USERNAME, in box 1.

(ab) Select the Tail configuration, by entering the Tail number, in box 2. Then click "OK".

1) Select "615A LOADER STANDARD (IP ADDRESS = 192.168.1.2)".

(ac) Select Continue.

(ad) Select Operator Defined Download in the select operation box.

(ae) Wait for the screen to show that a target has been found, then select that Target.

NOTE: You should see the target "GEC-EECCPU_B". It may take 1 to 2 minutes to appear on the screen.

(af) Select Continue.

(ag) Select the file by using Select All in the Download List window.

(ah) Do these steps to download CPU NVM:

1) Select directory EECCPU_NVM_B.RTV as shown in Table 204.

Table 204/73-21-05-993-804-G00 Channel B CPU File

File Name(s)	Data type
When connected to Channel B	
EECCPU_NVM_B.RTV	Fault history and EEC control learning data

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2) Select Start, after a few seconds a progress bar should appear and throughput indication should begin updating (in KBytes/Sec).

NOTE: While selecting the directory, if this step is not completed within 10 seconds, the PMAT will have a timeout error. You will need select the operator defined download in the select operation box again. Wait for the screen to show that a target has been found and then select that target (you should see the target "GEC-EECCPU_B" in 1 to 2 minutes). Then select continue.

3) The progress status bar will turn green after the download is completed successfully.

(ai) Click Close on each of the individual status window to properly terminate the download session.

(aj) Click Go Back to exit the current window and display the PMAT Main Menu.

(ak) Set the EEC MAINT POWER on the MISC SYSTEM CTRLS to NORM (TASK 73-21-00-800-801-G00).

(al) Disconnect the reprogramming box cable from the J9 connector on EEC and connect the dust cap.

SUBTASK 73-21-05-970-003-G00

(3) Do these steps to access the downloaded files from the EEC:

(a) Go to the ARINC 615/615A data loader, COM-4963, or ARINC 645-1 compliant portable data loader, COM-21223, Main Menu.

(b) Select Archived Data.

NOTE: This will open a new screen with header, Archived Data.

(c) On the Archived Data screen, select 615A Downloads.

(d) On the 615A Downloads screen, select relevant files to be transferred.

(e) Select the icon Transfer to Removable Media.

NOTE: This will open a new screen.

(f) The transfer from 615A Downloads to Removal Media screen will show the Removable Media Drive when connected to the PMAT.

(g) Select the location on the Media Drive to save the files using Browse.

(h) Select the appropriate Transfer type.

(i) Select appropriate options in Downloads to Transfer.

(j) Click on the Transfer Download(s) to initiate the transfer of files.

NOTE: The file transfer progress can be seen from the transfer status.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 73-21-05-410-007-G00

(1) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

(a) On the applicable engine, close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

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SUBTASK 73-21-05-860-020-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

SUBTASK 73-21-05-860-044-G00

(3) Set the GND TESTS switch on the auxiliary panel, P61, to the NORM position.

SUBTASK 73-21-05-750-001-G00

(4) Do a check on the Onboard Maintenance Function (OMF) for the following EEC maintenance messages (TASK 73-00-00-700-803-G00): 73-41170, 73-41175, 73-42701, 73-42702, 73-42703, 73-42704, 73-42711, 73-42712, 73-42713, and 73-42714.

NOTE: No fault isolation is required for these maintenance messages when set due to EEC connection to PMAT.

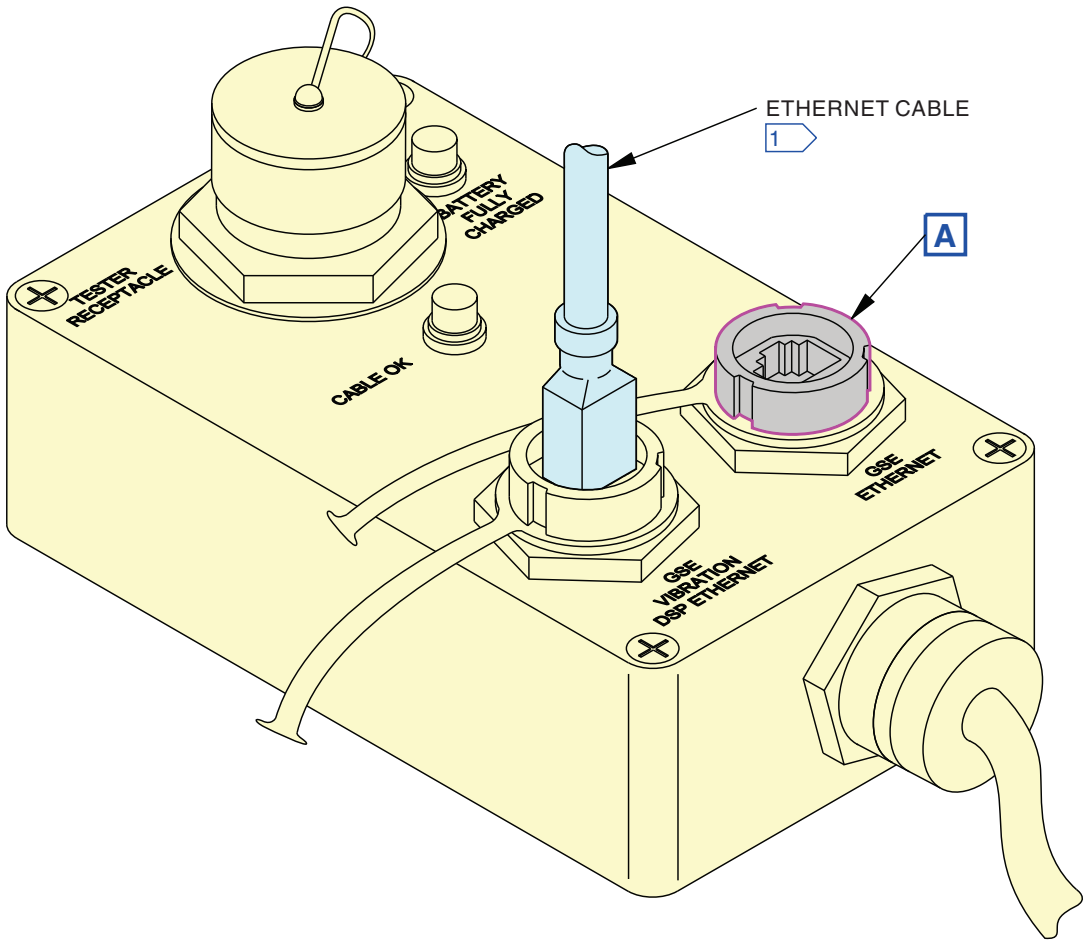
(a) If the maintenance messages are present, cycle power to the EECs using the EEC MAINT POWER switch on the MISC SYSTEM CTRLS page on the Multi Function Display (MFD) (TASK 73-21-00-800-801-G00).

1) If the maintenance messages clear, confirm no STATUS or SMT messages are present.

a) Do this task: Fault Interrogation for a Maintenance Message, TASK 31-65-00-750-801.

——— END OF TASK ———

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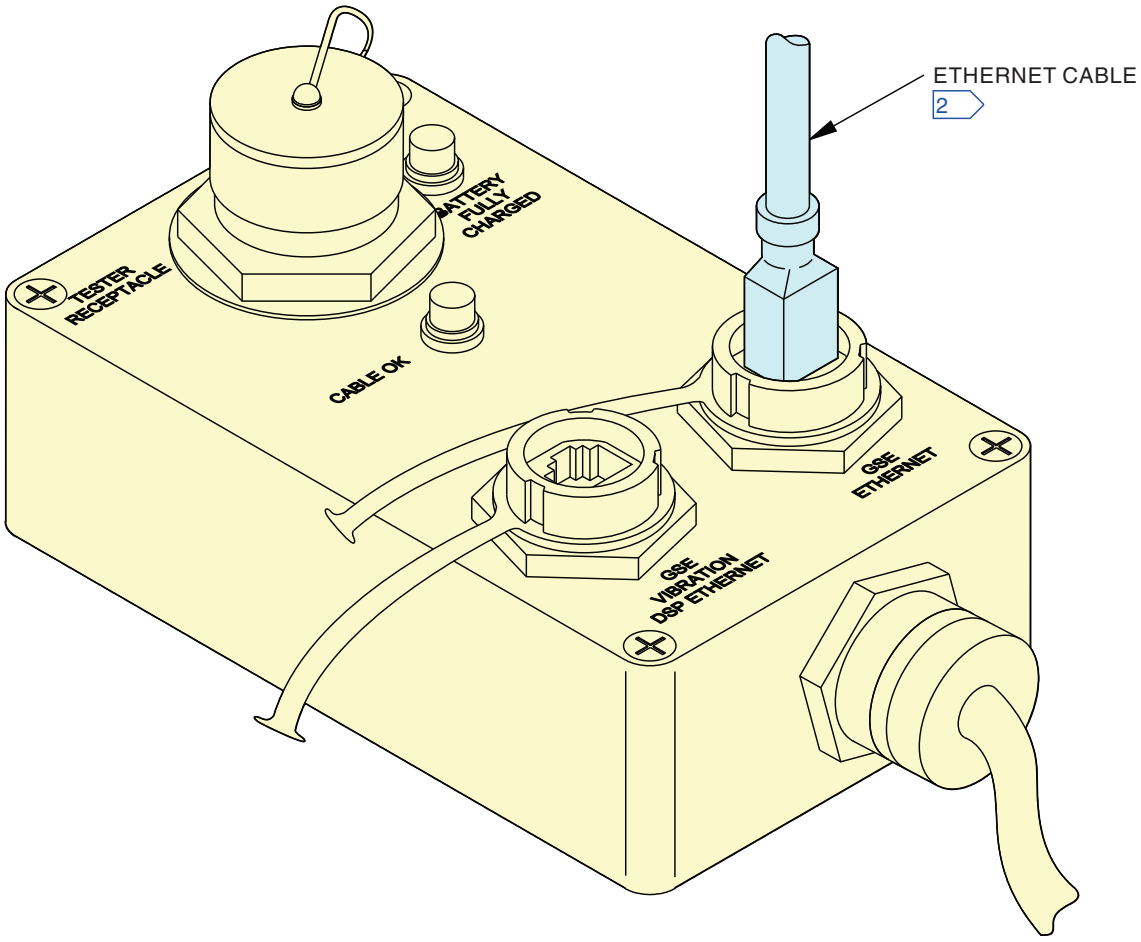


1 ETHERNET CABLE CONNECTED TO GSE VIBRATION DSP ETHERNET PORT FOR DOWNLOADING CONTINUOUS ENGINE OPERATIONAL DATA (CEOD), EEC GENERATED SNAPSHOT DATA, EVENT HISTORY AND BEARING DETAILED REPORTS.

5031436-00
2532204 S0000599744_V1

Reprogramming Box
Figure 202/73-21-05-990-813-G00 (Sheet 1 of 2)

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2 ETHERNET CABLE CONNECTED TO GSE ETHERNET PORT FOR DOWNLOADING FLIGHT HISTORY VIBRATION, FAULT HISTORY, AND ENGINE TRIM BALANCING DATA.

5031437-00
2532208 S0000599746_V1

Reprogramming Box
Figure 202/73-21-05-990-813-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 73-21-05-970-804-G00

5. Electronic Engine Control (EEC) Data Download (On Wing) Using the Engine Health Management (EHM) Function

A. General

(1) This task gives the instructions on how to transfer and then download data from the EEC memory to the Onboard Network System (ONS).

(2) You will use the Engine Health Management (EHM) data download function and Onboard Maintenance Function (OMF) download manager to transfer engine data files to the portable universal maintenance device, SPL-11321.

(3) The EHM data download function lets personnel select engine data files to transfer to the ONS from the EEC.

(4) Engine data files available to transfer to the ONS are as follows:

(a) System log of EEC communication with aircraft ONS

(b) Detailed snapshot of EEC fault information, including specific engine parameters when the fault was detected (EEC CPU NVM, detailed fault history)

(c) Detailed vibration data used for event investigation and maturing vibration analytics

(d) EEC-generated snapshot report data.

(5) Do this task when you get access to the Onboard Maintenance Function (OMF) with the maintenance laptop or other type of approved Portable Maintenance Device (PMD).

(6) This task is not available when you access the OMF with the two inboard Multi-Function Displays (MFD's).

B. References

Reference	Title
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)
73-21-00-800-801-G00	EEC Maintenance Power Selection (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Prepare for the Download

SUBTASK 73-21-05-480-002-G00

(1) Connect the PMD to the ONS (TASK 46-13-00-480-801).

SUBTASK 73-21-05-860-036-G00

(2) Rotate the SELECTOR knob to C or R on the multi-function panel.

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SUBTASK 73-21-05-860-037-G00

(3) Push SEL button to activate the MFD screen.

SUBTASK 73-21-05-860-038-G00

(4) Set the applicable EEC MAINT POWER on the MISC SYSTEM CTRLS page to TEST (TASK 73-21-00-800-801-G00).

NOTE: The EEC will do a self-test when you set the EEC MAINT POWER switch to the TEST position. Wait for a minimum of 30 seconds to let the EEC complete the test.

SUBTASK 73-21-05-860-039-G00

(5) Set the GND TESTS switch on the auxiliary panel, P61, to the ENABLE position.

SUBTASK 73-21-05-210-004-G00

(6) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) On the P5 overhead panel, make sure that the applicable ENGINE START switch is in the OFF position.

(b) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(c) On the P10 control stand, make sure that the applicable START LEVER switch is in the CUTOFF position.

(d) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-21-05-860-040-G00

(7) Set the applicable thrust lever to idle.

F. Electronic Engine Control (EEC) Data Download (On Wing) Using the Engine Health Management (EHM) Function

SUBTASK 73-21-05-480-003-G00

(1) Make these selections on the PMD main function menu:

(a) OTHER FUNCTIONS.

(b) ENGINE HEALTH MANAGEMENT (EHM).

(c) ENGINE 1 or ENGINE 2.

(d) FUNCTIONS > EEC DATA DOWNLOAD.

SUBTASK 73-21-05-970-011-G00

(2) Select the file type to be transferred from EEC non-volatile memory (NVM) to ONS File Download Manager from the table below:

NOTE: It is recommended to select only one file type at a time.

File Type	File Content	Expected Transfer Time
EEC Syslog Channel A & B	Log of EEC communication with aircraft ONS	7 minutes
EEC Shop Fault Channel A & B	Detailed snapshot of EEC fault information, including specific engine parameters when fault was detected	7 minutes
Advanced Vibration Data Channel A & B	EEC flight history of PHM vibration indicators	135 minutes
ACARS Channel A & B	EEC-generated snapshot report data	7 minutes

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(Continued)

File Type	File Content	Expected Transfer Time
Full Flight Data Channel A & B *[1]	Continuous Engine Operation Data (CEOD)	inhibited

*[1] Full Flight Data Channel A & B file is prevented from being downloaded, as the large file size would result in a very long transfer time.

SUBTASK 73-21-05-970-012-G00

(3) Select REPORT.

SUBTASK 73-21-05-970-013-G00

(4) Select SEND.

NOTE: The ONS will automatically display an error message 60 seconds after selecting SEND. If the transfer error appears, wait for the remaining expected transfer time before you check Download Manager for the transferred file. If you turn off power to the EECs or change the GND TESTS switch position, it will interrupt the file transfer.

SUBTASK 73-21-05-970-014-G00

(5) Download the transferred files from the applicable engine EECx:ENxSHOPFLT directory in Download Manager (TASK 73-21-05-970-802-G00).

NOTE: The x in EECx:ENxSHOPFLT corresponds to engine position (1 for left, 2 for right).

G. Put the Airplane Back to Its Usual Condition

SUBTASK 73-21-05-860-041-G00

(1) Set the applicable EEC MAINT POWER switch on the MISC SYSTEM CTRLS page on the MFD to the NORM position (TASK 73-21-00-800-801-G00).

SUBTASK 73-21-05-860-042-G00

(2) Set the GND TESTS switch on the auxiliary panel, P61, to the NORM position.

SUBTASK 73-21-05-080-004-G00

(3) Remove the DO NOT OPERATE tag, STD-858, from the ENGINE START and the START LEVER switches.

SUBTASK 73-21-05-080-005-G00

(4) Disconnect the PMD from the ONS.

END OF TASK

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ELECTRONIC ENGINE CONTROL (EEC) - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Electronic Engine Control (EEC) Channel A Removal

(2) Electronic Engine Control (EEC) Channel A Installation

(3) Electronic Engine Control (EEC) Channel B Removal

(4) Electronic Engine Control (EEC) Channel B Installation.

B. If you remove and replace the two EECs at the same time, make a record of the current engine trim balance weight configuration before you remove the EECs.

TASK 73-21-05-000-801-G00

2. Electronic Engine Control (EEC) Channel A Removal

(Figure 401 and Figure 402)

A. General

(1) This task gives the instructions to remove the Electronic Engine Control (EEC) Channel A.

(2) A minimum of two persons are required to handle the EEC if the jack stand and EEC adapter support equipment are not used.

B. References

Reference	Title
46-13-02-970-802	Download Manager (P/B 201)
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-00-00-720-805-G00	Test No. 11 - Low Pressure System (Fan) Trim Balance (With ONS) (P/B 501)
71-00-00-970-801-G00	Maintenance Data Pages from the Engine Propulsion Control System (EPCS) (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14500	Stand - Jack Part #: 9471M17P01 Supplier: 58828
SPL-14529	Adapter - LRU Part #: 956A3616G01 Supplier: 58828
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

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E. Access Panels

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

F. Prepare for the Removal

SUBTASK 73-21-05-860-006-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-21-05-970-006-G00

(2) If you will replace the two EECs at the same time, do these steps before you remove the EECs:

(a) Make a record of the current engine trim balance weight configuration (TASK 71-00-00-720-805-G00).

(b) Make a record of the current engine cumulative oil debris chip count ("DMS") on EPCS page 2 (TASK 71-00-00-970-801-G00).

SUBTASK 73-21-05-970-009-G00

(3) If you will remove the EEC because of a fault isolation procedure, do this step:

(a) Download the EEC Non Volatile Memory (NVM) from the two EEC channels before you remove the EEC (TASK 46-13-02-970-802).

NOTE: Downloading the CEOD and EEC NVM data is optional. The EEC NVM and CEOD download is recommended only for cases where repetitive maintenance actions or component replacements have failed to identify the root cause of the fault condition. The EEC NVM and CEOD can be helpful to identify the root cause of the fault condition. However, when the EEC is removed for a known cause, it is not necessary to do the EEC NVM and CEOD download.

NOTE: The EEC NVM is listed in Download Manager as EEC1:EN1SHOPFLT for Engine 1 and EEC2:EN2SHOPFLT for Engine 2.

NOTE: You can also use the EEC DATA DOWNLOAD function on the OTHER FUNCTIONS > EHM menu on the Portable Maintenance Device (PMD) to download the EEC NVM. The EEC NVM is listed as 'EEC Shop Fault Channel A & B' in the EEC DATA DOWNLOAD menu.

SUBTASK 73-21-05-040-001-G00

(4) Open these applicable circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A
B	3	C01312	ENGINE 1 RUN/PWR

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F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	5	C01313	ENGINE 2 RUN/PWR
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	3	C04018	ENG 1 STBY ALTN PWR RELAY CHAN A
E	4	C04019	ENG 1 STBY ALTN PWR CHAN A

SUBTASK 73-21-05-860-012-G00

(5) On the applicable engine, open the right fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

G. Electronic Engine Control (EEC) Channel A Removal

SUBTASK 73-21-05-030-001-G00

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

CAUTION MAKE SURE TO FOLLOW THE PROCEDURE FOR DEVICES THAT ARE SENSITIVE TO ELECTROSTATIC DISCHARGE. ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE EEC.

(1) Use the teflon-jawed pliers, STD-664, to disconnect these electrical connectors from the EEC CH A [1] receptacles (TASK 70-00-01-910-803-G00):

(a) The electrical connector (HJ8A) [3]

(b) The electrical connector (HJ7) [4]

(c) The electrical connector (HJ6A) [5]

(d) The electrical connector (HJ5A) [6]

(e) The electrical connector (W4A) [7]

(f) The electrical connector (HJ3A) [8]

(g) The electrical connector (W2) [9]

(h) The electrical connector (W4A) [7].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 73-21-05-480-005-G00

(2) Install the protection covers on the electrical connectors and the electrical receptacles on the EEC CH A [1].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 75-0010 OR PRE LEAP-1B SERVICE BULLETIN 75-0011

SUBTASK 73-21-05-030-002-G00

(3) Remove the EEC cooling duct [2] from the EEC CH A [1] as follows:

(a) Remove the four bolts [10] that attach the EEC cooling duct [2] to the EEC CH A [1].

(b) Disconnect the EEC cooling duct [2] from the EEC CH A [1].

(c) Remove and discard the gasket seal [13].

SUBTASK 73-21-05-480-004-G00

(4) Install the protective covers on the EEC cooling duct [2].

PTF ALL

SUBTASK 73-21-05-020-001-G00

WARNING THE EEC IS HEAVY AND TWO PERSONS ARE NECESSARY TO LIFT IT. IF YOU LIFT THE EEC INCORRECTLY, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(5) Do these steps to remove the EEC CH A [1] from the fan case mounting brackets [12] as follows:

NOTE: The EEC weighs approximately 37 lb (17 kg).

NOTE: If the two-person alternative method is used, you do not have to install the EEC to the adapter, SPL-14529, and jack stand, SPL-14500.

(a) Attach the adapter, SPL-14529, to the EEC CH A [1].

(b) Move the jack stand, SPL-14500, to the correct position.

(c) Install the adapter, SPL-14529, on the jack stand, SPL-14500.

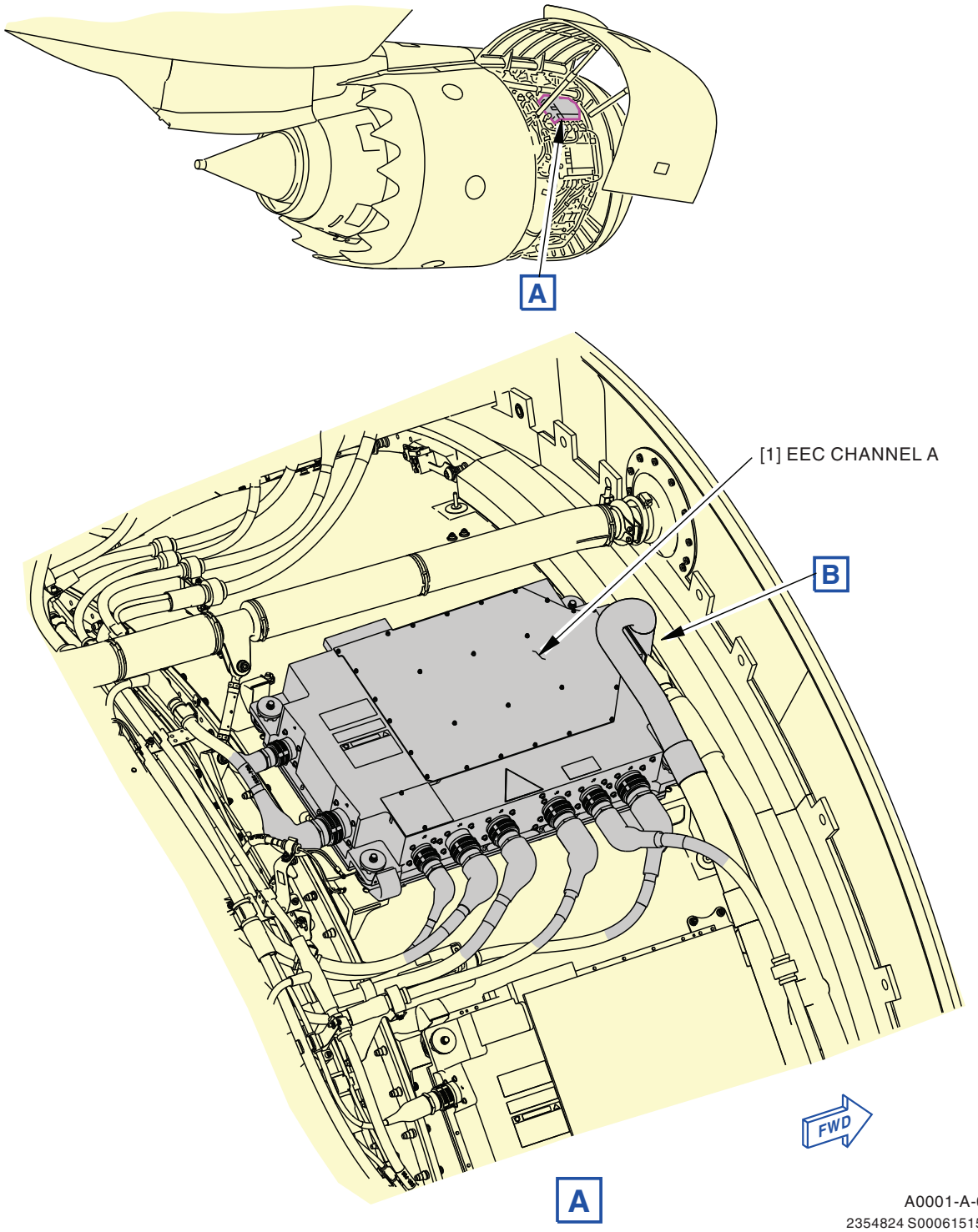
(d) Remove the four screws [11] that attach the EEC CH A [1] to the four mounting brackets [12].

NOTE: Do not remove the captive screws.

(e) Remove the EEC CH A [1] from the four mounting brackets [12].

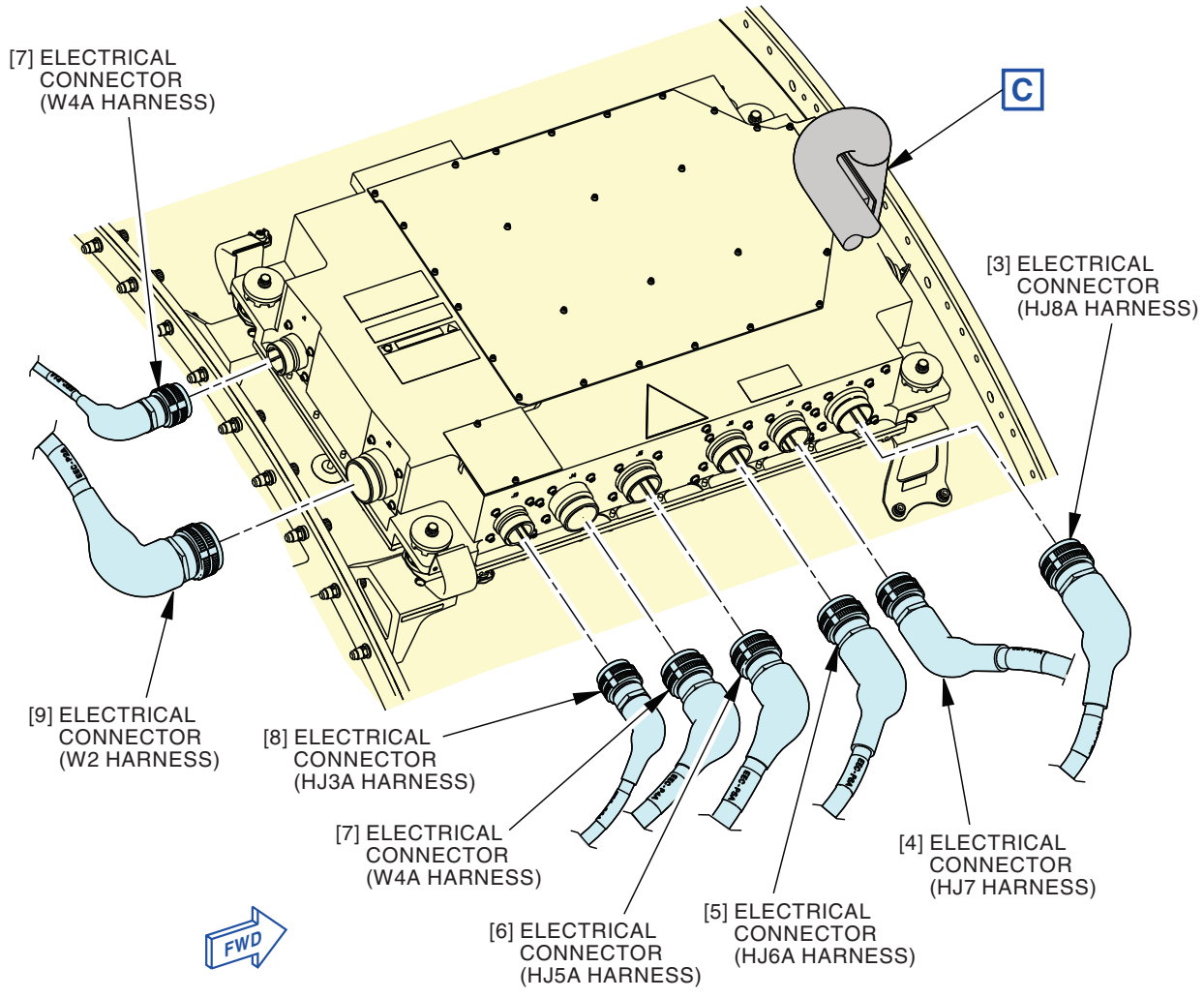
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Electronic Engine Control (EEC) Channel A Installation
Figure 401/73-21-05-990-808-G00 (Sheet 1 of 5)

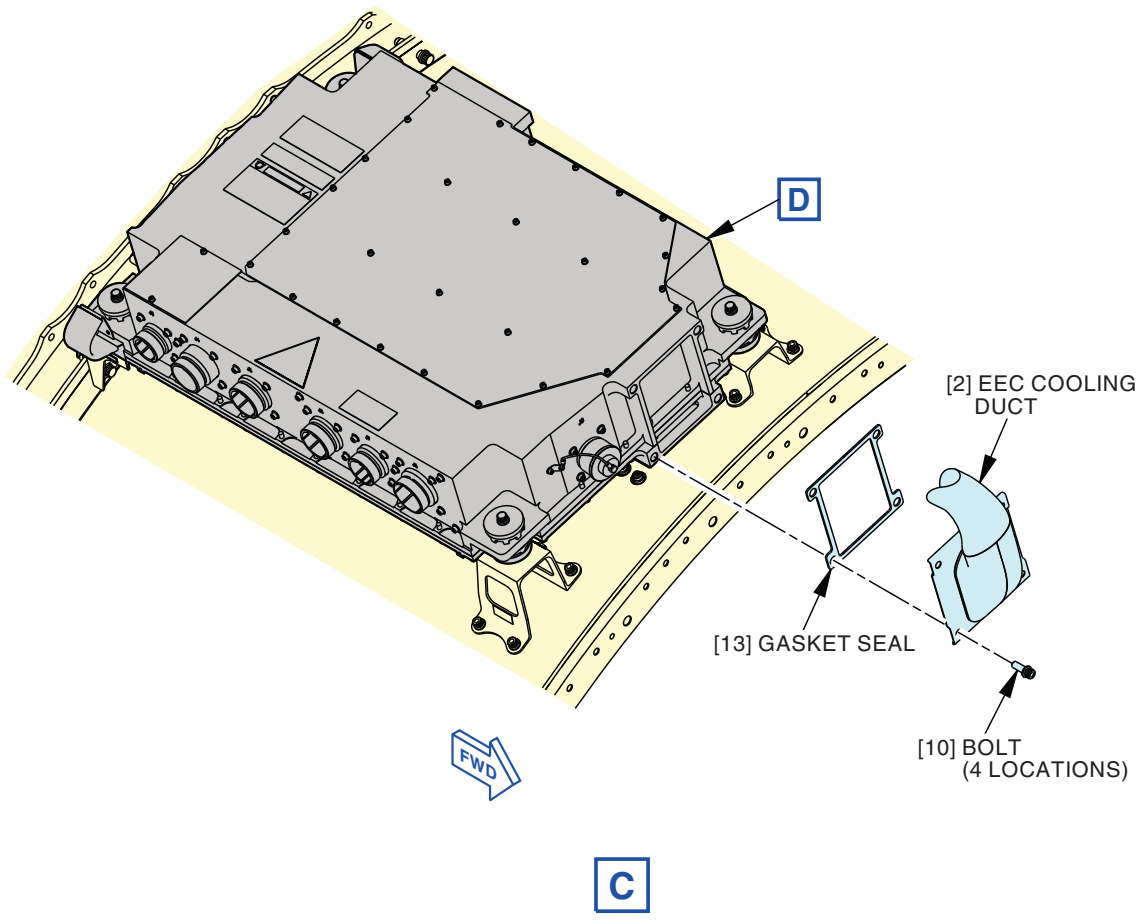
ECCN 9E991 BOEING PROPRIETARY - See title page for details



A0007-A-001-01
2354825 S00061515993_V2

Electronic Engine Control (EEC) Channel A Installation
Figure 401/73-21-05-990-808-G00 (Sheet 2 of 5)

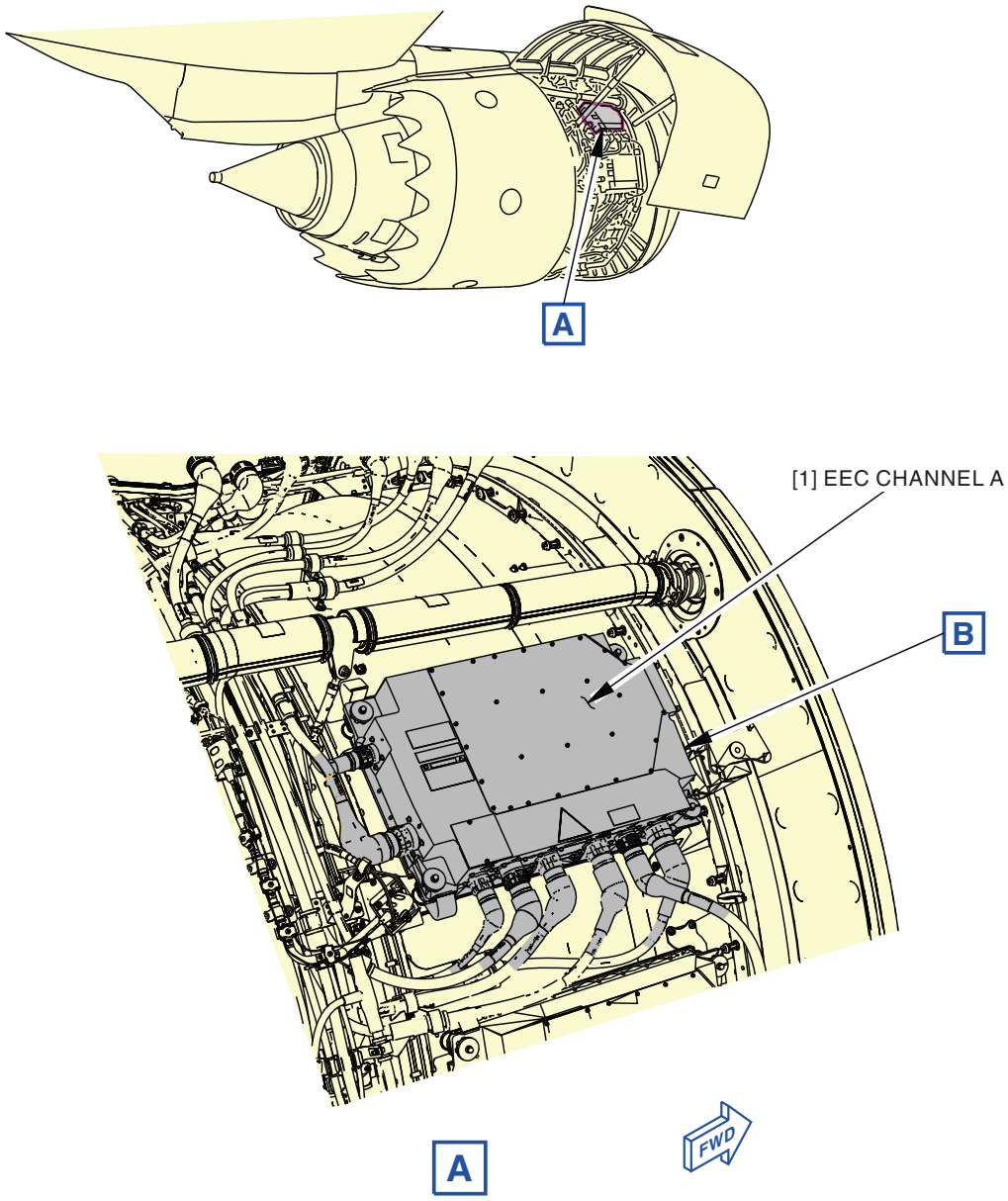
ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICN-LEAP-1B-732105-B-F0301-A0006-A-001-01
2398488 S0000553117_V2

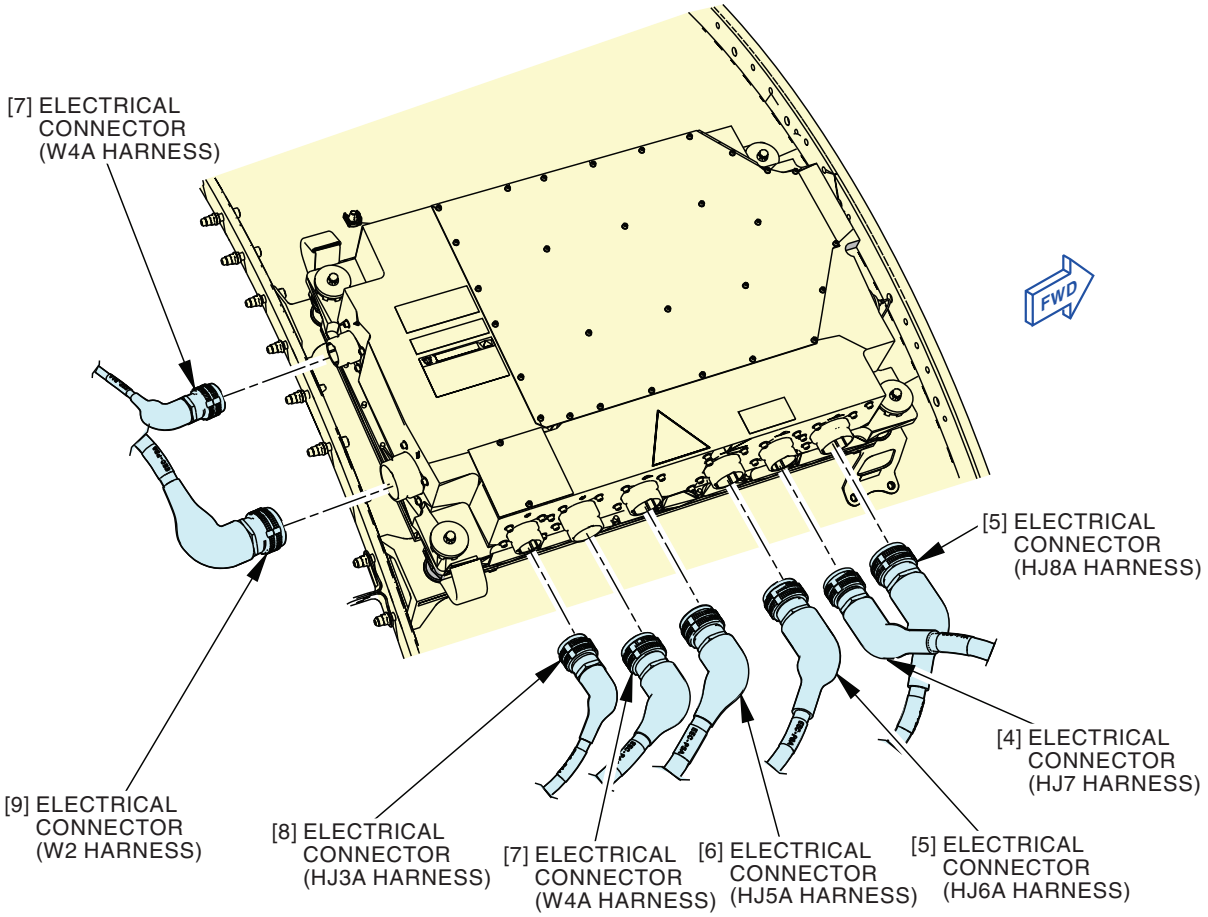
Electronic Engine Control (EEC) Channel A Installation
Figure 401/73-21-05-990-808-G00 (Sheet 3 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Electronic Engine Control (EEC) Channel A Installation
Figure 401/73-21-05-990-808-G00 (Sheet 4 of 5)

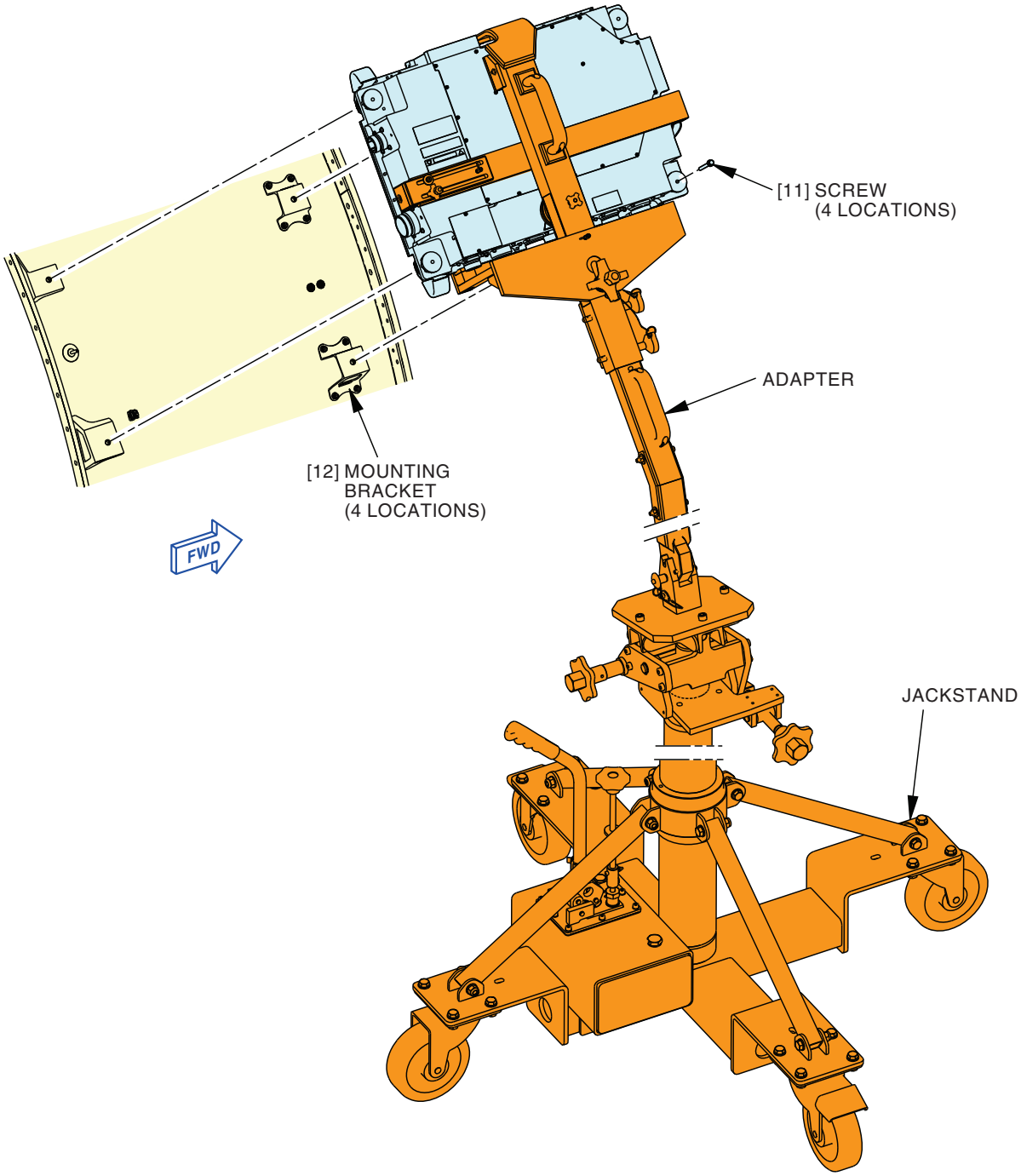
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2831276 S0000657419_V1

Electronic Engine Control (EEC) Channel A Installation
Figure 401/73-21-05-990-808-G00 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



EEC LRU Fixture Installation
Figure 402/73-21-05-990-810-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 73-21-05-400-801-G00

3. Electronic Engine Control (EEC) Channel A Installation

(Figure 401 and Figure 402)

A. General

(1) This task gives the instructions to install the Electronic Engine Control (EEC) channel A.

(2) A minimum of two persons are required to handle the EEC if the jack stand and EEC adapter support equipment are not used.

B. References

Reference	Title
46-13-00-860-803	Client Credentials Check (P/B 201)
70-00-01-400-801-G00	Electrical Bonding Straps Installation (P/B 201)
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
73-21-00-470-803-G00	EEC Onboard Software Installation (P/B 201)
73-21-00-470-804-G00	EEC Onboard Software Installation Verification (P/B 201)
73-21-05-970-801-G00	Electronic Engine Control (EEC) Data Download (On-Wing) Using a PMAT (P/B 201)
73-21-05-970-802-G00	Electronic Engine Control (EEC) Data Download (On Wing) Using the ONS (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14500	Stand - Jack Part #: 9471M17P01 Supplier: 58828
SPL-14529	Adapter - LRU Part #: 956A3616G01 Supplier: 58828
STD-592	Meter - Milliohms, Range from 0.001 to 100 milliohms
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00601 [CP2101]	Vaseline - Graphite Mineral	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	EEC CH A	73-21-05-01-010	PTF ALL
13	Gasket seal	75-24-08-01-140	PTF ALL
		75-24-08-01-180	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Zone	Area
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

H. EEC CH A Installation

SUBTASK 73-21-05-420-001-G00

WARNING THE EEC IS HEAVY AND TWO PERSONS ARE NECESSARY TO LIFT IT. IF YOU LIFT THE EEC INCORRECTLY, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

CAUTION MAKE SURE TO FOLLOW THE PROCEDURE FOR DEVICES THAT ARE SENSITIVE TO ELECTROSTATIC DISCHARGE. ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE EEC.

(1) Do these steps to install the EEC CH A [1] on the fan case mounting brackets [12] as follows:

NOTE: The EEC weighs approximately 37 lb (17 kg).

NOTE: If the two-person alternative method is used, you do not have to install the EEC to the adapter, SPL-14529 and jack stand, SPL-14500.

(a) Install the EEC CH A [1] in the jack stand, SPL-14500.

1) Make sure that the mating faces are clean and clear of unwanted materials.
2) Install the adapter, SPL-14529 on the jack stand, SPL-14500.
3) Attach the adapter, SPL-14529 on the EEC CH A [1].

(b) Install the EEC CH A [1] in the correct position on the mounting brackets [12].

(c) Install the four screws [11] that attach the EEC CH A [1] to the four mounting brackets [12].

NOTE: There are captive screws.

(d) Move the jack stand, SPL-14500 and adapter, SPL-14529 away from the engine.

(e) Tighten the screws [11] to 109.31 in-lb (12.35 N·m) - 120.81 in-lb (13.65 N·m).

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 75-0010 OR PRE LEAP-1B SERVICE BULLETIN 75-0011

(f) Install the EEC cooling duct [2] on the EEC CH A [1] as follows:

1) Remove the protective covers from the EEC cooling duct [2].
2) Make sure that the mating faces are clean and clear of unwanted materials.
3) Lubricate the four bolts [10] with graphite mineral vaseline, D00601 [CP2101].
4) Install the EEC cooling duct [2] with the new gasket seal [13] on the EEC CH A [1].
5) Install the four bolts [10] that attach the EEC cooling duct [2] to the EEC CH A [1].
6) Tighten the bolts [10] to 109.31 in-lb (12.35 N·m) - 120.81 in-lb (13.65 N·m).

PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 73-21-05-760-001-G00

(2) Check the electrical bonding between the EEC CH A [1] and the fan case:

NOTE: You must do this bonding check before you connect the electrical harness connectors to the EEC.

(a) Use the milliohm meter .001 to 100 milliohms, STD-592 to measure the electrical resistance between the EEC mounting foot next to the EEC bond strap on the EEC CH A [1] and the alodine treated area on the aft fan case flange near the EEC CH A [1].

NOTE: Look for the locations to put the ohmmeter probes when you measure the electrical resistance.

NOTE: The fan case flange has bolt holes with circular alodine patches at several locations. The alodine patch is gold in color.

1) Push the tip of the ohmmeter probe through the alodine patch until you reach the bare metal surface.

(b) Make sure that the resistance is less than 5 milliohms (0.005 ohms).

(c) If the resistance is more than 5 milliohms (0.005 ohms), do this step:

1) Make sure that the installation of the EEC CH A [1] electrical bonding strap is correct (TASK 70-00-01-400-801-G00).

SUBTASK 73-21-05-080-002-G00

(3) Remove the protective covers from these connectors and receptacles:

(a) The electrical connector (HJ8A) [3]

(b) The electrical connector (HJ7) [4]

(c) The electrical connector (HJ6A) [5]

(d) The electrical connector (HJ5A) [6]

(e) The two electrical connector (W4A)s [7]

(f) The electrical connector (HJ3A) [8]

(g) The electrical connector (W2) [9]

(h) The electrical receptacles on the EEC CH A [1].

SUBTASK 73-21-05-430-001-G00

CAUTION

USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTOR AND RECEPTACLE ARE CLEAN WHEN YOU CONNECT THEM. DIRTY CONNECTORS CAN CAUSE DAMAGE TO EQUIPMENT.

(4) Use the teflon-jawed pliers, STD-664 to connect and tighten these electrical connectors to the EEC CH A [1] (TASK 70-00-01-910-803-G00):

(a) The electrical connector (HJ8A) [3]

(b) The electrical connector (HJ7) [4]

(c) The electrical connector (HJ6A) [5]

(d) The electrical connector (HJ5A) [6]

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) The electrical connector (W4A) [7]

(f) The electrical connector (HJ3A) [8]

(g) The electrical connector (W2) [9]

(h) The electrical connector (W4A) [7].

I. Put the Airplane Back to Its Usual Condition

SUBTASK 73-21-05-860-013-G00

(1) On the applicable engine, close the right fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

SUBTASK 73-21-05-860-007-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

SUBTASK 73-21-05-922-001-G00

(3) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A
B	3	C01312	ENGINE 1 RUN/PWR

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	5	C01313	ENGINE 2 RUN/PWR
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	3	C04018	ENG 1 STBY ALTN PWR RELAY CHAN A
E	4	C04019	ENG 1 STBY ALTN PWR CHAN A

SUBTASK 73-21-05-470-001-G00

(4) To determine whether the Electronic Engine Control (EEC)s have preloaded credentials, conduct Method 3 in this task: Client Credentials Check, TASK 46-13-00-860-803.

NOTE: This is only applicable when the ONS OS9.1 EEC is new and needs to generate client credential for the first time.

(a) If EEC client credentials generate with a current date and time, the EECs have preloaded credentials and no further action is needed to generate client credentials.

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(b) If EEC client credentials do not generate with a current date and time, the EECs do not have preloaded credentials.

1) If the installed EEC does not have preloaded credentials, generate the EEC client credential by performing Method 1 or Method 2 in this task: Client Credentials Check, TASK 46-13-00-860-803.

NOTE: If neither Method 1 or 2 can be performed, this task may be deferred. EEC data will not store to the Network File Server (NFS) until the EEC client credential is generated, but the data can be pulled via a Portable Data Loader (PDL), Portable Maintenance Access Terminal (PMAT), or Universal Maintenance Device (UMD) from the EEC. These are the tasks: Electronic Engine Control (EEC) Data Download (On-Wing) Using a PMAT, TASK 73-21-05-970-801-G00 or Electronic Engine Control (EEC) Data Download (On Wing) Using the ONS, TASK 73-21-05-970-802-G00.

NOTE: Ensure all EEC credentials have a date and time stamp that occur after the date and time of the EEC installation.

SUBTASK 73-21-05-470-009-G00

(5) Make sure that the correct software configuration is in the EEC as follows:

(a) Do this task: EEC Onboard Software Installation Verification, TASK 73-21-00-470-804-G00.

(b) If the software version is not correct, reprogram the EEC software (TASK 73-21-00-470-803-G00).

J. EEC CH A Installation Test

SUBTASK 73-21-05-730-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

TASK 73-21-05-000-802-G00

4. Electronic Engine Control (EEC) Channel B Removal

(Figure 403 and Figure 404)

A. General

(1) This task gives the instructions to remove the Electronic Engine Control (EEC) channel B.

(2) A minimum of two persons are required to handle the EEC if the jack stand and EEC adapter support equipment are not used.

B. References

Reference	Title
46-13-02-970-802	Download Manager (P/B 201)
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-00-00-720-805-G00	Test No. 11 - Low Pressure System (Fan) Trim Balance (With ONS) (P/B 501)
71-00-00-970-801-G00	Maintenance Data Pages from the Engine Propulsion Control System (EPCS) (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14500	Stand - Jack Part #: 9471M17P01 Supplier: 58828
SPL-14529	Adapter - LRU Part #: 956A3616G01 Supplier: 58828
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

F. Prepare for the Removal

SUBTASK 73-21-05-860-008-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-21-05-970-007-G00

(2) If you will replace the two EECs at the same time, do these steps before you remove the EECs:

(a) Make a record of the current engine trim balance weight configuration (TASK 71-00-00-720-805-G00).

(b) Make a record of the current engine cumulative oil debris chip count ("DMS") on EPCS page 2 (TASK 71-00-00-970-801-G00).

SUBTASK 73-21-05-970-010-G00

(3) If you will remove the EEC because of a fault isolation procedure, do this step:

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(a) Download the EEC Non Volatile Memory (NVM) from the two EEC channels before you remove the EEC (TASK 46-13-02-970-802).

NOTE: Downloading the CEOD and EEC NVM data is optional. The EEC NVM and CEOD download is recommended only for cases where repetitive maintenance actions or component replacements have failed to identify the root cause of the fault condition. The EEC NVM and CEOD can be helpful to identify the root cause of the fault condition. However, when the EEC is removed for a known cause, it is not necessary to do the EEC NVM and CEOD download.

NOTE: The EEC NVM is listed in Download Manager as EEC1:EN1SHOPFLT for Engine 1 and EEC2:EN2SHOPFLT for Engine 2.

NOTE: You can also use the EEC DATA DOWNLOAD function on the OTHER FUNCTIONS > EHM menu on the Portable Maintenance Device (PMD) to download the EEC NVM. The EEC NVM is listed as 'EEC Shop Fault Channel A & B' in the EEC DATA DOWNLOAD menu.

SUBTASK 73-21-05-860-009-G00

(4) Open these applicable circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A
B	3	C01312	ENGINE 1 RUN/PWR

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	5	C01313	ENGINE 2 RUN/PWR
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	3	C04018	ENG 1 STBY ALTN PWR RELAY CHAN A
E	4	C04019	ENG 1 STBY ALTN PWR CHAN A

SUBTASK 73-21-05-860-014-G00

(5) On the applicable engine, open the right fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

G. EEC CH B Removal

SUBTASK 73-21-05-010-003-G00

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION

USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

CAUTION

MAKE SURE TO FOLLOW THE PROCEDURE FOR DEVICES THAT ARE SENSITIVE TO ELECTROSTATIC DISCHARGE. ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE EEC.

(1) Use the teflon-jawed pliers, STD-664, to disconnect these electrical connectors from the EEC CH B [1] (TASK 70-00-01-910-803-G00):

(a) The electrical connector (HJ8B) [3]

(b) The electrical connector (HJ7) [4]

(c) The electrical connector (HJ6B) [5]

(d) The electrical connector (HJ5B) [6]

(e) The electrical connector (W4B) [7]

(f) The electrical connector (HJ3B) [8]

(g) The electrical connector (W2) [9]

(h) The electrical connector (W4B) [7].

SUBTASK 73-21-05-480-007-G00

(2) Install the protection covers on the electrical connectors and the electrical receptacles on the EEC CH B [1].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 75-0010 OR PRE LEAP-1B SERVICE BULLETIN 75-0011

SUBTASK 73-21-05-010-004-G00

(3) Remove the EEC cooling duct [2] from the EEC CH B [1] as follows:

(a) Remove the four bolts [10] that attach the EEC cooling duct [2] to the EEC CH B [1].

(b) Disconnect the EEC cooling duct [2] from the EEC CH B [1].

(c) Remove and discard the gasket seal [13].

(d) Loosen the screw of the upper flexible duct.

(e) Disconnect the upper flexible duct of the cooling duct.

SUBTASK 73-21-05-480-006-G00

(4) Install the protective covers on the EEC cooling duct [2].

PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 73-21-05-010-005-G00

WARNING THE EEC IS HEAVY AND TWO PERSONS ARE NECESSARY TO LIFT IT. IF YOU LIFT THE EEC INCORRECTLY, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(5) Do these steps to remove the EEC CH B [1] from the fan case mounting brackets [12] as follows:

NOTE: The EEC weighs approximately 37 lb (17 kg).

NOTE: If the two-person alternative method is used, you do not have to install the EEC to the adapter, SPL-14529, and jack stand, SPL-14500.

(a) Attach the adapter, SPL-14529, to the EEC CH B [1].

(b) Move the jack stand, SPL-14500, to the correct position.

(c) Install the adapter, SPL-14529, on the jack stand, SPL-14500.

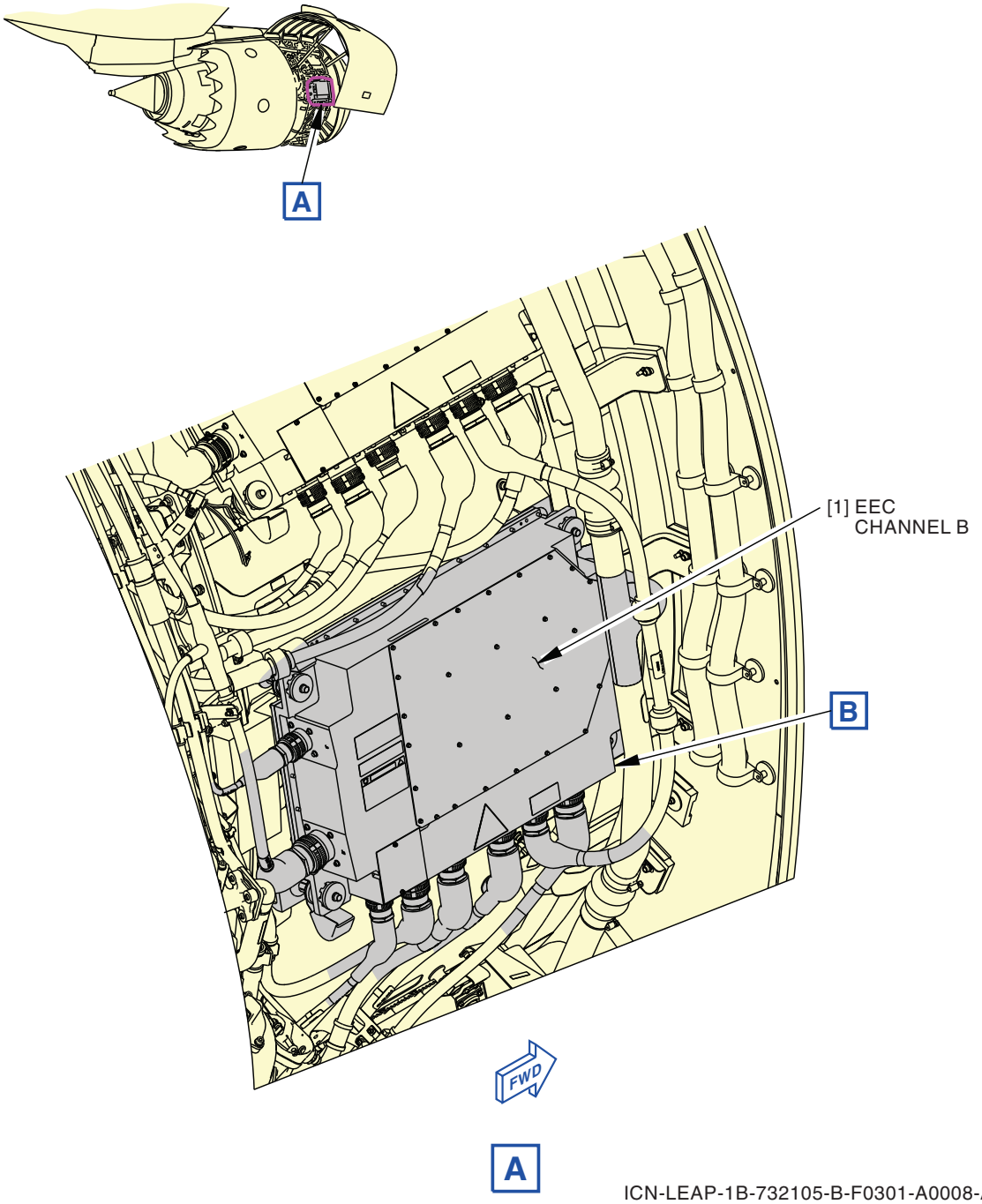
(d) Remove the four screws [11] that attach the EEC CH B [1] on the four mounting brackets [12].

NOTE: Do not remove the captive screws.

(e) Remove the EEC CH B [1] from the four mounting brackets [12].

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

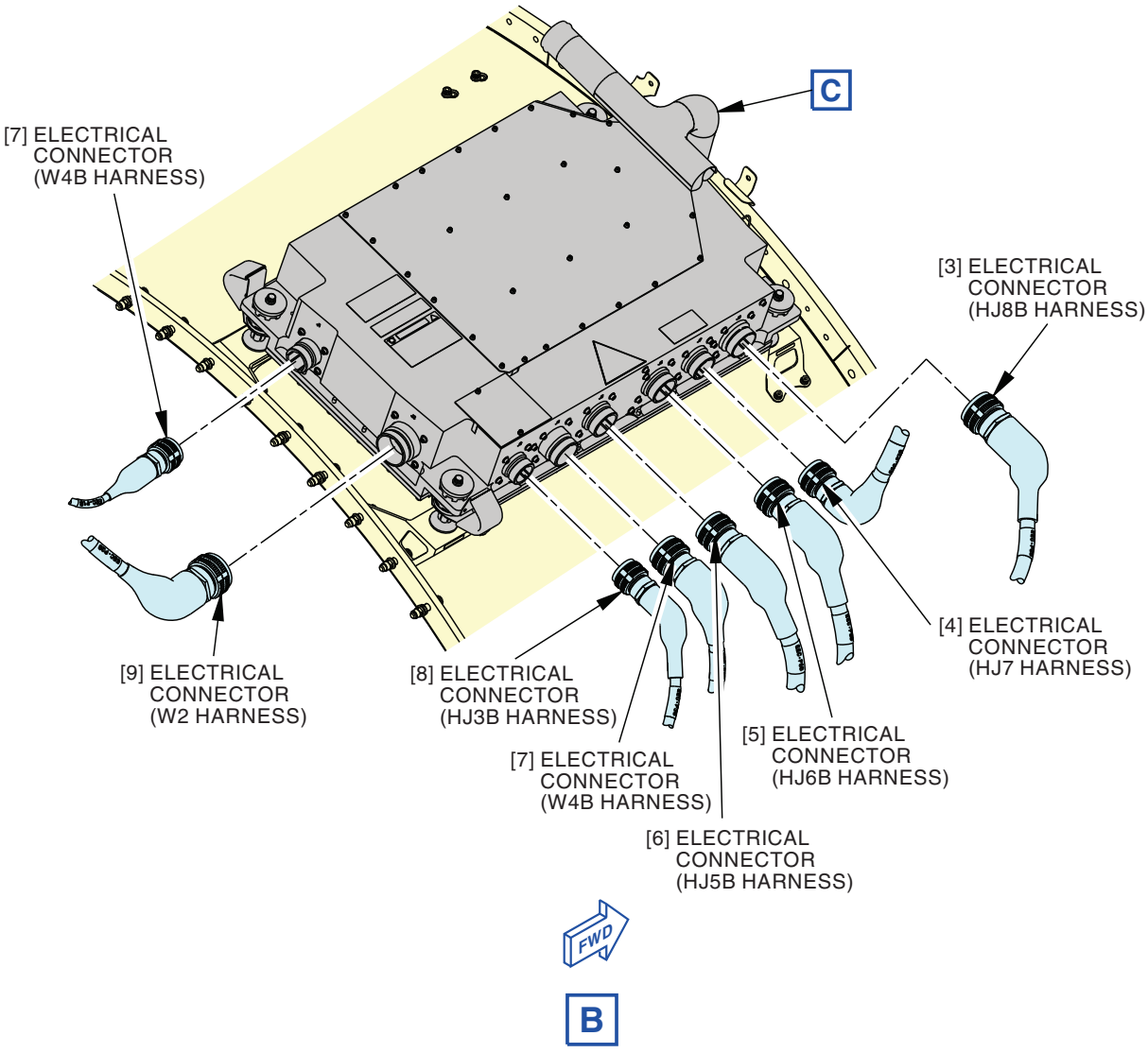


ICN-LEAP-1B-732105-B-F0301-A0008-A-001-01

2398544 S0000553118_V2

Electronic Engine Control (EEC) Channel B Installation
Figure 403/73-21-05-990-811-G00 (Sheet 1 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

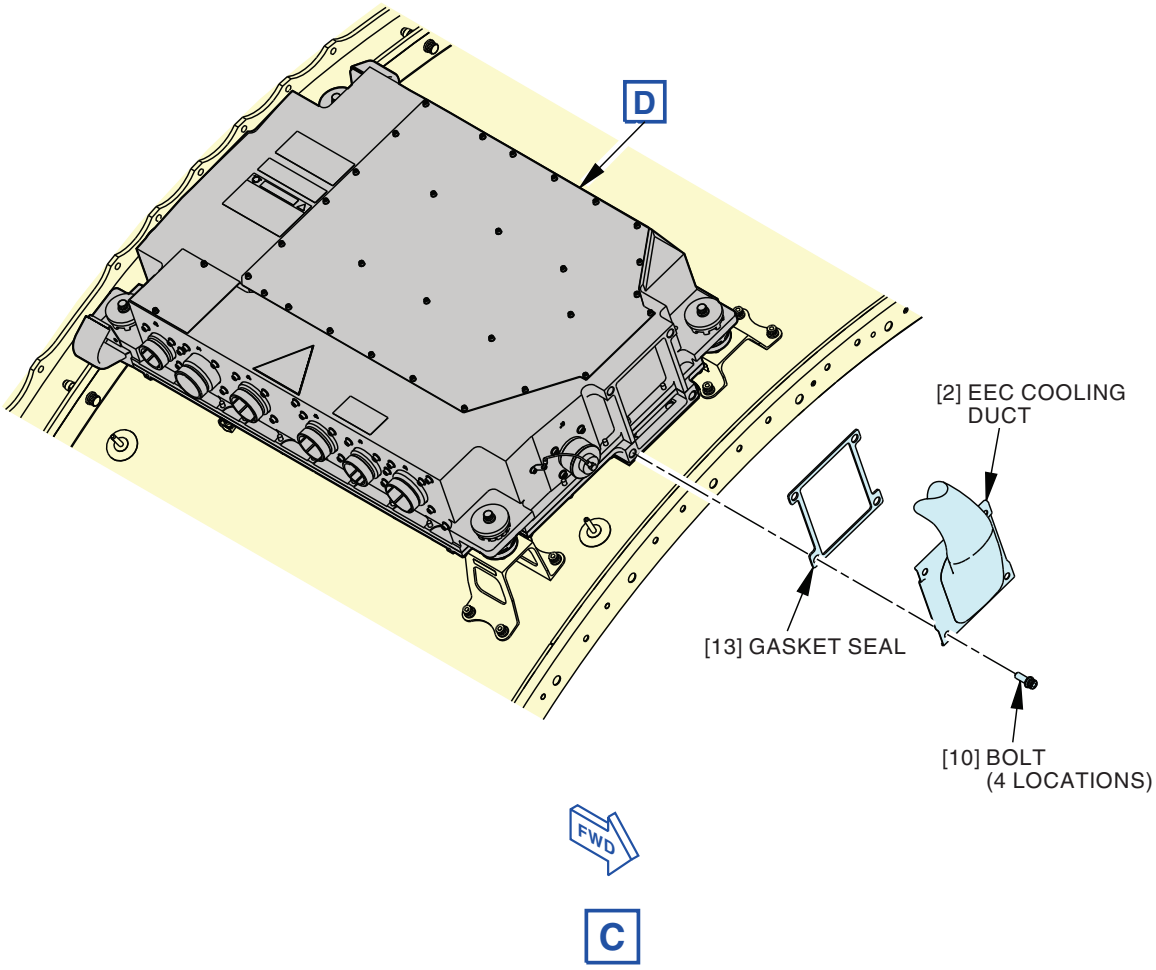


ICN-LEAP-1B-732105-B-F0301-A0009-A-001-01

2398546 S0000553120_V2

Electronic Engine Control (EEC) Channel B Installation
Figure 403/73-21-05-990-811-G00 (Sheet 2 of 5)

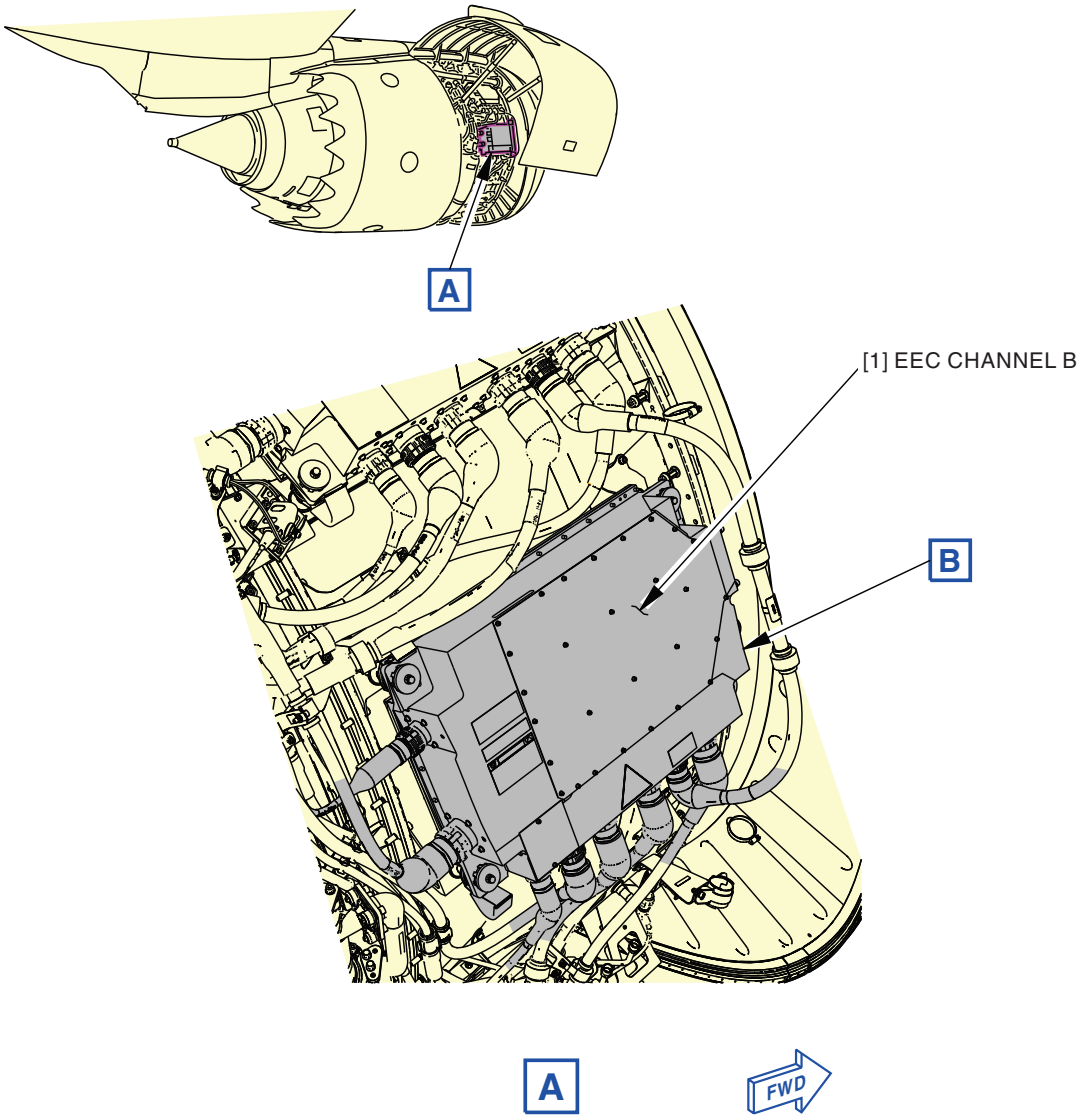
ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICN-LEAP-1B-732105-B-F0301-A0010-A-001-01
2398547 S0000553121_V3

Electronic Engine Control (EEC) Channel B Installation
Figure 403/73-21-05-990-811-G00 (Sheet 3 of 5)

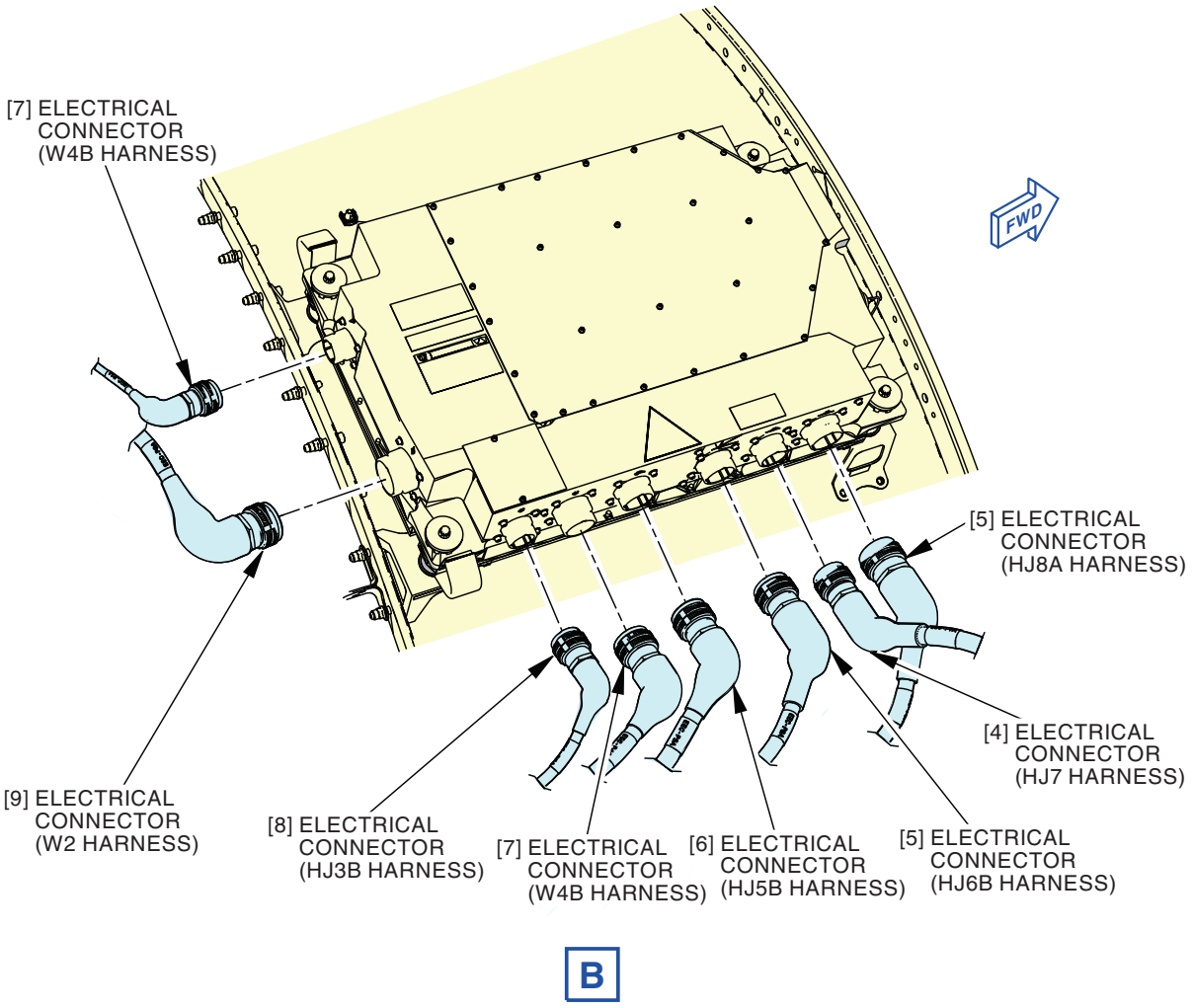
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2831340 S0000657420_V1

Electronic Engine Control (EEC) Channel B Installation
Figure 403/73-21-05-990-811-G00 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

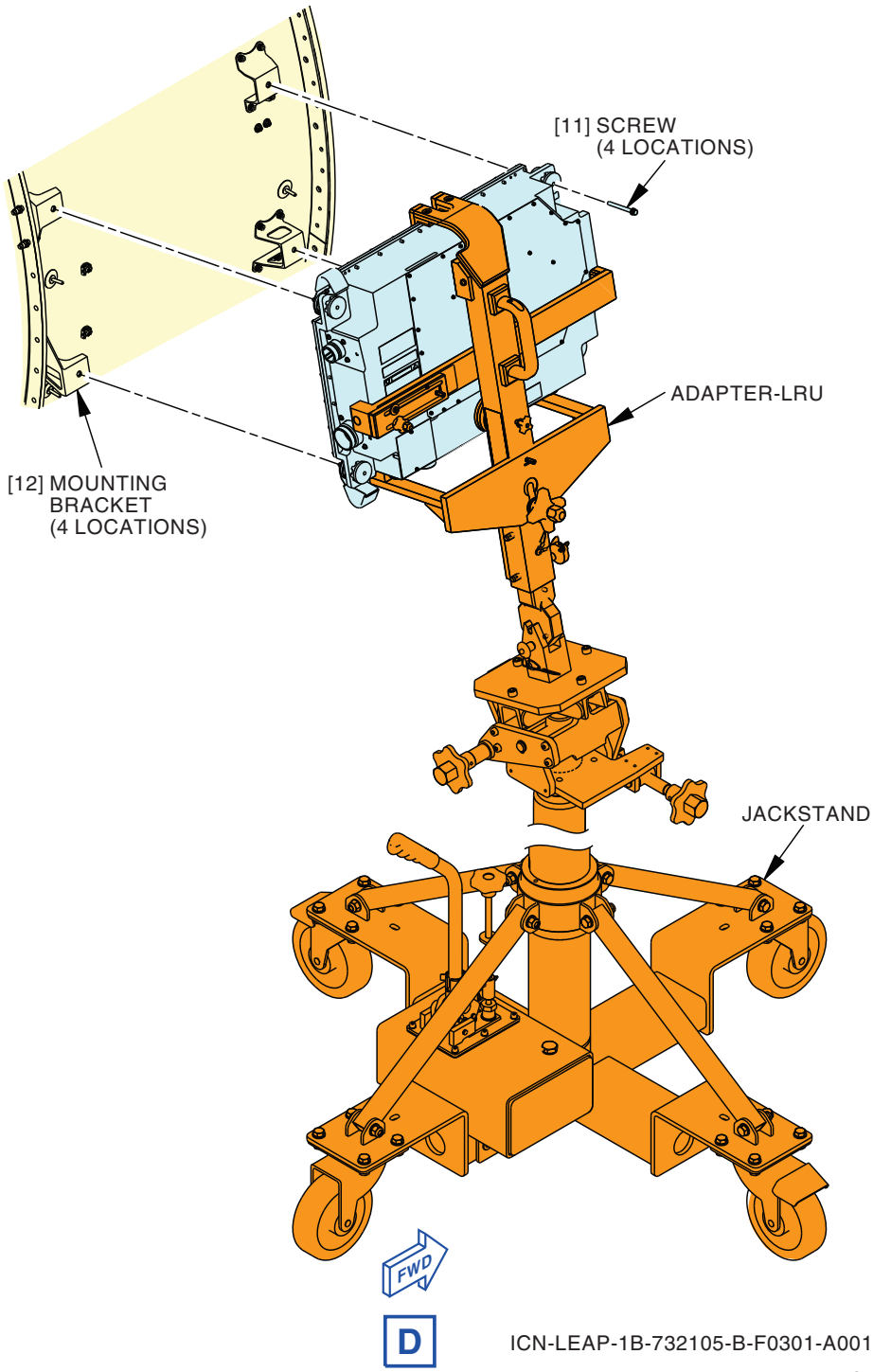


5055173-00

2831353 S0000657421_V1

Electronic Engine Control (EEC) Channel B Installation
Figure 403/73-21-05-990-811-G00 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



EEC LRU Fixture Installation
Figure 404/73-21-05-990-812-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 73-21-05-400-802-G00

5. Electronic Engine Control (EEC) Channel B Installation

(Figure 403 and Figure 404)

A. General

(1) This task gives the instructions to install the Electronic Engine Control (EEC) channel B.

(2) A minimum of two persons are required to handle the EEC if the jack stand and EEC adapter support equipment are not used.

B. References

Reference	Title
46-13-00-860-803	Client Credentials Check (P/B 201)
70-00-01-400-801-G00	Electrical Bonding Straps Installation (P/B 201)
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
73-21-00-470-803-G00	EEC Onboard Software Installation (P/B 201)
73-21-00-470-804-G00	EEC Onboard Software Installation Verification (P/B 201)
73-21-05-970-801-G00	Electronic Engine Control (EEC) Data Download (On-Wing) Using a PMAT (P/B 201)
73-21-05-970-802-G00	Electronic Engine Control (EEC) Data Download (On Wing) Using the ONS (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14500	Stand - Jack Part #: 9471M17P01 Supplier: 58828
SPL-14529	Adapter - LRU Part #: 956A3616G01 Supplier: 58828
STD-592	Meter - Milliohms, Range from 0.001 to 100 milliohms
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00601 [CP2101]	Vaseline - Graphite Mineral	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	EEC CH B	73-21-05-01-020	PTF ALL
13	Gasket seal	75-24-08-01-140	PTF ALL
		75-24-08-01-180	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Zone	Area
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

H. EEC CH B Installation

SUBTASK 73-21-05-410-002-G00

WARNING THE EEC IS HEAVY AND TWO PERSONS ARE NECESSARY TO LIFT IT. IF YOU LIFT THE EEC INCORRECTLY, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

CAUTION MAKE SURE TO FOLLOW THE PROCEDURE FOR DEVICES THAT ARE SENSITIVE TO ELECTROSTATIC DISCHARGE. ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE EEC.

(1) Do these steps to install the EEC CH B [1] on the fan case mounting brackets [12] as follows:

NOTE: The EEC weighs approximately 37 lb (17 kg).

NOTE: If the two-person alternative method is used, you do not have to install the EEC to the adapter, SPL-14529, and jack stand, SPL-14500.

(a) Install the EEC CH B [1] in the jack stand, SPL-14500.

1) Make sure that the mating faces are clean and clear of unwanted materials.
2) Install the adapter, SPL-14529, on the jack stand, SPL-14500.
3) Attach the adapter, SPL-14529, on the EEC CH B [1].

(b) Install the EEC CH B [1] in the correct position on the mounting brackets [12].

(c) Install the four screws [11] to attach the EEC CH B [1] on the four mounting brackets [12].

(d) Move the jack stand, SPL-14500, and adapter, SPL-14529, away from the engine.

(e) Tighten the four screws [11] to 109.31 in-lb (12.35 N·m) - 120.81 in-lb (13.65 N·m).

NOTE: There are captive screws.

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 75-0010 OR PRE LEAP-1B SERVICE BULLETIN 75-0011

(f) Install the EEC cooling duct [2] on the EEC CH B [1] as follows:

1) Remove the protection covers from the EEC cooling duct [2].
2) Make sure that the mating faces are clean and clear of unwanted materials.
3) Lubricate the four bolts [10] with graphite mineral vaseline, D00601 [CP2101].
4) Connect the upper flexible duct of the cooling duct.
5) Tighten the screw of the upper flexible duct.
6) Install the EEC cooling duct [2] with the new gasket seal [13] on the EEC CH B [1].
7) Install the four bolts [10] to attach the EEC cooling duct [2] to the EEC CH B [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 75-0010 OR PRE LEAP-1B SERVICE BULLETIN 75-0011 (Continued)

8) Tighten the bolts [10] to 109.31 in-lb (12.35 N·m) - 120.81 in-lb (13.65 N·m).

PTF ALL

SUBTASK 73-21-05-760-002-G00

(2) Examine the electrical bonding between the EEC CH B [1] and the fan case as follows:

NOTE: You must do this bonding check before you connect the electrical harness connectors to the EEC.

(a) Use the milliohm meter .001 to 100 milliohms, STD-592, to measure the electrical resistance between the EEC mounting foot next to the EEC bond strap on the EEC CH B [1] and the alodine treated area on the aft fan case flange near the EEC CH B [1].

NOTE: Look for the locations to put the ohmmeter probes when you measure the electrical resistance.

NOTE: The fan case flange has bolt holes with circular alodine patches at several locations. The alodine patch is gold in color.

1) Push the tip of the ohmmeter probe through the alodine patch until you reach the bare metal surface.

(b) Make sure that the resistance is less than 5 milliohms (0.005 ohms).

(c) If the resistance is more than 5 milliohms (0.005 ohms), do this step:

1) Make sure that the installation of the EEC CH B [1] electrical bonding strap is correct (TASK 70-00-01-400-801-G00).

SUBTASK 73-21-05-080-003-G00

(3) Remove the protective covers from these connectors and receptacles:

(a) The electrical connector (HJ8B) [3]

(b) The electrical connector (HJ7) [4]

(c) The electrical connector (HJ6B) [5]

(d) The electrical connector (HJ5B) [6]

(e) The two electrical connector (W4B)s [7]

(f) The electrical connector (HJ3B) [8]

(g) The electrical connector (W2) [9]

(h) The electrical receptacles on the EEC CH B [1].

SUBTASK 73-21-05-410-003-G00

CAUTION

USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTOR AND RECEPTACLE ARE CLEAN WHEN YOU CONNECT THEM. DIRTY CONNECTORS CAN CAUSE DAMAGE TO EQUIPMENT.

(4) Use the teflon-jawed pliers, STD-664, to connect and tighten these electrical connectors to the EEC CH B [1] (TASK 70-00-01-910-803-G00):

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) The electrical connector (HJ8B) [3]

(b) The electrical connector (HJ7) [4]

(c) The electrical connector (HJ6B) [5]

(d) The electrical connector (HJ5B) [6]

(e) The electrical connector (W4B) [7]

(f) The electrical connector (HJ3B) [8]

(g) The electrical connector (W2) [9]

(h) The electrical connector (W4B) [7].

I. Put the Airplane Back to Its Usual Condition

SUBTASK 73-21-05-860-015-G00

(1) On the applicable engine, close the right fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

SUBTASK 73-21-05-860-010-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

SUBTASK 73-21-05-860-011-G00

(3) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A
B	3	C01312	ENGINE 1 RUN/PWR

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	5	C01313	ENGINE 2 RUN/PWR
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	3	C04018	ENG 1 STBY ALTN PWR RELAY CHAN A
E	4	C04019	ENG 1 STBY ALTN PWR CHAN A

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 73-21-05-211-001-G00

(4) To determine whether the EECs have preloaded credentials, conduct Method 3 in this task: Client Credentials Check, TASK 46-13-00-860-803.

NOTE: This is only applicable when the ONS OS9.1 EEC is new and needs to generate client credential for the first time.

(a) If EEC client credentials generate with a current date and time, the EECs have preloaded credentials and no further action is needed to generate client credentials.

(b) If EEC client credentials do not generate with a current date and time, the EECs do not have preloaded credentials.

1) If the installed EEC does not have preloaded credentials, generate the EEC client credential by performing Method 1 or Method 2 in this task: Client Credentials Check, TASK 46-13-00-860-803.

NOTE: If neither Method 1 or 2 can be performed, this task may be deferred. EEC data will not store to the Network File Server (NFS) until the EEC client credential is generated, but the data can be pulled via a Portable Data Loader (PDL), Portable Maintenance Access Terminal (PMAT), or Universal Maintenance Device (UMD) from the EEC. These are the tasks: Electronic Engine Control (EEC) Data Download (On-Wing) Using a PMAT, TASK 73-21-05-970-801-G00 or Electronic Engine Control (EEC) Data Download (On Wing) Using the ONS, TASK 73-21-05-970-802-G00.

NOTE: Ensure all EEC credentials have a date and time stamp that occur after the date and time of the EEC installation.

SUBTASK 73-21-05-211-002-G00

(5) Do the EEC software configuration check as follows:

(a) Make sure that the correct software is installed in the EEC as follows:

1) Do this task: EEC Onboard Software Installation Verification, TASK 73-21-00-470-804-G00.

2) If incorrect software version is installed, reprogram the EEC software (TASK 73-21-00-470-803-G00).

J. EEC CH B Installation Test

SUBTASK 73-21-05-730-002-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ELECTRONIC ENGINE CONTROL - INSPECTION/CHECK

1. General

A. This procedure has one task:

(1) Electronic Engine Control (EEC) Inspection.

TASK 73-21-05-210-801-G00

2. Electronic Engine Control Inspection

(Figure 601)

A. General

(1) This task gives the instructions to inspect the Electronic Engine Controls (EEC).

(a) The two EECs are located on the right side of the fan case.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
73-21-05-000-801-G00	Electronic Engine Control (EEC) Channel A Removal (P/B 401)
73-21-05-000-802-G00	Electronic Engine Control (EEC) Channel B Removal (P/B 401)
73-21-05-400-801-G00	Electronic Engine Control (EEC) Channel A Installation (P/B 401)
73-21-05-400-802-G00	Electronic Engine Control (EEC) Channel B Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

F. Prepare for the Inspection

SUBTASK 73-21-05-860-032-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 73-21-05-860-033-G00

(2) Open these applicable circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A
B	3	C01312	ENGINE 1 RUN/PWR

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	5	C01313	ENGINE 2 RUN/PWR
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	3	C04018	ENG 1 STBY ALTN PWR RELAY CHAN A
E	4	C04019	ENG 1 STBY ALTN PWR CHAN A

SUBTASK 73-21-05-010-010-G00

(3) On the applicable engine, open the right fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

G. EEC Inspection

SUBTASK 73-21-05-210-003-G00

(1) Inspect the Electronic Engine Controls (EECs):

(a) Make sure that the EECs are not loose.

1) If an EEC is loose, tighten the bolts that attaches the EEC to the mount brackets (Electronic Engine Control (EEC) Channel A Installation, TASK 73-21-05-400-801-G00 or Electronic Engine Control (EEC) Channel B Installation, TASK 73-21-05-400-802-G00).

(b) Look for cracks in the EEC housings.

1) If you see cracks, do these steps:

a) If the crack is on EEC channel A, replace the EEC (Electronic Engine Control (EEC) Channel A Removal, TASK 73-21-05-000-801-G00 and Electronic Engine Control (EEC) Channel A Installation, TASK 73-21-05-400-801-G00).

b) If the crack is on EEC channel B, replace the EEC (Electronic Engine Control (EEC) Channel B Removal, TASK 73-21-05-000-802-G00 and Electronic Engine Control (EEC) Channel B Installation, TASK 73-21-05-400-802-G00).

(c) Make sure that the electrical connectors are not loose.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

1) If an electrical connector is loose, use a teflon-jawed pliers, STD-664 to tighten the electrical connector.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 73-21-05-410-009-G00

(1) On the applicable engine, close the right fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

SUBTASK 73-21-05-860-034-G00

(2) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A
B	3	C01312	ENGINE 1 RUN/PWR

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	5	C01313	ENGINE 2 RUN/PWR
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	3	C04018	ENG 1 STBY ALTN PWR RELAY CHAN A
E	4	C04019	ENG 1 STBY ALTN PWR CHAN A

SUBTASK 73-21-05-860-035-G00

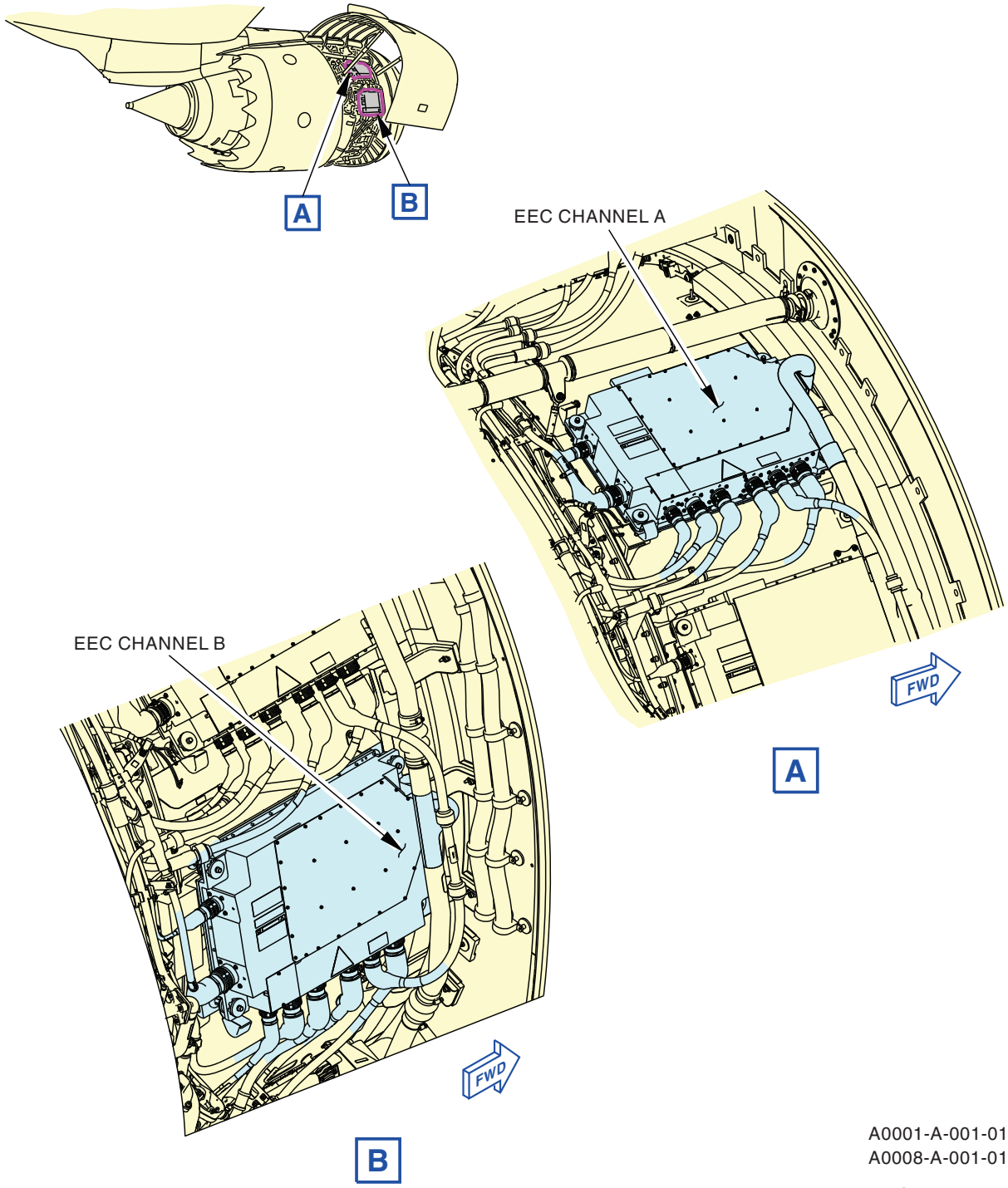
(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



A0001-A-001-01
A0008-A-001-01

2581338 S0000621802_V1

Electronic Engine Control Inspection
Figure 601/73-21-05-990-816-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

T12 SENSOR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the T12 sensor

(2) An installation of the T12 sensor.

TASK 73-21-06-000-801-G00

2. T12 Sensor Removal

(Figure 401)

A. General

(1) This task provides the instructions on how to remove the T12 sensor.

(2) The T12 sensor is located at the 11:00 o'clock position on the inlet cowl from the aft looking forward.

(3) You must remove the T12 access panel on the outer surface of the inlet cowl to get access to the T12 sensor.

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Reference	Description
COM-2481	Sealant Removal Tool (Meets BSS7384 Requirements) Part #: 1-6390-A Supplier: 63318 Part #: 10810 Supplier: $0855 Part #: 10811 Supplier: $0855 Part #: 10812 Supplier: $0855 Part #: 234350 Supplier: 5HCF1 Part #: 235072 Supplier: 5HCF1 Part #: 235073 Supplier: 5HCF1 Part #: 235074 Supplier: 5HCF1 Part #: 235075 Supplier: 5HCF1 Part #: 235076 Supplier: 5HCF1 Part #: 311/03 Supplier: F6892 Part #: 311/14 Supplier: F6892 Part #: 311/25 Supplier: F6892 Part #: 311/37 Supplier: F6892 Part #: AS1 Supplier: $1351 Part #: AS2 Supplier: $1351 Part #: AS3 Supplier: $1351 Part #: DAD5013 Supplier: 7RKH2 Part #: DFD5019 Supplier: 7RKH2 Part #: JNT411B60 Supplier: 3DN12 Part #: JNT411B90 Supplier: 3DN12 Part #: PT6529-S Supplier: 81205 Part #: SCD5019 Supplier: 7RKH2 Part #: ST982LF-9 Supplier: 81205 Part #: TS1275-4 Supplier: 22975
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
412AT	T12 Access Door, Engine 1
413	Left Fan Cowl, Engine 1
422AT	T12 Access Door, Engine 2
423	Left Fan Cowl, Engine 2

F. Prepare for the Removal

SUBTASK 73-21-06-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 73-21-06-860-002-G00

(2) For the left engine, open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A

SUBTASK 73-21-06-860-003-G00

(3) For the right engine, open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

SUBTASK 73-21-06-040-001-G00

(4) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 73-21-06-020-001-G00

(5) Remove the T12 access panel [2] as follows:

(a) Open these applicable access panels:

Number	Name/Location
412AT	T12 Access Door, Engine 1
422AT	T12 Access Door, Engine 2

PTF ALL; AIRPLANES WITH T-12 PANEL WITH 17 FASTENERS

(b) Remove the 17 screws [7] that attach the T12 access panel [2] to the inlet cowl [3].

PTF ALL; AIRPLANES WITH T-12 PANEL WITH 9 FASTENERS

(c) Remove the nine screws [7] that attach the T12 access panel [2] to the inlet cowl [3].

PTF ALL

(d) Remove the T12 access panel [2] from the engine.

SUBTASK 73-21-06-010-002-G00

(6) On the applicable engine, open the left fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

G. T12 Sensor Removal

SUBTASK 73-21-06-020-002-G00

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS OR RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION

USE TEFLON-JAWED PLIERS TO LOOSEN THE ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(1) Use a teflon-jawed pliers, STD-664, to disconnect the T12-A electrical connector [5] and T12-B electrical connector [6] from the T12 sensor [1] as follows (TASK 70-00-01-910-803-G00):

(a) Disconnect the T12-A electrical connector [5].

(b) Disconnect the T12-B electrical connector [6].

(c) Install the protective covers on the T12-A electrical connector [5], T12-B electrical connector [6], T12-A and T12-B electrical receptacles on the T12 sensor [1].

SUBTASK 73-21-06-420-001-G00

(2) Remove the T12 sensor [1] as follows:

(a) Remove the nut [12] and washer [15] that attach the bonding strap [11] to the T12 sensor [1].

(b) Remove the four bolts [8] and four washers [13] that attach the T12-A and T12-B electrical receptacles to the aft bulk head [4].

CAUTION

MAKE SURE THAT YOU USE THE CORRECT TOOLS FOR THE AFT BULK-HEAD AND T12 SENSOR. IF YOU DO NOT OBEY, DAMAGE TO THE AFT BULK-HEAD AND T12 SENSOR CAN OCCUR.

CAUTION

BE CAREFUL WHEN YOU REMOVE THE SEALANTS FROM THE AFT BULK-HEAD AND T12 CONNECTOR HOUSING. IF YOU ARE NOT CAREFUL, DAMAGE TO THE AFT BULK-HEAD AND T12 CONNECTOR HOUSING CAN OCCUR.

(c) Use a sealant removal tool, COM-2481, to remove cured sealant from surfaces.

PTF ALL; AIRPLANES WITH PROBE PAN HELD BY BOLTS

(d) Remove the probe pan [10] with the T12 sensor [1] attached.

1) Remove the four bolts [16] and washers [17] from the sides of the mounting brackets that attach the probe pan [10].

(e) Remove the four bolts [9], four washers [18], and four nuts [19] that attach the T12 sensor [1] to the probe pan [10].

PTF ALL; AIRPLANES WITH PROBE PAN HELD BY STUDS

(f) Remove the four nuts [19] and six washers [18] that attach the T12 sensor [1] to the probe pan [10].

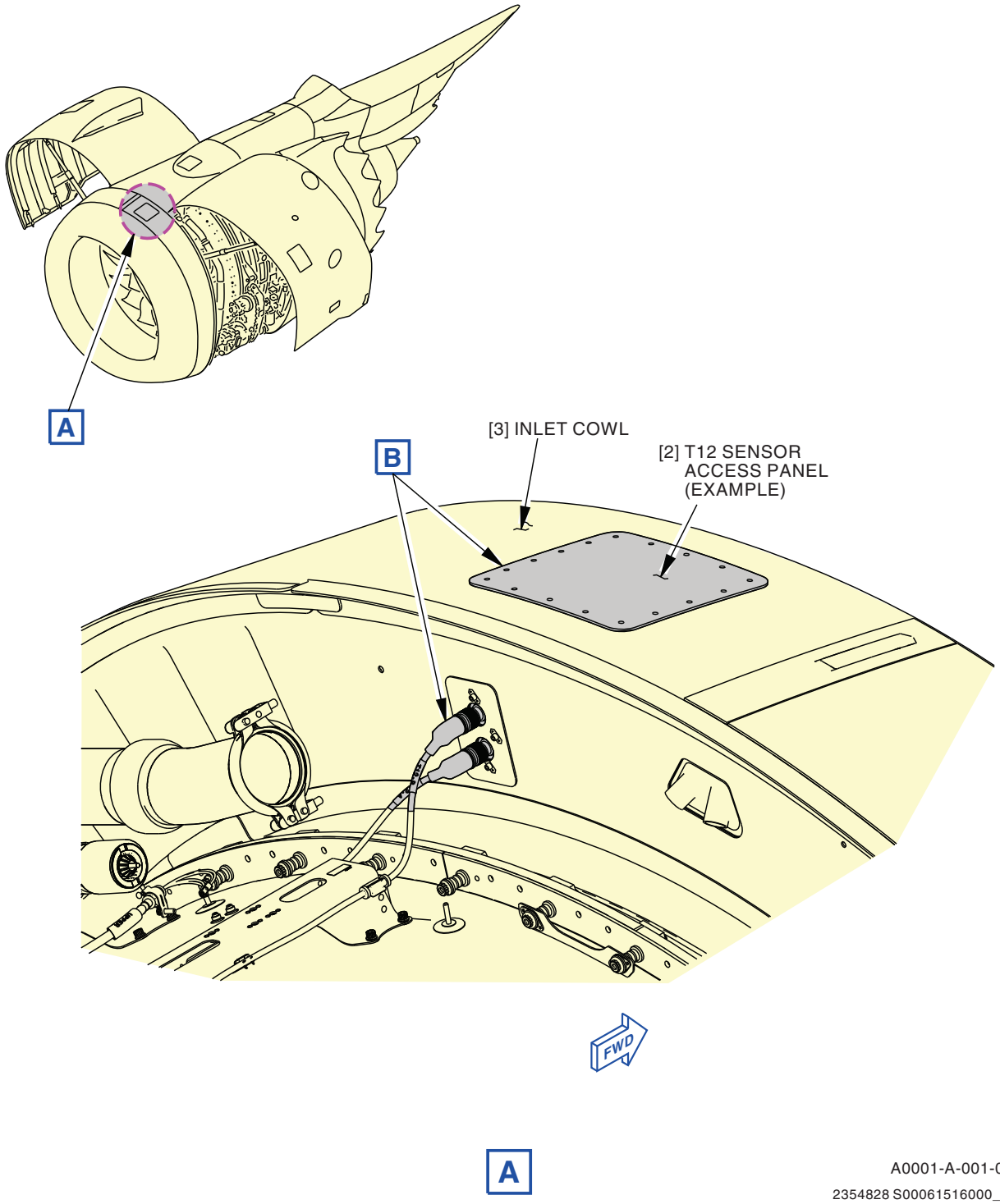
PTF ALL

(g) Remove the T12 sensor [1] from the probe pan [10].

(h) Install the protective covers on the openings.

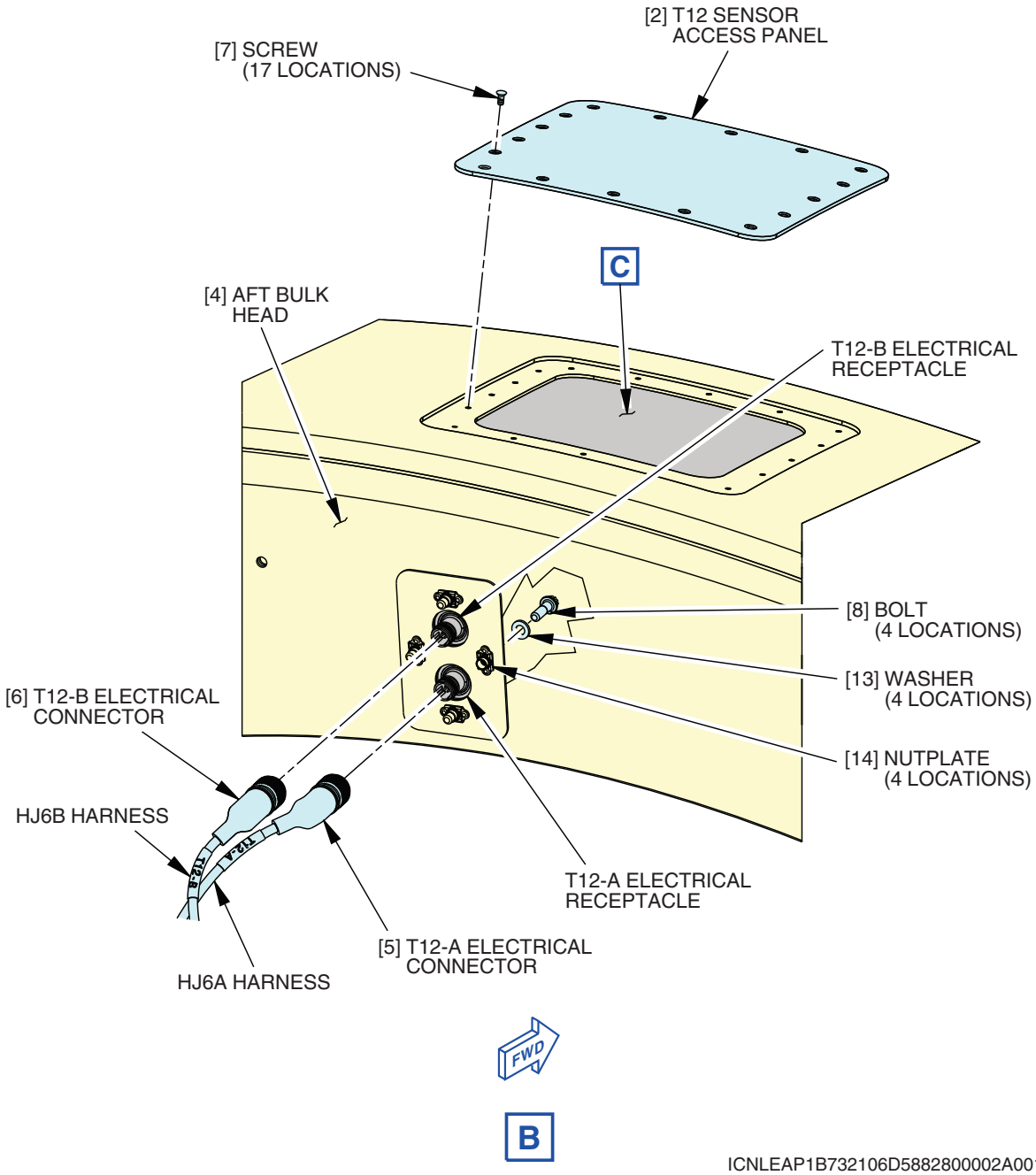
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



T12 Sensor Installation
Figure 401/73-21-06-990-801-G00 (Sheet 1 of 5)

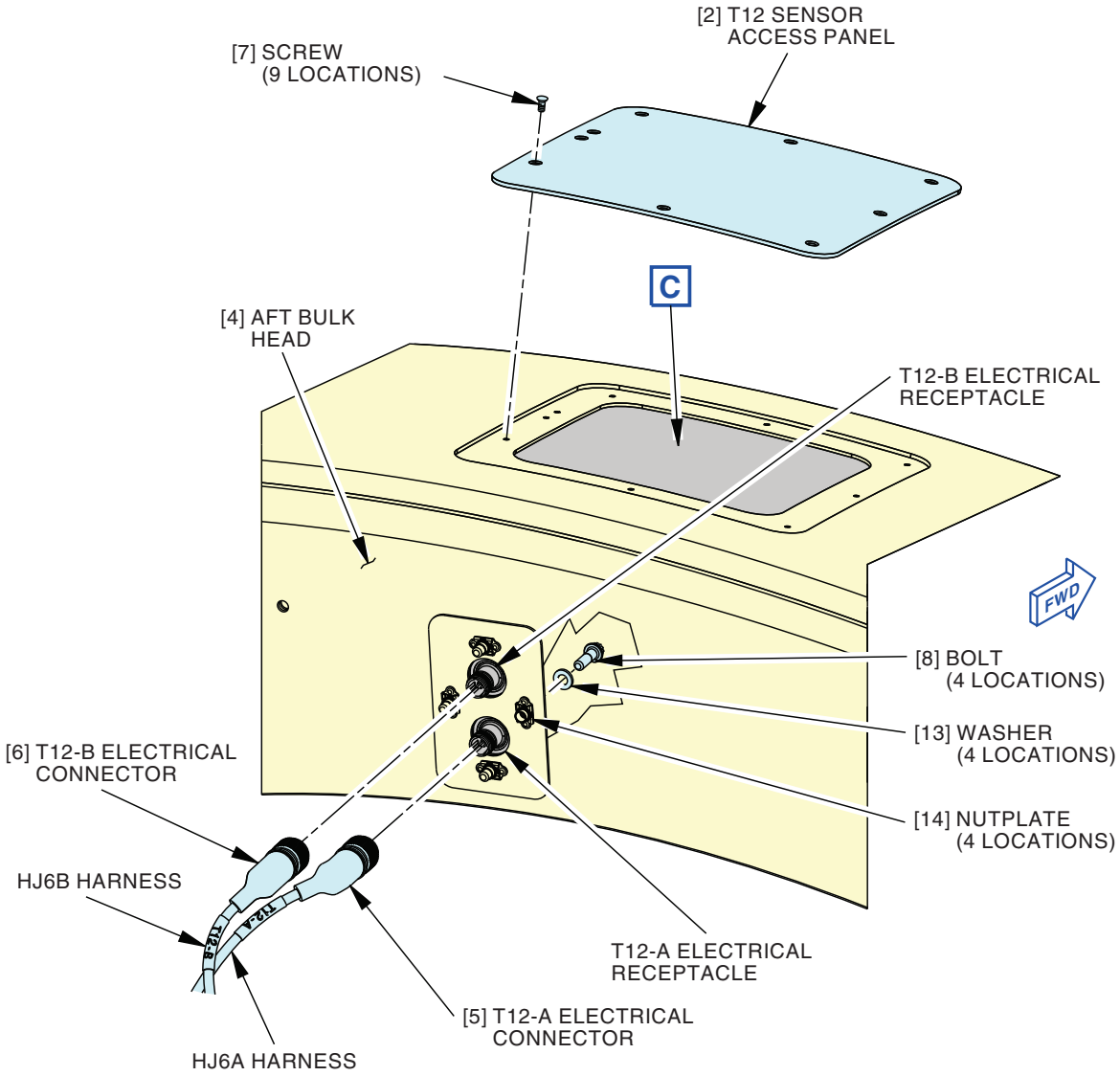
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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2354829 S00061516001_V6

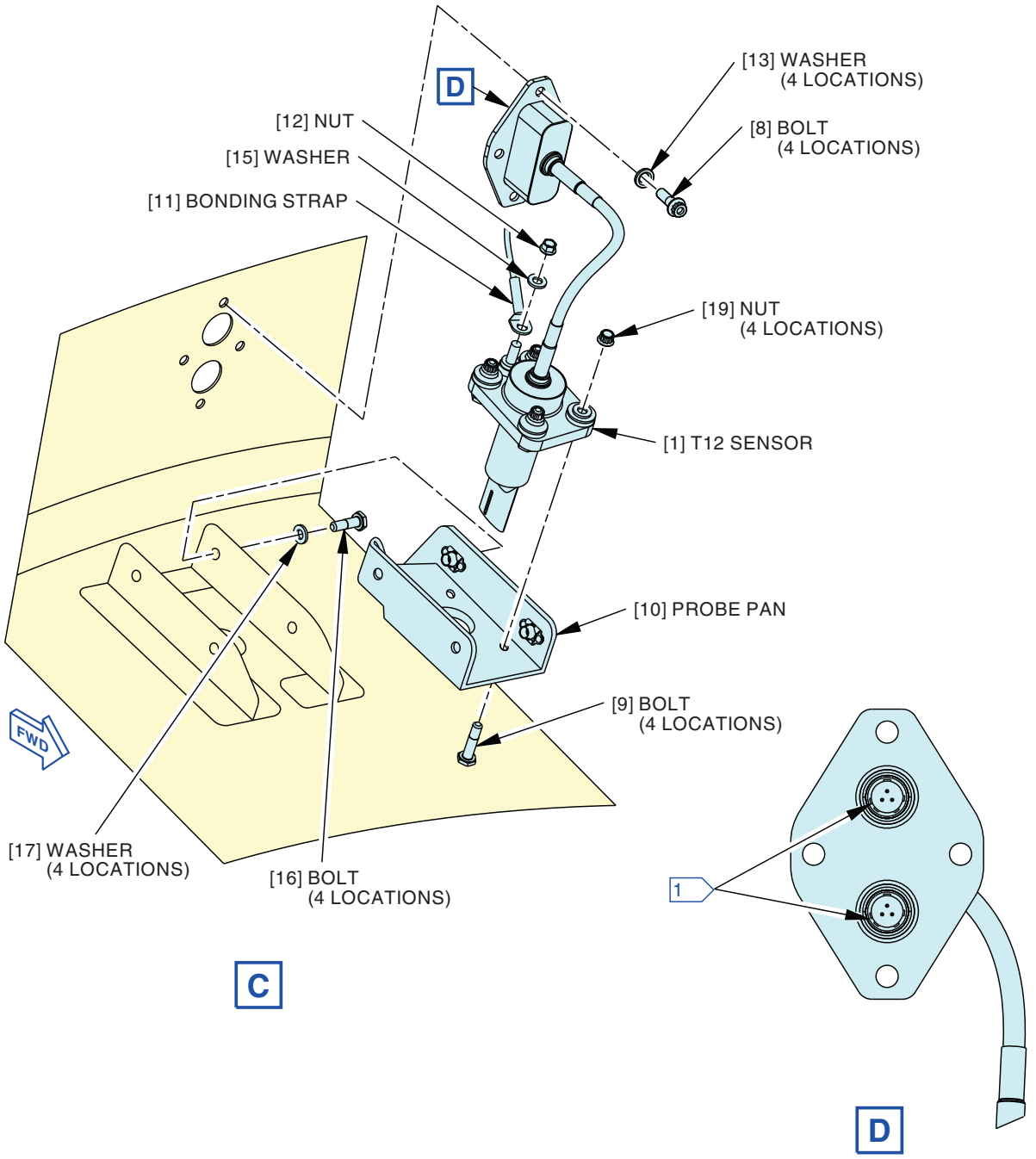
T12 Sensor Installation
Figure 401/73-21-06-990-801-G00 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



T12 Sensor Installation
Figure 401/73-21-06-990-801-G00 (Sheet 3 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



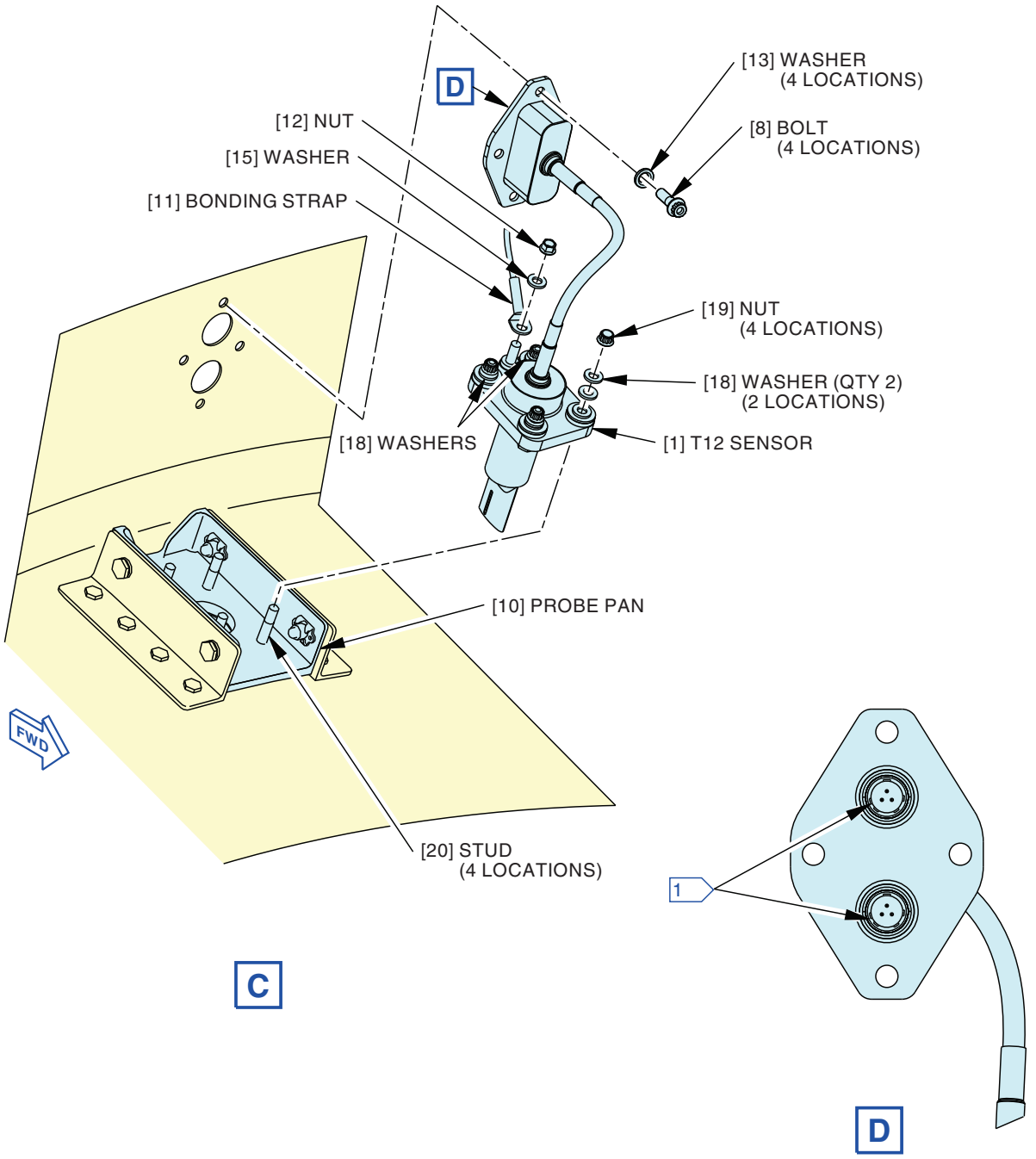
1 T12-A ELECTRICAL RECEPTACLE MASTERKEY AND T12-B ELECTRICAL RECEPTACLE MASTERKEY ARE ORIENTED UPWARD.

ICNLEAP1B732106D5882800003A00101

2402415 S0000555196_V8

T12 Sensor Installation
Figure 401/73-21-06-990-801-G00 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



1 T12-A ELECTRICAL RECEPTACLE MASTERKEY AND T12-B ELECTRICAL RECEPTACLE MASTERKEY ARE ORIENTED UPWARD.

ICNLEAP1B732106D5882800003A00101

3058915 S0000822203_V1

T12 Sensor Installation
Figure 401/73-21-06-990-801-G00 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 73-21-06-400-801-G00

3. T12 Sensor Installation

(Figure 401)

A. General

(1) This task provides the instructions on how to install the T12 sensor.

(2) You must do the tests listed in the power plant test reference table after you replace the T12 sensor.

B. References

Reference	Title
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
SWPM 20-20-00 Paragraph 14.C	Category 1 Unsealed Fay Surface Bonds

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
A50483 [CP2927]	Silicone Paste - Primerless, For Fire Resistance application (Dapco 2100 Firewall Sealant)	
B50073	Alcohol - Isopropyl	ASTM D 770
D00599 [CP2442]	Oil - Engine	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	T12 sensor	73-21-06-01-010	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

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G. Access Panels

Number	Name/Location
412AT	T12 Access Door, Engine 1
413	Left Fan Cowl, Engine 1
422AT	T12 Access Door, Engine 2
423	Left Fan Cowl, Engine 2

H. T12 Sensor Installation

SUBTASK 73-21-06-420-002-G00

(1) Install the T12 sensor [1] as follows:

(a) Remove the protective covers from the openings.

(b) Make sure that the mating faces are clean and clear of unwanted materials.

(c) Put the T12 sensor [1] on the probe pan [10].

PTF ALL; AIRPLANES WITH PROBE PAN HELD BY STUDS

(d) Install the six washers [18] and four nuts [19] on the four studs [20] to attach the T12 sensor [1] to the probe pan [10].

1) Install the two washers [18] under each of the two forward nuts [19] and one washer [18] under each of the two aft nuts [19].

(e) Tighten the four nuts [19] to 82 in-lb (9.3 N·m) - 88 in-lb (9.9 N·m).

PTF ALL; AIRPLANES WITH PROBE PAN HELD BY BOLTS

(f) Install the four bolts [9], four washers [18], and four nuts [19] that attach the T12 sensor [1] to the probe pan [10].

(g) Tighten the four bolts [9] to 67.0 in-lb (7.6 N·m) - 73.0 in-lb (8.2 N·m).

(h) Apply Dapco 2100 sealant, A50483 [CP2927], to the under-head of the four bolts [16].

(i) Install the probe pan [10] between the brackets on the cowl with four bolts [16] and four washers [17].

(j) Tighten the four bolts [16] to 72.0 in-lb (8.1 N·m) - 78.0 in-lb (8.8 N·m).

PTF ALL

(k) Do these steps to attach the bonding strap [11] to the T12 sensor [1] (Category 1 Unsealed Fay Surface Bonds, SWPM 20-20-00 Paragraph 14.C):

1) Clean the interfacing surfaces with alcohol, B50073.

2) Install the nut [12] and washer [15] that attach the bonding strap [11] to the T12 sensor [1].

3) Tighten the nut [12] to 94.0 in-lb (10.6 N·m) - 100.0 in-lb (11.3 N·m).

4) Use a intrinsically safe approved bonding meter, COM-1550, to measure the resistance in the following three locations:

- Aft bulkhead to stud
- Stud to probe
- Stud to stud.

5) Make sure that sum of readings from the three locations is 0.00582 ohm (5.82 milliohms) or less.

(l) Apply faying surface seal between the T12 connector assembly and bulkhead with Dapco 2100 sealant, A50483 [CP2927].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(m) Apply Dapco 2100 sealant, A50483 [CP2927], to the shanks and threads of the four bolts [8].

(n) Install the four bolts [8] and four washers [13] that attach the T12-A and T12-B electrical receptacles to the aft bulk head [4] (View C and View D, Figure 401).

1) Make sure that the T12-A electrical receptacle masterkey and T12-B electrical receptacle masterkey are oriented upward.

(o) Tighten the four bolts [8] to 77.0 in-lb (8.7 N·m) - 83.0 in-lb (9.4 N·m).

SUBTASK 73-21-06-420-004-G00

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS OR RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(2) Use a teflon-jawed pliers, STD-664, to connect the T12-A electrical connector [5] and T12-B electrical connector [6] to the T12 sensor [1] as follows (TASK 70-00-01-910-803-G00):

(a) Remove the protective covers from the T12-A electrical connector [5], T12-B electrical connector [6], and T12-A and T12-B electrical receptacles on the T12 sensor [1].

(b) Connect the T12-A electrical connector [5] and T12-B electrical connector [6] to the T12 sensor [1] (View B, Figure 401).

(c) Turn the coupling rings on the electrical connectors 1/8 turn or until the teflon-jawed pliers, STD-664, slippage occurs.

NOTE: Do not over tighten the coupling ring.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 73-21-06-420-003-G00

(1) Install the T12 access panel [2] as follows:

(a) Close these applicable access panels:

Number	Name/Location
412AT	T12 Access Door, Engine 1
422AT	T12 Access Door, Engine 2

PTF ALL; AIRPLANES WITH T-12 PANEL WITH 17 FASTENERS

(b) Lubricate the threads of the 17 screws [7] with oil, D00599 [CP2442].

(c) Install the 17 screws [7] that attach the T12 access panel [2] to the inlet cowl [3].

(d) Tighten the 17 screws [7] to 29.0 in-lb (3.3 N·m) to 31.0 in-lb (3.5 N·m).

PTF ALL; AIRPLANES WITH T-12 PANEL WITH 9 FASTENERS

(e) Lubricate the threads of the nine screws [7] with oil, D00599 [CP2442].

(f) Install the nine screws [7] that attach the T12 access panel [2] to the inlet cowl [3].

(g) Tighten the nine screws [7] to 29.0 in-lb (3.3 N·m) to 31.0 in-lb (3.5 N·m).

PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 73-21-06-440-001-G00

(2) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

SUBTASK 73-21-06-860-004-G00

(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

SUBTASK 73-21-06-860-005-G00

(4) For the left engine, remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A

SUBTASK 73-21-06-860-006-G00

(5) For the right engine, remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

SUBTASK 73-21-06-860-007-G00

(6) On the applicable engine, close the left fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

J. T12 Sensor Installation Test

SUBTASK 73-21-06-730-002-G00

(1) Do the applicable test(s) given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

T12 SENSOR - INSPECTION/CHECK

1. General

A. This procedure has one task:

(1) T12 Sensor Inspection.

TASK 73-21-06-210-801-G00

2. T12 Sensor Inspection

(Figure 601)

A. General

(1) This task provides the instructions to inspect the T12 Sensor.

(2) The T12 sensor is located at the 11:00 o'clock position on the inlet cowl from the aft looking forward.

(3) You must remove the T12 access panel on the outer surface of the inlet cowl to get access to the T12 sensor.

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
73-21-06-000-801-G00	T12 Sensor Removal (P/B 401)
73-21-06-400-801-G00	T12 Sensor Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
412AT	T12 Access Door, Engine 1
413	Left Fan Cowl, Engine 1
422AT	T12 Access Door, Engine 2
423	Left Fan Cowl, Engine 2

G. Prepare for the Inspection

SUBTASK 73-21-06-860-008-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-21-06-860-009-G00

(2) For the left engine, open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A

SUBTASK 73-21-06-860-010-G00

(3) For the right engine, open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

SUBTASK 73-21-06-010-003-G00

(4) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 73-21-06-010-004-G00

(5) Remove the T12 access panel:

(a) Open these applicable access panels:

Number	Name/Location
412AT	T12 Access Door, Engine 1
422AT	T12 Access Door, Engine 2

PTF ALL; AIRPLANES WITH T-12 PANEL WITH 17 FASTENERS

(b) Remove the 17 screws that attach the T12 access panel to the inlet cowl.

PTF ALL; AIRPLANES WITH T-12 PANEL WITH 9 FASTENERS

(c) Remove the nine screws that attach the T12 access panel to the inlet cowl.

PTF ALL

(d) Remove the T12 access panel from the engine.

SUBTASK 73-21-06-010-005-G00

(6) On the applicable engine, open the left fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

H. T12 Sensor Inspection

SUBTASK 73-21-06-210-001-G00

(1) Inspect the T12 sensor:

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(a) Make sure that the T12 sensor is not loose.

1) If the T12 sensor is loose, tighten the bolts that attach the sensor to the mounting bracket (TASK 73-21-06-400-801-G00).

(b) Look for cracks in the mounting flange of the T12 sensor.

1) If you see cracks, replace the T12 sensor (TASK 73-21-06-000-801-G00 and TASK 73-21-06-400-801-G00).

(c) Make sure that the electrical connectors are not loose.

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

1) If an electrical connector is loose, use a teflon-jawed pliers, STD-664, to tighten the electrical connector.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 73-21-06-410-002-G00

(1) On the applicable engine, close the left fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

SUBTASK 73-21-06-410-003-G00

(2) Install the T12 access panel:

(a) Close these access panels:

Number	Name/Location
412AT	T12 Access Door, Engine 1
422AT	T12 Access Door, Engine 2

(b) Close these applicable access panels:

Number	Name/Location
412AT	T12 Access Door, Engine 1
422AT	T12 Access Door, Engine 2

PTF ALL; AIRPLANES WITH T-12 PANEL WITH 17 FASTENERS

(c) Lubricate the threads of the 17 screws that attach the T12 access panel to the inlet cowl with oil, D00599 [CP2442].

(d) Install the 17 screws that attach the T12 access panel to the inlet cowl.

(e) Tighten the 17 screws to 29.0 in-lb (3.3 N·m) to 31.0 in-lb (3.5 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH T-12 PANEL WITH 9 FASTENERS

(f) Lubricate the threads of the nine screws that attach the T12 access panel to the inlet cowl with oil, D00599 [CP2442].

(g) Install the nine screws that attach the T12 access panel to the inlet cowl.

(h) Tighten the nine screws to 29.0 in-lb (3.3 N·m) to 31.0 in-lb (3.5 N·m).

PTF ALL

SUBTASK 73-21-06-860-011-G00

(3) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

SUBTASK 73-21-06-860-012-G00

(4) For the left engine, remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A

SUBTASK 73-21-06-860-013-G00

(5) For the right engine, remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

SUBTASK 73-21-06-860-014-G00

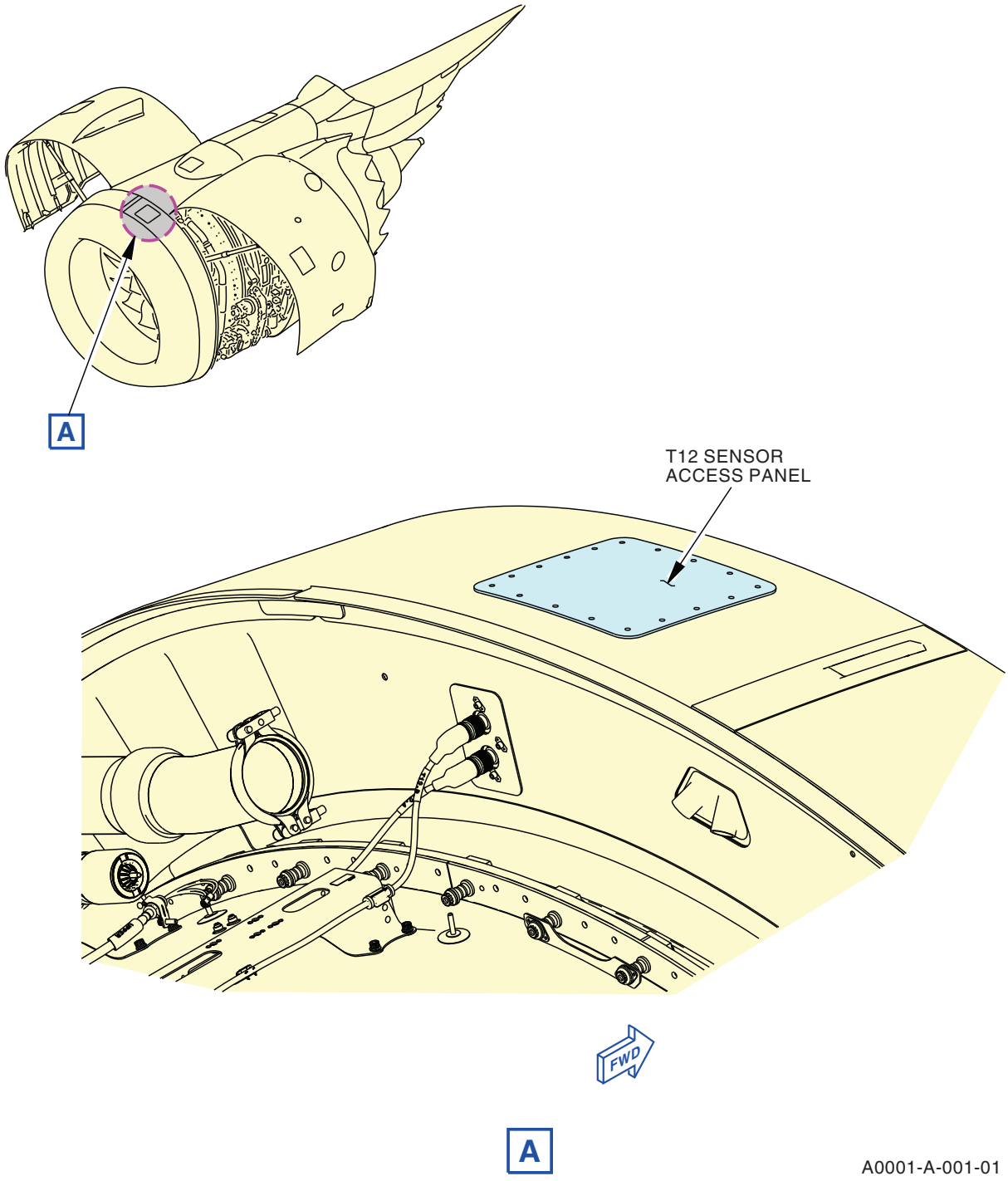
(6) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



T12 Sensor Inspection
Figure 601/73-21-06-990-802-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PERMANENT MAGNET ALTERNATOR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the Permanent Magnet Alternator (PMA)

(2) An installation of the PMA.

TASK 73-21-08-000-801-G00

2. Permanent Magnet Alternator Removal

(Figure 401 and Figure 402)

A. General

(1) This task gives the instructions to remove the Permanent Magnet Alternator (PMA).

(2) The PMA consists of the PMA stator and the PMA rotor.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
70-00-04-000-803-G00	Gasket Seal (with Imbedded Flexible Seal Material) Removal (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
72-63-04-000-801-G00	Accessory Gearbox Manual Control Cover Removal (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14905	Guide Pin Set - Jackscrews (PMA R/I) Part #: 956A3506G01 Supplier: 58828
SPL-14920	Adapter - Core Engine Lock Part #: 956A8679G01 Supplier: 58828
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
9	Gasket seal	73-21-08-01-040	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

H. Prepare for the Removal

SUBTASK 73-21-08-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-21-08-010-001-G00

(2) On the applicable engine, open the left fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

I. Permanent Magnet Alternator Removal

SUBTASK 73-21-08-030-001-G00

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(1) Use the teflon-jawed pliers, STD-664, to disconnect the electrical connector (HJ3A harness) [4] and the electrical connector (HJ3B harness) [5] from the PMA stator [2] as follows:

(a) Disconnect the electrical connector (HJ3A harness) [4] and the electrical connector (HJ3B harness) [5] from the PMA stator [2] (TASK 70-00-01-910-803-G00).

(b) Install the protections covers on the electrical connector (HJ3A harness) [4], electrical connector (HJ3B harness) [5], and PMA stator [2] receptacles.

SUBTASK 73-21-08-020-001-G00

WARNING DO NOT TOUCH THE COMPONENTS OF THE OIL SYSTEM IF THE ENGINE IS HOT. THESE COMPONENTS STAY HOTTER THAN OTHER COMPONENTS. HOT COMPONENTS CAN BURN YOU.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE STOPS. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.

WARNING DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

CAUTION DO NOT LET OIL GET ON THE ENGINE, OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE AREAS THAT OIL FALLS ON. OIL CAN CAUSE DAMAGE TO PAINT AND RUBBER.

CAUTION DO NOT HIT THE PERMANENT MAGNET ALTERNATOR HOUSING WITH A HAMMER OR OTHER OBJECT TO REMOVE IT FROM THE GEARBOX. DAMAGE TO THE WINDINGS CAN OCCUR IF THE PERMANENT MAGNET ALTERNATOR HITS THE GUIDE PINS OR STUDS DURING THE REMOVAL.

CAUTION MAKE SURE THAT YOU USE THE GUIDE PINS. IF YOU DO NOT USE THE GUIDE PINS TO REMOVE OR INSTALL THE ALTERNATOR, DAMAGE TO THE ROTOR AND THE STATOR OF THE ALTERNATOR CAN OCCUR.

(2) Remove the PMA stator [2] as follows:

NOTE: The PMA consists of the PMA stator and the PMA rotor.

(a) Lubricate the threads of the guide pins from the guide pin set, SPL-14905, with oil, D00599 [CP2442].

(b) Remove the four bolts [6] from the PMA stator [2].

(c) Turn the PMA stator [2] clockwise 90 degrees and install the two guide pins through the bolt holes of the PMA stator [2] to the AGB pad [8].

1) Make sure that the PMA stator [2] does not pull outward when you turn it.

(d) Carefully remove the PMA stator [2] along the guide pins.

(e) Examine the gasket seal [9] for damage (TASK 70-00-04-000-803-G00).

1) Remove the two screws [12] that attach the gasket seal [9] to the PMA stator [2].

2) If the gasket seal [9] is not damaged, re-use the gasket seal [9].

3) If the gasket seal [9] is damaged, replace the gasket seal [9].

a) Remove and discard the gasket seal [9].

b) Use the new gasket seal [9].

<1> Examine the surfaces of the gasket seal [9] for nicks, cuts or abrasions.

<2> Make sure that there is no damage on the gasket seal [9].

4) Remove the two guide pins from the AGB pad [8].

SUBTASK 73-21-08-020-002-G00

(3) Remove the PMA rotor [3] as follows:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Remove the manual control cover [26] on the accessory gearbox (TASK 72-63-04-000-801-G00).

(b) Install the locking adapter [23] on the core motoring mount on the accessory gearbox as follows:

NOTE: The Engine core lock adapter, SPL-14920, installed through the core motoring mount is to lock the rotation of the N2 rotor.

1) Install the three bolts [21] on the locking adapter [23].

2) Tighten the bolts [21] to 93 in-lb (11 N·m) - 102 in-lb (12 N·m).

(c) Remove the rotor nut [10] that attaches the PMA rotor [3] to the AGB gearshaft [11].

NOTE: The rotor nut is loosened clockwise.

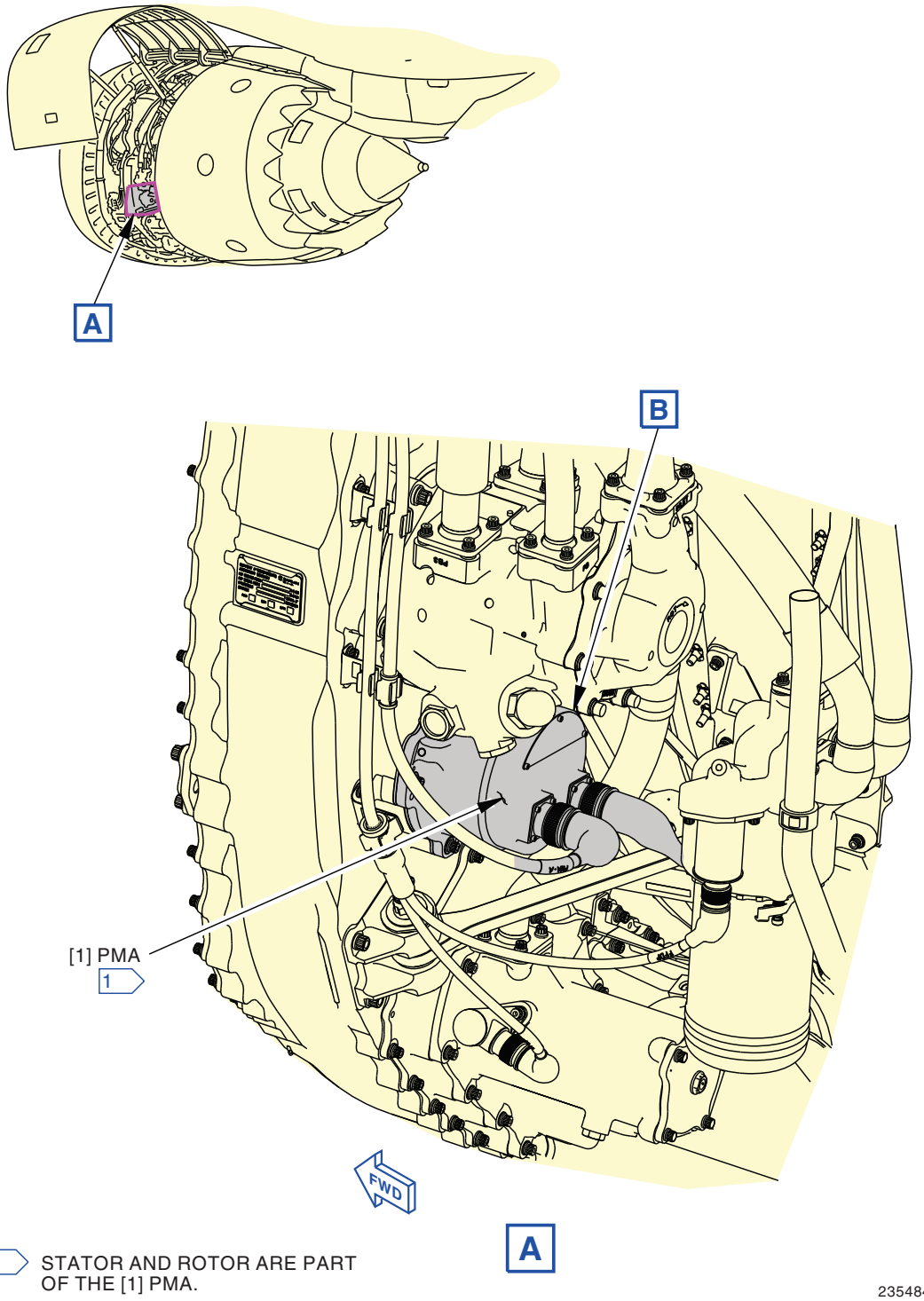
(d) Carefully remove the PMA rotor [3] from the AGB gearshaft [11].

(e) Remove the locking adapter [23] from the core motoring mount.

(f) Install the protection covers on the AGB gearshaft [11] and the PMA rotor [3].

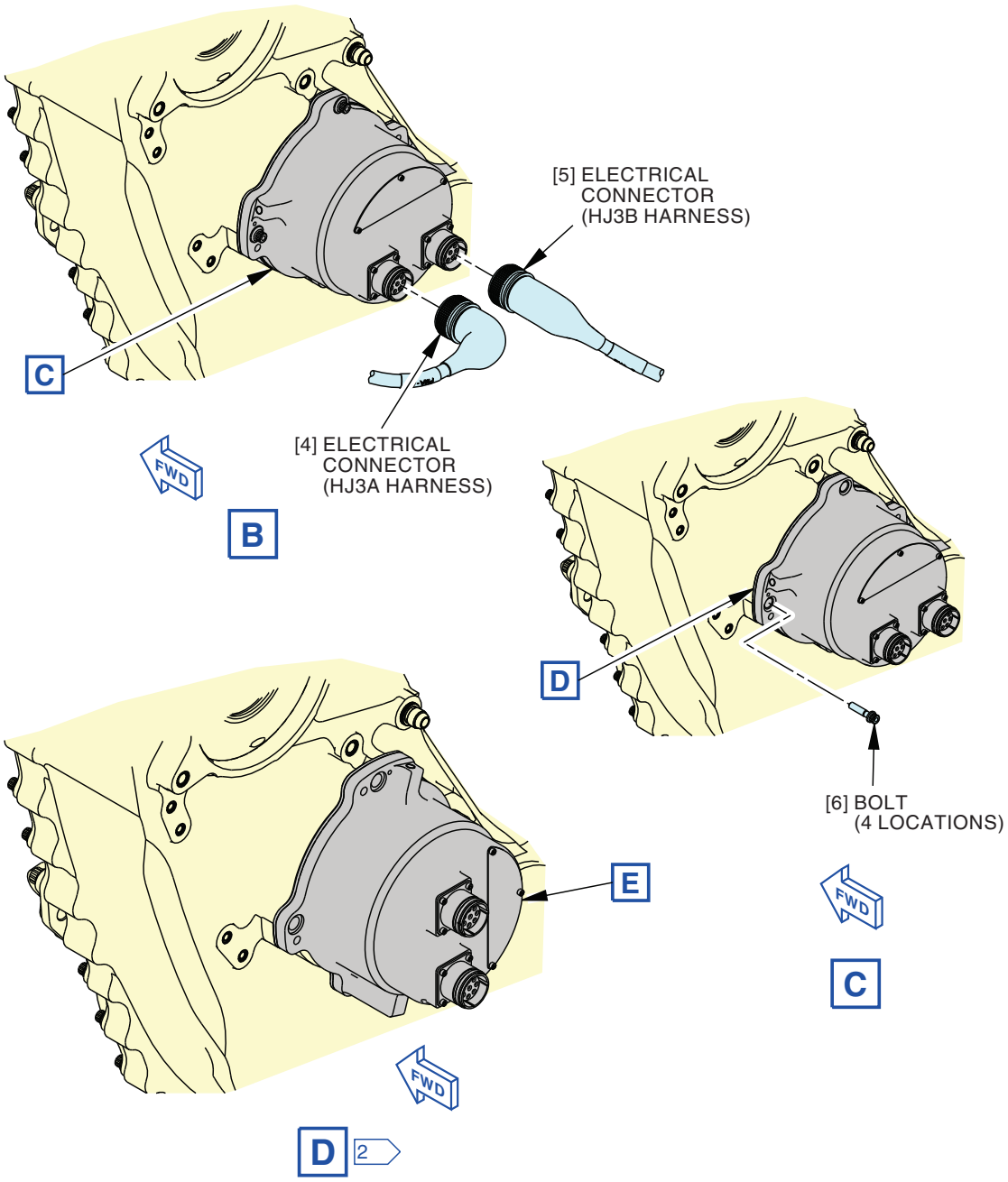
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Permanent Magnet Alternator Installation
Figure 401/73-21-08-990-801-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



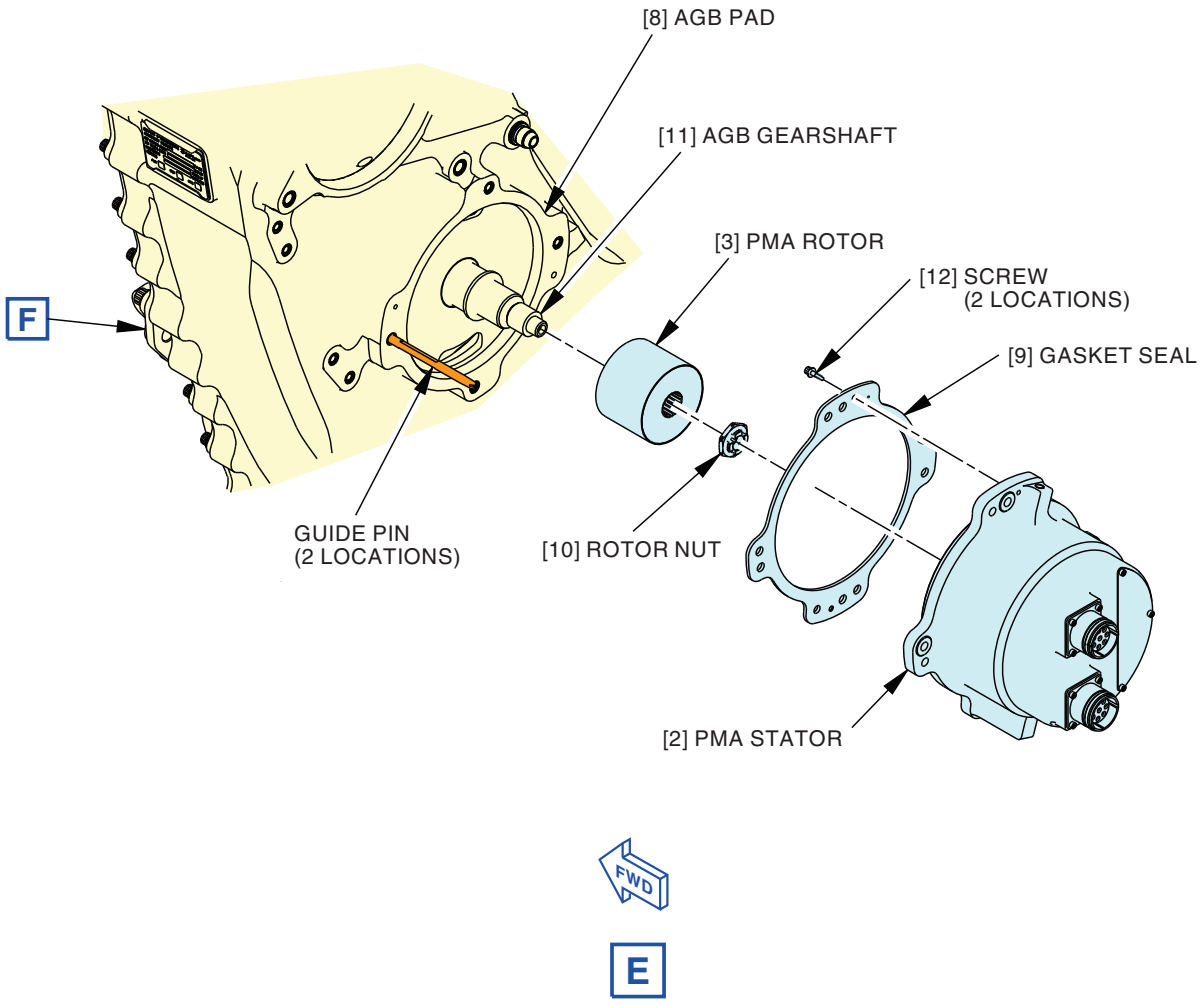
2 PMA STATOR IS TURNED CLOCKWISE 90 DEGREES.

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Permanent Magnet Alternator Installation
Figure 401/73-21-08-990-801-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

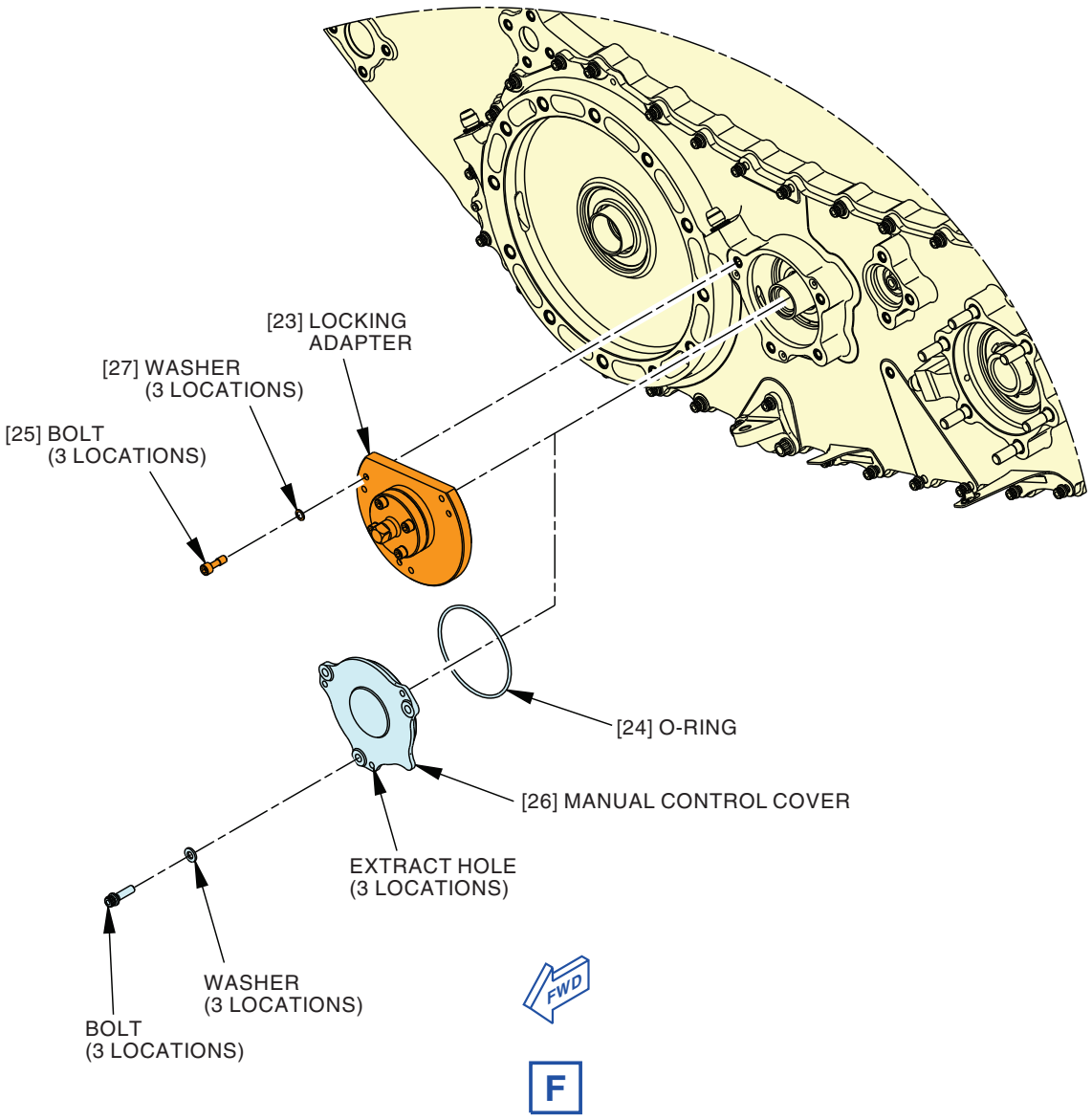


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2568027 S0000615182_V3

Permanent Magnet Alternator Installation
Figure 401/73-21-08-990-801-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICN-LEAP-1B-732108-B-F0301-A0004-A-001-01

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Accessory Gearbox Manual Control Cover Installation
Figure 402/73-21-08-990-803-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 73-21-08-400-801-G00

3. Permanent Magnet Alternator Installation

(Figure 401 and Figure 402)

A. General

(1) This task gives the instructions to install the Permanent Magnet Alternator (PMA).

(2) The PMA consists of the PMA stator and the PMA rotor.

(3) You must do the tests that are listed in the power plant test reference table after you install the PMA.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-63-04-000-801-G00	Accessory Gearbox Manual Control Cover Removal (P/B 401)
72-63-04-400-801-G00	Accessory Gearbox Manual Control Cover Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14905	Guide Pin Set - Jackscrews (PMA R/I) Part #: 956A3506G01 Supplier: 58828
SPL-14920	Adapter - Core Engine Lock Part #: 956A8679G01 Supplier: 58828
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
2	PMA stator	73-21-08-01-020	PTF ALL
9	Gasket seal	73-21-08-01-040	PTF ALL
24	O-ring	72-63-01-04-180	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Permanent Magnet Alternator Installation

SUBTASK 73-21-08-420-001-G00

(1) Install the PMA rotor [3] as follows:

NOTE: The PMA [1] consists of the PMA stator [2] and the PMA rotor [3].

(a) Remove the protection covers from the AGB gearshaft [11] and the PMA rotor [3].

(b) Make sure that the mating faces are clean and clear of unwanted materials.

(c) If it has not done so, remove the manual control cover [26] (TASK 72-63-04-000-801-G00).

1) Install the locking adapter [23] on the core motoring mount on the accessory gearbox as follows:

NOTE: The Engine core lock adapter, SPL-14920 installed through the core motoring mount is to lock the rotation of the N2 rotor.

a) Install the three bolts [21] on the locking adapter [23].

b) Tighten the bolts [21] to 93 in-lb (11 N·m) to 102 in-lb (12 N·m).

(d) Lubricate the threads of the AGB gearshaft [11] with the oil, D00599 [CP2442].

(e) Align the PMA rotor [3] with the AGB gearshaft [11].

(f) Put the PMA rotor [3] on the drive shaft.

(g) Install the rotor nut [10] on the AGB gearshaft [11].

(h) Tighten the rotor nut [10] to 454.1 in-lb (51.3 N·m) - 501.7 in-lb (56.7 N·m).

NOTE: The rotor nut [10] is tightened counterclockwise.

1) Use the reference oil to lubricate the rotor nut [10].

2) Tighten the rotor nut [10] until it contacts the PMA rotor [3].

3) Loosen the rotor nut [10] about one turn, then tighten the rotor nut [10] again and record the self-locking torque value.

NOTE: Do not take the breakaway torque into account when recording the self-locking torque value.

4) Add the recorded self-locking torque value to the tightening torque value specified above.

5) Tighten the rotor nut [10] to the calculated value.

(i) Remove the locking adapter [23] from the core motoring mount.

SUBTASK 73-21-08-420-002-G00

(2) Install the manual control cover [26] on the accessory gearbox (TASK 72-63-04-400-801-G00).

(a) Make sure that you install a new o-ring [24].

SUBTASK 73-21-08-420-003-G00

WARNING

DO NOT TOUCH THE COMPONENTS OF THE OIL SYSTEM IF THE ENGINE IS HOT. THESE COMPONENTS STAY HOTTER THAN OTHER COMPONENTS. HOT COMPONENTS CAN BURN YOU.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE STOPS. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.

WARNING DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

CAUTION DO NOT LET OIL GET ON THE ENGINE, OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE AREAS THAT OIL FALLS ON. OIL CAN CAUSE DAMAGE TO PAINT AND RUBBER.

(3) Install the PMA stator [2] as follows:

(a) Make sure that the mating faces are clean and clear of unwanted materials.

(b) Lubricate the threads of the two guide pins from the guide pin set, SPL-14905, and the four bolts [6] with the oil, D00599 [CP2442].

(c) Install the new gasket seal [9] on the PMA stator [2], if it is necessary.

1) Install the two screws [12] that attach the gasket seal [9] to the PMA stator [2].

2) Tighten the two screws [12] to 11 in-lb (1.24 N·m) - 13 in-lb (1.47 N·m).

(d) Install the two guide pins through the bolt holes of the PMA stator [2] to the AGB pad [8], in the same position as the removal task.

(e) Install the PMA stator [2] on the AGB pad [8] with its flange seated correctly on the AGB pad [8].

(f) Remove the two guide pins from the AGB pad [8].

(g) Turn the PMA stator [2] 90 degree counter-clockwise and align it in the correct position on the AGB pad [8].

(h) Install the four bolts [6] on the PMA stator [2].

1) Tighten the bolts [6] to 109.31 in-lb (12.35 N·m) - 120.81 in-lb (13.65 N·m).

SUBTASK 73-21-08-430-001-G00

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION DO NOT USE PLIERS THAT HAVE METAL JAWS TO TIGHTEN THE ELECTRICAL CONNECTOR. DAMAGE TO THE ELECTRICAL CONNECTOR CAN OCCUR.

(4) Use the teflon-jawed pliers, STD-664 to connect the electrical connector (HJ3A harness) [4] and the electrical connector (HJ3B harness) [5] to the PMA stator [2] as follows :

(a) Remove the protection covers from the electrical connector (HJ3A harness) [4], the electrical connector (HJ3B harness) [5], and the PMA stator [2] receptacles.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Connect the electrical connector (HJ3A harness) [4] and the electrical connector (HJ3B harness) [5] to the PMA stator [2] (TASK 70-00-01-910-803-G00).

(c) Tighten the electrical connector (HJ3A harness) [4] and the electrical connector (HJ3B harness) [5].

SUBTASK 73-21-08-860-002-G00

(5) Clean the work area and clear the tools and other items.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 73-21-08-860-003-G00

(1) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

SUBTASK 73-21-08-410-001-G00

(2) On the applicable engine, close the left fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

J. Permanent Magnet Alternator Installation Test

SUBTASK 73-21-08-730-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PERMANENT MAGNET ALTERNATOR - INSPECTION/CHECK

1. General

A. This procedure has one task:

(1) Permanent Magnet Alternator (PMA) inspection.

TASK 73-21-08-210-801-G00

2. Permanent Magnet Alternator (PMA) Inspection

(Figure 601)

A. General

(1) This task provides instructions to inspect the Permanent Magnet Alternator (PMA).

(a) The PMA is located on the aft face of the accessory gearbox on the left side of the fan case.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
73-21-08-000-801-G00	Permanent Magnet Alternator Removal (P/B 401)
73-21-08-400-801-G00	Permanent Magnet Alternator Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

F. Prepare for the Inspection

SUBTASK 73-21-08-860-004-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-21-08-010-002-G00

(2) On the applicable engine, open the left fan cowl panel (TASK 71-11-04-010-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

G. Permanent Magnet Alternator Inspection

SUBTASK 73-21-08-212-001-G00

(1) Inspect the PMA:

(a) Look for signs of oil leaks.

1) If you see signs of oil leak between the permanent magnet alternator and the AGB pad replace the gasket seal (Permanent Magnet Alternator Removal, TASK 73-21-08-000-801-G00 and Permanent Magnet Alternator Installation, TASK 73-21-08-400-801-G00).

(b) Look for cracks in the mounting flange:

1) If you see cracks, replace the permanent magnet alternator (Permanent Magnet Alternator Removal, TASK 73-21-08-000-801-G00 and Permanent Magnet Alternator Installation, TASK 73-21-08-400-801-G00).

(c) Look for dents in the permanent magnet alternator.

1) If you see dents, replace the permanent magnet alternator (Permanent Magnet Alternator Removal, TASK 73-21-08-000-801-G00 and Permanent Magnet Alternator Installation, TASK 73-21-08-400-801-G00).

(d) Make sure that the electrical connectors are not loose.

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

1) If an electrical connector is loose, use a teflon-jawed pliers, STD-664 to tighten the electrical connector.

(e) Look for ablative coating that is loose on the housing, does not have a good bond, or is missing.

NOTE: Loose ablative coating or ablative coating that does not have a good bond is not permitted.

1) If you find ablative coating that is loose on the housing or does not have a good bond, replace the PMA (Permanent Magnet Alternator Removal, TASK 73-21-08-000-801-G00 and Permanent Magnet Alternator Installation, TASK 73-21-08-400-801-G00).

(f) Look for ablative coating that is cut, peeling, flaking, missing coating or conditions that decrease the coating thickness.

1) If you find ablative coating that is cut, peeling, flaking, missing coating, or have conditions that decrease the coating thickness, do these steps:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

a) If the area with the damage is less than 10% of the total ablative coating, repair the ablative coating.

NOTE: You can repair a maximum of 10% of the total ablative coating.

NOTE: Contact CFM for the “Permanent Magnet Alternator - Specific Repair” procedure.

b) If the area with the damage is more than 10% of the total ablative coating, or there is any sign of the ablative coating separated from the housing (signs of a weak or defective bond), replace the PMA (Permanent Magnet Alternator Removal, TASK 73-21-08-000-801-G00 and Permanent Magnet Alternator Installation, TASK 73-21-08-400-801-G00).

H. Put the Airplane Back to Its Usual Condition

SUBTASK 73-21-08-410-002-G00

(1) On the applicable engine, close the left fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

SUBTASK 73-21-08-860-005-G00

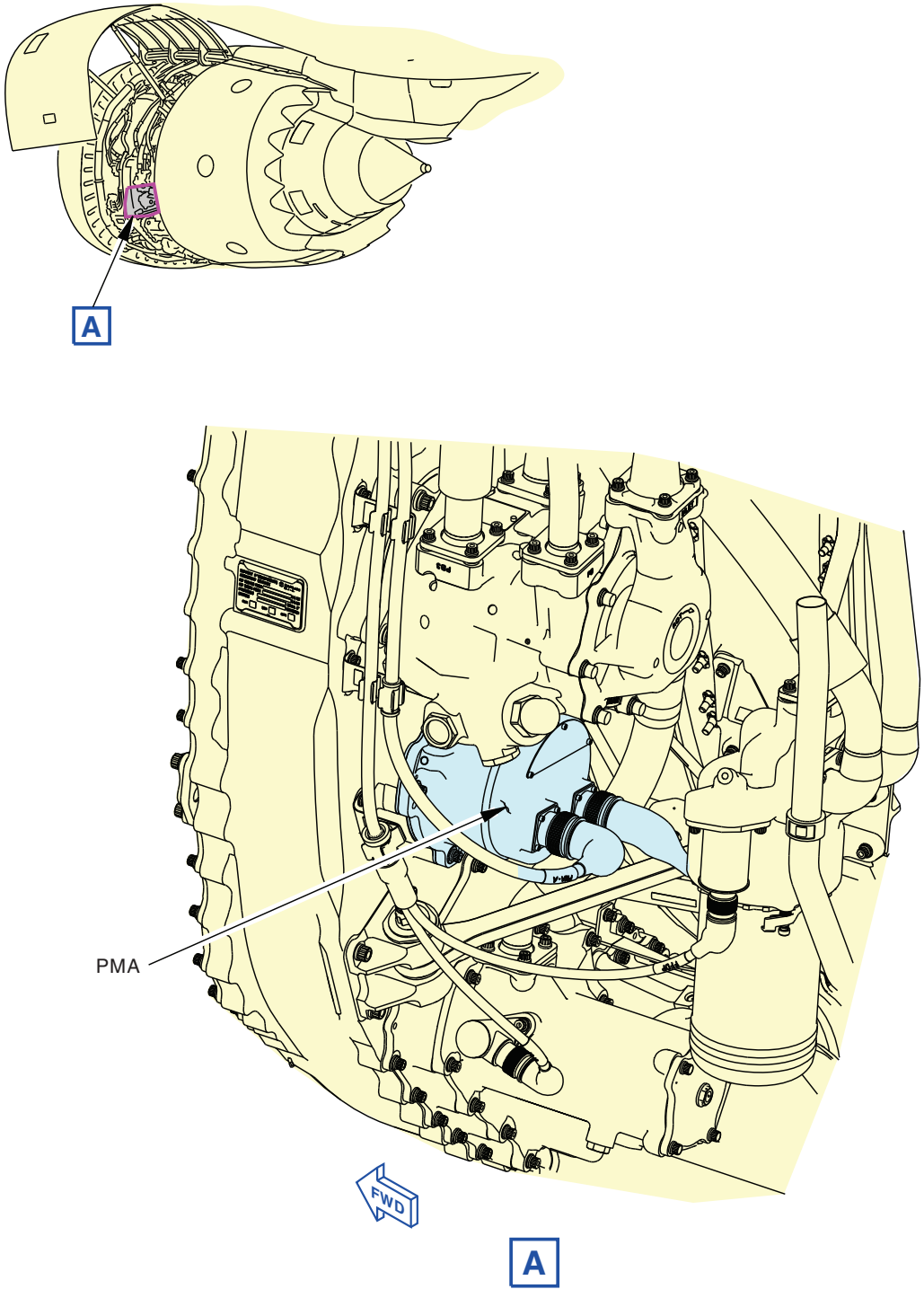
(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Permanent Magnet Alternator Inspection
Figure 601/73-21-08-990-804-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

DATA ENTRY PLUG - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Data entry plug removal

(2) Data entry plug installation.

TASK 73-21-09-000-801-G00

2. Data Entry Plug Removal

(Figure 401)

A. General

(1) This task provides the instructions on how to remove the data entry plug.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

F. Prepare for the Removal

SUBTASK 73-21-09-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-21-09-860-002-G00

(2) Open these applicable circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B

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(Continued)

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	5	C04007	ENGINE 1 ALTN PWR CHAN A
B	3	C01312	ENGINE 1 RUN/PWR

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	5	C01313	ENGINE 2 RUN/PWR
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	3	C04018	ENG 1 STBY ALTN PWR RELAY CHAN A
E	4	C04019	ENG 1 STBY ALTN PWR CHAN A

SUBTASK 73-21-09-010-001-G00

(3) On the applicable engine, open the right fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

G. Data Entry Plug Removal

SUBTASK 73-21-09-030-001-G00

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(1) Use the teflon-jawed pliers, STD-664 to disconnect the data entry plug [1] from the pressure sub-system (PSS) [2].

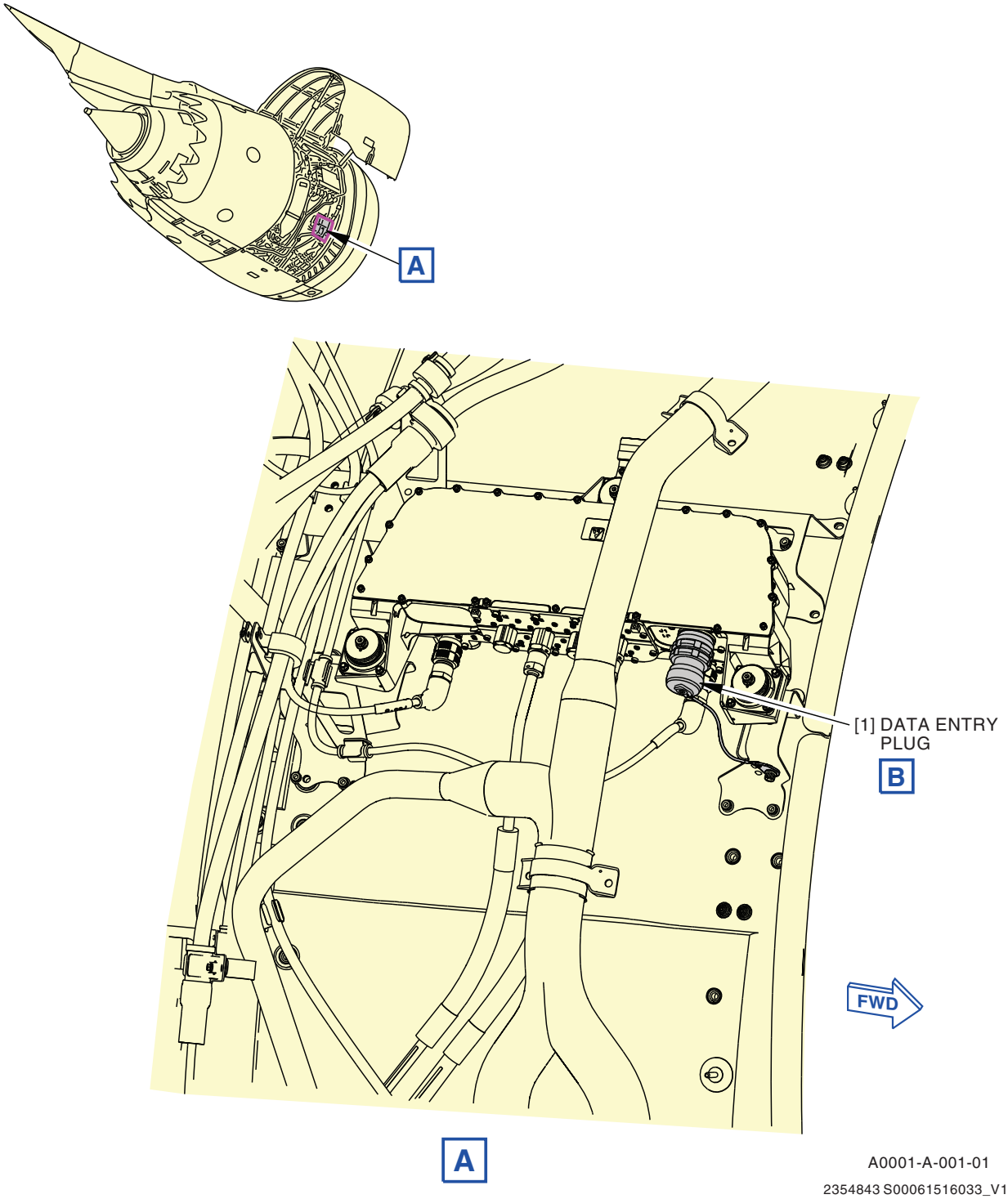
(a) Disconnect the data entry plug [1] from the receptacle of the pressure sub-system (PSS) [2].

(b) Remove the bolt [3] that attaches the metal braid [5] to the bracket [4].

(c) Install the protective covers on the data entry plug [1] and the pressure sub-system (PSS) [2] receptacle.

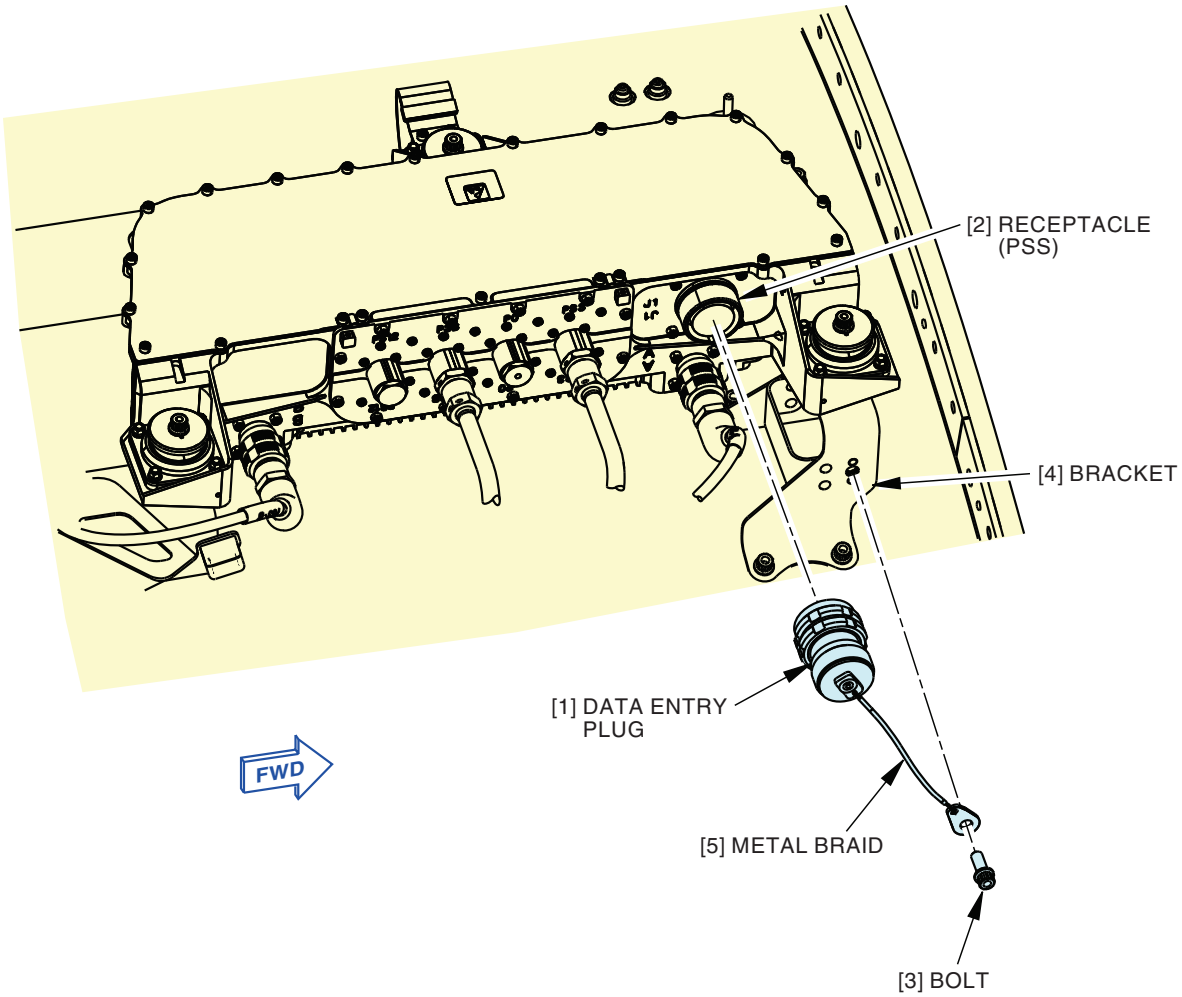
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Data Entry Plug Installation
Figure 401/73-21-09-990-802-G00 (Sheet 1 of 2)

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2402375 S0000555193_V1

Data Entry Plug Installation
Figure 401/73-21-09-990-802-G00 (Sheet 2 of 2)

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TASK 73-21-09-400-801-G00

3. Data Entry Plug Installation

(Figure 401)

A. General

(1) This task provides the instructions on how to install the data entry plug.

(2) You must do the tests that are listed in the power plant test reference table after you install the data entry plug.

B. References

Reference	Title
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

F. Data Entry Plug Installation

SUBTASK 73-21-09-540-001-G00

(1) Remove the protective covers from the data entry plug [1] and the pressure sub-system (PSS) [2] receptacle.

SUBTASK 73-21-09-860-003-G00

(2) Make sure that the mating faces are clean and clear of unwanted materials.

SUBTASK 73-21-09-430-001-G00

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

DO NOT USE PLIERS THAT HAVE METAL JAWS TO TIGHTEN THE ELECTRICAL CONNECTOR. DAMAGE TO THE ELECTRICAL CONNECTOR CAN OCCUR.

(3) Use the teflon-jawed pliers, STD-664 to connect the data entry plug [1] to the pressure sub-system (PSS) [2].

(a) Connect the data entry plug [1] to the pressure sub-system (PSS) [2] receptacle.

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(b) Make sure that the data entry plug [1] does not move after you fully engage them.

(c) Install the bolt [3] which attaches the metal braid [5] to the bracket [4].

1) Tighten the bolt [3] to 67.45 in-lb (7.6 N·m) to 74.55 in-lb (8.4 N·m).

(d) Clean the work area and clear the tools and other items.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 73-21-09-860-005-G00

(1) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

SUBTASK 73-21-09-860-006-G00

(2) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A
B	3	C01312	ENGINE 1 RUN/PWR

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	5	C01313	ENGINE 2 RUN/PWR
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	3	C04018	ENG 1 STBY ALTN PWR RELAY CHAN A
E	4	C04019	ENG 1 STBY ALTN PWR CHAN A

SUBTASK 73-21-09-410-002-G00

(3) On the applicable engine, close the right fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

H. Data Entry Plug Installation Test

SUBTASK 73-21-09-730-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

DATA ENTRY PLUG - INSPECTION/CHECK

1. General

A. This procedure has these tasks.

(1) Data Entry Plug Configuration Check

(2) Data Entry Plug Inspection.

TASK 73-21-09-700-801-G00

2. Data Entry Plug Configuration Check

(Figure 601 and Figure 602)

A. General

(1) This task gives the instructions to examine the program on the data entry plug.

(2) The data entry plug is referred to as DEP.

(3) The DEP is a one piece of hardware with two functions. The DEP defines the engine thrust rating via fused links and cannot be re-programmed. The DEP also defines the engine hardware configuration and N1 trim via programmable “push-pull” pins.

(4) The DEP is installed on the Pressure Sub System (PSS) below the EECs. The DEP “push-pull” pins are protected by a cover via a lanyard installed on the fan case.

(5) The “push-pull” pin configuration must be programmed to align with the N1 trim defined from engine test results or documented in as-delivered Engine Data Submittal package as well as the engine hardware configuration defined by Service Bulletin incorporation.

(6) When the “push-pull” pins are set to a combination that defines a specific trim value and/or engine configuration, the engine data plate (nameplate) is then re-marked with the new N1 trim level and configuration.

NOTE: Replacement pre-programmed data entry plug thrust rating must match the engine data plate’s rated thrust, the opposite engine's rated thrust, and be compatible with the airplane model.

B. References

Reference	Title
46-13-00-710-802	System Configuration Check (P/B 201)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
73-21-00-470-801-G00	EEC System Configuration Data Functions Check (P/B 201)
73-21-00-800-801-G00	EEC Maintenance Power Selection (P/B 201)
73-21-09-000-801-G00	Data Entry Plug Removal (P/B 401)
73-21-09-400-801-G00	Data Entry Plug Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE
STD-14055	Tweezers, Flat Nose Industrial

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D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

F. Prepare for the Check

SUBTASK 73-21-09-860-007-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-21-09-970-002-G00

(2) Make a record of the applicable engine EEC system configuration page (TASK 46-13-00-710-802).

NOTE: The EEC system configuration page displays the engine thrust rating, N1 modifier, and engine configuration information that are referenced later in this task. For explanation of the EEC system configuration page refer to this task: EEC System Configuration Data Functions Check, TASK 73-21-00-470-801-G00.

SUBTASK 73-21-09-010-004-G00

(3) On the applicable engine, open the right fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

G. Data Entry Plug Configuration Check

SUBTASK 73-21-09-970-001-G00

(1) Make a record of the current engine thrust rating, N1 trim, and hardware configuration displayed on the EEC system configuration page (TASK 73-21-00-470-801-G00).

SUBTASK 73-21-09-010-002-G00

CAUTION

DO NOT REMOVE THE COVER FROM THE ENGINE. FIRST, REMOVE COVER FROM THE DATA ENTRY PLUG, THEN REMOVE DATA ENTRY PLUG FROM PRESSURE SUB SYSTEM. DAMAGE TO THE DATA ENTRY PLUG CAN OCCUR.

(2) Remove the cover from the data entry plug and the data entry plug from the Pressure Sub System (PSS) as follows (View B, Figure 601):

(a) Use the teflon-jawed pliers, STD-664, to remove the cover from the data plug.

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(b) Remove the data entry plug from the PSS (TASK 73-21-09-000-801-G00).

NOTE: You can now get access to the pins of the data entry plug to make sure that there are "push"- "pull" pin configurations.

SUBTASK 73-21-09-860-015-G00

(3) Make sure that the DEP engine thrust rating part number matches the engine thrust rating on the engine data plate.

(a) Locate the engine thrust rating on the engine data plate.

(b) Locate the engine thrust rating part number on the DEP (View C, Figure 601).

(c) Make sure that the engine thrust rating matches the appropriate part number (Table 601).

1) If the DEP engine thrust rating part number does not match the engine data plate, replace the DEP with a P/N that matches the engine data plate engine thrust rating.

Table 601/73-21-09-993-806-G00 Engine Thrust Rating

Engine Data Plate Engine Thrust Rating	Data Entry Plug (DEP) Engine Thrust Rating P/N
LEAP-1B21	2531M61P02
LEAP-1B23	2531M61P06
LEAP 1B25	2531M61P10
LEAP-1B27	2531M61P18
LEAP-1B28C2	2531M61P24
LEAP-1B28	2531M61P26
LEAP-1B28B1	2531M61P27
LEAP-1B28B2	2531M61P28
LEAP-1B28B3	2531M61P29
LEAP-1B28BBJ1	2531M61P14
LEAP-1B28BBJ2	2531M61P30

SUBTASK 73-21-09-820-001-G00

CAUTION YOU MUST INSTALL A PROTECTIVE COVER ON THE TOP OF THE DATA ENTRY PLUG DURING HANDLING. THE COVER PROTECTS SENSITIVE PROGRAMMING SWITCHES. IF YOU DO NOT OBEY THIS CAUTION, DAMAGE TO THE DATA ENTRY PLUG CAN OCCUR.

(4) If it is necessary, program the DEP to the specified configuration as follows:

(a) Do the steps below to program the N1 trim configuration:

1) Locate the N1 trim level.

a) If you will not change the thrust rating, the location of the N1 trim level is on the engine data plate.

b) If you will change the engine thrust rating, use the N1 trim level for the new thrust rating from the engine test summary data sheet.

NOTE: Refer to CFM LEAP-1B Service Bulletin 72-0018 for more information.

2) Locate the affected "push-pull" pins.

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3) With a flat nose tweezer, STD-14055, put the affected push-pull pins in the correct position (Table 602).

Table 602/73-21-09-993-802-G00 1-N1 Trim Pin Configuration

N1 Modifier	N1 Trim Pin*[1]			
Level	28	21	20	30 (Parity)
No Modifier	Pull	Pull	Pull	Push
Modifier 1	Pull	Pull	Push	Pull
Modifier 2	Pull	Push	Pull	Pull
Modifier 3	Pull	Push	Push	Push
Modifier 4	Push	Pull	Pull	Pull
Modifier 5	Push	Pull	Push	Push
Modifier 6	Push	Push	Pull	Push
Modifier 7	Push	Push	Push	Pull
Invalid	All other combinations			

*[1] Pins 19 and 22 are unused spares and are to remain in the closed (pushed-in) position.

(b) If it is necessary, do the steps below to program the engine hardware configuration:

1) Locate the engine hardware configuration number.

a) Look for the engine hardware configuration located on the EEC System configuration page recorded prior to removing the DEP from the PSS.

b) If the engine hardware configuration number is not known, refer to your airline's records of engine Service Bulletin incorporation to determine the correct engine hardware configuration number.

2) Locate the affected push-pull pins.

3) With a flat nose tweezer, STD-14055, put the affected push-pull pins in the correct position (Table 603).

Table 603/73-21-09-993-803-G00 Hardware Configuration Data Entry Plug Pin Configuration

Engine Model	Engine Configuration*[1]	Engine Configuration Pin*[2]						
		36	54	31	27	47	46	37
LEAP-1B Baseline*[3]	0001	Push	Push	Pull	Pull	Pull	Pull	Pull
LEAP-1B with Reverse-Sense LPTACC Actuator (Post LEAP-1B Service Bulletin 73-0013R0)*[3]	0002	Push	Pull	Pull	Pull	Push	Pull	Pull
LEAP-1B with Expanded EGT Capability (Post LEAP-1B Service Bulletin 72-0169R0)	0003	Push	Pull	Pull	Pull	Pull	Push	Pull
LEAP-1B with Reverse-Sense LPTACC Actuator (Post-LEAP-1B Service Bulletin 73-0013R0) and Expanded EGT Capability (Post LEAP-1B Service Bulletin 72-0169R0)	0004	Push	Push	Pull	Pull	Push	Push	Pull

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Table 603/73-21-09-993-803-G00 Hardware Configuration Data Entry Plug Pin Configuration (Continued)

Engine Model	Engine Configuration*[1]	Engine Configuration Pin*[2]						
LEAP-1B with 21 Count HPC Stage 1 Blisk and Associated Booster IGV (Post LEAP-1B Service Bulletin 72-0182R0), Reverse-Sense LPTACC Actuator (Post LEAP-1B Service Bulletin 73-0013R0), and Expanded EGT Capability (Post LEAP-1B Service Bulletin 72-0169R0)	0010	Pull	Pull	Push	Pull	Push	Pull	Pull
-	Invalid	All other combinations						

*[1] Make sure that you program the correct engine configuration. If the incorrect selection is made for pre or post-Service Bulletin 73-0013, LPTACC position disagree faults will be set at the next engine start (maintenance messages 75-4263X or 75-4264X where X is specific to the EEC channel reporting the fault).

*[2] Pins 19 and 22 are unused spares and are to remain in the closed (pushed-in) position.

*[3] These engine configurations are no longer valid for EEC software version 7.1 and later (post-CFM LEAP-1B SB 73-0040, EEC OPS P/N 2628M86P13 or later). If software version 7.1 or later is installed, selection of these configurations will result in a no dispatch fault (ENGINE CONTROL light) and maintenance message 73-4120X where X is specific to the EEC channels that report the fault.

(c) Install the data entry plug on the PSS (TASK 73-21-09-400-801-G00).

(d) Use the teflon-jawed pliers, STD-664, to install the cover on the data entry plug.

1) Make sure that the green line is covered (View D, Figure 601).

H. Put the Airplane Back to Its Usual Condition

SUBTASK 73-21-09-410-004-G00

(1) On the applicable engine, close the right fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

SUBTASK 73-21-09-860-010-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

I. Data Entry Plug Configuration Post Installation Test

SUBTASK 73-21-09-860-020-G00

(1) Energize the EECs with the EEC MAINT POWER switch that is on the MISC SYSTEM CTRLS page on the Multi-Function Display (MFD) (TASK 73-21-00-800-801-G00).

NOTE: You must energize the EECs to make sure that the data entry plug information is read and kept correctly in the EEC NVM.

NOTE: The next engine start after one of the tasks that follow can have a longer pre-start motoring time than typical of cold engines. This is because the engine off time will be erased from the EEC NVM when you do these:

- Install new EEC software

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- Enter new Engine Serial Number (ESN)
- Install the engine on a new aircraft tail number
- Install the engine on the opposite engine position on the same aircraft tail number
- Install a data entry plug with new thrust rating, N1 trim, or engine configuration.

SUBTASK 73-21-09-200-001-G00

(2) Make sure that the engine thrust rating, N1 trim, and hardware configuration displayed on the EEC system configuration page matches the intended data entry plug configuration (TASK 73-21-00-470-801-G00).

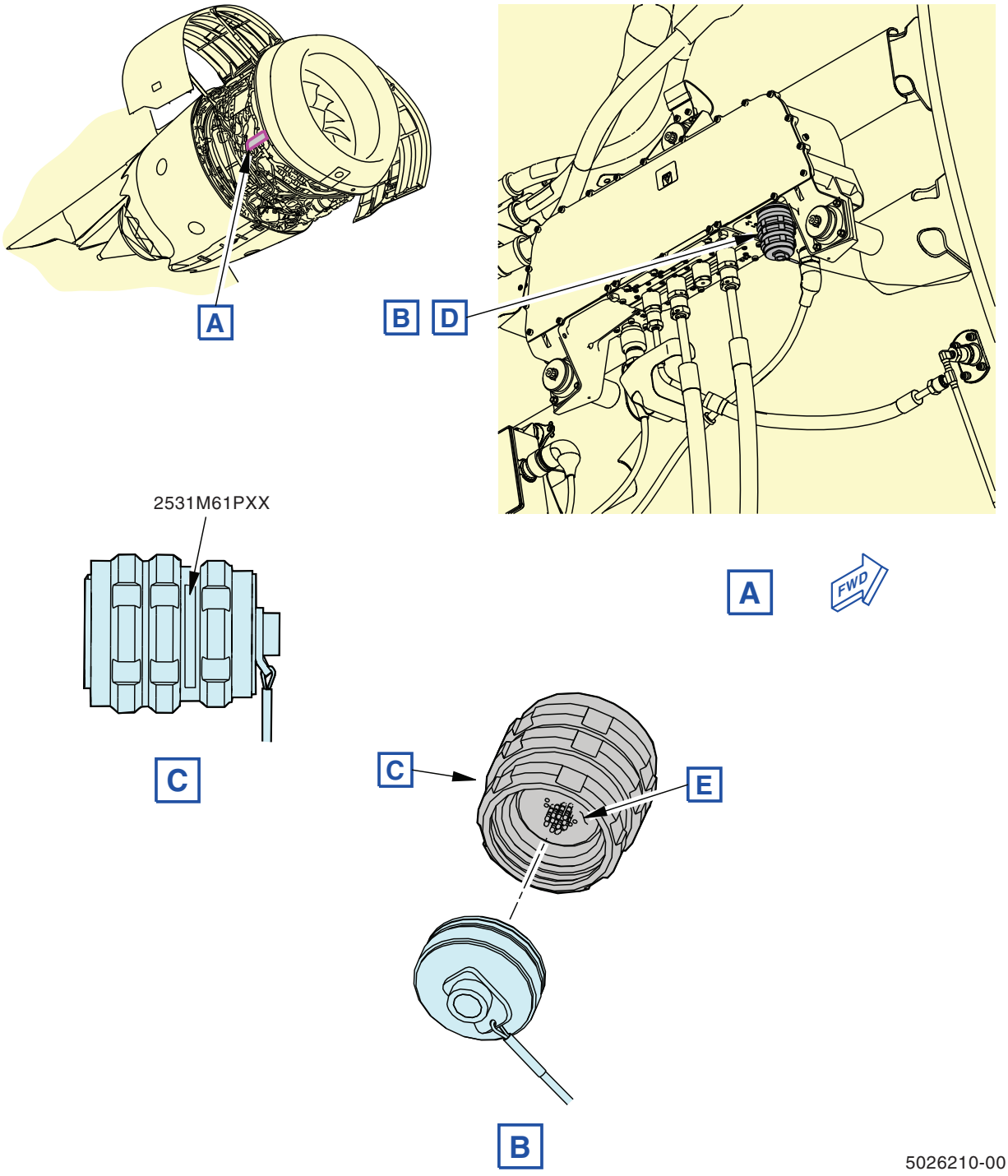
NOTE: The engine thrust rating as shown on the EEC System Configuration Page must match the engine data plate's rated thrust, the opposite engine's rated thrust, and be compatible with the airplane model.

SUBTASK 73-21-09-710-001-G00

(3) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

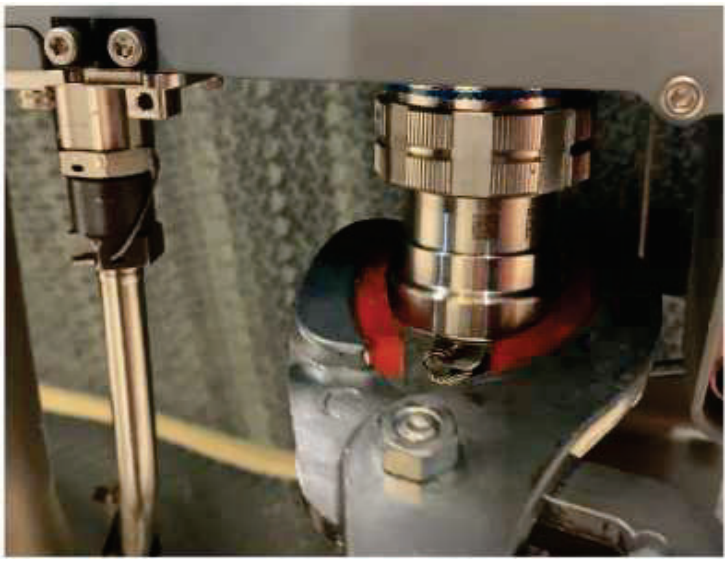
——— END OF TASK ———

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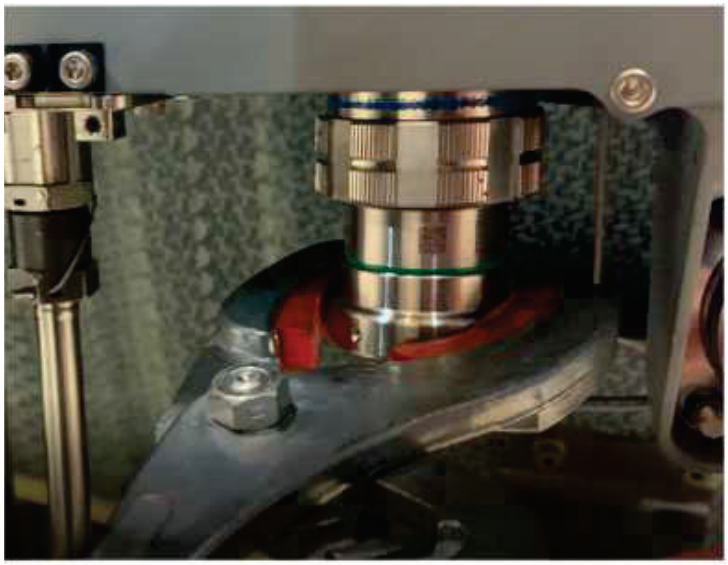
Data Entry Plug Configuration Check
Figure 601/73-21-09-990-803-G00 (Sheet 1 of 3)

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(CORRECT POSITION OF COVER)

D



(INCORRECT POSITION OF COVER)

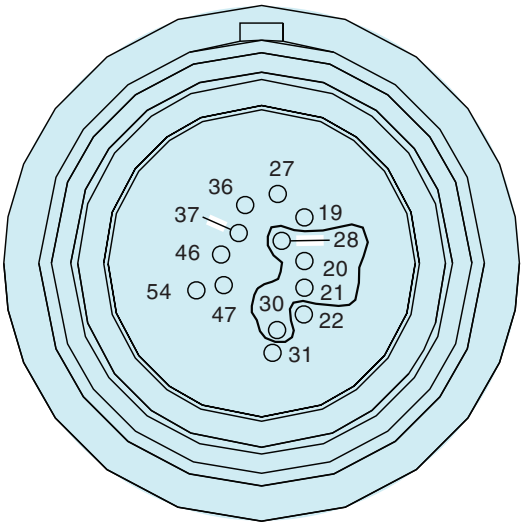
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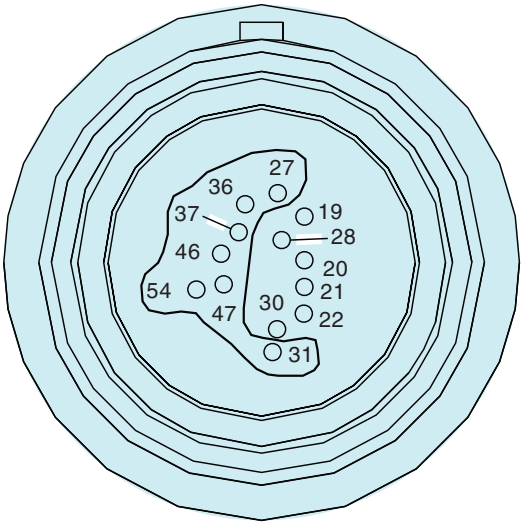
Data Entry Plug Configuration Check
Figure 601/73-21-09-990-803-G00 (Sheet 2 of 3)

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N1 TRIM PIN CONFIGURATION

E 1



HARDWARE CONFIGURATION

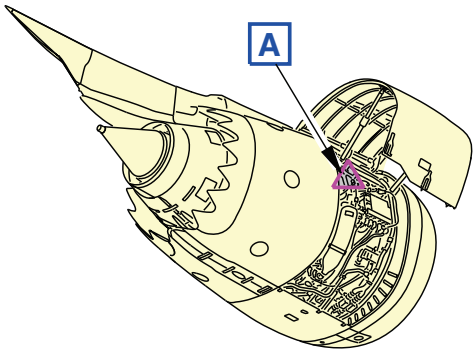
E 1

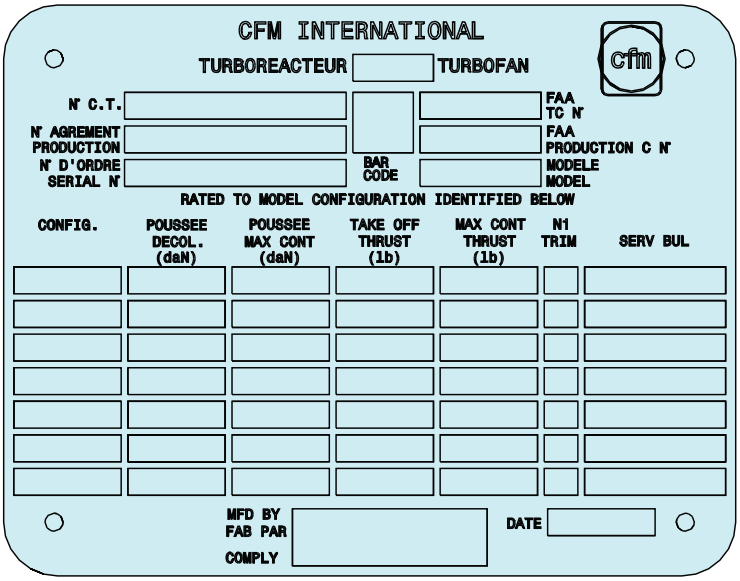
1 PINS 19 AND 22 ARE UNUSED SPARES AND ARE TO REMAIN IN THE CLOSED (PUSHED-IN) POSITION.

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2474533 S0000579631_V3

Data Entry Plug Configuration Check
Figure 601/73-21-09-990-803-G00 (Sheet 3 of 3)

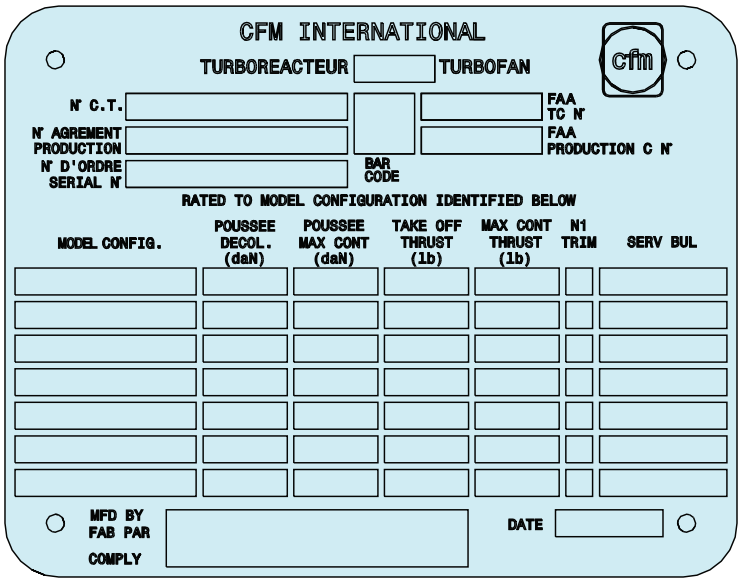
ECCN 9E991 BOEING PROPRIETARY - See title page for details





DATA (NAME) PLATE OLD CONFIGURATION
(PRE LEAP-1B S/B 72-0160)
P/N 649-804-037-0

A



DATA (NAME) PLATE NEW CONFIGURATION
(POST LEAP-1B S/B 72-0160)
P/N 649-804-037-0

A

6008678-00
2474538 S0000579632_V2

Data Plate Reference
Figure 602/73-21-09-990-804-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 73-21-09-211-801-G00

3. Data Entry Plug Inspection

(Figure 603)

A. General

(1) This task provides the instructions to inspect the condition of the Data Entry Plug (DEP).

(2) The Data Entry Plug is attached to the Pressure Sub-System (PSS) located on the right side of the fan case.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
73-21-09-000-801-G00	Data Entry Plug Removal (P/B 401)
73-21-09-400-801-G00	Data Entry Plug Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

F. Prepare for the Inspection

SUBTASK 73-21-09-860-016-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-21-09-860-017-G00

(2) Open these applicable circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A
B	3	C01312	ENGINE 1 RUN/PWR

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F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	5	C01313	ENGINE 2 RUN/PWR
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	3	C04018	ENG 1 STBY ALTN PWR RELAY CHAN A
E	4	C04019	ENG 1 STBY ALTN PWR CHAN A

SUBTASK 73-21-09-010-003-G00

(3) On the applicable engine, open the right fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

G. Data Entry Plug Inspection

SUBTASK 73-21-09-210-001-G00

(1) Inspect the data entry plug as follows:

(a) Look for signs of arcing.

1) If you see signs of arcing, replace the data entry plug (TASK 73-21-09-000-801-G00 and TASK 73-21-09-400-801-G00).

(b) Make sure that the electrical connector is not loose.

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

1) If an electrical connector is loose, use a teflon-jawed pliers, STD-664 to tighten it.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 73-21-09-410-003-G00

(1) On the applicable engine, close the right fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

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SUBTASK 73-21-09-860-018-G00

(2) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A
B	3	C01312	ENGINE 1 RUN/PWR

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	5	C01313	ENGINE 2 RUN/PWR
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	3	C04018	ENG 1 STBY ALTN PWR RELAY CHAN A
E	4	C04019	ENG 1 STBY ALTN PWR CHAN A

SUBTASK 73-21-09-860-019-G00

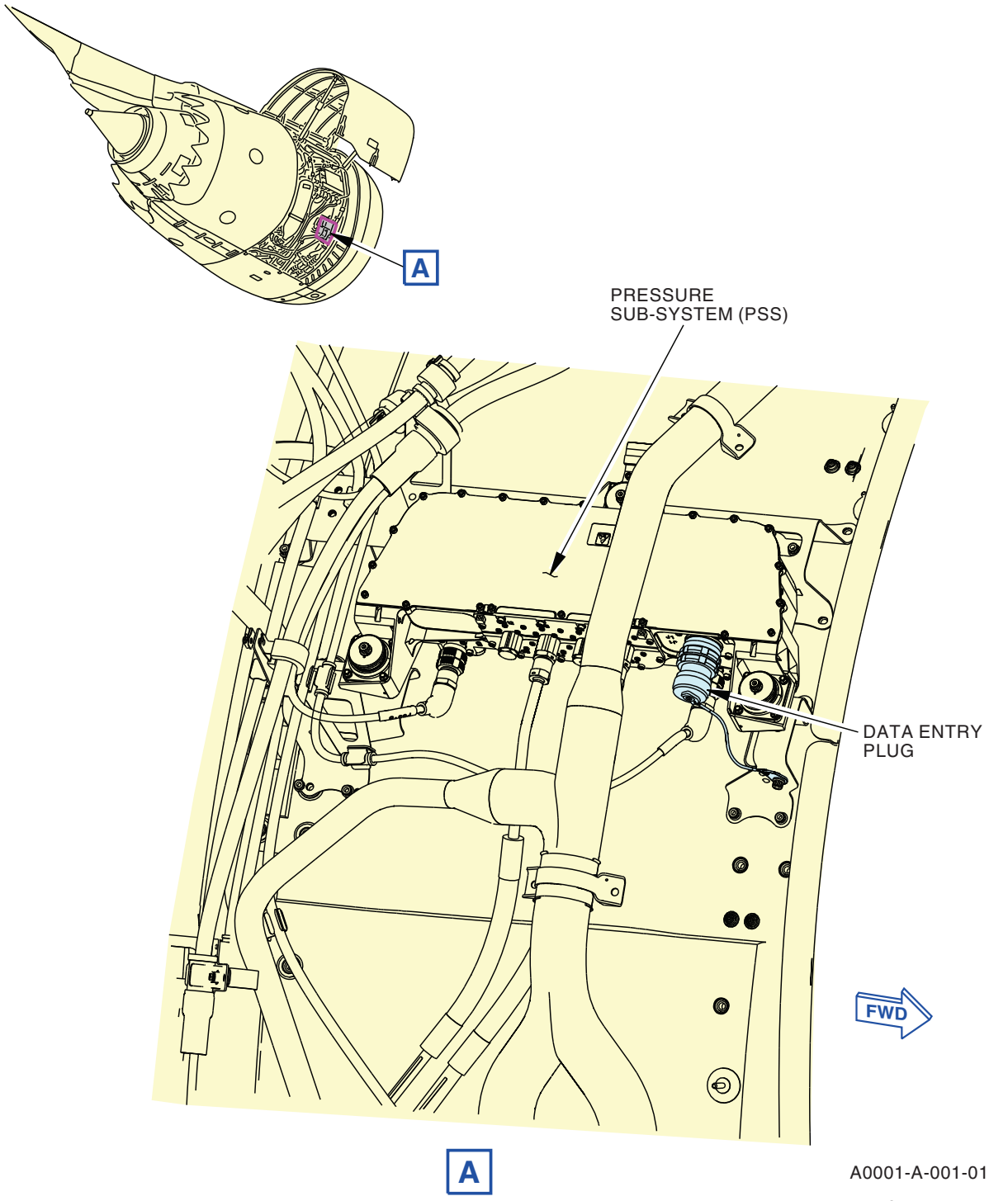
(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

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Data Entry Plug Inspection
Figure 603/73-21-09-990-807-G00

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PRESSURE SUB-SYSTEM - MAINTENANCE PRACTICES

1. General

A. This procedure has this task:

(1) Pressure Sub-System (PSS) and Sense Line Dry Out.

TASK 73-21-11-160-801-G00

2. Pressure Sub-System and Sense Line Dry Out

(Figure 201, Figure 202, and Figure 203)

A. General

(1) This task gives the instructions to remove accumulated moisture from the Pressure Sub-System (PSS) sensing system.

(2) Over time and under certain operating conditions, moisture can accumulate inside the PSS sensor air manifolds and air tubes [compressor discharge pressure (PS3) and compressor bleed pressure (P3B) sense lines]. This accumulated moisture can lead to a PS3 and/or P3B sense line disagree fault and possible loss of thrust control, if that moisture were to freeze and disrupt the pressure measurement.

B. References

Reference	Title
71-00-00-700-808-G00	Test No. 32 - Engine Idle System Test (P/B 501)
71-00-00-790-804-G00	Test No. 4 - Idle Leak Test (P/B 501)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-15157	Heat Gun - Thermogun
SPL-14895	Dryout Kit - EEC Part #: RC331931G01 Supplier: 62340 Part #: RC331931G02 Supplier: 62340 Part #: RC331931G03 Supplier: 62340 Part #: RC331931G04 Supplier: 62340
STD-858	Tag - DO NOT OPERATE
STD-1067	Dropper - Medicine
STD-3940	Air Source - Regulated, Dry Filtered, 0-150 psig

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	
G02345 [CP8001]	Wire - Safety, 0.032 Inch (0.8 mm) Diameter	AMS 5687

E. Location Zones

Zone	Area
411	Engine 1 - Engine

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(Continued)

Zone	Area
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

G. Prepare for the Dry Out

SUBTASK 73-21-11-860-009-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-21-11-860-010-G00

(2) Open these applicable circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A
B	3	C01312	ENGINE 1 RUN/PWR

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	5	C01313	ENGINE 2 RUN/PWR
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	3	C04018	ENG 1 STBY ALTN PWR RELAY CHAN A
E	4	C04019	ENG 1 STBY ALTN PWR CHAN A

SUBTASK 73-21-11-860-011-G00

(3) On the applicable engine, open the right fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

H. Pressure Sub-System and Sense Line Dry Out

SUBTASK 73-21-11-160-003-G00

(1) If the ambient temperature is less than 65°F (18.3°C), do this step:

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